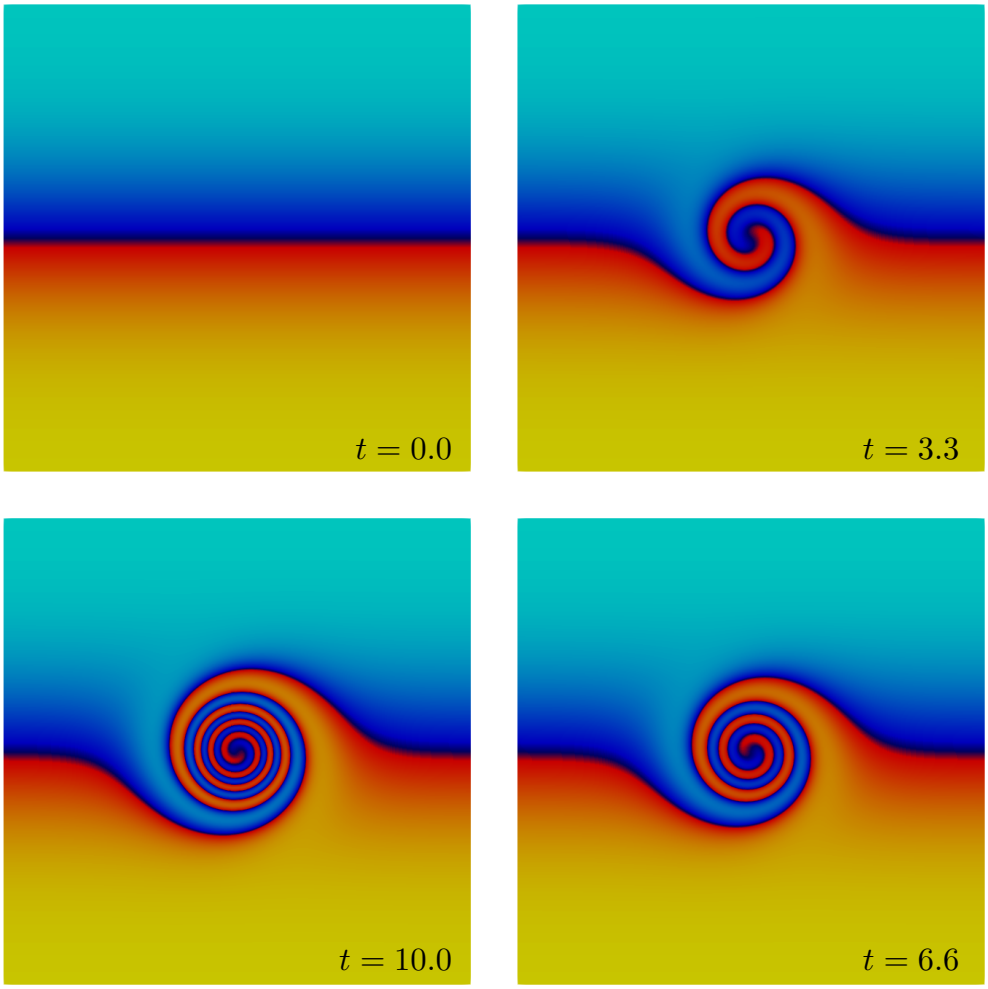
Multi-Moment Method

for the Convection-Diffusion Equation

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Preface

This book presents a systematic introduction to the theoretical foundations and practical formulation of the *multi-moment method*, a relatively new class of numerical discretization methods that builds upon conventional finite-difference (FDM) and finite-volume (FVM) approaches. By employing different types of discrete information to represent and evolve a physical field, the multi-moment framework provides an alternative perspective on the construction of numerical approximations for partial differential equations. The formulation developed in this book reflects more than two decades of the author's research in numerical methods for fluid-flow problems. The presentation is primarily aimed at graduate students and researchers with a background in numerical analysis and some familiarity with numerical PDE methods, particularly finite-difference, finite-volume, and finite-element methods, while remaining accessible to advanced undergraduate students through the inclusion of essential introductory material.

The term *multi-moment* was introduced by Prof. Feng Xiao of the Tokyo Institute of Technology [35] to describe and generalize numerical concepts originating from the Cubic Interpolated Pseudoparticle (CIP) method, originally proposed by Takewaki, Nishiguchi, and Yabe [32]. In its original formulation, the CIP method incorporated the first spatial derivative of a field variable as an additional dependent variable and evolved it within a semi-Lagrangian framework. The method subsequently became widely known as the Constrained Interpolation Profile method, particularly in formulations employing local discrete data. As the methodology developed, additional types of discrete information, including cell-, face-, and line-averaged quantities, were introduced into the reconstruction process. Prof. Xiao recognized that these developments shared a common underlying principle: a local representation of the solution could be constructed using different types of discrete information, collectively referred to as "moments." This observation provided the foundation for the broader concept of the multi-moment method.

The author independently developed a closely related concept in 2016 during the development of the finite-surface method [11]. During this research, Prof. Feng Xiao drew the author's attention to several related numerical methods, particularly the CIP-CSL2 scheme of Nakamura *et al.* [29] and the IDO-CF scheme of Imai *et al.* [15]. The former employs a semi-Lagrangian formulation, whereas the latter is based on an Eulerian formulation. Among these approaches, the multi-moment formulation presented in this book is most closely related to the Interpolated Differential Operator (IDO) scheme, as both methods evolve multiple types of discrete data within an Eulerian framework.

A fundamental principle of the IDO approach is that field variables are represented by a common underlying function reconstructed from different types of supporting data, such as pointwise values, pointwise derivatives, and integral or averaged quantities. The approach adopted in this book differs somewhat in its conceptual interpretation. Each type of discrete data is treated as an independent discretization of the governing equation, and an approximation appropriate to that data type is developed for its corresponding discrete evolution equation. These distinct discrete representations are subsequently coupled through a common local reconstruction. In this sense, the term *multi-moment method* is adopted to describe the framework developed and presented in this book.

Although the conceptual foundations of the present formulation and those of the IDO-CF scheme are not entirely identical, many of their resulting discrete equations are closely related. The principal differences lie in the approximation procedures, the construction of local reconstruction functions, and the treatment of nonlinear fluxes. Accordingly, this book

is intended not only to introduce the fundamental principles of the multi-moment method but also to present approximation techniques that may be applicable to the broader class of IDO-type and related high-order discretization methods.

To support the learning process, the book is accompanied by software designed to help readers connect mathematical formulations and numerical algorithms with their practical implementation. The computational examples and pseudocode presented throughout the text are intended to guide readers from an understanding of the underlying theoretical concepts to the development of working numerical programs and, ultimately, their own multi-moment solvers.

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Chapter 1
Introduction

"Divide each difficulty into as many parts as is feasible and necessary to resolve it."
René Descartes

Partial Differential Equations (PDEs) constitute the mathematical foundation of modern science and engineering. Alongside Ordinary Differential Equations (ODEs), they provide the definitive framework for analyzing dynamical systems and physical processes. ODEs govern systems characterized by a single independent variable—typically time—and are ideally suited for lumped-parameter models where spatial variations are negligible. Familiar examples include classical Newtonian mechanics, electrical circuits governed by Kirchhoff's laws, and the fundamental rate equations of chemical kinetics.

However, physical reality is rarely perfectly uniform. When the simplifying assumption of spatial homogeneity is removed, models must transition from discrete ODEs to continuous PDEs. A discrete mass-spring oscillator, for example, naturally extends into the elastodynamic PDEs that describe vibrating beams and complex structural deformations. Likewise, a lumped-capacitance thermal model expands into the multidimensional heat equation the moment internal temperature gradients are resolved. PDEs are therefore the natural generalization of ODEs, bridging the gap between discrete approximations and true continuum mechanics.

Ultimately, PDEs serve as the universal language for field variables that evolve across multiple dimensions. Whether describing the turbulent flow of fluids, the transmission of electromagnetic waves, or the probability amplitudes in quantum mechanics, PDEs provide the rigorous mathematical architecture required to capture the full spatial and temporal complexity of the physical world.

Many governing laws of physics naturally lead to PDEs. Fundamental examples include:

- The Navier–Stokes equations for fluid dynamics,
- The heat equation for thermal diffusion,
- Maxwell's equations for electromagnetics,
- The wave equation for vibration and acoustics,
- The Schrödinger equation in quantum mechanics.

Although certain ODEs and PDEs possess analytical solutions, these exact solutions are generally limited to highly idealized problems with simplified assumptions, geometries, and boundary conditions. While analytical techniques remain fundamental for understanding the qualitative behavior of differential equations, real-world engineering systems are often characterized by nonlinearities, multidimensional interactions, irregular geometries, moving interfaces, and multiscale physical processes that render exact solutions unattainable. Consequently, numerical methods have become indispensable tools for solving differential equations in practical scientific and engineering applications.

Partial Differential Equations (PDEs) form the mathematical foundation of modern science and engineering. They are the universal language used to describe physical phenomena involving multiple independent variables, most commonly space and time.

1.1 Numerical Solution to ODEs and PDEs

The objective of numerical analysis for differential equations is to construct accurate approximations to solutions that satisfy prescribed relationships among dependent variables and their derivatives. In the context of ODEs, numerical methods seek approximations to functions evolving primarily in time. At the same time, for PDEs, the solution must simultaneously satisfy spatial and temporal relationships throughout a multidimensional computational domain. For many simple ODEs, analytical solutions can be derived explicitly using classical techniques such as separation of variables, integrating factors, or Laplace transforms. Similarly, certain PDEs admit closed-form solutions under highly restrictive assumptions involving simple geometries, linear governing equations, and ideal boundary conditions. These analytical solutions remain extremely valuable because they provide physical insight, establish theoretical understanding, and serve as benchmark problems for validating numerical algorithms.

However, practical scientific and engineering systems rarely conform to such idealized conditions. Realistic models often involve strong nonlinearities, discontinuities, turbulent transport, moving boundaries, coupled multiphysics interactions, and highly irregular computational domains. In many cases, the governing equations may not even possess analytical solutions in a closed mathematical form. Even when exact solutions theoretically exist, they are frequently too complicated to derive or evaluate efficiently for practical use. As a consequence, numerical computation has become an essential pillar of modern computational science and engineering. Rather than solving the governing differential equations directly in their continuous form, numerical methods approximate the unknown solution over a discrete set of spatial and temporal locations. Derivatives are replaced by algebraic approximations, transforming the original continuous problem into a finite system of algebraic equations that can be solved computationally.

The increasing availability of high-performance computing (HPC) resources has further accelerated the importance of numerical methods. Modern numerical simulations now enable researchers and engineers to investigate physical phenomena that would otherwise be prohibitively expensive, dangerous, or physically impossible to study experimentally. Applications range from aerodynamic optimization and weather prediction to seismic analysis, plasma physics, and biomedical simulations.

Under such conditions, numerical approximation—discretizing the continuous governing equations into a finite algebraic system that a computer can solve —becomes the only feasible approach. Common foundational numerical methods include: i.) Finite Difference Method, ii.) Finite Volume Method, iii.) Finite Element Method, and iv.) Spectral Method

Among these methods, finite-difference and finite-volume methods are widely used and form the foundation of the multi-moment method. Thus, in the following sections we briefly describe the finite difference and finite volume methods.

Method	Main Idea	Typical Applications
Finite Difference Method (FDM)	Approximates spatial and temporal derivatives using nodal grid differences.	Heat transfer, simple wave equations, structured-grid solvers.
Finite Volume Method (FVM)	Integrates equations over control volumes to strictly enforce exact conservation laws.	Computational Fluid Dynamics (CFD), aerodynamic flows.
Finite Element Method (FEM)	Divides the domain into elements and minimizes an error function using global basis functions.	Structural mechanics, solid deformation, thermal analysis.
Spectral Methods	Uses global continuous basis functions (e.g., Fourier series) for approximation.	Highly accurate simulations, fundamental turbulence research.

Table 1.1: Common numerical methods for PDEs and their characteristics.

1.1.1 Finite Difference Method

Consider the convection-diffusion equation:

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = K\frac{\partial^2 \phi}{\partial x^2} \quad \text{in one-dimensional domain } \Omega \tag{1.1}$$

with boundary conditions $\phi(0) = \phi_0$ and $\phi(L) = \phi_L$. This one-dimensional Partial Differential Equation (PDE) describes the change of the scalar ϕ under convection by a constant velocity u and diffusion with mass-diffusivity coefficient K. From an engineering perspective, the above equation is referred to as the *governing equation* of the said processes. It is valid everywhere inside the domain, requiring the continuous solution ϕ to be at least twice differentiable. To simplify the concept, let us consider the steady-state form:

$$u\frac{\partial \phi}{\partial x} = K\frac{\partial^2 \phi}{\partial x^2} \quad \text{in one-dimensional domain } \Omega \tag{1.2}$$

This reduces the PDE to an Ordinary Differential Equation (ODE). To solve this equation numerically using the Finite Difference Method (FDM), we must discretize the one-dimensional line by placing grid points in the x-direction at $x_i = ih$, $i = 0, 1, \ldots, N_x - 1$ (see Fig.1.1). These points are known as *nodal points*. The FDM applies the governing equation at these discrete nodes and utilizes numerical approximations to replace the continuous derivatives. For example, applying central difference approximations for the first and second derivatives yields:

$$\left.\frac{\partial \phi}{\partial x}\right|_i = \frac{\phi_{i+1} - \phi_{i-1}}{2h}, \quad \text{and} \tag{1.3}$$

$$\left.\frac{\partial^2 \phi}{\partial x^2}\right|_i = \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2}. \tag{1.4}$$

Substituting these into Eq.(1.2) forms the following discrete approximation:

$$u\frac{\phi_{i+1} - \phi_{i-1}}{2h} - K\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2} = 0. \tag{1.5}$$

By grouping the terms associated with each nodal value, this can be further simplified to:

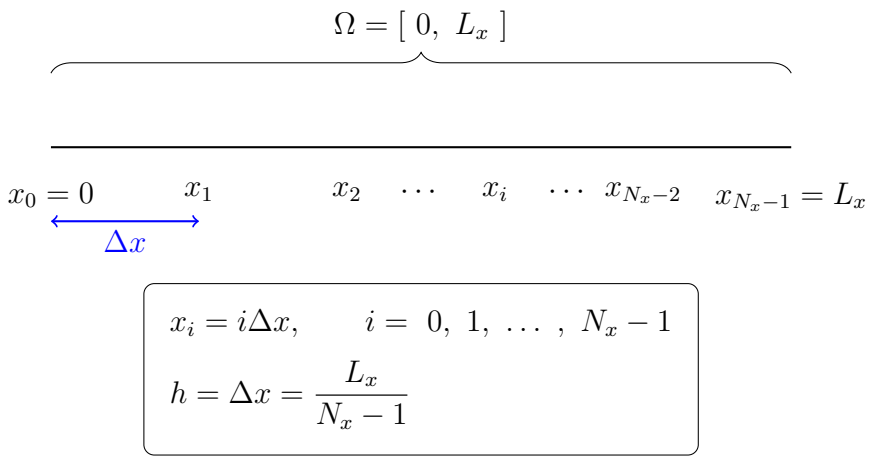


Fig. 1.1: Discretization of the continuous domain into a set of discrete points in the finite-difference method.

$$\left(-\frac{u}{2h} - \frac{K}{h^2}\right)\phi_{i-1} + \frac{2K}{h^2}\phi_i + \left(\frac{u}{2h} - \frac{K}{h^2}\right)\phi_{i+1} = 0, \tag{1.6}$$

which applies to all inner nodes ϕ_i within the domain.

As a minimal example, let us divide the domain into 4 parts, which corresponds to using 5 nodes ($N_x = 5$). With this discretization, we have ϕ_0 on the left boundary and ϕ_4 on the right boundary. These are known boundary conditions that do not need to be solved. This leaves us with three unknowns in the middle (ϕ_1, ϕ_2, ϕ_3). Applying the discretized equation to these three inner points leads to the following *linear system of equations (LSE)*:

$$\begin{aligned}
\left(\tfrac{2K}{h^2}\right)\phi_1 + \left(\tfrac{u}{2h} - \tfrac{K}{h^2}\right)\phi_2 &= -\left(-\tfrac{u}{2h} - \tfrac{K}{h^2}\right)\phi_0 \\
\left(-\tfrac{u}{2h} - \tfrac{K}{h^2}\right)\phi_1 + \left(\tfrac{2K}{h^2}\right)\phi_2 + \left(\tfrac{u}{2h} - \tfrac{K}{h^2}\right)\phi_3 &= 0 \\
\left(-\tfrac{u}{2h} - \tfrac{K}{h^2}\right)\phi_2 + \left(\tfrac{2K}{h^2}\right)\phi_3 &= -\left(\tfrac{u}{2h} - \tfrac{K}{h^2}\right)\phi_L
\end{aligned}$$

This system can be rearranged into a matrix-vector equation:

$$\left[\begin{array}{c|c|c}
\frac{2K}{h^2} & \frac{u}{2h} - \frac{K}{h^2} & 0 \\ \hline
-\frac{u}{2h} - \frac{K}{h^2} & \frac{2K}{h^2} & \frac{u}{2h} - \frac{K}{h^2} \\ \hline
0 & -\frac{u}{2h} - \frac{K}{h^2} & \frac{2K}{h^2}
\end{array}\right]
\left[\begin{array}{c} \phi_1 \\ \phi_2 \\ \phi_3 \end{array}\right] =
\left[\begin{array}{c} -\left(-\frac{u}{2h} - \frac{K}{h^2}\right)\phi_0 \\ 0 \\ -\left(\frac{u}{2h} - \frac{K}{h^2}\right)\phi_L \end{array}\right] \tag{1.7}$$

Or in short form,

$$\mathbf{A}\boldsymbol{\phi} = \mathbf{b}. \tag{1.8}$$

Solving this linear system gives the inner-node solution. Provided the discrete formulation is consistent, the numerical solution will converge toward the exact solution as the number of grid points increases. The finite-difference solution using five nodes, with $u = 1.0$ and $K = 0.25$, is shown in Fig.1.2 and agrees reasonably well with the exact solution. In three-dimensional problems, it is common to encounter linear systems with hundreds of millions of unknowns. Furthermore, in frontier research or large-scale industrial simulations, the degrees

of freedom can easily scale into the billions. Consequently, developing algorithms to solve these massive linear systems efficiently and accurately is paramount, constituting a highly active field of research in its own right.

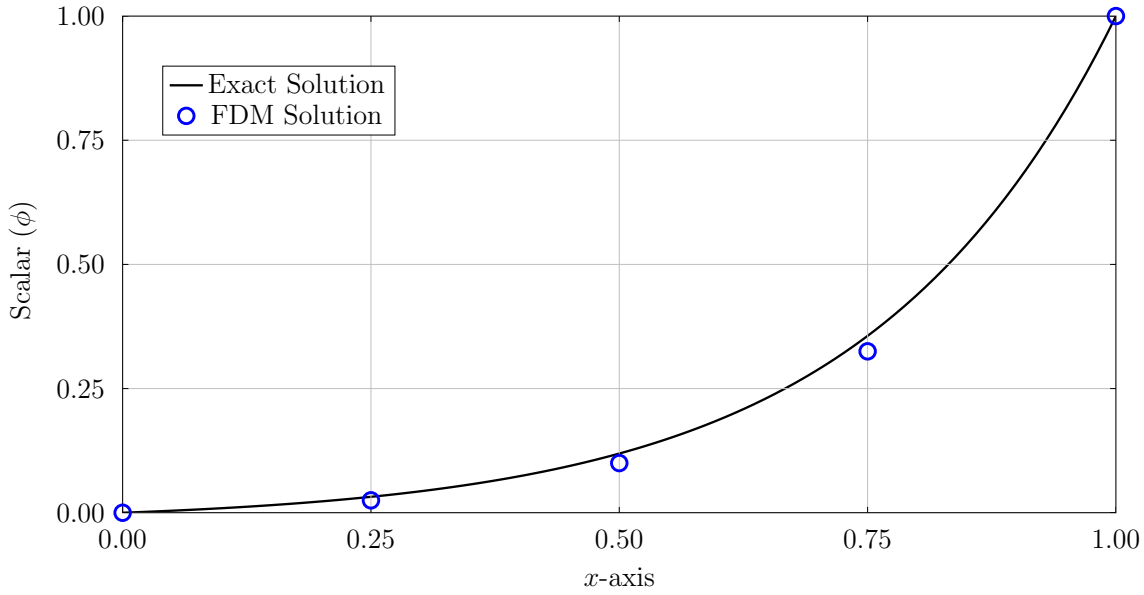


Fig. 1.2: Finite-difference solution with 5 nodes, $u = 1.0$, $K = 0.25$.

1.1.2 Finite Volume Method

Let us recall the time-dependent convection-diffusion equation:

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = K\frac{\partial^2 \phi}{\partial x^2}$$

From a mathematical perspective, the finite-volume method integrates this equation over a discrete control volume (or cell). In fact, most partial differential equations describing physical processes are originally derived from such integral conservation laws before being reduced to differential forms.

At this stage, it is sufficient to consider the integration of the above equation over a 1D domain. Let us integrate from x_i to x_{i+1}:

$$\int_{x_i}^{x_{i+1}} \left(\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x}\right) dx = \int_{x_i}^{x_{i+1}} K\frac{\partial^2 \phi}{\partial x^2} dx \tag{1.9}$$

By linearity of integration and interchange of integration with differentiation in time, we obtain:

$$\frac{\partial}{\partial t}\left(\int_{x_i}^{x_{i+1}} \phi\, dx\right) + u\int_{x_i}^{x_{i+1}} \frac{\partial \phi}{\partial x} dx = K\int_{x_i}^{x_{i+1}} \frac{\partial^2 \phi}{\partial x^2} dx \tag{1.10}$$

$$\frac{\partial}{\partial t}\left(\int_{x_i}^{x_{i+1}} \phi\, dx\right) + u\,(\phi_{i+1} - \phi_i) = K\left(\left.\frac{\partial \phi}{\partial x}\right|_{x_{i+1}} - \left.\frac{\partial \phi}{\partial x}\right|_{x_i}\right) \tag{1.11}$$

Next, we invoke the mean value theorem for integrals:

$$\overline{\phi}_{[x_i,x_{i+1}]} = \frac{1}{h}\int_{x_i}^{x_{i+1}} \phi \, dx \qquad \text{with } h = \Delta x = x_{i+1} - x_i. \tag{1.12}$$

By replacing the integral inside the time derivative with $h\overline{\phi}_{[x_i,x_{i+1}]}$, we obtain:

$$h\frac{\partial}{\partial t}\left(\overline{\phi}_{[x_i,x_{i+1}]}\right) + u\left(\phi_{i+1} - \phi_i\right) = K\left(\left.\frac{\partial \phi}{\partial x}\right|_{x_{i+1}} - \left.\frac{\partial \phi}{\partial x}\right|_{x_i}\right). \tag{1.13}$$

This equation is the crux of the finite-volume method. It describes the temporal change of the *cell-averaged* variable due to the imbalance of convective and diffusive fluxes at the *cell boundaries*. Because fluxes exiting one cell strictly enter the adjacent cell, the total quantity is conserved automatically, which is a fundamental strength of the FVM.

In the early development of higher-order finite-volume methods, many researchers mistakenly treated $\overline{\phi}_{[x_i,x_{i+1}]}$ as the point value $\phi_{i+1/2}$ due to a reliance on the midpoint rule. Such treatment is strictly second-order accurate and led to significant confusion and failures in developing higher-order schemes before the 21st century. The seminal work of Kobayashi [21] cleared up this confusion, laying the foundation for modern higher-order FVM. To avoid this ambiguity, let us use the bar notation to strictly denote an average quantity, defining $\overline{\phi}_{i+1/2}$ as:

$$\overline{\phi}_{i+\frac{1}{2}} = \frac{1}{x_{i+1} - x_i}\int_{x_i}^{x_{i+1}} \phi(x)\, dx. \tag{1.14}$$

The half-integer index indicates that the variable is averaged over the spatial interval bounded by its adjacent integer indices. Crucially, this cell average does not equal the exact pointwise function value at that location; specifically, $\overline{\phi}_{i+1/2} \neq \phi_{i+1/2}$. This common misconception has led some researchers to mistakenly conclude that the Finite Difference Method (FDM) and the Finite Volume Method (FVM) are identical, despite their fundamentally different derivations.

Let us consider, as an example, the discretization of the one-dimensional domain $x \in [0, 2\pi]$, as shown in Fig. 1.3. The figure illustrates the physical arrangement of the point values and cell values resulting from a discretization in which the continuous domain is divided into five equal segments, or cells. From the finite-difference method (FDM) point of view, the domain is discretized using six grid points. Example 1.1 discusses the discrete values associated with the two discretizations and the need to distinguish a cell value from cell-centered pointwise data.

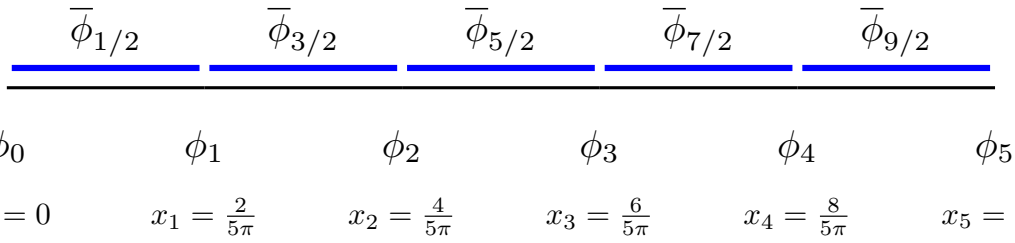


Fig. 1.3: Arrangement of point value (PV) and cell value (CV) of a real line $x \in [0, 2\pi]$ discretized by 5 equal segments.

Example 1.1 (The Difference Between Point Values and Cell Values)
To clarify the meaning of the discrete variables used in FDM and FVM, consider the simple function

$$\phi(x) = \sin(x),$$

defined over

$$\Omega = [0, 2\pi].$$

Divide the domain into five uniform cells. The grid spacing is

$$h = \frac{2\pi}{5},$$

and the six grid points are

$$x_i = ih, \qquad i = 0, 1, \ldots, 5.$$

From the FDM point of view, the discrete data are obtained simply by evaluating the function at these grid points:

$$\phi_i = \phi(x_i) = \sin(x_i).$$

Thus,

i	0	1	2	3	4	5
x_i	0	$\frac{2\pi}{5}$	$\frac{4\pi}{5}$	$\frac{6\pi}{5}$	$\frac{8\pi}{5}$	2π
ϕ_i	0.0000	0.9511	0.5878	−0.5878	−0.9511	0.0000

Each ϕ_i is therefore a *point value*: it represents the value of the continuous function at one particular spatial location.
The interpretation is different in FVM. For the cell

$$I_{i+1/2} = [x_i, x_{i+1}],$$

the discrete variable is usually the cell average

$$\overline{\phi}_{i+1/2} = \frac{1}{h} \int_{x_i}^{x_{i+1}} \phi(x)\, dx.$$

For the present function,

$$\overline{\phi}_{i+1/2} = \frac{1}{h} \int_{x_i}^{x_{i+1}} \sin(x)\, dx = \frac{\cos(x_i) - \cos(x_{i+1})}{h}.$$

Now consider the geometric center of the same cell,

$$x_{i+1/2} = \frac{x_i + x_{i+1}}{2}.$$

Evaluating the function at this point gives

$$\phi_{i+1/2} = \phi(x_{i+1/2}) = \sin(x_{i+1/2}).$$

This is a *pointwise* quantity and is not, in general, the same as the cell average. For the five cells considered here,

Cell	$I_{1/2}$	$I_{3/2}$	$I_{5/2}$	$I_{7/2}$	$I_{9/2}$
$\overline{\phi}_{i+1/2}$	0.5499	0.8897	0.0000	−0.8897	−0.5499
$\phi_{i+1/2}$	0.5878	0.9511	0.0000	−0.9511	−0.5878

For example, over the first cell

$$I_{1/2} = \left[0, \frac{2\pi}{5}\right],$$

the FVM cell average is

$$\overline{\phi}_{1/2} = 0.5499,$$

whereas the pointwise value at the center of the cell is

$$\phi_{1/2} = \phi\left(\frac{\pi}{5}\right) = 0.5878.$$

Although both quantities are associated with the first cell, they contain different information. This example highlights an important distinction in numerical discretization:

$$\underbrace{\phi_i}_{\text{nodal point value}}, \qquad \underbrace{\phi_{i+1/2}}_{\text{cell-center point value}}, \qquad \underbrace{\overline{\phi}_{i+1/2}}_{\text{cell-average value}}$$

are three different discrete quantities. In particular,

$$\overline{\phi}_{i+1/2} \neq \phi_{i+1/2}$$

in general. If a piecewise-constant FVM reconstruction is used, the cell-average value may be assigned throughout the corresponding cell for reconstructing an approximate solution. This does not mean that the underlying continuous function itself is constant within that cell.

The ability to distinguish and, when appropriate, simultaneously evolve different types of discrete information is one of the central ideas underlying multi-moment discretization methods which we will introduce in the next section.

We have now distinguished the data type of FVM from that of FDM. The next question is how we can obtain the fluxes at the cell interfaces. This problem may tempt the reader to use an interpolation and obtain ϕ_i from the average of the two nearest cell values, i.e.

$$\phi_i = \frac{\overline{\phi}_{i+1/2} + \overline{\phi}_{i-1/2}}{2}. \tag{1.15}$$

and likewise, compute the derivative by

$$\left.\frac{\partial \phi}{\partial x}\right|_i = \frac{\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}}{h}. \tag{1.16}$$

This approach is acceptable if one is satisfied with the second-order accuracy and if we use these approximations, we will eventually arrive at the same discrete system in Eq.(1.7). Thus, let us avoid the pitfall and use the following fourth-order approximations instead:

$$\phi_i = -\frac{1}{12}\overline{\phi}_{i-\frac{3}{2}} + \frac{7}{12}\overline{\phi}_{i-\frac{1}{2}} + \frac{7}{12}\overline{\phi}_{i+\frac{1}{2}} - \frac{1}{12}\overline{\phi}_{i+\frac{3}{2}} \tag{1.17}$$

$$\left.\frac{\partial \phi}{\partial x}\right|_i = \frac{1}{12h}\overline{\phi}_{i-\frac{3}{2}} - \frac{5}{4h}\overline{\phi}_{i-\frac{1}{2}} + \frac{5}{4h}\overline{\phi}_{i+\frac{1}{2}} - \frac{1}{12h}\overline{\phi}_{i+\frac{3}{2}} \tag{1.18}$$

The first approximation employs a *fourth-order deconvolution* to reconstruct pointwise data from cell-averaged values. If standard *Lagrange interpolation* were used to obtain ϕ_i from pointwise data at the half-integer indices, e.g. $\phi_{i\pm1/2}$ and $\phi_{i\pm3/2}$, the coefficients would instead be $[-1/12,\ 7/12,\ 7/12, -1/12]$. Therefore, it is mathematically inaccurate to classify this flux approximation as an "interpolation". Similarly, the preceding differentiation formula approximates the pointwise derivative directly from the cell-averaged field.

An accuracy comparison between the second- and fourth-order approximations is detailed in Example 1.2. The formal order of accuracy, denoted as p, is given by the asymptotic slope of the error norm with respect to the grid spacing h on a logarithmic scale as $h \to 0$. In practice, it can be computed directly using the error norms (E_1 and E_2) from two distinct grid spacings (h_1 and h_2):

$$p = \frac{\ln(E_1) - \ln(E_2)}{\ln(h_1) - \ln(h_2)} = \frac{\ln(E_1/E_2)}{\ln(h_1/h_2)}. \tag{1.19}$$

with either L_2 or L_∞ error norm.

Example 1.2 (Accuracy of Numerical Approximations) We can determine the accuracy of numerical approximations by applying them to a known function and checking how the errors change as the grid spacing decreases. Let us check the accuracy of the central difference for the first derivative and the second- and fourth-order deconvolutions. To this end, we compare the error of $sin(x)$ at $x = \pi/4$ in the table below

h	**Second-Order Flux**	**Fourth-Order Flux**
2π	$7.0711 \cdot 10^{-1}$	$7.0711 \cdot 10^{-1}$
$3\pi/2$	$2.6562 \cdot 10^{-2}$	$1.1922 \cdot 10^{-3}$
π	$7.0711 \cdot 10^{-1}$	$7.0711 \cdot 10^{-1}$
$\pi/2$	$2.5695 \cdot 10^{-1}$	$1.0690 \cdot 10^{-1}$
$\pi/4$	$7.0487 \cdot 10^{-2}$	$8.3331 \cdot 10^{-3}$
$\pi/16$	$4.5348 \cdot 10^{-3}$	$3.4873 \cdot 10^{-5}$
$\pi/64$	$2.8394 \cdot 10^{-4}$	$1.3681 \cdot 10^{-7}$

Table 1.2: Numerical error comparison of the second- and fourth-order finite volume flux reconstructions under grid refinement.

Beyond $h = \pi$, it is impossible to resolve the function $sin(x)$ whose period is 2π, this limitation is called Nyquist–Shannon Sampling Theorem, which dictates that a computational grid must have at least two points per wavelength ($h < \lambda/2$) to actually resolve a physical wave. Thus, coarse grids where $h \geq \pi$ violate this limit, making the numerical schemes completely *blind* to the wave and causing the error to stagnate regardless of how sophisticated the math is. Once the grid is sufficiently fine, the higher-order terms become significantly smaller than

the leading term, and the second- and fourth-order methods converge at their expected rates. A method is considered n-th order accurate if its numerical error is proportional to h^n.

Once the grid is sufficiently fine, the higher-order truncation-error terms become negligible relative to the leading term, and the numerical errors of the second- and fourth-order methods scale as $O(h^2)$ and $O(h^4)$, respectively. The observed convergence rates are 1.999 and 3.997, which closely match the theoretical values of 2 and 4, respectively.

Next, let us remove the time derivative and consider the steady-state form of Eq.(1.13):

$$u\left(\phi_{i+1}-\phi_i\right)=K\left(\left.\frac{\partial\phi}{\partial x}\right|_{x_{i+1}}-\left.\frac{\partial\phi}{\partial x}\right|_{x_i}\right). \tag{1.20}$$

then apply those two approximations (Eq.(1.17) and Eq.(1.18)) to the above equation to obtain the discrete equation for the conservation balance over the control volume of $\overline{\phi}_{i+\frac{1}{2}}$.

$$\begin{aligned} u\left(\frac{1}{12}\overline{\phi}_{i-\frac{3}{2}}-\frac{2}{3}\overline{\phi}_{i-\frac{1}{2}}+\frac{2}{3}\overline{\phi}_{i+\frac{3}{2}}-\frac{1}{12}\overline{\phi}_{i+\frac{5}{2}}\right) \\ =\frac{K}{h}\left(-\frac{1}{12}\overline{\phi}_{i-\frac{3}{2}}+\frac{4}{3}\overline{\phi}_{i-\frac{1}{2}}-\frac{5}{2}\overline{\phi}_{i+\frac{1}{2}}+\frac{4}{3}\overline{\phi}_{i+\frac{3}{2}}-\frac{1}{12}\overline{\phi}_{i+\frac{5}{2}}\right). \end{aligned} \tag{1.21}$$

This equation expresses the conservation balance over the control volume: the net convective flux equals the net diffusive flux through the control-volume boundaries.

Having obtained the discrete equation, we can formulate the linear system for the same discrete domain used earlier for FDM (5-node system). By applying the above equation for $\overline{\phi}_{\frac{1}{2}}$, $\overline{\phi}_{\frac{3}{2}}$, $\overline{\phi}_{\frac{5}{2}}$, and $\overline{\phi}_{\frac{7}{2}}$ and then moving the terms involving exterior unknowns outside the domain to the right-hand side, we obtain:

$$\left[\begin{array}{c|c|c|c} \frac{5K}{2h} & \frac{2u}{3}-\frac{4K}{3h} & -\frac{u}{12}+\frac{K}{12h} & 0 \\ \hline -\frac{2u}{3}-\frac{4K}{3h} & \frac{5K}{2h} & \frac{2u}{3}-\frac{4K}{3h} & -\frac{u}{12}+\frac{K}{12h} \\ \hline \frac{u}{12}+\frac{K}{12h} & -\frac{2u}{3}-\frac{4K}{3h} & \frac{5K}{2h} & \frac{2u}{3}-\frac{4K}{3h} \\ \hline 0 & \frac{u}{12}+\frac{K}{12h} & -\frac{2u}{3}-\frac{4K}{3h} & \frac{5K}{2h} \end{array}\right]\left[\begin{array}{c} \overline{\phi}_{\frac{1}{2}} \\ \overline{\phi}_{\frac{3}{2}} \\ \overline{\phi}_{\frac{5}{2}} \\ \overline{\phi}_{\frac{7}{2}} \end{array}\right]=\left[\begin{array}{c} -\left(\frac{u}{12}+\frac{K}{12h}\right)\bar{\phi}_{-\frac{3}{2}}-\left(-\frac{2u}{3}-\frac{4K}{3h}\right)\bar{\phi}_{-\frac{1}{2}} \\ -\left(\frac{u}{12}+\frac{K}{12h}\right)\bar{\phi}_{-\frac{1}{2}} \\ -\left(-\frac{u}{12}+\frac{K}{12h}\right)\bar{\phi}_{\frac{9}{2}} \\ -\left(\frac{2u}{3}-\frac{4K}{3h}\right)\bar{\phi}_{\frac{9}{2}}-\left(-\frac{u}{12}+\frac{K}{12h}\right)\bar{\phi}_{\frac{11}{2}} \end{array}\right] \tag{1.22}$$

The resulting matrix structure is similar to that obtained with FDM:

$$\mathbf{A}\overline{\boldsymbol{\phi}}=\mathbf{b}. \tag{1.23}$$

However, the interpretation is fundamentally different. In FDM, the unknowns are pointwise nodal values satisfying the governing equation locally. In contrast, FVM solves for line-averaged quantities obtained from integral conservation laws over finite control volumes. Consequently, conservation is enforced naturally over every discrete volume, which is one of the primary reasons why FVM is extensively used in computational fluid dynamics and large-scale engineering simulations. In addition, FVM and FDM generally produce different discrete systems for approximations of order higher than second. Moreover, the underlying

data representation in each method is fundamentally different: FDM is based on nodal values, whereas FVM operates on cell-averaged quantities. Consequently, it is essential to employ reconstruction and approximation procedures that are consistent with the nature of the underlying data.

In the above system, we have used the quantities $\overline{\phi}_{-\frac{3}{2}}$, $\overline{\phi}_{-\frac{1}{2}}$, $\overline{\phi}_{\frac{9}{2}}$, and $\overline{\phi}_{\frac{11}{2}}$ which do not physically exist within the computational domain. In fact, such information is generally not available. This renders the interior stencil not applicable near the boundaries. The procedure used to modify the approximation near the boundaries to obtain a solvable system is referred to as *boundary closure*. To avoid interrupting the main discussion with technical details, the treatment of boundary closure is deferred to Section 3.3.3, where it will be discussed separately and in greater detail.

1.2 Introduction to Multi-moment Method

Despite their massive industrial success, classical methods face significant bottlenecks when pushed to modern extremes. FVM and FDM often suffer from numerical dissipation and dispersion errors that smear sharp discontinuities. High-order extensions of FDM and FVM require large, complex stencil structures that hinder parallel computing efficiency. FEM, while excellent for complex geometries, can become computationally prohibitive for highly transient fluid problems. These limitations motivate the development of advanced numerical frameworks—such as Discontinuous Galerkin (DG) and Multi-Moment methods—which aim to achieve higher accuracy, extreme compactness, and HPC efficiency simultaneously. The former excels with body-fitted meshes and strong mathematical development. The latter offers a much more efficient solution process.

The multi-moment method extends the traditional discretization frameworks by employing multiple types of local information, or "moments," within each computational cell or grid element. Instead of using only a single dependent variable, such as point values or cell averages, multi-moment methods simultaneously utilize additional quantities including spatial derivatives, interface values, or higher-order moments. By incorporating richer local representations of the solution, these methods can achieve high-order accuracy while maintaining compact computational stencils and strong conservation properties. Consequently, multi-moment methods provide a powerful framework for solving partial differential equations involving complex wave propagation, sharp gradients, and nonlinear interactions. Their ability to combine high resolution, numerical robustness, and computational efficiency has made them increasingly important in modern computational fluid dynamics and scientific computing.

The multi-moment method discretizes the domain into a collection of geometric entities that form the mesh. This includes the volume cell, the planar faces that enclose the cell, the boundary edges that bound the faces, and, finally, the nodal points that define the edges. This data set is shown in the exploded view of a Cartesian cell in Fig.1.4. We can use a suitable form of the partial differential equation to solve for each solution simultaneously. These methods are the Finite Volume, Finite Surface (FSM), Finite Line (FLM), and Finite Difference Methods. Each method solves the cell-average (CV), surface-average, line-average (LV), and point values (PV), respectively. Xiao [35] coins the term multi-moment method, defining the moment as the datum used to construct the approximation function. They can be cell-average, edge-average, point data, or node-derivative, among others. With this definition,

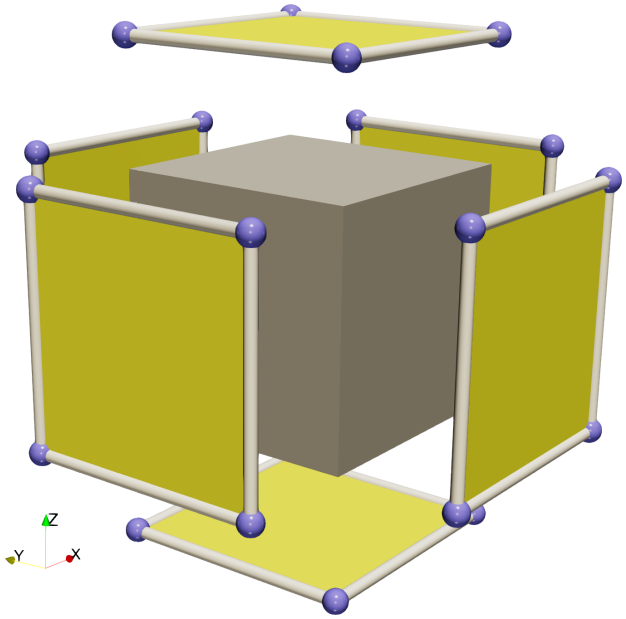


Fig. 1.4: Multi-moment discretization defines the dependent variables on every geometric entity: volumes (dark gray box), surfaces (green faces), edges (light gray cylinders), and nodes (purple sphere).

the CIP and IDO-CF schemes can be classified as multi-moment methods. In contrast, we view the multi-moment method as a multi-discretization approach in which different data types are solved simultaneously and reinforce each other's approximations. Each data type must have its own physical evolution equation; however, the mathematical discretization and numerical methods can be entirely different.

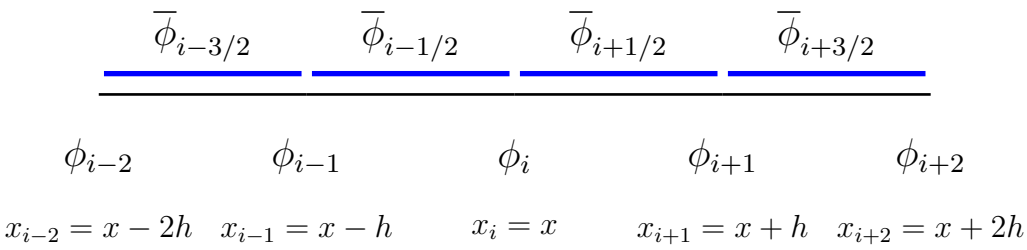


Fig. 1.5: An arrangement of one-dimensional multi-moment data. The pointwise data is defined on nodes. Each ϕ_i is defined on the corresponding x_i. The cell-averaged data is defined over the volume enclosed by the two consecutive nodes.

Recall the one-dimensional convection–diffusion equation:

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = K\frac{\partial^2 \phi}{\partial x^2}. \tag{1.24}$$

The multi-moment method employs a dual discretization in which both the strong and integral forms of Eq. (1.24) are solved simultaneously:

$$\frac{\partial \phi_i}{\partial t} + u \left(\frac{\partial \phi}{\partial x} \right)_i = K \left(\frac{\partial^2 \phi}{\partial x^2} \right)_i, \tag{1.25}$$

$$\Omega_{i+\frac{1}{2}} \frac{\partial \overline{\phi}^x_{i+\frac{1}{2}}}{\partial t} + u \left(\phi_{i+1} - \phi_i \right) = K \left[\left(\frac{\partial \phi}{\partial x} \right)_{i+1} - \left(\frac{\partial \phi}{\partial x} \right)_i \right]. \tag{1.26}$$

Equation (1.25) represents the finite-difference formulation for the nodal (pointwise) values, whereas Eq. (1.26) corresponds to the finite-volume formulation for the line-averaged quantities. The variable $\overline{\phi}^x_{i-1/2}$ denotes the line-averaged value in the x-direction over the control volume $\Omega_{i-1/2}$. In the following, this quantity is referred to as the *cell value* (CV), in contrast to the *point value* (PV) defined at the grid nodes. These terms are used interchangeably with *cell-averaged* and *pointwise* values, respectively.

When we solve the problem at steady state ($\partial\phi/\partial t = 0$), we usually form the linear system of equations as we have discussed previously for the FD and the FV methods. In time-dependent problems, the solution is evolved in time successively from the initial conditions until the final time.

When the two equations are solved independently, their computational costs are comparable. The finite-difference method (FDM) requires evaluation of the first derivative, $\partial\phi_i/\partial x$, to approximate the convective term. In contrast, the finite-volume method (FVM) requires reconstruction (or deconvolution) of the pointwise values ϕ_i from the cell-averaged data. These operations have similar computational complexity. For example, Lele's compact FDM [24] and Kobayashi's compact FVM [21] exhibit nearly identical computational costs.

However, when the two formulations are coupled within the multi-moment framework, the FVM directly gains access to the pointwise data (ϕ_i) provided by the FDM solution. As a result, the dominant computational expense arises from solving the FD equation, while the evaluation of the net flux in the FV equation incurs only negligible additional cost.

From a finite-difference perspective, the availability of cell averages creates a new possibility to compute the derivatives. Previously, we used the second-order central difference to compute the first derivative in Eq.(1.3) using the point value:

$$\left. \frac{\partial \phi}{\partial x} \right|_i = \frac{\phi_{i+1} - \phi_{i-1}}{2h}.$$

The availability of cell averages allows us to compute the first derivative by

$$\left. \frac{\partial \phi}{\partial x} \right|_i = \frac{\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}}{h}. \tag{1.27}$$

This approximation is called *the second-order multi-moment scheme* for the first derivative. It looks very similar to the central-difference scheme of the FDM, but their performance differs significantly. The former can resolve only signals with wavelength greater than 2π, while the latter can resolve signals with wavelength greater than π because the cell average is physically closer to x_i.

Likewise, the second-order derivative can be approximated using both data types by

$$\phi''_i = \frac{3}{h^2} \overline{\phi}_{i-\frac{1}{2}} + \frac{3}{h^2} \overline{\phi}_{i+\frac{1}{2}} - \frac{6}{h^2} \phi_i. \tag{1.28}$$

After determining the approximation of the FDM equation, we can construct the discrete system for the multi-moment method for the steady-state convection-diffusion equation:
1. FDM discrete equation of Equation (1.25):

$$\frac{u}{h}\left(\overline{\phi}_{i+\frac{1}{2}} - \overline{\phi}_{i-\frac{1}{2}}\right) - \frac{3K}{h^2}\left(\overline{\phi}_{i+\frac{1}{2}} + \overline{\phi}_{i-\frac{1}{2}}\right) + \frac{6K}{h^2}\phi_i = 0 \tag{1.29}$$

2. FVM discrete equation of Equation (1.26):

$$u\left(\phi_{i+1} - \phi_i\right) - \frac{K}{h}\left[\left(\overline{\phi}_{i+\frac{3}{2}} - \overline{\phi}_{i+\frac{1}{2}}\right) - \left(\overline{\phi}_{i+\frac{1}{2}} - \overline{\phi}_{i-\frac{1}{2}}\right)\right] = 0. \tag{1.30}$$

The FDM equation relies on the value of the neighboring CV and itself to compute the derivatives. Thus, if the functional value of the boundary is defined (Dirichlet BCs), the FDM does not need to solve for the value at the boundaries. On the other hand, the FVM equation needs the first derivatives at the boundaries to compute the diffusive flux. At this point, let us defer the treatment of the boundary closure and assume that the derivatives are given. Then, we can reorganize the global matrix by grouping the equations by variable type. This creates a highly structured block matrix system that clearly separates the Point-Value (PV) interactions from the Cell-Value (CV) interactions.

$$\left[\begin{array}{ccc|cccc}
\beta & 0 & 0 & \alpha & \gamma & 0 & 0 \\
0 & \beta & 0 & 0 & \alpha & \gamma & 0 \\
0 & 0 & \beta & 0 & 0 & \alpha & \gamma \\
\hline
u & 0 & 0 & a & -a & 0 & 0 \\
-u & u & 0 & -a & 2a & -a & 0 \\
0 & -u & u & 0 & -a & 2a & -a \\
0 & 0 & -u & 0 & 0 & -a & a
\end{array}\right]
\left[\begin{array}{c}
\phi_1 \\ \phi_2 \\ \phi_3 \\ \hline \overline{\phi}_{\frac{1}{2}} \\ \overline{\phi}_{\frac{3}{2}} \\ \overline{\phi}_{\frac{5}{2}} \\ \overline{\phi}_{\frac{7}{2}}
\end{array}\right]
=
\left[\begin{array}{c}
0 \\ 0 \\ 0 \\ \hline u\phi_0 - ah\phi'_0 \\ 0 \\ 0 \\ -u\phi_L + ah\phi'_L
\end{array}\right]$$

with $a = K/h$, $\alpha = -\frac{u}{h} - \frac{3K}{h^2}$, $\beta = \frac{6K}{h^2}$, and $\gamma = \frac{u}{h} - \frac{3K}{h^2}$. This system can be rewritten in a block matrix form as

$$\left[\begin{array}{c|c}
\boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \hline \boldsymbol{A}_{21} & \boldsymbol{A}_{22}
\end{array}\right]
\left[\begin{array}{c}
\boldsymbol{\phi} \\ \hline \overline{\boldsymbol{\phi}}
\end{array}\right]
=
\left[\begin{array}{c}
\mathbf{0} \\ \hline \boldsymbol{b}_{CV}
\end{array}\right] \tag{1.31}$$

If the Finite Difference Method (FDM) and Finite Volume Method (FVM) equations were solved simultaneously—but restricted to utilizing only the values known to their respective discretizations—the global matrix would be block diagonal (i.e., $\boldsymbol{A}_{12} = \boldsymbol{A}_{21} = \mathbf{0}$), containing no interaction between the two data types. However, the multi-moment approximation establishes a critical link between them. Specifically, the block matrix $\boldsymbol{A}_{21}$ represents the exact convective flux obtained from the point-value data at the cell boundaries, effectively injecting FDM information into the FVM framework. Conversely, $\boldsymbol{A}_{12}$ injects FVM cell-average information into the approximation of the FDM. The multi-moment method has twice the degrees of freedom as applying either FDM or FVM independently on the same grid. Consequently, for a given spatial discretization, the multi-moment method requires twice as much memory to store the state variables. Its ability to resolve signals with twice the frequency of standard FDM is fundamentally attributed to this increased local data density.

1.3 Truncation Error

The conventional approach for analyzing the error of finite-difference and finite-volume approximations is to expand the function ϕ in a Taylor series about the point x:

$$\phi(x+h) = \phi(x) + h\phi'(x) + \frac{h^2}{2!}\phi''(x) + \frac{h^3}{3!}\phi'''(x) + \cdots + \frac{h^n}{n!}\phi^{(n)}(x). \tag{1.32}$$

This series expresses the function value $\phi(x+h)$ as a polynomial in the step size h, whose coefficients are determined by the local derivatives of ϕ evaluated at x. For sufficiently small step sizes ($|h| \ll 1$), the higher-order terms become progressively smaller, allowing the series to be truncated after a finite number of terms. Using Big-O notation, the expansion may be written as

$$\phi(x+h) = \phi(x) + h\phi'(x) + \frac{h^2}{2!}\phi''(x) + \frac{h^3}{3!}\phi'''(x) + O(h^4). \tag{1.33}$$

The term $O(h^4)$ represents the local truncation error (LTE) arising from the neglected higher-order terms in the Taylor-series expansion. Specifically, it indicates that the magnitude of the omitted terms is bounded by a quantity proportional to h^4, i.e.,

$$|O(h^4)| \leq C|h|^4,$$

for some constant C independent of h as $h \to 0$. In the above case, the neglected terms include contributions proportional to h^4, h^5, h^6, and higher powers of h. As $h \to 0$, the h^4 term becomes the dominant contribution among the neglected terms, while the higher-order terms diminish more rapidly. Therefore, the lowest-order neglected term governs the asymptotic behavior of the truncation error and is referred to as the *leading truncation error term*.

In the finite-volume method, the primary variable is the cell-averaged quantity rather than the pointwise value. The cell-average over the interval $[x_i, x_{i+1}]$ is defined as

$$\overline{\phi}_{i+\frac{1}{2}} = \overline{\phi}_{[x_i,x_{i+1}]} = \frac{1}{h}\int_{x_i}^{x_i+h} \phi(\xi)\, d\xi, \qquad h = x_{i+1} - x_i. \tag{1.34}$$

Substituting the Taylor expansion into the cell-average definition yields

$$\overline{\phi}_{[x_i,x_{i+1}]} = \frac{1}{h}\int_0^h \left[\phi(x_i) + \xi\phi'(x_i) + \frac{(\xi)^2}{2!}\phi''(x_i) + \frac{(\xi)^3}{3!}\phi'''(x_i) + O(h^4)\right] d\xi. \tag{1.35}$$

Performing the integration term by term yields

$$\overline{\phi}_{i+\frac{1}{2}} = \phi(x_i) + \frac{h}{2}\phi'(x_i) + \frac{h^2}{6}\phi''(x_i) + \frac{h^3}{24}\phi'''(x_i) + O(h^4). \tag{1.36}$$

This expression relates the cell-averaged quantity to the pointwise value and its derivatives evaluated at x_i. It also illustrates that the averaging operation naturally introduces higher-order smoothing effects through the derivative terms.

Similarly, the expansion for the neighboring cell average $\overline{\phi}_{i-1/2}$ can be obtained as

$$\overline{\phi}_{i-\frac{1}{2}} = \phi(x_i) - \frac{h}{2}\phi'(x_i) + \frac{h^2}{6}\phi''(x_i) - \frac{h^3}{24}\phi'''(x_i) + O(h^4). \tag{1.37}$$

Using these two expansions, the accuracy of the second-order multi-moment scheme (Eq. (1.27)) can be analyzed by taking their difference:

$$\phi_i' = \frac{\overline{\phi}^x_{i+\frac{1}{2}} - \overline{\phi}^x_{i-\frac{1}{2}}}{h} - \frac{h^2}{12}\phi_i''' + O(h^4). \tag{1.38}$$

Next, applying the Taylor expansion in Eq. (1.32) to ϕ_{i+1} and ϕ_{i-1} by substituting h and $-h$, respectively, and substituting the resulting expressions into Eq. (1.3), gives

$$\frac{\partial \phi}{\partial x} = \phi'(x) = \frac{\phi_{i+1} - \phi_{i-1}}{2h} - \frac{h^2}{6}\phi'''(x) + O(h^4). \tag{1.39}$$

Thus, if the grid spacing is sufficiently small, we should expect the second-order multi-moment approximation to produce half the error of the second-order central difference in FDM. Furthermore, the leading truncation error (LTE) terms in the two schemes can be eliminated by forming the linear combination.

$$2 \times \text{Eq. (1.27)} - \text{Eq. (1.39)}$$

which yields

$$\phi_i' = \frac{2}{h}\left(\overline{\phi}^x_{i+\frac{1}{2}} - \overline{\phi}^x_{i-\frac{1}{2}}\right) - \frac{1}{2h}\left(\phi_{i+1} - \phi_{i-1}\right) + O(h^4). \tag{1.40}$$

This expression corresponds to the fourth-order multi-moment scheme used previously. Note that the two Big-O terms are combined into a single truncation term. To determine the exact leading truncation error, additional terms must be retained in the Taylor expansion, which reveals

$$\frac{h^4}{360}\phi_i^{(5)} + O(h^6).$$

The Taylor expansion presented above can be used not only to analyze local truncation errors, but also to systematically construct approximations for derivatives, function values, and integrals at desired locations. As such, it is an exceptionally powerful mathematical tool and serves as one of the fundamental pillars in the analysis and development of numerical methods.

The approximation of the second-order finite difference methods for the first derivative can be elevated to fourth-order accuracy by including the next two data points:

$$\left.\frac{\partial \phi}{\partial x}\right|_i = \frac{8}{12h}(\phi_{i+1} - \phi_{i-1}) - \frac{1}{12h}(\phi_{i+2} - \phi_{i-2}). \tag{1.41}$$

This approximation gathers the data from $[x_{i-2},\ x_{i+2}]$. In contrast, the fourth-order multi-moment scheme operates within a spatial footprint bounded only by $[x_{i-1}, x_{i+1}]$, effectively halving the stencil size required by the FDM. The practical accuracy of these schemes can be evaluated through a convergence study analogous to Ex. 1.2. To this end, the following example presents an accuracy assessment comparing the second- and fourth-order formulations of both the finite-difference and multi-moment schemes, evaluated using the test function $\phi(x) = \sin(x)$ at $x = \pi/4$.

Example 1.3 (Accuracy of Multi-Moment Approximation of the First Derivative)

In this example, we evaluate the accuracy of the second-order (EX2) and fourth-order (EX4) approximations of the first derivative obtained using the multi-moment method, and compare them with their finite-difference counterparts of equivalent formal order. As in the previous example, the function $\sin(x)$ is used as the reference signal, and the derivative is evaluated at $x = \pi/4$. The errors of the numerical approximations relative to the exact solution are summarized in the table below.

h	**FDEX2**	**FDEX4**	**MMEX2**	**MMEX4**
2π	$7.0711 \cdot 10^{-1}$	$7.0711 \cdot 10^{-1}$	$7.0711 \cdot 10^{-1}$	$7.0711 \cdot 10^{-1}$
$8\pi/5$	$8.4090 \cdot 10^{-1}$	$8.7171 \cdot 10^{-1}$	$6.6843 \cdot 10^{-1}$	$4.9597 \cdot 10^{-1}$
$3\pi/2$	$8.5716 \cdot 10^{-1}$	$9.0718 \cdot 10^{-1}$	$6.4342 \cdot 10^{-1}$	$4.2969 \cdot 10^{-1}$
π	$7.0711 \cdot 10^{-1}$	$7.0711 \cdot 10^{-1}$	$4.2053 \cdot 10^{-1}$	$1.3395 \cdot 10^{-1}$
$\pi/2$	$2.5695 \cdot 10^{-1}$	$1.0690 \cdot 10^{-1}$	$1.3395 \cdot 10^{-1}$	$1.0947 \cdot 10^{-2}$
$\pi/4$	$7.0487 \cdot 10^{-2}$	$8.3331 \cdot 10^{-3}$	$3.5609 \cdot 10^{-2}$	$7.3108 \cdot 10^{-4}$
$\pi/16$	$4.5348 \cdot 10^{-3}$	$3.4873 \cdot 10^{-5}$	$2.2688 \cdot 10^{-3}$	$2.9154 \cdot 10^{-6}$
$\pi/64$	$2.8394 \cdot 10^{-4}$	$1.3681 \cdot 10^{-7}$	$1.4197 \cdot 10^{-4}$	$1.1403 \cdot 10^{-8}$

Table 1.3: Numerical error comparison of FDM and MM methods under grid refinement.

At the theoretical boundary $h = 2\pi$, all evaluated methods fail to resolve the wave, compute a zero gradient, and yield an exact error equal to the true gradient ($\sqrt{2}/2$). Within the coarse regime of $\pi \leq h \leq 2\pi$, the finite-difference method (FDM) fails to produce meaningful results due to the Shannon-Nyquist limit discussed previously. Conversely, the multi-moment method demonstrates the unique capability to resolve the wave within the $\pi \leq h < 2\pi$ range. While the second-order multi-moment scheme (MMEX2) lacks practical accuracy at exactly $h = \pi$, the fourth-order scheme (MMEX4) successfully captures approximately 81% of the exact gradient, providing a viable approximation for specific applications. As the grid spacing decreases below π, the multi-moment methods exhibit a continuous reduction in error, whereas the FDM error paradoxically increases. This divergence is attributed to the fact that cell-averaged data provide a more robust local representation than discrete point data. In these finer resolutions, the second-order multi-moment scheme is consistently twice as accurate as its FDM counterpart. This structural advantage is significantly amplified in the fourth-order scheme, which reduces the error by a factor of 9 to 12—yielding nearly an order-of-magnitude improvement in accuracy without any additional computational expense.

These results highlight two distinct advantages of the multi-moment method. In the resolved regime, where the numerical errors approach their theoretical asymptotic convergence rates, multi-moment schemes achieve substantially higher accuracy than conventional FDM schemes of equivalent formal order. In contrast, when the grid becomes under-resolved, particularly for $h \geq \pi$, conventional FDM schemes fail to represent the wave adequately. In contrast, multi-moment schemes retain some ability to capture its spatial variation. However, this does not imply that low-order multi-moment schemes can accurately resolve such severely under-resolved waves; substantially higher-order approximations are required. The multi-moment method therefore offers both improved accuracy in the resolved regime and greater potential for resolving waves on coarse physical grids.

1.4 Deriving Multi-Moment Approximation

Although manually deriving approximation coefficients is manageable for low-order approximations, the process becomes increasingly tedious when extending to very high-order schemes or non-uniform grids. In these cases, expressing the derivation in the form of a linear system is substantially more effective, since solving the system directly provides the required coefficients.

To demonstrate this procedure, we derive the approximation for the second-order derivative within the multi-moment framework. From Eq. (1.4), we see that a second-order accurate formulation requires a three-point stencil. Hence, the approximation is constructed from the three nearest discrete quantities around ϕ_i''.

$$\begin{aligned}
\overline{\phi}_{i+\frac{1}{2}} &= \phi(x_i) + \frac{h}{2}\phi'(x_i) + \frac{h^2}{6}\phi''(x_i) + \frac{h^3}{24}\phi'''(x_i) + O(h^4),\\
\phi_i &= \phi(x_i),\\
\overline{\phi}_{i-\frac{1}{2}} &= \phi(x_i) - \frac{h}{2}\phi'(x_i) + \frac{h^2}{6}\phi''(x_i) - \frac{h^3}{24}\phi'''(x_i) + O(h^4).
\end{aligned}$$

The objective is to determine a linear combination of these three expansions that isolates ϕ_i''. This requirement can be reformulated as the following matrix–vector system:

$$a_1\overline{\phi}_{i-\frac{1}{2}} + a_2\overline{\phi}_{i+\frac{1}{2}} + a_3\phi_i = \phi_i'', \tag{1.42}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ -\frac{h}{2} & \frac{h}{2} & 0 \\ \frac{h^2}{6} & \frac{h^2}{6} & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \tag{1.43}$$

To obtain a unique solution, the system must be square, meaning that the number of unknown coefficients must match the number of imposed constraints. In the present case, the first row enforces the cancellation of the zeroth-order term, the second row eliminates the first-derivative contribution, and the third row normalizes the coefficient of ϕ_i'' to unity. Solving the system directly yields

$$[a_1,\ a_2,\ a_3] = \frac{1}{h^2}[3,\ 3,\ -6].$$

Once these coefficients are determined, they can be substituted back into the Taylor expansions to derive both the resulting approximation and its corresponding local truncation errors.

More generally, this procedure for constructing approximation coefficients belongs to the class of techniques known as the *method of undetermined coefficients*. In Numerical Analysis, it is commonly called *Taylor Table Method*.

Definition 1.4 (Taylor Table Method) The *Taylor Table Method* is a systematic procedure for constructing finite-difference, finite-volume, and multi-moment approximations by matching the coefficients of Taylor-series expansions.

Given a set of discrete quantities surrounding a target location, each quantity is expressed as a Taylor-series expansion about a common reference point. These quantities may include pointwise values, cell-averaged values, or spatial derivatives. While pointwise values are

expanded directly, cell-averaged quantities are expanded by integrating the Taylor series over their corresponding control volumes and dividing by the volume measure.

The desired approximation is constructed as a linear combination of the available discrete quantities,

$$\mathcal{D}\phi \approx \sum_{j=1}^{N} a_j M_j[\phi],$$

where $\mathcal{D}$ denotes the target differential operator, $M_j[\phi]$ represents the j-th discrete quantity (or moment), and a_j are unknown approximation coefficients. Substituting the Taylor expansions into this expression yields a system of algebraic constraints. The coefficients associated with unwanted lower-order derivatives are eliminated, while the coefficient corresponding to the target derivative is normalized to unity. The resulting linear system can be written as

$$\mathbf{Ta} = \mathbf{b},$$

where $\mathbf{T}$ contains the Taylor-series coefficients of the discrete quantities, $\mathbf{a}$ is the vector of unknown approximation coefficients, and $\mathbf{b}$ specifies the required consistency conditions. Solving this system determines the approximation coefficients. The remaining unmatched higher-order terms determine the truncation error and hence the formal order of accuracy of the resulting scheme.

Problems

Problem 1.1 Explain the physical conditions that necessitate the transition from an Ordinary Differential Equation (ODE) to a Partial Differential Equation (PDE) in engineering models. Provide one example.

Problem 1.2 Contrast the fundamental data representation between the Finite Difference Method (FDM) and the Finite Volume Method (FVM). Explain why FVM is uniquely suited for conservation laws.

Problem 1.3 What was the critical misconception regarding cell-averaged quantities ($\overline{\phi}_{i+1/2}$) and pointwise values ($\phi_{i+1/2}$) in early finite-volume development? How did this ambiguity affect the formulation of higher-order schemes?

Problem 1.4 State the Nyquist-Shannon Sampling Theorem in the context of grid-based numerical methods. For a test function like $\sin(x)$, explain why the numerical error stagnates when the grid spacing is $h \geq \pi$.

Problem 1.5 Define the term "moment" within the context of the multi-moment method as classified by Xiao. How does the multi-discretization approach expand upon this definition?

Problem 1.6 In the multi-moment framework coupling FDM and FVM, explain the physical and mathematical significance of the off-diagonal block matrices (e.g., $\boldsymbol{A}_{12}$ and $\boldsymbol{A}_{21}$) in the global linear system.

Problem 1.7 Given the error norms E_1 and E_2 obtained from two distinct grid spacings h_1 and h_2, state the mathematical expression used to calculate the practical formal order of accuracy, p.

Problem 1.8 Using Taylor series expansions about a node x_i, demonstrate how the integration of the function over the cell interval $[x_i, x_{i+1}]$ naturally introduces higher-order derivative smoothing effects into the cell-average expression $\overline{\phi}_{i+1/2}$.

Problem 1.9 Compare the spatial footprint of the fourth-order multi-moment scheme for the first derivative to its fourth-order FDM counterpart. What computational advantage does this localized footprint confer?

Problem 1.10 Describe the primary objective of the Taylor Table Method (method of undetermined coefficients) in the derivation of high-order approximations. Why is it advantageous over manual algebraic substitution?

Solution Key

1. The transition occurs when the simplifying assumption of spatial homogeneity is removed, requiring variables to evolve across multiple dimensions rather than just time. For example, a lumped-capacitance thermal model expands into the multidimensional heat equation the moment internal temperature gradients are resolved.
2. FDM stores discrete pointwise nodal values and approximates derivatives locally, whereas FVM stores integral, cell-averaged quantities. FVM is suited for conservation laws because it directly tracks the integral fluxes entering and exiting a cell, ensuring the total quantity is conserved automatically.
3. Early researchers mistakenly equated the integral cell-average ($\overline{\phi}_{i+1/2}$) directly with the exact pointwise center value ($\phi_{i+1/2}$) due to a reliance on the midpoint rule. This ambiguity strictly limited schemes to second-order accuracy and caused significant failures when attempting to develop higher-order extensions.
4. The theorem dictates that a computational grid must possess at least two points per wavelength ($h < \lambda/2$) to actually resolve a physical wave. For $h \geq \pi$, the grid violates this limit and becomes completely "blind" to the wave, causing the error to stagnate regardless of the scheme's mathematical sophistication.
5. Xiao defines a "moment" as the specific datum (e.g., cell-average, point data, derivative) used to construct the approximation function. The multi-discretization approach expands this by treating the method as solving different data types simultaneously with their own physical evolution equations, allowing them to reinforce each other's approximations mutually.
6. The block matrices represent the direct interaction and coupling between the two data types. Specifically, $\boldsymbol{A}_{21}$ injects FDM point-value information into the FVM framework (representing the exact convective flux at cell boundaries). In contrast, $\boldsymbol{A}_{12}$ injects FVM cell-average information into the FDM approximation.
7. The theoretical order p is defined by the asymptotic slope of the error norm with respect to grid spacing on a logarithmic scale. It is calculated as $p = \ln(E_1/E_2)/\ln(h_1/h_2)$.
8. Integrating the standard Taylor expansion term-by-term from 0 to h and dividing by h yields $\overline{\phi}_{i+1/2} = \phi(x_i) + \frac{h}{2}\phi'(x_i) + \frac{h^2}{6}\phi''(x_i) + \frac{h^3}{24}\phi'''(x_i) + O(h^4)$. This relationship explicitly shows how the averaging operation incorporates higher-order derivative terms.
9. The fourth-order multi-moment scheme operates within a highly compact spatial footprint bounded only by $[x_{i-1}, x_{i+1}]$. In contrast, the fourth-order FDM requires a wider stencil extending to $[x_{i-2}, x_{i+2}]$. This effectively halves the required stencil size, leading to robust high-order accuracy with enhanced parallel computing efficiency.
10. The objective is to formulate the derivation of approximation coefficients as a linear system, which can be solved directly to yield the required constants. This is highly preferred because manual derivation becomes exceptionally tedious and unmanageable for very high-order schemes or non-uniform grids.

Chapter 2
Multi-Moment Method for the Convection Equation

"The water you touch in a river is the last of what has passed and the first of what is to come."
Leonardo da Vinci

The convection–diffusion equation (CDE) is fundamental to understanding heat and mass transfer in fluid flows. When combined with other relevant physical laws, it serves as a unified framework for modeling diverse transport phenomena, including momentum, chemical species, aerosols, and thermal energy. In convection-dominated regimes, the transported fields can develop structures of remarkable complexity, often surpassing that of the underlying flow field itself. Classic examples include scalar transport at high Schmidt numbers, turbulent mixing at high Reynolds numbers, and thermal transport at low Prandtl numbers.

Advances in high-performance computing have made practical steady-state simulations routine; however, high-fidelity transient simulations remain highly computationally intensive. Even with modern supercomputers, fully resolved simulations of realistic, high-Reynolds-number turbulent flows—such as the aerodynamic flow around an entire aircraft—are still prohibitively expensive. Between the two governing physical mechanisms, resolving convection numerically presents a significantly steeper challenge than resolving diffusion. This chapter, therefore, focuses primarily on numerical approaches tailored to convection-dominated transport problems.

To isolate the numerical challenges associated with fluid transport, we strip away the dissipative effects of diffusion. The resulting one-dimensional linear convection equation provides a canonical model for scalar transport and wave propagation:

$$\frac{\partial \phi}{\partial t} + \frac{\partial (u\phi)}{\partial x} = 0, \tag{2.1}$$

where ϕ represents the transported scalar quantity and u is the fluid velocity field. Assuming a constant, uniform velocity $u > 0$, Eq. (2.1) reduces to the linear advective form:

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = 0. \tag{2.2}$$

This equation models the pure transport of a scalar quantity driven entirely by the bulk motion of a fluid.

In most engineering and applied mathematics contexts, the terms *advection* and *convection* are used interchangeably. However, a distinct semantic line is drawn in disciplines like meteorology and oceanography, where *convection* is strictly reserved for buoyancy-driven vertical transport. In contrast, *advection* refers to horizontal or general transport by the bulk flow. Here, we follow the standard engineering convention, treating the two terms as equivalent.

The analytical solution to the linear advection equation is straightforward: an initial distribution $\phi(x, 0) = f(x)$ is translated downstream at a constant velocity u without any change to its shape or amplitude:

$$\phi(x,t) = f(x-ut). \tag{2.3}$$

Because the governing physics contains no mechanisms for physical dissipation (damping) or dispersion (phase shifting), the solution behaves as a pure, shape-preserving translation.

Due to this traveling-wave behavior, Eq. (2.2) is frequently referred to as the *unidirectional wave equation*. This first-order formulation is significantly more tractable than the full, second-order one-dimensional wave equation:

$$\frac{\partial^2\phi}{\partial t^2} = c^2\frac{\partial^2\phi}{\partial x^2}, \tag{2.4}$$

where physical disturbances propagate in both directions simultaneously at the wave speed c.

Consequently, the one-dimensional convection equation provides a natural starting point for the study of hyperbolic partial differential equations (PDEs), one of the three classical categories of second-order PDEs alongside parabolic and elliptic equations. A defining characteristic of hyperbolic equations is the finite speed at which information propagates through the physical domain. For the linear convection and wave equations considered here, this propagation can be described by characteristic lines in space-time, along which information is transported.

For the one-dimensional wave equation given in Eq. (2.4), information propagates along two families of characteristic lines, $x - ct =$ constant and $x + ct =$ constant, corresponding to waves traveling in the positive and negative x-directions, respectively. These characteristics define the boundaries of the characteristic cone illustrated in Fig. 2.1. The solution at a given space-time point (x_n, t_n) depends only on information contained within its past-facing *domain of dependence* (the red region). Conversely, a disturbance introduced at (x_n, t_n) can affect the future solution only within its future-facing domain of influence (the blue region).

In contrast, the one-dimensional linear convection equation,

$$\frac{\partial\phi}{\partial t} + u\frac{\partial\phi}{\partial x} = 0,$$

possesses only one family of characteristic lines, $x - ut =$ constant, along which the scalar field is transported with velocity u. For a constant positive velocity, information propagates exclusively in the positive x-direction; for a negative velocity, the direction is reversed. Unlike the wave equation, whose two characteristic families represent propagation in opposite directions, the convection equation transports information along a single characteristic direction. This distinction follows from the mathematical structure of the governing equations rather than from the placement of individual terms on either side of the equality.

Despite its simple mathematical form, the convection equation presents significant challenges for numerical approximation. Spatial discretization introduces errors that can alter the propagation speed and amplitude of the numerical solution. Classical central-difference approximations may exhibit numerical dispersion, producing phase errors and spurious oscillations near sharp gradients. First-order upwind schemes, although monotone under appropriate time-step restrictions, introduce numerical dissipation that smears sharp fronts and attenuates high-wavenumber components. Godunov's theorem [8] establishes a fundamental limitation: a linear monotonicity-preserving numerical scheme for linear advection cannot achieve an order of accuracy higher than one. To overcome this restriction, higher-order methods commonly employ nonlinear reconstruction or limiting procedures, including total variation diminishing (TVD) schemes [9] and weighted essentially non-oscillatory (WENO)

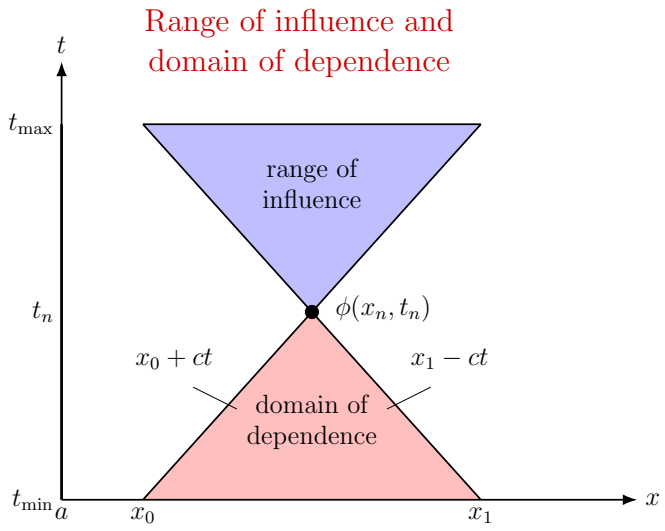


Fig. 2.1: Characteristic lines, domain of dependence (red), and domain of influence (blue) for the one-dimensional wave equation.

schemes [16]. These techniques improve the resolution of transported structures while controlling nonphysical oscillations, although they generally increase the complexity of the numerical algorithm.

In this chapter, we systematically investigate the numerical discretization of the one-dimensional convection equation, with particular emphasis on the multi-moment method. We begin by examining truncation errors and constructing multi-moment approximations of the first spatial derivative. We then analyze their dispersive and dissipative properties through Fourier analysis, investigate the stability of the resulting semi-discrete and fully discrete systems, and evaluate their accuracy and wave-resolving capabilities through representative numerical experiments.

2.1 Multi-Moment Discretization of the One-Dimensional Convection Equation

The multi-moment method solves the one-dimensional convection equation for a scalar quantity using the purely convective form of the coupled system introduced in Chapter 1 (Eq. (1.25) and Eq. (1.26)):

$$\frac{\partial \phi_i}{\partial t} + u \left(\frac{\partial \phi}{\partial x} \right)_i = 0, \tag{2.5}$$

$$\Omega_{i+\frac{1}{2}} \frac{\partial \overline{\phi}^x_{i+\frac{1}{2}}}{\partial t} + u \left(\phi_{i+1} - \phi_i \right) = 0. \tag{2.6}$$

The first equation corresponds to the finite-difference formulation governing the evolution of point values. In contrast, the second equation represents the finite-volume formulation governing the evolution of cell-averaged quantities. After spatial discretization, the governing system may be expressed in matrix–vector form as

$$\frac{\partial}{\partial t}\begin{bmatrix}\widetilde{\phi}\\ \widehat{\phi}\end{bmatrix}^n = -u\boldsymbol{C}\begin{bmatrix}\widetilde{\phi}\\ \widehat{\phi}\end{bmatrix}^n . \tag{2.7}$$

The matrix $\boldsymbol{C}$ is referred to as the *spatial differentiation matrix*, as it governs the evaluation of spatial derivatives and intercell fluxes. At the semi-discrete level, Eqs. (2.5) and (2.6) remain exact representations of the continuous governing equations. Numerical errors arise only when the spatial operators are approximated using a finite set of discrete moments and when temporal integration is subsequently applied.

In practice, the solution field is represented by localized point values and cell-averaged moments. Consequently, the spatial derivative appearing in Eq. (2.5) must be approximated numerically from the available degrees of freedom. By contrast, the finite-volume update in Eq. (2.6) requires only the interface values ϕ_i and ϕ_{i+1}, which are directly available from the point-value representation from the FDM. Therefore, no additional spatial reconstruction is required within the finite-volume equation itself.

Nevertheless, dispersive and dissipative errors arising from the derivative approximation in the finite-difference equation inevitably influence the evolution of the cell averages through the common interface fluxes. As a result, the point-value and cell-average equations are dynamically coupled, even though the derivative approximation enters explicitly only in the former. To isolate and better understand the contribution of these errors, we begin by analyzing the modified-wavenumber properties of the point-value update in isolation before considering the fully coupled system. Subsequently, we discuss the stability characteristics of the time-integration procedure and examine several practical issues and applications relevant to convection equations.

2.2 Explicit Approximation of the First Derivative

To solve the point-value convection equation, Eq. (2.5), the first spatial derivative must be approximated numerically. The most straightforward approach is to evaluate this derivative directly from the available data, namely the point values and cell values. Such an approximation is referred to here as an *explicit* scheme. This is in contrast to an *implicit* or *compact* scheme, in which the unknown derivatives are themselves coupled through the discretization equations.

Compact schemes are attractive because their implicit coupling can provide higher resolving power than explicit schemes with comparable stencil width. However, this coupling also introduces additional difficulty in parallel implementation, especially for very high-order schemes. In particular, the forward and backward elimination stages require sequential communication, and this two-way dependency can lead to substantial idle time during synchronization. Therefore, it is desirable to develop an alternative approach that improves accuracy while avoiding the global or semi-global coupling associated with compact schemes.

2.2.1 Explicit Schemes

In the previous chapter, we introduced two basic approximations for the first derivative under the multi-moment discretization. They belong to the class of *explicit schemes*, where the final

approximated value is obtained directly from a linear combination of existing local values. We can generalize the explicit scheme for an arbitrary order of accuracy using the following formulation:

$$\phi_i' = \sum_{\tau=1}^{m_1} \frac{\beta_\tau}{h} \left(\overline{\phi}_{i+\tau-\frac{1}{2}} - \overline{\phi}_{i-\tau+\frac{1}{2}} \right) - \sum_{\tau=1}^{m_2} \frac{\gamma_\tau}{h} \left(\phi_{i+\tau} - \phi_{i-\tau} \right). \tag{2.8}$$

To maximize resolving power, the stencil expansion should strictly adhere to data locality: incorporating the immediately adjacent cell averages first, followed by the enclosing point values, and repeating this process radiating outward. This dense data clustering provides significantly better spectral resolution than omitting intermediate inner data.

In this work, we follow the reconstruction method and assume that the data in the local neighborhood obey a Taylor series expansion. The coefficients of the approximation are derived using the Taylor table method described in Chapter 1. Because the stencil is dense (no data points are skipped), the underlying polynomial reconstruction forces the signs of the raw coefficients to alternate from one term to the next. Specifically, the cell-average coefficients (β_τ) are strictly positive, while the point-value coefficients are strictly negative. Factoring out the negative sign, as in Eq. (2.8), clarifies the physical mechanism of the multi-moment approximation: the finite difference of the integral cell averages inherently overestimates the local gradient, and the enclosing pointwise differences counterbalance this overestimation.

For example, the linear system for determining the fourth-order explicit scheme is constructed by equating the Taylor expansion of the first derivative to the linear combination of the selected local supports:

$$\phi_i' = \beta_1 \overline{\phi}_{i-\frac{1}{2}} + \beta_2 \overline{\phi}_{i+\frac{1}{2}} + \gamma_1 \phi_{i-1} + \gamma_2 \phi_{i+1} + O(h^4), \tag{2.9}$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -\frac{h}{2} & \frac{h}{2} & -h & h \\ \frac{h^2}{6} & \frac{h^2}{6} & \frac{h^2}{2} & \frac{h^2}{2} \\ -\frac{h^3}{24} & \frac{h^3}{24} & -\frac{h^3}{6} & \frac{h^3}{6} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \gamma_1 \\ \gamma_2 \end{bmatrix}. \tag{2.10}$$

This yields the solution vector $\left[-\frac{2}{h}, \frac{2}{h}, \frac{1}{2h}, -\frac{1}{2h}\right]^T$, which was utilized in the previous chapter. The Leading Truncation Error (LTE) of this specific approximation is $-h^4/360$. It is worth noting that this set of fourth-order coefficients matches those derived by Imai et al. [15]. The exact coefficients for higher-order explicit schemes up to the 24th order of accuracy are listed in Table 2.1, with their associated local truncation errors displayed in Table 2.2. For example, the sixth-order explicit multi-moment scheme is given by

$$\phi_i' = \frac{1}{h} \left[\frac{9}{4} \left(\overline{\phi}_{i+\frac{1}{2}} - \overline{\phi}_{i-\frac{1}{2}} \right) + \frac{1}{36} \left(\overline{\phi}_{i+\frac{3}{2}} - \overline{\phi}_{i-\frac{3}{2}} \right) \right] - \frac{1}{h} \left[\frac{2}{3} \left(\phi_{i+1} - \phi_{i-1} \right) \right], \tag{2.11}$$

and the eighth-order scheme is

$$\phi_i' = \frac{1}{h} \left[\frac{5}{2} \left(\overline{\phi}_{i+\frac{1}{2}} - \overline{\phi}_{i-\frac{1}{2}} \right) + \frac{7}{54} \left(\overline{\phi}_{i+\frac{3}{2}} - \overline{\phi}_{i-\frac{3}{2}} \right) \right] - \frac{1}{h} \left[\frac{8}{9} \left(\phi_{i+1} - \phi_{i-1} \right) + \frac{1}{36} \left(\phi_{i+2} - \phi_{i-2} \right) \right]. \tag{2.12}$$

Care must be taken when solving the linear system for higher orders. It is self-evident that the coefficient matrix is highly Vandermonde-like, meaning it becomes increasingly

Order	β_1	β_2	β_3	β_4	β_5	β_6	γ_1	γ_2	γ_3	γ_4	γ_5	γ_6
4	2	—	—	—	—	—	$\frac{1}{2}$	—	—	—	—	—
6	$\frac{9}{4}$	$\frac{1}{36}$	—	—	—	—	$\frac{2}{3}$	—	—	—	—	—
8	$\frac{5}{2}$	$\frac{7}{54}$	—	—	—	—	$\frac{8}{9}$	$\frac{1}{36}$	—	—	—	—
10	$\frac{47}{18}$	$\frac{7}{36}$	$\frac{1}{900}$	—	—	—	1	$\frac{1}{20}$	—	—	—	—
12	$\frac{49}{18}$	$\frac{41}{144}$	$\frac{157}{18000}$	—	—	—	$\frac{9}{8}$	$\frac{9}{100}$	$\frac{1}{600}$	—	—	—
14	$\frac{401}{144}$	$\frac{1241}{3600}$	$\frac{301}{18000}$	$\frac{1}{19600}$	—	—	$\frac{6}{5}$	$\frac{3}{25}$	$\frac{2}{525}$	—	—	—
16	$\frac{205}{72}$	$\frac{3737}{9000}$	$\frac{281}{9000}$	$\frac{199}{343000}$	—	—	$\frac{32}{25}$	$\frac{4}{25}$	$\frac{32}{3675}$	$\frac{1}{9800}$	—	—
20	$\frac{5269}{1800}$	$\frac{2807}{5400}$	$\frac{38267}{617400}$	$\frac{157}{50400}$	$\frac{7633}{200037600}$	—	$\frac{25}{18}$	$\frac{100}{441}$	$\frac{25}{1176}$	$\frac{25}{31752}$	$\frac{1}{158760}$	—
22	$\frac{591}{200}$	$\frac{16477}{29400}$	$\frac{10601}{137200}$	$\frac{56431}{11113200}$	$\frac{1181}{11113200}$	$\frac{1}{7683984}$	$\frac{10}{7}$	$\frac{25}{98}$	$\frac{25}{882}$	$\frac{5}{3528}$	$\frac{1}{48510}$	—
24	$\frac{5369}{1800}$	$\frac{372767}{617400}$	$\frac{474511}{4939200}$	$\frac{1095797}{133358400}$	$\frac{38477}{133358400}$	$\frac{88331}{35500006080}$	$\frac{72}{49}$	$\frac{225}{784}$	$\frac{50}{1323}$	$\frac{1}{392}$	$\frac{2}{29645}$	$\frac{1}{2561328}$

Table 2.1: Coefficients of the central explicit multi-moment approximation to the first derivative (Eq.(2.8)) .

Order	LTE	Order	LTE	Order	LTE
4	$-\frac{1}{360}h^4$	12	$-\frac{1}{33633600}h^{12}$	20	$-\frac{1}{2710264100544}h^{20}$
6	$\frac{1}{5040}h^6$	14	$\frac{1}{504504000}h^{14}$	22	$\frac{1}{41557382875008}h^{22}$
8	$-\frac{1}{113400}h^8$	16	$-\frac{1}{9648639000}h^{16}$	24	$-\frac{1}{750341635243200}h^{24}$
10	$\frac{1}{1663200}h^{10}$	18	$\frac{1}{146659312800}h^{18}$	—	—

Table 2.2: Local Truncation Error (LTE) of the central explicit multi-moment approximation to the first derivative (Eq.(2.8)) .

ill-conditioned as the target order of the scheme increases. This numerical stiffness becomes particularly pronounced when calculating coefficients for very high-order schemes on non-uniform grids. To mitigate round-off errors in such situations, we recommend solving the matrix system in quadruple precision during pre-processing and storing the final coefficients in single or double precision for the main flow solver.

Unlike standard explicit Finite-Difference Methods (FDM), explicit multi-moment schemes achieve high-order convergence with significantly narrower physical stencils—rivaling the spatial compactness of implicit compact schemes [24], but without the computational overhead of solving a global linear system. For instance, a fourth-order explicit FDM requires a $4h$-wide stencil, whereas the fourth-order explicit multi-moment scheme requires a footprint of only $2h$. This scaling advantage persists across all approximation orders. Thus, when matched by the number of degrees of freedom (nDoF), the multi-moment scheme is consistently more accurate and more efficient than an explicit FDM of the same formal order of accuracy.

The primary architectural advantage of explicit schemes lies in their simplicity and excellent potential for massive parallelization. Because the numerical updates are purely localized, they utilize contiguous memory access. In contrast, implicit compact schemes—which will be discussed shortly—require global forward and backward substitution sweeps when solving their linear systems of equations along the direction of approximation. On modern cache-based hardware, this necessity results in frequent access to strided memory locations, which can lead to increased cache misses and severely degraded computational performance.

2.3 Fourier Analysis of Multi-Moment Method

Previously, we utilized Taylor series expansions to approximate the local behavior of functions. This approach allows us to derive schemes for differentiation, interpolation, and integration. While Taylor series analysis is invaluable for establishing formal grid convergence, Fourier analysis provides another powerful framework—one that is particularly suited for analyzing wave propagation and resolving efficiency.

The fundamental premise of applying Fourier analysis to numerical schemes is that any periodic, sufficiently smooth flow field can be represented as a superposition of distinct harmonic waves via Fourier decomposition:

$$\phi(x) = \sum_k \hat{\phi}_k e^{jkx} \tag{2.13}$$

where $j = \sqrt{-1}$ is the imaginary unit, and $\hat{\phi}_k$ represents the complex Fourier coefficient associated with the spatial wavenumber k.

Because spatial discretization schemes act as linear operators, the principle of superposition dictates that the global numerical error can be exactly quantified by examining the scheme's action on a single, arbitrary Fourier mode. The exact analytical first derivative of the full field is purely imaginary with respect to its coefficients:

$$\frac{\partial \phi}{\partial x} = \sum_k \mathrm{j}k\hat{\phi}_k e^{\mathrm{j}kx} \tag{2.14}$$

For an individual harmonic mode, computing the first derivative is straightforward:

$$\frac{\partial}{\partial x}\left(\hat{\phi}_k e^{\mathrm{j}kx}\right) = \mathrm{j}k\hat{\phi}_k e^{\mathrm{j}kx} \tag{2.15}$$

The Fourier series is orthogonal, meaning that each mode is independent of the others. This property allows us to reduce a linear partial differential equation (PDE) into a set of independent ordinary differential equations (ODEs). To find the solution for a single Fourier mode (for example, under the linear advection equation), we assume the initial condition at $t = 0$ is a harmonic wave with a spatial wavenumber k and an initial complex amplitude $\hat{\phi}_0$:

$$\phi(x, 0) = \hat{\phi}_0 e^{\mathrm{j}kx} \tag{2.16}$$

We then postulate that the solution at any time t maintains its spatial harmonic shape, but its amplitude and phase evolve over time. We separate the variables as follows:

$$\phi(x, t) = \hat{\phi}(t) e^{\mathrm{j}kx} \tag{2.17}$$

Substituting this assumed form into the governing advection equation yields:

$$\frac{d\hat{\phi}}{dt} e^{\mathrm{j}kx} + u\hat{\phi}(t)\left(\mathrm{j}k e^{jkx}\right) = 0 \tag{2.18}$$

Dividing through by the non-zero exponential term e^{jkx} reduces the PDE to a first-order ODE for the time-dependent amplitude:

$$\frac{d\hat{\phi}}{dt} = -\mathrm{j}ku\hat{\phi} \tag{2.19}$$

Integrating this ODE with respect to time from $t = 0$ gives the exact amplification factor:

$$\hat{\phi}(t) = \hat{\phi}_0 e^{-\mathrm{j}kut} \tag{2.20}$$

Finally, substituting $\hat{\phi}(t)$ back into our assumed solution form gives the exact analytical solution for the single Fourier mode:

$$\phi(x,t) = \hat{\phi}_0 e^{-\mathrm{j}kut} e^{\mathrm{j}kx} = \hat{\phi}_0 e^{\mathrm{j}k(x-ut)} \tag{2.21}$$

This result demonstrates that the wave undergoes pure spatial translation at a phase velocity u. Because the time-dependent exponential term is purely imaginary, the evolution is strictly oscillatory; the initial signal propagates with a perfectly conserved amplitude, experiencing neither attenuation (damping) nor amplification over time.

Unlike the exact wavenumber k, the modified wavenumber $k^*(kh)$ is generally a complex number. It represents the *effective* wavenumber that the numerical scheme actually sees and propagates. By defining the modified wavenumber with real and imaginary components ($k^* = k^*_R + \mathrm{j}k^*_I$), the exact wave solution is altered to:

$$\phi(x,t) = \hat{\phi}_0 e^{uk^*_I t} e^{\mathrm{j}(kx - uk^*_R t)} \tag{2.22}$$

Thus, the real and imaginary parts of the modified wavenumber play distinct physical roles in the numerical error:

- **Wave speed Error:** The real part, k^*_R, governs the numerical phase speed. Because k^*_R typically varies non-linearly with k, waves of different frequencies travel at speeds different from the exact advection velocity u. This generates dispersive, non-physical separation of the signal, leading to spurious oscillations (wiggles) in the solution. This type of error is commonly called *dispersive error*.
- **Amplitude Error:** The imaginary component, k^*_I, controls the amplitude of the signal. Through the exponential term $e^{uk^*_I t}$, it introduces numerical diffusion that can unphysically damp (attenuate) or—if the scheme is unstable—amplify the signal over time. This type of error is commonly called *dissipative error*.

It is typically evaluated as a function of the non-dimensional exact wavenumber, $\phi = k\Delta x$, which spans the resolvable range of the grid from $\phi = 0$ (long, highly resolved waves) to the Nyquist limit $\phi = \pi$ (the shortest resolvable two-point wave).

2.3.1 Initial Modified Wavenumber Analysis of the Multi-Moment Method

In this section, we examine the initial response of the multi-moment (MM) method to harmonic waves. We first review the standard modified wavenumber analysis used for finite-difference (FD) schemes, and then extend this methodology to a multi-moment discretization.

Because Fourier modes are orthogonal, the response of a linear numerical scheme can be analyzed mode by mode. Therefore, rather than evaluating the full Fourier expansion in Eq. (2.13), we consider a single harmonic mode:

$$\phi(x) = e^{\mathrm{j}kx}, \tag{2.23}$$

where k is the wavenumber, and we set the amplitude to unity because the spatial derivative operator acts linearly on each Fourier component. The convection speed is also omitted, as the present analysis focuses strictly on the spatial discretization.

Using this ansatz, the nodal values at x_i and its immediate neighbors are:

$$\phi_{i-1} = e^{\mathrm{j}k(x_i-h)}, \tag{2.24}$$

$$\phi_i = e^{\mathrm{j}kx_i}, \tag{2.25}$$

$$\phi_{i+1} = e^{\mathrm{j}k(x_i+h)}. \tag{2.26}$$

This notation is mathematically equivalent to evaluating the full Fourier series at neighboring points, but it is significantly more compact. For example, the full series expansion:

$$\phi_{i+1} = \phi(x_i + h) = \sum_k \widehat{\phi}_k e^{\mathrm{j}k(x_i+h)}, \tag{2.27}$$

reduces to the single-mode expression above when isolating a single Fourier component. This reduction highlights why analyzing one Fourier mode is sufficient for linear schemes.

For the MM formulation, we must also establish the Fourier representation of the cell average. Over the interval $[x_i, x_{i+1}]$:

$$\begin{aligned}
\overline{\phi}_{i+1/2} &= \frac{1}{h}\int_{x_i}^{x_i+h} e^{\mathrm{j}kx}\,dx \\
&= \frac{1}{\mathrm{j}kh}\left(e^{\mathrm{j}k(x_i+h)} - e^{\mathrm{j}kx_i}\right) \\
&= \frac{e^{\mathrm{j}kh}-1}{\mathrm{j}kh} e^{\mathrm{j}kx_i}.
\end{aligned} \tag{2.28}$$

We first revisit the standard second-order central FD approximation. Substituting the single Fourier mode yields:

$$\begin{aligned}
\left.\frac{\partial\phi}{\partial x}\right|_i &= \frac{\phi_{i+1}-\phi_{i-1}}{2h} \\
&= \frac{1}{2h}\left(e^{jk(x_i+h)} - e^{\mathrm{j}k(x_i-h)}\right) \\
&= \frac{e^{\mathrm{j}kh}-e^{-\mathrm{j}kh}}{2h} e^{\mathrm{j}kx_i} \\
&= \mathrm{j}\frac{\sin(kh)}{h} e^{\mathrm{j}kx_i}.
\end{aligned} \tag{2.29}$$

Comparing this to the exact analytical derivative:

$$\frac{d\phi}{dx} = \mathrm{j}k e^{\mathrm{j}kx_i}, \tag{2.30}$$

We extract the modified wavenumber for the second-order FD scheme:

$$k^*_{FDEX2} = \frac{\sin(kh)}{h}. \tag{2.31}$$

The comparison between k^* and the exact wavenumber k forms the foundation of modified wavenumber analysis, quantifying how accurately a spatial discretization propagates the phase of each Fourier mode.

We now apply this same methodology to the second-order MM approximation. Using the cell-averaged quantities:

$$\begin{aligned}\left.\frac{\partial \phi}{\partial x}\right|_i &= \frac{\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}}{h} \\ &= \frac{1}{h}\left[\frac{e^{jkh}-1}{jkh}e^{jkx_i} - \frac{1-e^{-jkh}}{jkh}e^{jkx_i}\right] \\ &= \frac{e^{jkh}+e^{-jkh}-2}{jkh^2}e^{jkx_i} \\ &= \frac{2\left(\cos(kh)-1\right)}{jkh^2}e^{jkx_i}. \end{aligned} \tag{2.32}$$

Rearranging this expression into the standard form of the exact derivative gives:

$$\left.\frac{\partial \phi}{\partial x}\right|_i = j\left[\frac{2\left(1-\cos(kh)\right)}{kh^2}\right]e^{jkx_i}. \tag{2.33}$$

Applying the half-angle identity, $1-\cos(kh) = 2\sin^2(kh/2)$, we obtain the initial modified wavenumber for the second-order MM approximation:

$$k^*_{MMEX2} = \frac{4\sin^2(kh/2)}{kh^2}. \tag{2.34}$$

This result warrants careful interpretation. Evaluated on the same physical grid, the MM approximation inherently contains more information than the corresponding FD scheme because both point values and cell averages are utilized. Consequently, its initial modified wavenumber curve remains closer to the exact analytical curve over a broader range of wavenumbers. However, this superior spectral resolution is largely attributable to the increased number of degrees of freedom per cell.

Because this quantity is derived by assuming that all MM variables are prescribed exactly from the continuous Fourier mode at the initial instant, we term it the *initial modified wavenumber*. This procedure, which we refer to as *Initial Modified Wavenumber Analysis* (IMWA), accurately describes the initial spatial response of the MM discretization. However, it does not necessarily predict the asymptotic, long-term wave speed of the fully coupled MM system.

The initial modified wavenumbers for the second-, fourth-, and eighth-order MM and FD schemes are plotted in Fig. 2.2(a). On the same physical grid, the MM schemes clearly resolve higher-frequency modes more accurately than their FD counterparts. To provide a fair comparison at an equivalent resolution per degree of freedom, we normalize the wavenumber to define an effective nondimensional wavenumber:

$$kh_{\text{eff}} = k\frac{h}{\text{nDoFC}}, \tag{2.35}$$

where nDoFC denotes the number of degrees of freedom per cell. These normalized results are presented in Fig. 2.2(b). Under this normalization, the IMWA curves indicate that the spatial resolution of the MM method is actually slightly inferior to the corresponding FD method. This conclusion aligns with the truncation error findings in Ex. 1.3, which examined the first-derivative approximations.

The fundamental limitation of IMWA becomes apparent when the MM method is applied dynamically to the convection equation. For a standard FD scheme, modified wavenumber analysis reliably predicts the long-term numerical wave speed because each Fourier mode evolves via a simple scalar amplification factor. In contrast, the MM method simultaneously

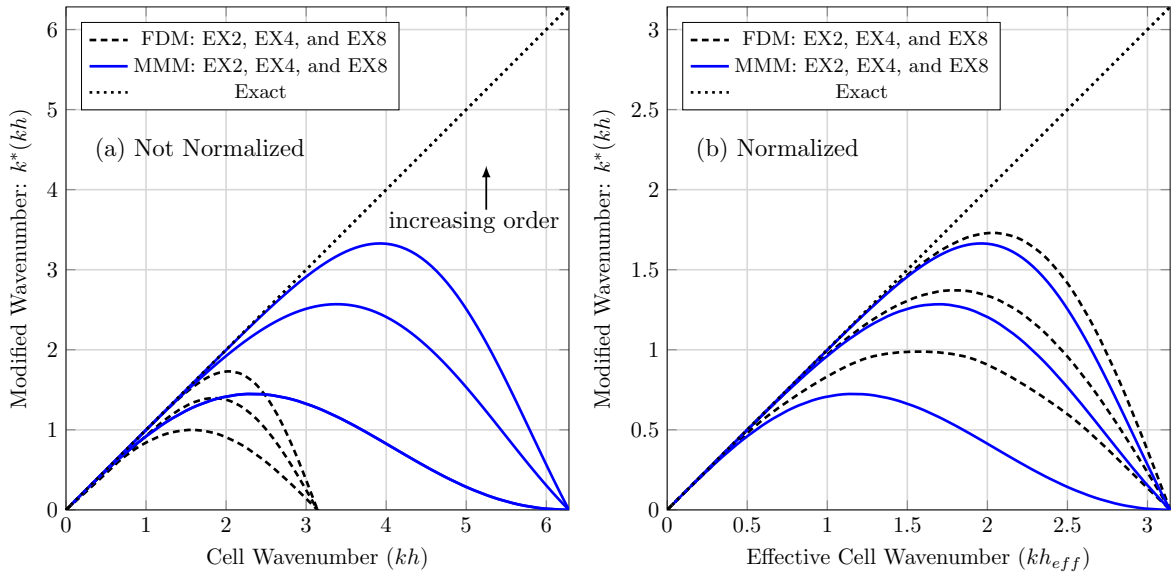


Fig. 2.2: Initial modified wavenumbers for multi-moment methods (MMM) and finite-difference methods (FDM) across second-, fourth-, and eighth-order schemes. (a) Results evaluated on the same physical grid. (b) Results normalized by the number of degrees of freedom (DoFs).

evolves two coupled quantities: point values and cell averages. Although they are initialized perfectly in phase from the same continuous Fourier mode, they do not remain phase-locked as the simulation progresses. Because IMWA cannot account for this dynamic phase separation, it only predicts the initial instantaneous response of the MM discretization. To determine the true asymptotic wave behavior, the Fourier response of the fully coupled system matrix must be evaluated.

2.3.2 Coupled Modified Wavenumber Analysis

In the initial modified-wavenumber analysis, we consider only the very first instant of the simulation, where both point values and cell averages are prescribed exactly from the analytical initial condition. Consequently, the cell-average evolution is exact at $t = 0$, since the flux difference across the cell boundaries is evaluated from exact point-value data. However, the finite-difference derivative of the point-value data yields a non-exact modified wavenumber, introducing numerical error into the point values after the first time step. This dispersive error then enters the finite-volume update through the boundary fluxes and is subsequently fed back into the point-value update, resulting in a dynamically coupled system.

For this reason, the point values and cell averages must be treated as distinct variables, even though they are initially sampled from the same continuous function. We denote variables associated with the finite-difference point value with a tilde, $\widetilde{(\cdot)}$, and variables associated with the finite-volume cell average with a wide-arc accent, $\widehat{(\cdot)}$. By applying the Fourier ansatz to every term in the coupled spatial discretization, the evolution of each Fourier mode k can be expressed as a linear system:

$$\frac{\partial}{\partial t}\begin{bmatrix} \widetilde{\phi_k} \\ \widehat{\phi_k} \end{bmatrix} = -juk \begin{bmatrix} \mathcal{B}_2 & \mathcal{B}_1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \widetilde{\phi_k} \\ \widehat{\phi_k} \end{bmatrix}, \tag{2.36}$$

where $\mathcal{B}_1$ and $\mathcal{B}_2$ represent the complex transfer coefficients derived from the multi-moment spatial derivative approximations.

The eigensolution of this system yields the Coupled Modified Wavenumber (CMWA) for the multi-moment data:

$$k^*_{CMWA}(kh) = \frac{k}{2}\left(\sqrt{\mathcal{B}_2^2 + 4\mathcal{B}_1} + \mathcal{B}_2\right), \tag{2.37}$$

in contrast to the Initial Modified Wavenumber Analysis (IMWA), which predicts the linear combination:

$$k^*_{IMWA}(kh) = k\left(\mathcal{B}_1 + \mathcal{B}_2\right). \tag{2.38}$$

The coupled modified wavenumbers of the second-, fourth-, and eighth-order schemes applied to the one-dimensional convection equation are plotted in Fig. 2.3.

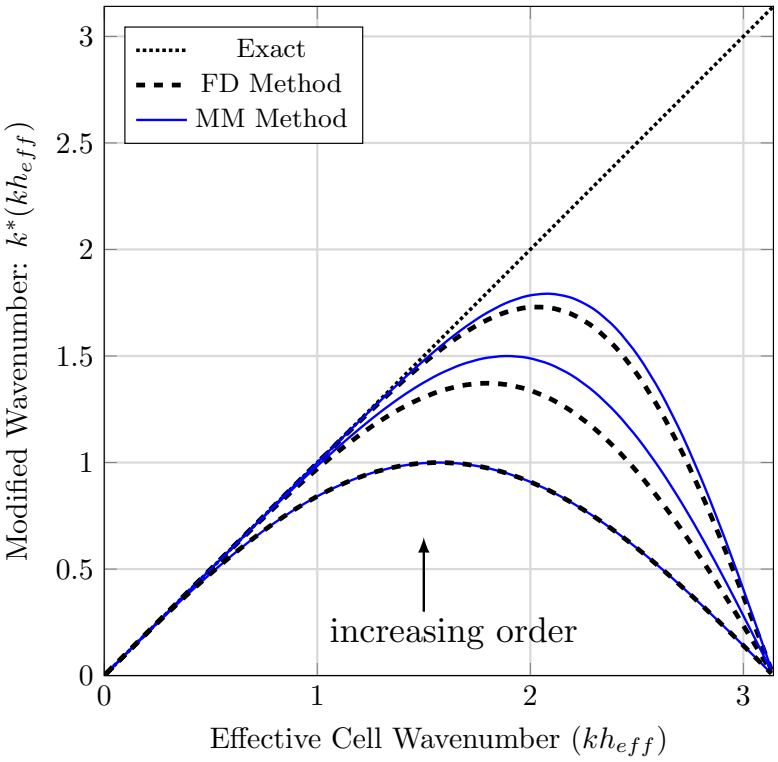


Fig. 2.3: Coupled Modified Wavenumber (CMWA) of explicit multi-moment methods and standard modified wavenumber of the finite-difference methods for second-, fourth-, and eighth-order schemes.

To establish a quantitative comparison of resolving efficiency between the two methodologies, we evaluate the ratio of the absolute wave-speed error in the multi-moment method to the corresponding error in the standard finite-difference method of the same formal order:

$$\text{Error Ratio} = \frac{\left|k - k^*_{MM}(kh_{eff})\right|}{\left|k - k^*_{FD}(kh_{eff})\right|}. \tag{2.39}$$

These error ratios are plotted in Fig. 2.4. The solid lines show the error ratio of the fully coupled system (CMWA). In contrast, the dashed lines show the theoretical error ratio if one considers only the uncoupled initial response (IMWA). The fully coupled system shows a much lower dispersive error than the uncoupled initial response alone. For the second-order

scheme, the CMWA numerical dispersion properties are identical to those of the finite-difference method after normalization by the number of degrees of freedom. However, for the high-order schemes, the multi-moment method demonstrates superior spectral resolution. Specifically, the solid lines indicate that the asymptotic dispersive error of the fourth- and eighth-order MMM schemes is less than half that of their corresponding FDM counterparts at low wavenumbers. Theorem 2.1 guarantees that the asymptotic wave speed is always faster than the initial wave speed under the multi-moment method.

Theorem 2.1 (Monotonic Dispersion Bound of Coupled Multi-Moment Systems) *Let $\mathcal{B}_1(kh)$ and $\mathcal{B}_2(kh)$ denote the complex transfer functions representing a consistent multi-moment spatial discretization for a first-order linear convection equation. If the discretization satisfies the fundamental consistency and stability requirements such that the composite spatial transfer function magnitude is bounded by unity, i.e.,*

$$\mathcal{B}_1(kh) + \mathcal{B}_2(kh) \leq 1, \tag{2.40}$$

then the Initial Modified Wavenumber $k^*_{\mathrm{IMWA}}(kh)$ serves as a lower bound for the asymptotic Coupled Modified Wavenumber $k^*_{\mathrm{CMWA}}(kh)$ across the stable resolvable spectrum:

$$k^*_{\mathrm{IMWA}}(kh) \leq k^*_{\mathrm{CMWA}}(kh). \tag{2.41}$$

Proof. By definition, the analytical expressions for the initial and coupled modified wavenumbers derived from the spatial transfer functions are given by:

$$k^*_{\mathrm{IMWA}}(kh) = k\left(\mathcal{B}_1 + \mathcal{B}_2\right), \tag{2.42}$$

$$k^*_{\mathrm{CMWA}}(kh) = \frac{k}{2}\left(\sqrt{\mathcal{B}_2^2 + 4\mathcal{B}_1} + \mathcal{B}_2\right). \tag{2.43}$$

For a forward-propagating wave mode where the physical wavenumber is strictly positive ($k > 0$), we construct the target algebraic inequality:

$$k\left(\mathcal{B}_1 + \mathcal{B}_2\right) \leq \frac{k}{2}\left(\sqrt{\mathcal{B}_2^2 + 4\mathcal{B}_1} + \mathcal{B}_2\right). \tag{2.44}$$

$$2\mathcal{B}_1 + \mathcal{B}_2 \leq \sqrt{\mathcal{B}_2^2 + 4\mathcal{B}_1}. \tag{2.45}$$

Because the transfer functions represent a physically stable, non-amplifying spatial derivative operator, both the left-hand and right-hand expressions are non-negative. Squaring both sides preserves the direction of the inequality:

$$(2\mathcal{B}_1 + \mathcal{B}_2)^2 \leq \mathcal{B}_2^2 + 4\mathcal{B}_1. \tag{2.46}$$

$$4\mathcal{B}_1^2 + 4\mathcal{B}_1\mathcal{B}_2 \leq 4\mathcal{B}_1. \tag{2.47}$$

Factoring out $4\mathcal{B}_1$ (where $\mathcal{B}_1 > 0$ represents the primary, forward-consistent finite-volume flux transfer component) reduces the inequality to:

$$\mathcal{B}_1 + \mathcal{B}_2 \leq 1. \tag{2.48}$$

The condition $\mathcal{B}_1 + \mathcal{B}_2 \leq 1$ is satisfied identically by the consistency condition of any stable spatial derivative approximation, ensuring the scheme does not yield unphysical, super-inductive phase velocities. Thus, the premise holds, proving that $k^*_{\mathrm{IMWA}}(kh) \leq k^*_{\mathrm{CMWA}}(kh)$ across the entire stable spectrum. □

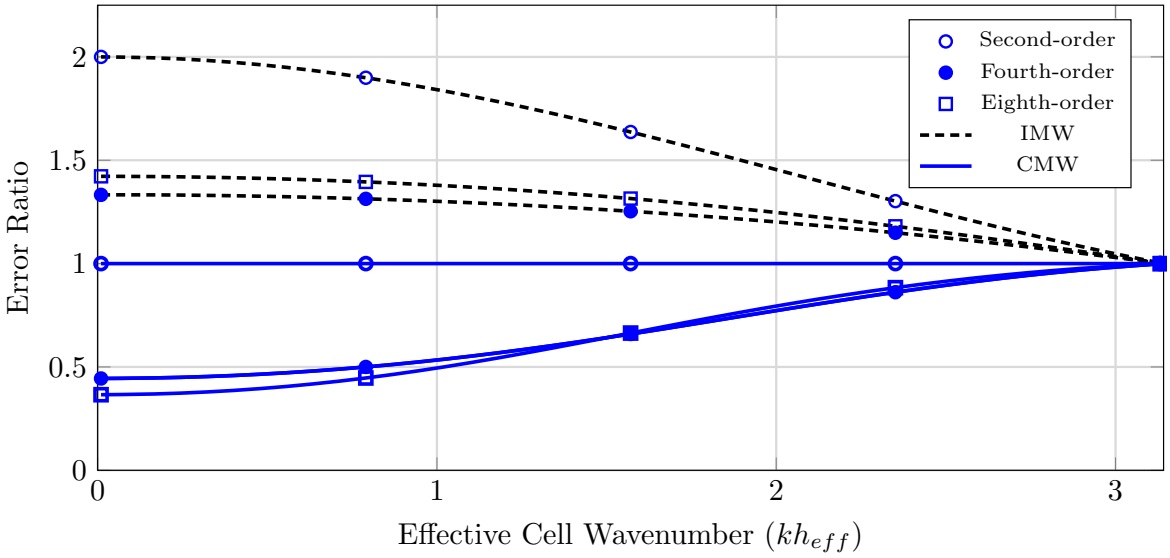


Fig. 2.4: The error ratio between the multi-moment methods and the finite-difference methods as defined in Eq. (2.39). The dashed lines represent the ratio of the initial modified wavenumber (IMWA) error to the FD error. In contrast, the solid lines represent the ratio using the fully coupled modified wavenumber (CMWA).

2.4 Compact Multi-Moment Scheme

While explicit numerical schemes excel in their simplicity of implementation and straightforward parallelization, they typically require progressively wider spatial stencils to achieve high-order accuracy. To circumvent the physical non-localities and boundary-treatment complexities associated with wide explicit stencils, Lele [24] introduced a seminal family of compact finite-difference schemes. By implicitly coupling the unknown derivative values at adjacent grid points within the discretization matrix, these compact formulations significantly reduce the required spatial stencil width for a target order of accuracy, and thereby lessen the boundary closure problem.

For instance, the author implemented a fourth-order compact finite-volume scheme for turbulent flow simulations [12] and observed a tenfold increase in computational efficiency compared to classical second-order methods. This dramatic performance gain arises from a multi-dimensional scaling advantage: while the localized evaluation cost of higher-order stencils scales linearly with the number of coupling points, the structural advantages gained from grid-point reduction scale cubically in three spatial dimensions. Furthermore, when factoring in the larger permissible time steps allowed by a coarser mesh under the Courant-Friedrichs-Lewy (CFL) condition, the total computational savings scale to the fourth power.

The fourth-order explicit multi-moment scheme can be improved to the sixth-order scheme by adding two implicit derivatives:

$$-\frac{1}{8}\phi'_{i-1} + \phi'_i - \frac{1}{8}\phi'_{i+1} = \frac{3}{h}\left(\overline{\phi}_{i+\frac{1}{2}} - \overline{\phi}_{i-\frac{1}{2}}\right) - \frac{9}{8h}\left(\phi_{i+1} - \phi_{i-1}\right). \tag{2.49}$$

Alternatively, this can be viewed as an enhancement of the Pade scheme by adding two cell average values. The final size of this stencil is still just two cells wide — *the exact size of the second-order central difference in FDM*. The general form of the compact multi-moment method is given below:

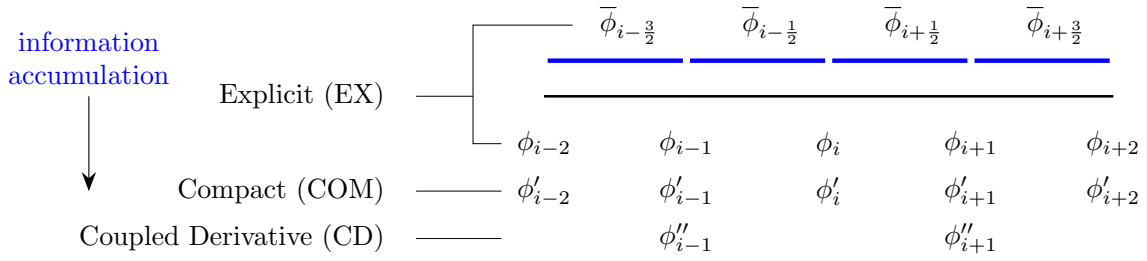


Fig. 2.5: Illustration of one-dimensional multi-moment data arrangement centered at a nodal point ϕ_i. The cell-averaged value $\overline{\phi}$ is defined over the control volume between two adjacent nodes. In the multi-moment method, both nodal and cell-averaged data are utilized to compute spatial derivatives, enhancing the accuracy of numerical schemes.

$$\phi'_i + \sum_{\tau=1}^{m_3} a_\tau \phi'_{i\pm\tau} = \sum_{\tau=1}^{m_1} c_\tau (\overline{\phi}_{i+\tau-\frac{1}{2}} - \overline{\phi}_{i-\tau+\frac{1}{2}}) + \sum_{\tau=1}^{m_2} b_\tau (\phi_{i+\tau} - \phi_{i-\tau}). \tag{2.50}$$

On the LHS, we have the implicit first derivatives, and on the RHS, we have the two known data types—the cell average and the pointwise values. This type of implicit scheme is referred to as a compact scheme, following the tradition of finite difference methods.

We recommend following the data-locality principle: using the cell average first (m_1-sum), then the pointwise data (m_2-sum), then the implicit term (m_3-sum), and cycle back. The coefficients of the implicit method up to the 24th order of convergence are listed in Tab.2.3. The rightmost column shows the ratios between the compact and explicit multi-moment schemes. At intermediate order of accuracy, the compact scheme is only slightly better than the explicit counterpart inside the convergence regime. As the order of accuracy grows, the error of the compact scheme is eventually three orders of magnitude smaller than that of the explicit scheme. However, this particular ratio is only meaningful when the grid size is extremely fine. The true value of the compact scheme lies in the powerful resolving power on coarse grids.

Order	a_1	a_2	a_3	a_4	b_1	b_2	b_3	b_4	c_1	c_2	c_3	c_4	$LTE_{COM/EX}$
6	$-\frac{1}{8}$	—	—	—	3	—	—	—	$-\frac{9}{8}$	—	—	—	$\frac{1}{4}$
8	$-\frac{1}{6}$	—	—	—	$\frac{13}{4}$	$-\frac{1}{108}$	—	—	$-\frac{23}{18}$	—	—	—	$\frac{1}{4}$
10	$-\frac{2}{9}$	—	—	—	$\frac{7}{2}$	$-\frac{1}{18}$	—	—	$-\frac{38}{27}$	$\frac{1}{108}$	—	—	$\frac{1}{9}$
12	$-\frac{8}{27}$	$\frac{1}{216}$	—	—	$\frac{15}{4}$	$-\frac{65}{324}$	—	—	$-\frac{40}{27}$	$\frac{25}{432}$	—	—	$\frac{4}{81}$
14	$-\frac{1}{3}$	$\frac{1}{120}$	—	—	$\frac{139}{36}$	$-\frac{7}{24}$	$\frac{1}{9000}$	—	$-\frac{3}{2}$	$\frac{109}{1200}$	—	—	$\frac{1}{36}$
16	$-\frac{3}{8}$	$\frac{3}{200}$	—	—	$\frac{143}{36}$	$-\frac{241}{576}$	$\frac{431}{360000}$	—	$-\frac{3}{2}$	$\frac{279}{2000}$	$-\frac{1}{6000}$	—	$\frac{1}{64}$
18	$-\frac{27}{64}$	$\frac{27}{1000}$	$-\frac{1}{8000}$	—	$\frac{49}{12}$	$-\frac{455}{768}$	$\frac{17549}{2400000}$	—	$-\frac{189}{128}$	$\frac{2079}{10000}$	$-\frac{147}{80000}$	—	$\frac{9}{1600}$
20	$-\frac{9}{20}$	$\frac{9}{250}$	$-\frac{1}{3500}$	—	$\frac{199}{48}$	$-\frac{2123}{3000}$	$\frac{1021}{75000}$	$-\frac{1}{686000}$	$-\frac{291}{200}$	$\frac{633}{2500}$	$-\frac{909}{245000}$	—	$\frac{9}{2500}$
22	$-\frac{12}{25}$	$\frac{6}{125}$	$-\frac{4}{6125}$	—	$\frac{101}{24}$	$-\frac{12643}{15000}$	$\frac{1873}{75000}$	$-\frac{281}{12005000}$	$-\frac{178}{125}$	$\frac{191}{625}$	$-\frac{1578}{214375}$	$\frac{1}{343000}$	$\frac{1}{625}$
24	$-\frac{64}{125}$	$\frac{8}{125}$	$-\frac{64}{42875}$	$\frac{1}{343000}$	$\frac{205}{48}$	$-\frac{50053}{50000}$	$\frac{6769}{150000}$	$-\frac{522313}{2521050000}$	$-\frac{864}{625}$	$\frac{228}{625}$	$-\frac{21408}{1500625}$	$\frac{2283}{48020000}$	$\frac{4}{5625}$

Table 2.3: Coefficients of the central compact multi-moment approximation of the first derivative (Eq.(2.49)) and the ratio between the local truncation errors of the compact scheme and the explicit scheme ($LTE_{COM/EX}$).

2.5 Optimality of the Multi-Moment Method

The fundamental concept of using multiple data types to reinforce the accuracy of numerical approximations is not entirely new. Perhaps the earliest proposal for approximating partial differential equations can be attributed to van Leer [23], who proposed improving the accuracy of numerical simulations by introducing cell boundary values as primary flow variables alongside the original cell averages. This concept remains highly popular in modern compressible flow solvers, with contemporary adaptations known as active flux (AF) schemes [7, 31]. However, in most conventional frameworks, tracking these additional moments requires complex approximations—either to evolve them through the governing equations or to interpolate them from surrounding quantities. This evolution and interpolation must be performed with accuracy comparable to the original moments; otherwise, truncation errors will rapidly propagate and degrade the scheme's overall accuracy.

To overcome this challenge without inflating the computational burden, the proposed multi-moment method (MMM) strategically couples a finite-difference method (FDM) for the point values with a finite-volume method (FVM) for the cell averages. This dual-scheme architecture not only ensures that both moments are evolved with the requisite high-order accuracy but also aggressively minimizes algorithmic overhead. By delegating the conservative update entirely to the FVM framework and evolving the interface value by the FDM framework, the traditional costs associated with sub-grid polynomial reconstruction are bypassed. This structural advantage is formalized in the following theorem.

Theorem 2.2 (Minimality of the Finite-Volume Conservation Cost) *Consider the finite-volume (FV) update of a cell-averaged moment $\widehat{\phi}_{i-1/2}$ for the one-dimensional convection equation. The computational cost of this update reaches its absolute theoretical minimum if and only if the exact boundary interface fluxes are supplied explicitly by an independent, concurrent solver step (such as a finite-difference collocation).*

Proof. In a standard high-order finite-volume framework, updating the cell average requires evaluating the numerical fluxes at the cell boundaries. To achieve p-th order accuracy, one must apply a sub-grid reconstruction operator $\mathcal{R} : \widehat{\phi} \to \widetilde{\phi}_{i\pm1/2}$ to determine these interface values from a surrounding stencil of cell averages. The operation count for this reconstruction scales polynomially with the order p in one-dimensional space.

If the interface state variables $\widetilde{\phi}_i$ are instead tracked and evolved as primary unknowns by a concurrent algebraic step, the reconstruction operator becomes the identity matrix, and its associated computational cost vanishes identically ($C_{\text{reconstruct}} \equiv 0$). The finite-volume update then simplifies to a single linear subtraction of these pre-evaluated nodal fluxes:

$$\frac{\partial \widehat{\phi}_{i-1/2}}{\partial t} + \frac{u}{\Delta x}\left(\widetilde{\phi}_i - \widetilde{\phi}_{i-1}\right) = 0. \tag{2.51}$$

Because this operation requires only a single subtraction per cell, the incremental complexity of the conservation step is strictly $O(1)$, completely independent of the formal order of accuracy p. This represents the absolute lower bound of computational complexity for a conservative update. □

While Theorem 2.2 establishes the local efficiency of the finite-volume step, the overarching goal of any high-order scheme is to minimize the total computational effort required to achieve a target spatial resolution across the entire physical domain. Because the multi-moment framework embeds multiple degrees of freedom within a single computational cell,

it inherently reduces the total number of cells required to resolve a given wave spectrum. This structural compactness directly translates into a global reduction in floating-point operation (FLOP) count, as demonstrated below.

Theorem 2.3 (Computational Complexity Reduction of High-Order Multi-Moment Methods) *To achieve a p-th order ($p > 2$) spatial resolution of a one-dimensional convection equation for a given total number of degrees of freedom N, the multi-moment method (MMM) requires strictly fewer floating-point operations over the global domain than a standard finite-difference or finite-volume scheme of equivalent order, provided the computational cost per degree of freedom of the standard scheme exceeds the cost of a localized conservative flux difference.*

Proof. Let M denote the computational cost (in floating-point operations) to evaluate a p-th order accurate spatial derivative per degree of freedom. To resolve the continuous domain with N total degrees of freedom, a standard numerical scheme incurs a global computational cost of:

$$C_{\text{FDM}} = M \cdot N \tag{2.52}$$

The multi-moment framework evolves two distinct moments per cell: the point value $\widetilde{\phi}_i$ and the cell average $\widehat{\phi}_{i+1/2}$. Therefore, to achieve the required N total degrees of freedom, the MMM requires only $N_{cell} = N/2$ computational cells.

The global cost of the MMM is the direct sum of the point-value updates evaluated at the $N/2$ cell boundaries and the updates applied to the $N/2$ cell averages. Drawing upon Theorem 2.2, the finite-volume update incurs a localized cost of exactly one operation (a single subtraction). Defining c as this minimal constant cost ($c \leq 2$) required to compute the net flux difference, the global operational cost of the coupled multi-moment method is formulated as:

$$\begin{aligned} C_{\text{MMM}} &= C_{\text{FDM}} + C_{\text{FVM}} \\ &= \left(M \cdot \frac{N}{2} \right) + \left(c \cdot \frac{N}{2} \right) \\ &= \frac{M + c}{2} N \end{aligned} \tag{2.53}$$

For the multi-moment method to be computationally cheaper, the condition $C_{\text{MMM}} < C_{\text{Standard}}$ requires that $\frac{M+c}{2} < M$, which simplifies to $c < M$.

In any practical high-order scheme ($p > 2$), the baseline evaluation cost M involves multiple stencil operations and strictly exceeds the minimal cost of a flux difference ($M > 2 \geq c$). Consequently, the inequality holds. This demonstrates that the multi-moment architecture reduces global operational complexity by exploiting the minimal $O(1)$ cost of the cell update. □

Example 2.4 (Computational Cost of an Explicit Multi-Moment Discretization) Consider an explicit finite-difference discretization employing a spatial stencil of width p. Evaluating the discrete derivative for a single degree of freedom requires p multiplications and $p - 1$ additions, yielding a local operational cost of:

$$M = 2p - 1. \tag{2.54}$$

For a conventional single-moment discretization containing N total degrees of freedom, the resulting global computational cost is:

$$C_{\text{FDM}} = (2p - 1)N. \tag{2.55}$$

In the multi-moment framework, the simultaneous evolution of nodal and cell-averaged moments within each computational cell dictates that only $N/2$ cells are needed to represent the same N degrees of freedom. Because the finite-volume update requires only a minor, constant-cost flux difference operation (denoted as c), the dominant computational expense remains the finite-difference evaluations at the nodes. The total operational cost is thus:

$$C_{\text{MMM}} = ((2p - 1) + c)\,\frac{N}{2} = \left(\frac{2p + c - 1}{2}\right) N. \tag{2.56}$$

Comparing the two costs, the multi-moment method is strictly more efficient ($C_{\text{MMM}} < C_{\text{FDM}}$) for all stencil widths satisfying $p > (c + 1)/2$. Furthermore, the ratio of their global computational expenses is given by:

$$\frac{C_{\text{MMM}}}{C_{\text{FDM}}} = \frac{2p + c - 1}{4p - 2}. \tag{2.57}$$

As $p \to \infty$, this ratio approaches 0.5. Because the conservative flux cost c is typically much smaller than the cost of high-order schemes ($c \ll 2p$), the multi-moment formulation fundamentally reduces the global complexity of explicit discretizations. For high-order schemes, this method approximately halves the required floating-point operations, offering near-proportional savings in computational time.

2.5.1 Theoretical Relative Performance of the Multi-Moment Method

In Theorem 2.3, we established that for the same order of convergence, the multi-moment scheme costs half as much as the standard FDM in terms of global operations. However, different approximations behave differently across spectral resolutions. Local truncation error only explains the error at the asymptotic limit, which may be hard to achieve in practice. In realistic cases, such as Direct Numerical Simulation (DNS), we usually resolve the smallest relevant scales down to several percent. In other classes of simulation, such as Large-Eddy Simulations (LES), we may discard these small scales completely and model their effects instead. Thus, to quantify the accuracy of numerical schemes at different spectral resolutions, Lele [24] proposed the *resolving efficiency*:

$$\mathcal{E}(\epsilon) = \frac{k^* h_{\max}}{\pi}, \quad \text{where} \quad \frac{|k^* h - kh|}{kh} \leq \epsilon. \tag{2.58}$$

This metric indicates the fraction of the wave space a scheme can capture with a relative error less than or equal to ϵ. To normalize this, we define the effective resolving efficiency:

$$\mathcal{E}_{\text{eff}}(\epsilon) = \frac{k^* h_{\text{eff,max}}}{\pi}, \quad \text{where} \quad \frac{|k^* h_{\text{eff}} - kh|}{k h_{\text{eff}}} \leq \epsilon, \tag{2.59}$$

using the effective wavenumber defined previously. We can convert this effective resolving efficiency into the number of cells per wavelength (CPW) required to resolve the wave for a target relative error:

$$\text{CPW}(\epsilon) = \frac{2}{\mathcal{E}_{\text{eff}}(\epsilon) \times N_{\text{DoF/C}}}, \tag{2.60}$$

where $N_{\text{DoF/C}}$ is the number of degrees of freedom per cell, representing the data density.

Next, we determine the theoretical computational complexity to solve a time-dependent one-dimensional linear convection equation while meeting the desired accuracy requirement:

$$\mathbb{C}_\epsilon = N_{\text{DoF/C}} \times (M + 1) \times \text{CPW}^2(\mathcal{E}_{\text{eff}}(\epsilon)), \tag{2.61}$$

which represents the number of floating-point operations per wavelength per unit time. In general, computing time is proportional to this complexity, assuming memory hierarchy access costs (such as registers, cache levels, and main memory) remain consistent.

Having determined the complexity, we can define a performance metric for a numerical method A solving a 1D convection equation by evaluating its cost relative to the second-order finite difference (FDEX2) baseline:

$$\mathcal{S}_\epsilon(A) = \frac{\mathbb{C}_\epsilon(\text{FDEX2})}{\mathbb{C}_\epsilon(A)}. \tag{2.62}$$

Strictly speaking, this is the ratio of the computational cost of the baseline FDEX2 scheme to that of method A. Assuming that computing time scales linearly with computational complexity, this ratio indicates how much faster method A can solve the system to the desired accuracy compared to FDEX2. Thus, we refer to $\mathcal{S}_\epsilon(A)$ as the *theoretical speedup* relative to the second-order finite difference method. This metric provides an objective measure of algorithmic efficiency that is entirely independent of software implementation and hardware architecture.

We select eight multi-moment schemes—four explicit and four compact (implicit)—and report their computational cost per cell, resolving efficiency, required cells per wavelength (CPW), and theoretical speedup in Table 2.4. The reported computational cost for each multi-moment scheme includes contributions from both the finite-difference and finite-volume components. It should be emphasized that the present analysis considers only resolvable spectral modes; aliasing errors and nonlinear mode interactions are not included.

As expected, the required number of cells per wavelength decreases as the formal order and spectral resolving capability of the schemes increase. However, the computational cost per cell also increases with order, implying that higher-order schemes are not necessarily the most computationally efficient for all error tolerances. Among the explicit multi-moment schemes, the lowest-order method (MMEX4) is most efficient at the relaxed threshold ($\epsilon = 0.1$), whereas MMEX8 and MMEX10 become optimal at $\epsilon = 0.01$ and $\epsilon = 0.001$, respectively. Within the compact multi-moment family, MMCOM6 performs best at $\epsilon = 0.1$, while MMCOM12 and MMCOM10 yield the highest theoretical speedup at $\epsilon = 0.01$ and $\epsilon = 0.001$, respectively. These results indicate that, for a given spectral error tolerance, an optimal discretization exists that balances spectral accuracy against floating-point cost.

Method	Cost/Cell (M)	Resolving Efficiency ($\mathcal{E}(\epsilon)$)			$N_{\mathrm{DoF/C}}$	CPW(ϵ)			$\mathcal{S}_\epsilon$		
		0.1	0.01	0.001		0.1	0.01	0.001	0.1	0.01	0.001
FDEX2	3	0.251	0.078	0.025	1	7.99	25.61	81.10	1.00	1.00	1.00
FDEX4	7	0.444	0.240	0.133	1	4.51	8.35	15.02	1.35	4.04	12.16
FDCOM4	8	0.594	0.356	0.205	1	3.37	5.63	9.77	2.11	7.76	25.07
FDCOM6	12	0.702	0.503	0.352	1	2.85	3.98	5.69	1.97	10.33	50.41
MMEX4	9	0.501	0.285	0.162	2	2.00	3.51	6.19	**2.66**	8.87	28.44
MMEX6	13	0.567	0.376	0.255	2	1.77	2.66	3.92	2.35	10.69	49.17
MMEX8	17	0.630	0.457	0.340	2	1.59	2.19	2.94	2.23	**12.07**	67.02
MMEX10	21	0.660	0.499	0.390	2	1.51	2.00	2.57	1.99	11.69	**71.29**
MMCOM6	14	0.681	0.475	0.327	2	1.47	2.11	3.06	**3.17**	15.79	114.41
MMCOM8	18	0.731	0.551	0.412	2	1.37	1.82	2.43	2.83	16.51	154.49
MMCOM10	22	0.777	0.620	0.491	2	1.29	1.61	2.03	2.62	17.26	**179.82**
MMCOM12	26	0.819	0.684	0.567	2	1.22	1.46	1.76	2.47	**17.72**	121.88

Table 2.4: Computational cost, resolving efficiency $\mathcal{E}(\epsilon)$, number of degrees of freedom per cell ($N_{\mathrm{DoF/C}}$), required grid resolution CPW(ϵ), and theoretical speedup ($\mathcal{S}_\epsilon$) relative to the second-order FDM baseline (FDEX2) across different error thresholds. The best performers for explicit and compact multi-moment schemes are bolded. Here we assume the FVM flux difference cost c is 2 floating-point operations.

To better isolate the origin of the observed performance gains, it is instructive to compare MMEX4 against FDEX4 and MMCOM6 against FDCOM6. For the explicit schemes, the multi-moment formulation achieves a theoretical speedup ranging from approximately 1.97 to 2.34 relative to its conventional counterpart across the error thresholds considered. By comparison, the structural cost reduction predicted solely from the increased number of degrees of freedom per cell, as discussed in Theorem 2.3, suggests a speedup factor of only 1.56. The additional improvement therefore cannot be attributed solely to data-density effects; rather, it indicates that the multi-moment discretization also provides superior spectral resolving efficiency compared with the conventional explicit finite-difference formulation.

The compact schemes exhibit a different scaling behavior. Although MMCOM6 remains approximately 1.53 to 1.61 times faster than FDCOM6 at the two largest error thresholds, this improvement is smaller than the theoretical factor of 1.71 predicted from the structural advantage associated with increased data density. This reduction suggests that the compact finite-difference formulation already has excellent spectral resolution in under-resolved regimes, reducing the relative advantage of the multi-moment representation. At the highest accuracy threshold, however, the advantage of MMCOM6 over FDCOM6 increases to approximately 2.27, indicating that the multi-moment formulation regains superior resolving efficiency in the fully resolved regime.

This behavior is fundamentally governed by stencil compactness and the density of spatial data. For explicit discretizations, the multi-moment method occupies roughly the same spatial footprint as a conventional explicit finite-difference method at twice the grid resolution. In contrast, compact finite-difference schemes attain high spectral resolution via implicit coupling between neighboring derivative evaluations, yielding much narrower effective stencils. As a result, a compact finite-difference scheme operating at doubled grid resolution maintains a smaller stencil than its multi-moment counterpart. This relationship is directly

reflected in the required cells-per-wavelength values. For example, while the standard fourth-order explicit finite-difference scheme requires more than twice as many cells as MMEX4 to achieve equivalent error tolerance, the sixth-order compact baseline (FDCOM6) needs fewer than twice as many cells as MMCOM6. Thus, the relative advantage of the multi-moment approach diminishes for compact discretizations, where baseline schemes already exhibit highly optimized spectral properties. Nevertheless, multi-moment schemes of both classes remain substantially faster than their FDM counterparts across all error thresholds.

2.6 Basic Time Integration

After spatial discretization, the governing equation is reduced to a semi-discrete ordinary differential equation in time. Using the scalar solution variable ϕ, this semi-discrete form may be written as

$$\frac{d\phi}{dt} = \mathcal{L}(\phi), \tag{2.63}$$

where $\mathcal{L}$ denotes the spatial discretization operator. In the present context, $\mathcal{L}$ may represent either the conventional finite-difference operator or the multi-moment spatial discretization introduced previously. To solve for the future value of ϕ, we integrate the above equation

$$\int_t^{t+\tau} \frac{d\phi}{dt} dt = \int_t^{t+\tau} \mathcal{L}(\phi) dt. \tag{2.64}$$

for some positive τ. This results in

$$\int_t^{t_n+\tau} d\phi = \int_t^{t+\tau} \mathcal{L}(\phi) dt. \tag{2.65}$$

$$\phi(t+\tau) - \phi(t) = \int_t^{t+\tau} \mathcal{L}(\phi) dt. \tag{2.66}$$

Then we discretize the time to a collection of t_n, $n = 0, \cdots, N_t - 1$ with $t_n = n\triangle t$ and rewrite the above equation as

$$\phi^{n+1} = \phi^{n+1} + \int_{t_n}^{t_{n+1}} \mathcal{L}(\phi) dt. \tag{2.67}$$

The process of computing the integration on the RHS is called *time-integration*. In this section, three basic time integration methods are reviewed: the Explicit Euler method, the Implicit Euler method, and the Crank–Nicolson method.

2.6.1 Explicit Euler Method

Assuming that the spatial operator $\mathcal{L}(\phi)$ varies continuously in time, we can visualize its temporal evolution as a curve on a graph (see Fig. 2.6a). We can approximate the exact temporal integral of $\mathcal{L}(\phi)$ over a time step Δt by assuming it remains constant at its initial value, $\mathcal{L}(\phi^n)$. Graphically, this corresponds to drawing a rectangle *forward* from the known state to estimate the area under the curve. This yields the explicit update:

$$\phi^{n+1} = \phi^n + \Delta t\, \mathcal{L}(\phi^n). \tag{2.68}$$

In many textbooks, the explicit Euler scheme is instead introduced by approximating the continuous time derivative with a forward finite-difference stencil:

$$\frac{\phi^{n+1} - \phi^n}{\Delta t} = \mathcal{L}(\phi^n). \tag{2.69}$$

This derivative-based derivation yields the same discrete formulation as the integral approach. While approximating the time derivative works well for establishing fundamental first- and second-order methods, the concept of time integration provides a more rigorous foundation for developing higher-order evolution schemes.

The explicit Euler method is first-order accurate in time ($\mathcal{O}(\Delta t)$) and requires only a single evaluation of the spatial operator per time step. Consequently, it is computationally simple and inexpensive. However, its stability is conditionally restricted by a strict time-step limit, and for certain classes of purely dispersive wave equations, it is unconditionally unstable.

2.6.2 Implicit Euler Method

Following the integral perspective, we can alternatively approximate the area under the continuous curve $\mathcal{L}(\phi)$ by assuming it remains constant at its *future* value, $\mathcal{L}(\phi^{n+1})$, throughout the time step. Graphically, as shown in Fig. 2.6b, this corresponds to evaluating the function at the unknown future state and drawing a rectangle *backward*. This yields the implicit update:

$$\phi^{n+1} = \phi^n + \Delta t\, \mathcal{L}(\phi^{n+1}). \tag{2.70}$$

Equivalent to the explicit scheme, this can also be derived by approximating the time derivative, this time utilizing a backward finite-difference stencil evaluated at the future state:

$$\frac{\phi^{n+1} - \phi^n}{\Delta t} = \mathcal{L}(\phi^{n+1}). \tag{2.71}$$

Because of the directionality of their respective rectangular area approximations, the explicit and implicit Euler methods are frequently referred to as the *Forward Euler* and *Backward Euler* time-integration schemes.

Because the spatial operator $\mathcal{L}$ is evaluated at the unknown time level $n + 1$, the method strictly requires the solution of a coupled algebraic system at each time step. For example, if a linear spatial discretization is applied such that $\mathcal{L}(\phi)$ is represented by a discrete matrix operator $\boldsymbol{L}$, we obtain the following matrix-vector system:

$$\boldsymbol{\phi}^{n+1} = \boldsymbol{\phi}^n + \Delta t\, \boldsymbol{L}\boldsymbol{\phi}^{n+1}. \tag{2.72}$$

To advance the solution, this system must be rearranged and inverted algebraically:

$$(\boldsymbol{I} - \Delta t\, \boldsymbol{L})\, \boldsymbol{\phi}^{n+1} = \boldsymbol{\phi}^n, \tag{2.73}$$

where $\boldsymbol{I}$ is the identity matrix. Although solving this implicit system significantly increases the computational cost per time step compared to explicit methods, the implicit Euler method possesses robust, unconditional stability properties for linear diffusion problems (A-stability), making it highly effective for simulating stiff governing equations.

2.6.3 Crank–Nicolson Method

Returning to the graphical perspective (Fig. 2.6c), we can improve upon the rectangular area approximations by instead assuming that the spatial operator varies linearly between the current time t_n and the future time t_{n+1}. This corresponds to estimating the area under the curve $\mathcal{L}(\phi)$ using a trapezoid. Analytically, this is equivalent to averaging the values of the spatial operator at the known and unknown time levels, yielding the update form:

$$\phi^{n+1} = \phi^n + \frac{\Delta t}{2}\left[\mathcal{L}(\phi^n) + \mathcal{L}(\phi^{n+1})\right]. \tag{2.74}$$

In finite-difference terms, this scheme can also be derived by applying a central difference approximation of the time derivative evaluated at the temporal midpoint $t_{n+1/2}$:

$$\frac{\phi^{n+1} - \phi^n}{\Delta t} = \frac{1}{2}\left[\mathcal{L}(\phi^n) + \mathcal{L}(\phi^{n+1})\right]. \tag{2.75}$$

A visual comparison of the three basic time-integration methods illustrates why this averaging is so effective. As shown in Fig. 2.6, if the gradient of $\mathcal{L}(\phi)$ is monotonic across the integration interval, the forward Euler and backward Euler methods will systematically under-predict and over-predict the true integral (or vice versa). By averaging these two rectangular extremes into a trapezoid, the Crank–Nicolson method effectively cancels the leading-order truncation errors, achieving second-order accuracy in time ($O(\Delta t^2)$).

Because it retains the unknown future state ϕ^{n+1} inside the spatial operator on the right-hand side, the method is implicit. Like the implicit Euler scheme, it requires the solution of an algebraic system at each time step. However, the substantial gain in temporal accuracy often justifies this additional computational cost.

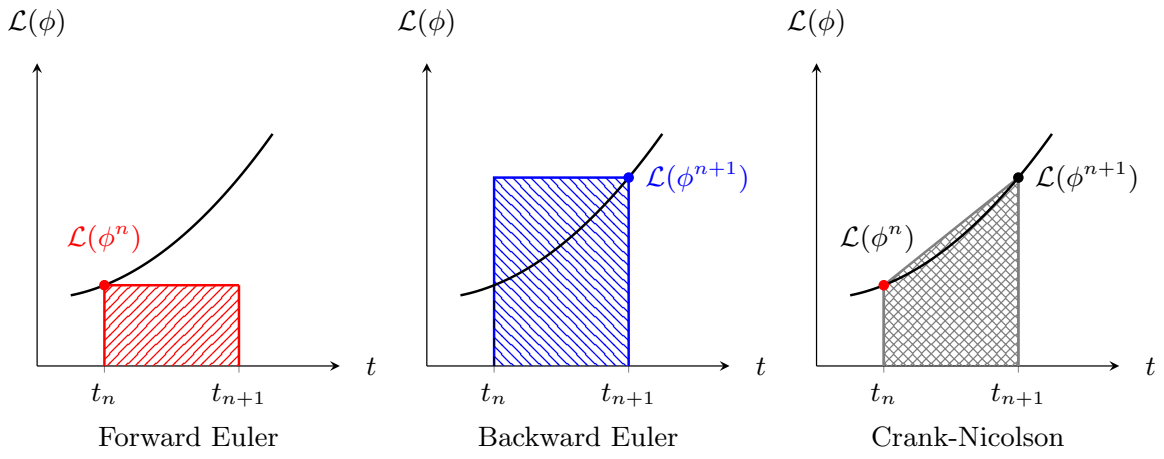


Fig. 2.6: Geometrical interpretation of the three fundamental time-integration schemes. To advance the solution from t_n to t_{n+1}, the exact temporal integral of the spatial operator $\mathcal{L}(\phi)$ is approximated by a forward rectangle (Explicit Euler, left), a backward rectangle (Implicit Euler, center), and a trapezoid (Crank–Nicolson, right).

2.6.4 Leapfrog Method

The Euler methods may be interpreted geometrically as first-order approximations of the temporal integral using either a forward or backward rectangular rule. A natural extension is to employ a centered approximation in time, thereby achieving second-order temporal accuracy. From an integral perspective, the spatial operator is evaluated at the intermediate time level (t_n), which serves as the midpoint of the interval $[t_{n-1}, t_{n+1}]$. The resulting midpoint approximation advances the solution directly from t_{n-1} to t_{n+1} over a time interval of length $2\Delta t$ (see Fig.2.7).

An equivalent derivation follows from approximating the time derivative with a central-difference stencil:

$$\left.\frac{\partial \phi}{\partial t}\right|_{t_n} \approx \frac{\phi^{n+1} - \phi^{n-1}}{2\Delta t}. \tag{2.76}$$

Substituting this approximation into the semi-discrete equation

$$\frac{\partial \phi}{\partial t} = \mathcal{L}(\phi) \tag{2.77}$$

yields

$$\frac{\phi^{n+1} - \phi^{n-1}}{2\Delta t} = \mathcal{L}(\phi^n). \tag{2.78}$$

Rearranging gives the update formula

$$\phi^{n+1} = \phi^{n-1} + 2\Delta t, \mathcal{L}(\phi^n). \tag{2.79}$$

Because the solution advances from (t_{n-1}) directly to (t_{n+1}), effectively "leaping over" the intermediate level (t_n), this scheme is known as the *Leapfrog method*. The method is second-order accurate in time, with a truncation error of $O(\Delta t^2)$, while requiring only a single evaluation of the spatial operator per time step. Consequently, its computational cost is comparable to that of the Explicit Euler method.

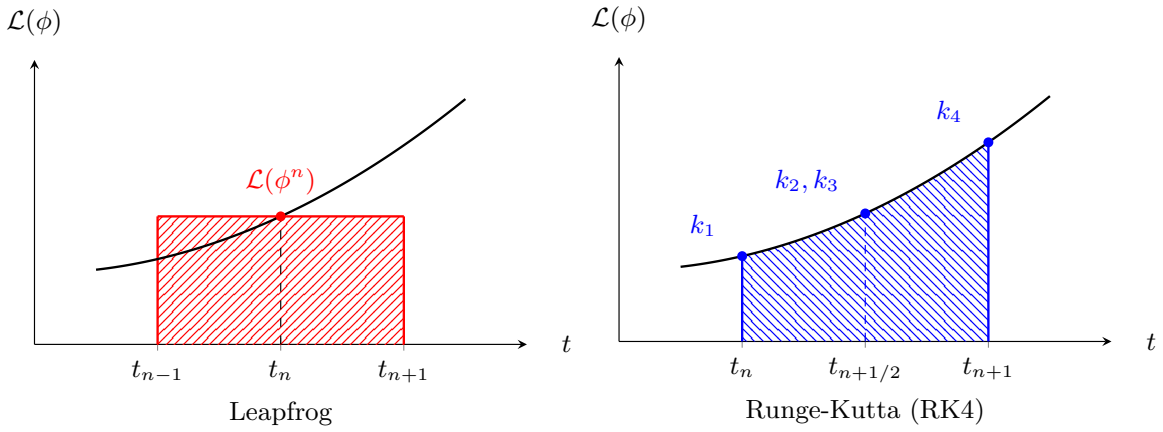


Fig. 2.7: Geometrical interpretation of Leapfrog (left) and fourth-order Runge-Kutta (right) time integration.

A notable disadvantage of the Leapfrog scheme is that it is not self-starting. Since the update formula requires data from two previous time levels, the computation of (ϕ^1) cannot

proceed directly from the initial condition (ϕ^0). An auxiliary one-step method, such as Forward Euler, Runge–Kutta, or Crank–Nicolson, must therefore be employed to generate the first intermediate solution before the Leapfrog recurrence can be applied.

The stability properties of the Leapfrog method are highly problem dependent. For linear hyperbolic equations, such as the advection equation, the method is conditionally stable under the Courant–Friedrichs–Lewy (CFL) restriction. Furthermore, it is non-dissipative, meaning that it introduces no artificial damping of wave amplitudes. This property makes the scheme particularly attractive for long-time wave propagation problems encountered in acoustics, atmospheric dynamics, and ocean modeling.

In contrast, the Leapfrog method performs poorly when applied to diffusive operators. For the diffusion equation, von Neumann analysis shows that the amplification factor exceeds unity for all nonzero wavenumbers, rendering the scheme unconditionally unstable. Consequently, pure Leapfrog discretizations are rarely employed for parabolic problems and are generally unsuitable for viscous terms without additional stabilization or operator-splitting techniques.

2.6.5 Runge–Kutta Methods

To achieve higher-order temporal accuracy without resorting to multi-step schemes (which are not self-starting) or implicit averaging (which requires computationally expensive matrix inversions), we can employ multi-stage methods. The Runge–Kutta (RK) family of time-integration schemes accomplishes this by evaluating the spatial operator at several intermediate states within a single time step Δt.

Geometrically, rather than assuming a single constant value for the spatial operator evaluated at the beginning of the step (as represented by the forward rectangle in the Explicit Euler method), RK methods take intermediate trial steps to sample the value of $\mathcal{L}(\phi)$ across the integration interval. These trial values are then weighted and averaged to compute a highly accurate final area under the curve, effectively capturing the nonlinear evolution of $\mathcal{L}(\phi)$ over time.

The most widely used variant in fluid dynamics and numerical analysis is the classical fourth-order Runge–Kutta (RK4) method. The scheme advances the solution from t_n to t_{n+1} through four sequential explicit stages:

$$
\begin{aligned}
k_1 &= \mathcal{L}(\phi^n), \\
k_2 &= \mathcal{L}\left(\phi^n + \frac{\Delta t}{2} k_1\right), \\
k_3 &= \mathcal{L}\left(\phi^n + \frac{\Delta t}{2} k_2\right), \\
k_4 &= \mathcal{L}\left(\phi^n + \Delta t\, k_3\right).
\end{aligned}
$$

Here, k_1 represents the initial value of the spatial operator at t_n, k_2 and k_3 provide two distinct estimates of the operator's value at the temporal midpoint $t_{n+1/2}$, and k_4 estimates the value at the endpoint t_{n+1}. The final update is computed as a weighted average of these four intermediate values, giving maximum mathematical weight to the midpoint evaluations (conceptually analogous to Simpson's rule for numerical integration):

$$
\phi^{n+1} = \phi^n + \frac{\Delta t}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right). \tag{2.80}
$$

If we regroup the integral part to

$$\frac{\Delta t}{6}\left(k_1 + 4((k_2 + 2k_3)) + k_4\right),$$

This shows a direct link to the 1/3-Simpson's rule.

Because the RK4 method is a single-step scheme that relies strictly on information from the current time level n, it is entirely self-starting, unlike the Leapfrog method. Furthermore, its purely explicit formulation entirely avoids the need to solve coupled algebraic systems. This method is fourth-order accurate in time ($O(\Delta t^4)$). Although it requires four separate evaluations of the spatial operator $\mathcal{L}$ per time step—effectively quadrupling the baseline computational expense relative to the Explicit Euler method—this local cost is generally justified. The superior temporal accuracy is complemented by a significantly expanded stability region that encompasses a large portion of the imaginary axis. This makes the RK4 scheme highly robust for advection-dominated wave propagation problems, allowing for the use of considerably larger time steps while maintaining strict error tolerances.

A comprehensive comparison of the fundamental time-integration methods discussed in this section is provided in Table 2.5. The method to determine the stability will be discussed in the next section.

Method	Temporal Order	Stability Characteristics	Implicit Solve	Self-Starting
Explicit Euler	1st	Conditionally stable	No	Yes
Implicit Euler	1st	Unconditionally stable (A-stable)	Yes	Yes
Crank–Nicolson	2nd	Unconditionally stable (A-stable)	Yes	Yes
Leapfrog	2nd	Conditionally stable (Neutral)	No	**No**
Runge–Kutta 4	4th	Conditionally stable (Expanded)	No	Yes

Table 2.5: Comparison of fundamental time-integration methods.

2.7 Stability of Time Integration

Numerical approximations of dynamical systems are not inherently stable. Even when the corresponding analytical solution remains bounded, a numerical solution may exhibit unphysical growth and eventually diverge. Understanding the mechanisms that drive such instabilities is therefore essential for designing reliable time-integration methods.

Before investigating the stability of fully discretized partial differential equations, it is useful to isolate the effects of temporal discretization alone. Studying time-integration schemes applied to ordinary differential equation initial value problems allows one to assess the intrinsic properties of a temporal scheme independently of any spatial discretization effects.

To this end, consider the classical linear scalar test equation

$$\frac{d\phi}{dt} = \lambda\phi, \qquad \phi(0) = \phi_0,$$

where λ is a constant complex number. Within the method of lines framework, λ may be interpreted as an eigenvalue of the spatial discretization operator $\mathcal{L}$. The exact solution is

$\phi(t) = \phi_0 e^{\lambda t}$, whose qualitative behavior is determined entirely by $\mathrm{Re}(\lambda)$. When $\mathrm{Re}(\lambda) > 0$, the solution grows without bound; when $\mathrm{Re}(\lambda) \leq 0$, it remains bounded or decays. Since physically stable systems are typically associated with $\mathrm{Re}(\lambda) \leq 0$, this regime is the primary focus of stability analysis.

A numerical scheme is said to be stable if it reproduces this bounded behavior in the discrete solution. The central objective of linear stability analysis is therefore to determine the conditions under which a scheme preserves the qualitative character of the exact solution, and to identify any restrictions on the time-step size required to guarantee stability. When a time-integration scheme is applied to the test equation, the discrete update takes the form

$$\phi^{n+1} = G(z)\,\phi^n,$$

where $z = \lambda \Delta t$ is a complex variable and $|G(z)|$ is called the *amplification factor* of the scheme because

$$|G(z)| = \frac{|\phi^n|}{|\phi^{n+1}|} \tag{2.81}$$

This factor governs how the amplitude evolves over a single time step. Since the exact solution satisfies $|e^z| \leq 1$ whenever $\mathrm{Re}(\lambda) \leq 0$, preventing unphysical growth requires

$$|G(z)| \leq 1.$$

The discrepancy between $|e^z|$ and $|G(z)|$ constitutes the *amplitude error*, or numerical dissipation. If $|G(z)| < |e^z|$, the scheme over-damps the solution; if $|G(z)| > |e^z|$, it artificially amplifies it.

Based on their behavior with respect to this criterion, time-integration schemes fall into three stability categories:

- **Unconditionally stable:** The numerical solution remains bounded for any choice of Δt. This robustness typically requires an implicit formulation, which carries a higher computational cost per step.
- **Unconditionally unstable:** The numerical solution grows without bound regardless of how small Δt is chosen. No matter what truncation accuracy such a scheme may possess in theory, it is useless in practice.
- **Conditionally stable:** The numerical solution remains bounded only when Δt satisfies specific constraints, such as a Courant–Friedrichs–Lewy condition. The practical utility of these schemes depends on whether the required time-step restriction is so severe that the overall simulation becomes computationally prohibitive.

2.7.1 Stability Regions in the Complex Plane

We are concerned with the stability of the numerical approximation when the analytical solution remains bounded.

This means that when $z = \lambda \Delta t$ is located to the left of the imaginary axis in the complex plane, the amplitude of the numerical solution should be bounded for all positive Δt. The region enclosed by this boundary is known as the *stability region* of the time-integration scheme. If the product of the time step Δt and the spatial operator's eigenvalue λ falls inside this region, the integration will remain stable.

2.7.2 Stability Analysis for the Euler Methods

To evaluate the stability of the Euler methods, we apply them to the linear test equation, $d\phi/dt = \lambda\phi$.

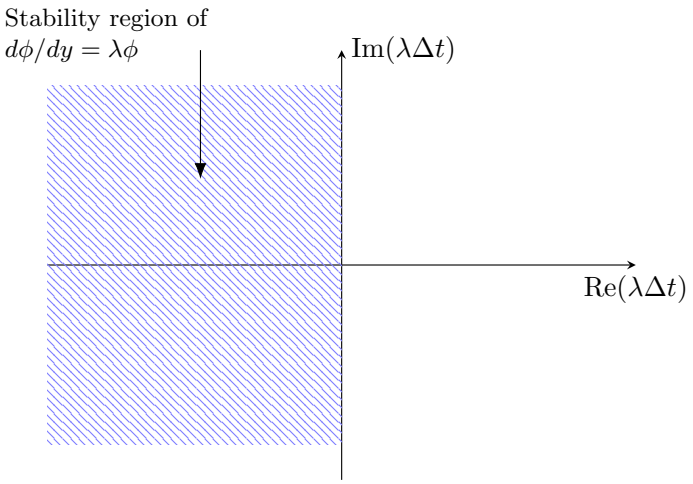


Fig. 2.8: Stability region of the model equation (Eq. (2.82)), illustrating the domain where the amplitude of the oscillatory solution remains bounded.

Explicit Euler Method

The explicit (forward) Euler method applied to the test equation reads:

$$\phi^{n+1} = \phi^n + \lambda\Delta t\phi^n. \tag{2.82}$$

Factoring out ϕ^n yields the amplification factor, G:

$$G_{\text{explicit}} = \frac{\phi^{n+1}}{\phi^n} = 1 + \lambda\Delta t. \tag{2.83}$$

For the numerical scheme to be stable, the magnitude of the amplification factor must not exceed unity ($|G| \leq 1$). This yields the stability condition:

$$|1 + \lambda\Delta t| \leq 1. \tag{2.84}$$

Unlike the exact analytical solution, which is stable across the entire left half of the complex $\lambda\Delta t$-plane, this inequality defines a restricted numerical stability region. Specifically, it is bounded by a solid circle of radius 1 centered at -1, as shown in Fig. 2.9. Consequently, if λ is purely real, the method is conditionally stable provided $-2 \leq \lambda\Delta t \leq 0$. However, if λ is a non-zero, purely imaginary number (representing a purely oscillatory system), the product $\lambda\Delta t$ lies on the imaginary axis. Because the stability circle only touches the imaginary axis at the origin, any non-zero time step pushes the solution outside the stability region. Thus, the explicit Euler method is *unconditionally unstable* for purely oscillatory problems.

Implicit Euler Method

Conversely, the implicit (backward) Euler method evaluates the right-hand side at the new time level:

$$\phi^{n+1} = \phi^n + \lambda \Delta t \phi^{n+1}. \tag{2.85}$$

Rearranging the terms to isolate ϕ^{n+1} gives:

$$(1 - \lambda \Delta t)\, \phi^{n+1} = \phi^n, \tag{2.86}$$

which results in the following amplification factor:

$$G_{\text{implicit}} = \frac{\phi^{n+1}}{\phi^n} = \frac{1}{1 - \lambda \Delta t}. \tag{2.87}$$

Applying the stability criterion ($|G| \leq 1$) yields:

$$|1 - \lambda \Delta t| \geq 1. \tag{2.88}$$

This inequality defines a stability region consisting of the entire complex plane *outside* a circle of radius 1 centered at +1. Because the entire left half-plane (where the exact continuous solution is physically stable) is fully encompassed by this region, the implicit Euler method is *unconditionally stable* (A-stable) for any physically stable system. This allows the use of much larger time steps compared to the explicit method, though it requires solving an implicit equation at each step and can introduce significant artificial damping.

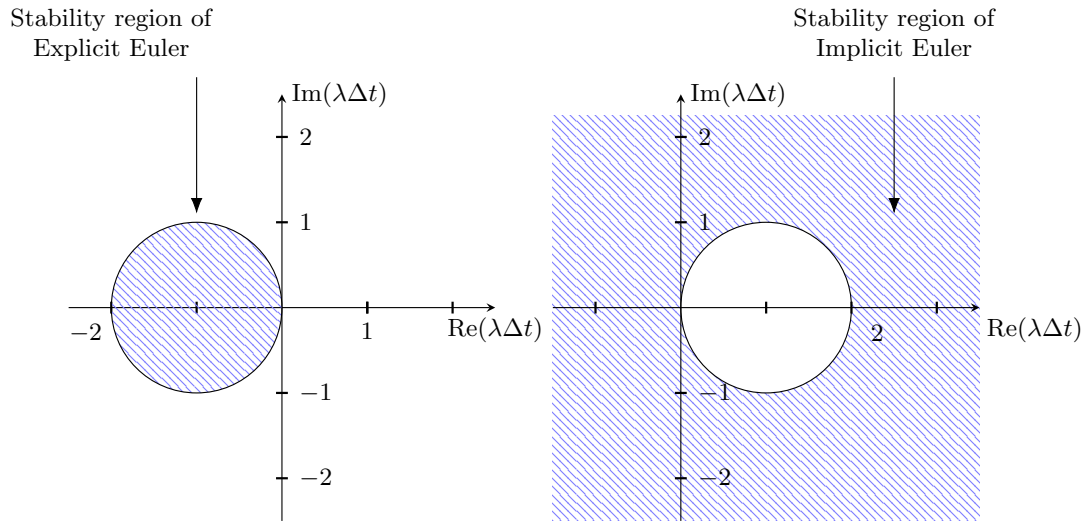


Fig. 2.9: Stability regions in the complex $\lambda \Delta t$-plane. The explicit Euler method (left) is stable only within the shaded circle, rendering it unconditionally unstable for purely oscillatory problems. Conversely, the implicit Euler method (right) is stable everywhere outside the unshaded circle.

Crank-Nicolson Method

The Crank-Nicolson (CN) method is an implicit scheme that improves upon the temporal accuracy of the Euler methods by effectively averaging the explicit (forward) and implicit

(backward) right-hand side evaluations:

$$\phi^{n+1} = \phi^n + \frac{\lambda \Delta t}{2}(\phi^n + \phi^{n+1}). \tag{2.89}$$

Rearranging the terms to isolate ϕ^{n+1} gives:

$$\left(1 - \frac{\lambda \Delta t}{2}\right) \phi^{n+1} = \left(1 + \frac{\lambda \Delta t}{2}\right) \phi^n, \tag{2.90}$$

which results in the following amplification factor:

$$G_{\text{CN}} = \frac{\phi^{n+1}}{\phi^n} = \frac{1 + \frac{\lambda \Delta t}{2}}{1 - \frac{\lambda \Delta t}{2}}. \tag{2.91}$$

Applying the stability criterion ($|G| \leq 1$) yields:

$$\left|1 + \frac{\lambda \Delta t}{2}\right| \leq \left|1 - \frac{\lambda \Delta t}{2}\right|. \tag{2.92}$$

This inequality is satisfied whenever the real part of $\lambda \Delta t$ is not greater than zero. Therefore, the stability region consists of the entire left half of the complex plane. Like the implicit Euler method, the Crank-Nicolson method is *unconditionally stable* (A-stable) for any physically stable system, allowing for the use of much larger time steps than explicit methods.

Furthermore, because the scheme is centered in time, it achieves second-order temporal accuracy. However, unlike the implicit Euler method, which introduces significant artificial damping, the Crank-Nicolson method yields an amplification factor magnitude of exactly 1 ($|G| = 1$) for purely imaginary eigenvalues. This lack of damping is advantageous for preserving wave amplitudes in pure advection problems. Thus, the CN method is *fully conserved*. It also means that the scheme does not damp high-frequency numerical errors at very large time steps, which can occasionally lead to spurious, non-decaying oscillations.

2.7.3 Fourth-order Runge-Kutta Methods

To further extend our stability analysis, we apply two widely used higher-order methods—the Leapfrog scheme and the fourth-order Runge-Kutta (RK4) method—to the linear test equation, $d\phi/dt = \lambda \phi$. For mathematical brevity and convention, we introduce the dimensionless complex parameter $z = \lambda \Delta t$.

Leapfrog Method (Explicit Midpoint)

The Leapfrog method is a two-step explicit scheme that utilizes a central difference in time. Applied to the test equation, it reads:

$$\phi^{n+1} = \phi^{n-1} + 2z\phi^n. \tag{2.93}$$

To find the amplification factor, G, we assume solutions of the form $\phi^n = G^n \phi^0$. Substituting this into the numerical scheme yields the characteristic equation:

$$G^2 - 2zG - 1 = 0. \tag{2.94}$$

Solving for G using the quadratic formula gives two roots:

$$G_{1,2} = z \pm \sqrt{z^2 + 1}. \tag{2.95}$$

For the scheme to be stable, the magnitude of both roots must not exceed unity ($\max(|G_1|, |G_2|) \leq 1$). Mathematically, this condition is satisfied *only* when z is a purely imaginary number and falls within the interval:

$$-1 \leq \mathrm{Im}(z) \leq 1, \quad \text{with} \quad \mathrm{Re}(z) = 0. \tag{2.96}$$

In the complex z-plane, this stability region contains strictly zero area; it collapses entirely to a line segment on the imaginary axis between $-i$ and i (see Fig.2.10). Consequently, the Leapfrog method is conditionally stable for purely oscillatory problems (such as pure advection), but it is *unconditionally unstable* if the physical system contains any dissipation or diffusion (which would introduce a non-zero real component to z).

Fourth-Order Runge-Kutta Method (RK4)

The standard fourth-order Runge-Kutta method is a multi-stage explicit scheme. Applying the four stages of RK4 to the test equation results in a series of evaluations that culminate in the update:

$$\phi^{n+1} = \left(1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}\right)\phi^n. \tag{2.97}$$

Factoring out ϕ^n reveals that the amplification factor for RK4 precisely matches the first five terms of the Taylor series expansion for the exact exponential solution, e^z:

$$G_{\mathrm{RK4}} = \frac{\phi^{n+1}}{\phi^n} = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}. \tag{2.98}$$

Applying the stability criterion yields the boundary inequality:

$$\left|1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}\right| \leq 1. \tag{2.99}$$

In the complex z-plane, this inequality defines a large, bulbous region in the left half-plane (see Fig.2.10). Notably, the region encompasses a significant portion of the negative real axis (extending to approximately -2.78) and covers a wide segment of the imaginary axis (extending to roughly $\pm 2.82i$). Because the RK4 stability region possesses both substantial area in the real left half-plane and a lengthy intersection with the imaginary axis, the method is highly versatile. It is conditionally stable for both strongly diffusive systems and purely oscillatory dynamics, provided the time step Δt is chosen such that z fits within these boundaries.

In this section, we have seen stability regions of several time-integration schemes. The region where the scheme conserves the amplitude of the signal perfectly is the borderline of the stability region. Inside it, the amplification factor is less than one. The exceptions are the Leapfrog and Crank-Nicolson time-integration schemes. In the next step, we will combine this study with the Modified Wave Number Analysis in the next section. Thus, most time integration is dissipative by nature. Note that while RK4's border looks like a straight vertical line near the origin, it is actually concave. Thus, the application of RK4 to the pure

oscillation problems introduces a very small ($O(h^5)$) diffusion at each time step, which sums up to ($O(h^4)$) at the end of the simulation. We can reduce the time-integration dissipation error by using a smaller time step. Later on, we will learn about the dissipation errors that originate from the spatial discretization, which cannot be improved by refining the time step size.

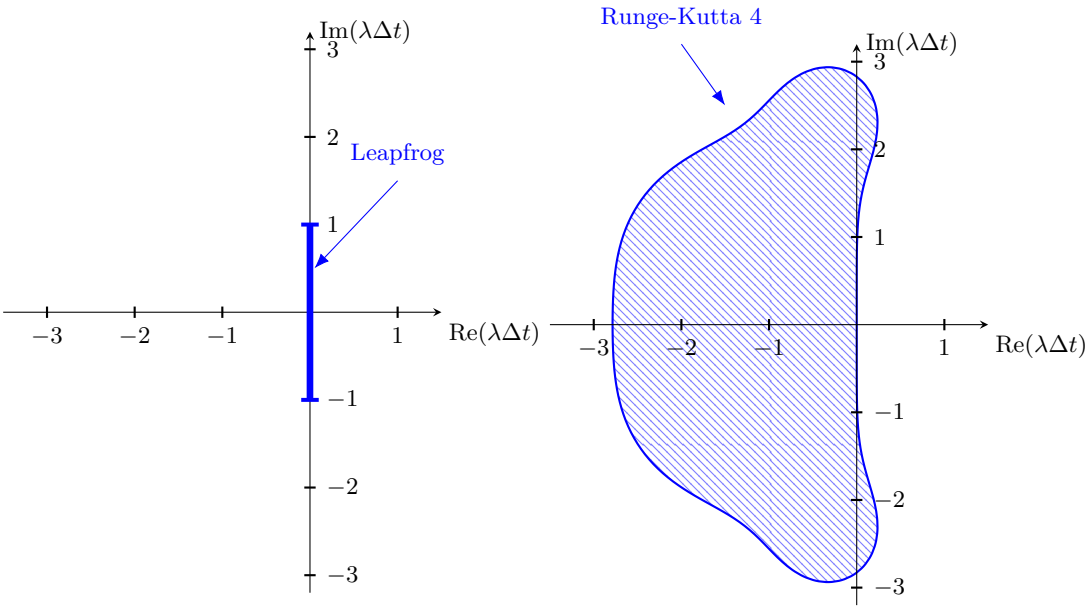


Fig. 2.10: Stability regions in the complex z-plane (where $z = \lambda\Delta t$). The Leapfrog method (left) possesses zero area, being stable only along the solid line segment on the imaginary axis between $-i$ and i. The fourth-order Runge-Kutta method (right) features a large stability region in the left half-plane, allowing conditional stability for both diffusive and oscillatory systems.

2.8 Stability of Discrete PDE Systems and von Neumann Analysis

While the ODE stability analysis evaluates how a time-integrator responds to a single eigenvalue, solving a PDE involves a vast matrix of coupled equations representing the entire spatial domain. The spatial discretization operator $\mathcal{L}$ possesses a complete spectrum of eigenvalues. For the fully coupled system to be stable, the scaled eigenvalues $\lambda\Delta t$ of *every resolvable spatial mode* must lie within the stability region of the chosen time integrator.

To evaluate this comprehensively for linear PDEs, we utilize *von Neumann stability analysis*. Rather than calculating discrete matrix eigenvalues, this method relies on Fourier decomposition. By assuming periodic boundary conditions, the numerical error is decomposed into a series of spatial harmonics (Fourier modes) of the form e^{jkx_m}, where k is the spatial wavenumber and $j = \sqrt{-1}$ is the imaginary unit.

By substituting a generic discrete Fourier mode $\phi_m^n = \hat{\phi}^n e^{jkx_m}$ into the fully discretized spatio-temporal scheme, we can derive a wavenumber-dependent amplification factor, $G(k\Delta x)$. For the overall scheme to be stable, the von Neumann criterion dictates that the amplification factor must not exceed unity for any resolvable wavenumber:

$$|G(k\Delta x)| \leq 1 \quad \text{for all} \quad -\pi \leq k\Delta x \leq \pi. \tag{2.100}$$

This analysis effectively maps the spatial footprint of the finite-difference or multi-moment discretization into the complex plane, allowing us to directly verify if it fits safely within the temporal integrator's stability bounds.

2.8.1 Coupling the Modified Wavenumber to the Temporal Parameter z

To understand how spatial discretization dictates the temporal stability limit, consider the 1D linear convection equation:

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = 0. \tag{2.101}$$

Substituting a continuous Fourier mode $\phi(x,t) = \hat{\phi}(t)e^{jkx}$ into the exact spatial derivative yields $-juk\hat{\phi}$. Thus, the exact continuous eigenvalue associated with the spatial operator is purely imaginary: $\lambda_{\text{exact}} = -juk$.

When the spatial derivative is approximated using a discrete grid (e.g., via finite differences), the numerical operator alters the wavenumber. We characterize this discrete dispersion error by introducing the **modified wavenumber**, k^*. The semi-discrete system in spectral space becomes:

$$\frac{d\hat{\phi}}{dt} = -juk^*\hat{\phi}, \tag{2.102}$$

yielding the discrete numerical eigenvalue $\lambda = -juk^*$.

We can now map the entire spatial spectrum directly into the complex stability parameter z used in our temporal ODE analysis:

$$z = \lambda\Delta t = -juk^*\Delta t = -j(k^*\Delta x)\text{CFL}, \tag{2.103}$$

where $\text{CFL} = u\Delta t/\Delta x$ is the Courant-Friedrichs-Lewy number. Because standard central difference schemes are skew-symmetric, k^* is strictly real. Consequently, the parameter z is purely imaginary for all spatial modes. We can then determine the maximum CFL number that maintains numerical stability via the inequality:

$$\text{CFL} \leq \frac{z_{\max}}{\max(k^*\Delta x)}. \tag{2.104}$$

For purely oscillatory problems, $z_{\max}$ represents the maximum magnitude of z along the imaginary axis that remains entirely within the time integrator's stability region. Determining the stability limit of the full PDE system now requires verifying that this locus of imaginary eigenvalues (z) fits entirely within the time integrator's stability region. The maximum value of the non-dimensional modified wavenumber across the resolvable spectrum, $\max(k^*\Delta x)$, ultimately dictates the maximum allowable CFL number.

2.8.2 Application to Specific Time Integrators

Explicit Euler Method

As established previously, the Explicit Euler method has an amplification factor of $G = 1 + z$. Substituting the convective parameter z yields:

$$G = 1 - juk^*\Delta t. \tag{2.105}$$

The magnitude of the amplification factor is:

$$|G|^2 = 1 + (uk^*\Delta t)^2. \tag{2.106}$$

For any non-trivial spatial mode ($k^* \neq 0$), $|G|^2$ is strictly greater than 1. Because the stability region of Explicit Euler only contacts the imaginary axis exactly at the origin ($z = 0$), every active wave mode is amplified at every time step. Thus, combining central spatial differences with the Explicit Euler method is unconditionally unstable, regardless of the time step size Δt.

Leapfrog Method (Explicit Midpoint)

The Leapfrog method perfectly complements purely convective problems because its stability region collapses to a line segment exactly along the imaginary axis, mathematically defined by $-1 \leq \mathrm{Im}(z) \leq 1$.

Substituting our purely imaginary $z = -juk^*\Delta t$, the stability criterion $|G| \leq 1$ translates directly to:

$$|uk^*\Delta t| \leq 1. \tag{2.107}$$

Dividing by Δx, we isolate the CFL number to define the stability limit:

$$(k^*\Delta x)\mathrm{CFL} \leq 1 \implies \mathrm{CFL} \leq \frac{1}{\max(k^*\Delta x)}. \tag{2.108}$$

For a standard second-order central difference scheme, the modified wavenumber is $k^*\Delta x = \sin(k\Delta x)$, yielding a maximum value of 1. Therefore, the formal stability limit for Leapfrog with second-order central differences is $\mathrm{CFL} \leq 1$. Higher-order schemes (such as 4th-order central differences) increase the maximum modified wavenumber, necessitating a proportionally stricter (lower) CFL limit to contain the spectrum within the stability boundary.

Fourth-Order Runge-Kutta (RK4)

The RK4 method is highly robust; its stability region covers a substantial portion of the left half-plane and intersects the imaginary axis at exactly $z = \pm j2\sqrt{2} \approx \pm 2.828j$. Requiring the entire discrete spatial spectrum to fit within this boundary yields:

$$|uk^*\Delta t| \leq 2\sqrt{2}. \tag{2.109}$$

Expressed in terms of the CFL number, the RK4 stability limit becomes:

$$\mathrm{CFL} \leq \frac{2\sqrt{2}}{\max(k^*\Delta x)}. \tag{2.110}$$

Coupled with a second-order central difference scheme ($\max(k^*\Delta x) = 1$), RK4 remains stable up to a generous CFL limit of approximately 2.828. This expansive imaginary boundary allows RK4 to process complex wave dynamics efficiently with relatively large time steps, overcoming the severe restrictions of simpler explicit schemes.

The theoretical stability limits, $z_{\max}/\max(k^*\Delta x)$, closely match those obtained from the spectral radius, $z_{\max}/\rho(\boldsymbol{C})$, where $\boldsymbol{C}$ is the spatial approximation of the derivative and fluxes of the MM method. This agreement confirms that the maximum relative wavenumber extracted from the Von Neumann analysis provides a reliable approximation of the true stability limit of the discretized system.

Although multi-moment methods exhibit increasingly restrictive time-step constraints as the spatial order increases, this behavior is a natural consequence of their enhanced spectral resolution. Higher-order spatial approximations resolve a broader range of high-frequency modes, which in turn requires smaller time steps to maintain temporal stability. Consequently, the stability limits of high-order multi-moment schemes gradually approach those of the spectral method at double resolution.

In this section, we perform a von Neumann analysis of the multi-moment method and compare it with the matrix method to determine whether the scheme has any unstable eigenvalues and how different time integration affects the method. To this end, we rewrite Eq.(1.25) and Eq.(1.26) for a periodic domain in the matrix form below

$$\mathrm{d}t\left(\begin{bmatrix}\widetilde{\boldsymbol{\phi}}\\ \widehat{\boldsymbol{\phi}}\end{bmatrix}^{n+1}\right)=\sigma_c\frac{\triangle x}{\triangle t}\boldsymbol{C}\begin{bmatrix}\widetilde{\boldsymbol{\phi}}\\ \widehat{\boldsymbol{\phi}}\end{bmatrix}^{n} \tag{2.111}$$

where σ_c is the CFL number $u\triangle t/\triangle x$, and

$$\boldsymbol{C}=\left[\begin{array}{c|c}\boldsymbol{B}_2 & \boldsymbol{B}_1\\ \hline \boldsymbol{E} & \boldsymbol{0}\end{array}\right]. \tag{2.112}$$

Matrix $\boldsymbol{B}_1$ and $\boldsymbol{B}_2$ are the approximation operator corresponding to m_1-sum and m_2-sum on the rhs of Eq.(2.8), respectively. Matrix $\boldsymbol{E}$ is a difference operator defined by two circulant Kronecker delta matrices $E_{ij}= -[\delta_{i,j}]/h + [\delta_{i,j+1}]/h$. The matrices $\boldsymbol{B}_1$ and $\boldsymbol{B}_2$ are skew-symmetric, but $\boldsymbol{E}$ is not. No method has yet been developed to predict the spectral radius of this matrix $\boldsymbol{C}$. Thus, we compute the spectral radius numerically and compare it with the Von Neumann analysis. *It should be noted that we have verified numerically that the eigenvalues of $\boldsymbol{C}$ are all imaginary and thus the method is inherently stable.*

When a Runge-Kutta method is applied to the decay initial value problem (IVP): $dy/dt = -\lambda$, with dt-time step size. We can determine the stability limit by solving for the complex root of the polynomial $z = \lambda dt$:

$$RK_m(z) = 1 + z + \frac{z^2}{2!} + \cdots + \frac{z^m}{m!}. \tag{2.113}$$

In a purely oscillating problem, where λ is imaginary, we can find the maximum stability limit λ such that $|RK_m(z)| \leq 1$. We call this value λdt_{max}. The value λdt here is treated as one variable. It can be shown easily through Von Neumann analysis that the numerical simulation of a convection problem with modified wave-number hk^* would be stable if

$$\frac{u\triangle t}{\triangle x} = CFL \leq \frac{\lambda dt_{max}}{hk^*_{max}}. \tag{2.114}$$

where hk^*_{max} is the highest value of the modified wave number obtained from the initial MWA. These values are compared with the spectral radius of the approximation matrix obtained from 1,000 cells (DOF=2,000). The values of λdt_{max} for each Runge-Kutta method up to 10th-order and hk^*_{max} for the selected schemes are displayed in Tab.3.3. The sixth-order

RK method is excluded because it is theoretically unstable for pure convection. In the table, we list the stability of the sixth-order FDM on the h-grid to emphasize its larger stability on the same grid. We also show the stability limit for the spectral method (SPM) on the double resolution ($h/2$-grid). Although the SPM can capture all π-waves, it requires more grids to resolve finer scales. The stability limits $\lambda h_{max}/hk^*_{max}$ are listed in the table. At first glance, the multi-moment method appears to require a smaller time step. However, this is a natural effect. The more accurate the approximation, the smaller the time step needed to accommodate the increased information. The stability limit of higher-order multi-moment schemes approaches that of the spectral method at double resolution. The stability limit according to the spectral radius can be easily obtained from the table using $\lambda dt_{max}/\rho(\boldsymbol{C})$. It can be seen that the maximum relative wave number from the von Neumann analysis is a reasonable approximation of this value.

Cell Per Wavelength	$[hk^*]_{max}$	$h\rho(\boldsymbol{C})$	CFL_{max}						
			RK3	RK4	RK5	RK7	RK8	RK9	RK10
λdt_{max}	—	—	1.73	2.83	3.39	1.76	3.39	4.57	5.26
FDCOM6	1.99	1.99	0.86	1.42	1.70	0.88	1.70	2.29	2.64
EX4	2.57	3.00	0.67	1.10	1.31	0.68	1.31	1.77	2.04
EX8	3.33	3.58	0.51	0.85	1.01	0.52	1.01	1.37	1.58
EX16	4.00	4.14	0.43	0.71	0.85	0.44	0.85	1.14	1.32
EX24	4.34	4.44	0.39	0.65	0.78	0.41	0.78	1.05	1.21
SPM (on $h/2$-grid)	6.28	6.28	0.27	0.45	0.53	0.28	0.53	0.72	0.83

Table 2.6: Stability limit $\lambda\Delta t_{max}/hk^*_{max}$ from the simplified von Neumann analysis. The limits shown are for the h-grid, except for the spectral method (SPM).

Finally, a comparison between the Initial Modified Wavenumber Analysis (IMWA) and the Coupled Modified Wavenumber Analysis (CMWA) reveals an important distinction. For MMEX4 through MMEX24, CMWA predicts maximum k^* values of 3.00, 3.58, 4.14, and 4.44, respectively, in exact agreement with those obtained from the spectral radius of the spatial approximation matrix. In contrast, IMWA consistently captures only the lower envelope of the spectral response and therefore systematically overestimates the corresponding CFL limits. As such, IMWA should be regarded as a preliminary indicator of stability rather than a rigorous tool for establishing stability bounds.

Furthermore, practical simulations rarely consist of pure linear advection. Physical source terms, nonlinear interactions, and complex boundary conditions can continuously generate unresolved high-frequency content that is not represented in the idealized Von Neumann analysis. For this reason, a conservative safety margin should always be maintained below the theoretical stability limit.

Regardless of the analysis method employed, operation near the absolute stability boundary is strongly discouraged. Under such conditions, particularly for very high-order multi-moment methods, the overall solution accuracy becomes dominated by temporal integration errors rather than spatial discretization errors. Moreover, to ensure accurate cell-to-cell information exchange—especially in multidimensional simulations—the CFL number should remain comfortably below unity. Since the CFL number represents the normalized distance traveled by information during a single time step, smaller values allow information from neighboring

cells to propagate and interact more effectively with orthogonal fluxes before the next temporal update, thereby improving the fidelity of multidimensional coupling.

2.9 Code Structure

Source code for this section is located in the folder ch01/MMM and ch01/FDM. The first folder contains the source files and input files of the multi-moment methods solving the convection equation. The second folder contains similar content for the finite-difference method. The MMM and FDM codes mirror each other, so we explain only the multi-moment codes here.

For a simple program like the one-dimensional convection equation, it is possible to implement everything in a single file. However, it will be difficult to expand and maintain the code. Thus, it is better to compartmentalize the code into several modules and identify common data communication so that changes in one small part do not unintentionally affect other parts. At this point, we have the main program *MMM_1Dconvect.f90* and the module file containing all other backend subroutines. The main computing activities are located in three positions in the code. The first part is the time-integration loop, which encapsulates the time-integration action within one subroutine. The main program calls *advance_one_step_inplace*, where each call advances the solution forward in time by *dt*. It is perfectly fine to solve backward in time by setting *dt* negative. But let us save the reverse problem for another time.

```
do i = 1, n_steps
call TimeIntegration(phi_pv, phi_cv, x, u, dx, dt, ncell,  &
           ng, work, spatial_tmp, method, scheme )
end do
```

Code 2.1: Main time integration loop

In large-scale computing, repeated memory allocation and deallocation can be costly. In some unfortunate cases, this can lead to memory leaks and memory management problems, especially when users do not deallocate memory properly. Thus, it is preferable to avoid creating and deleting large arrays at every time step. A common approach is to create work arrays that serve as scratch space for the subroutines to store temporary data. Once the program exits the subroutine, those work arrays remain as they are and can be passed to other subroutines to use in the same manner. These work arrays are created once at the start of the program and remain until the program terminates. However, using work arrays requires that the memory of the arrays be passed to the subroutine. This is not always true. In some programming languages, such as MATLAB, receiver subroutines always copy the data to a new memory area.

```
type:: work_arrays
        real(rk), allocatable :: tmp_pv(:), tmp_cv(:)
        real(rk), allocatable :: k1_pv(:),  k1_cv(:)
        real(rk), allocatable :: k2_pv(:),  k2_cv(:)
        ...
        ...
end type work_arrays
```

Code 2.2: Definition of work array for time integration

In some programming languages such as the C family, users can choose whether to send data by letting the receiver copy it and work with it, or by sending the actual data to the receiver. The first approach is called pass-by-value, and the latter is called pass-by-reference. In some programming languages such as FORTRAN, there is only pass-by-reference. In some languages, the user has no freedom at all, and the language designer dictates the mode of transfer. Python belongs to the last category. Thus, it needs more care to ensure that the data (object) is passed in a way similar to pass-by-reference.

In the past, when we could not define custom data types, we either passed a single large array and used pointers to slice it as needed inside the subroutine, or defined several work arrays. Both approaches make the code hard to maintain, and the interface must be updated if the number of work arrays changes. With this in mind, we create two types of work arrays: work and spatial_tmp. They are custom data types that encapsulate several work arrays. Code.2.2 shows a mini version of type work_arrays which is sufficient for up to fourth-order Runge-Kutta time integration. The advantage of this approach is that we do not need to redo the subroutine interface every time we change the number and size of the work array. It is passed as a type, as shown in line no.2 of Code.2.3. In this code, *allocate_work* allocates the memory. In this code, we enhance the program by introducing RK5 time integration. We added 4 new arrays to the type work_arrays and manage the allocation correctly. Once we complete those tasks, the interface is automatically handled, as the modules define it.

In this setting, if the user code implements RK9 time integration, which requires 16 stages. That means $k1$ to $k16$ must already be defined in work_arrays and allocated for use. What if a user wants to use RK4? Do we need to allocate all of $k1 - k16$? The answer is obviously No! And we should not do that in any case. Declaring each temporary k is $O(1)$. This virtually does not cost anything. However, the allocation cost is $O(N)$, and it needs exactly npx multiplied by the memory per datum to allocate it fully. In Code.2.3, we already saw the conditional array allocation where we only allocate what we need. The rest is virtual data waiting to be defined. They do not cost any CPU time or memory worth noting.

```
1  subroutine allocate_work(work, npx, method)
2
3          type(work_arrays), intent(inout) :: work
4          integer, intent(in) :: npx
5          character(len=*), intent(in) :: method
6          allocate(work%tmp_pv(npx), work%tmp_cv(npx-1))
7
8          select case (trim(adjustl(method)))
9          case ('euler')
10                     ! do nothing, explicit euler does not
     need work array
11         case ('rk4')
12         allocate(work%k1_pv(npx),  work%k1_cv(npx-1))
13                 allocate(work%k2_pv(npx),  work%k2_cv(npx-1))
14                 allocate(work%k3_pv(npx),  work%k3_cv(npx-1))
15                 allocate(work%k4_pv(npx),  work%k4_cv(npx-1))
16         case ('rk5')
17                 allocate(work%k1_pv(npx),  work%k1_cv(npx-1))
18                 allocate(work%k2_pv(npx),  work%k2_cv(npx-1))
19                 allocate(work%k3_pv(npx),  work%k3_cv(npx-1))
20                 allocate(work%k4_pv(npx),  work%k4_cv(npx-1))
21                 allocate(work%k5_pv(npx),  work%k5_cv(npx-1))
```

```
                allocate(work%k6_pv(npx),   work%k6_cv(npx-1))
        case ('rk5')
...
```

Code 2.3: Conditional data allocation (MMM_module_TimeINT.f90)

In the second main part of the code, we perform the actual time integration. Code 2.4 shows the kernel of the time integration, including a portion of the *advance_one_step_inplace* subroutine that performs the fourth-order Runge-Kutta time integration. All data in the multi-moment method must be synchronized because they rely on each other at every time step and within the Runge-Kutta substeps. For this simple periodic domain, we only update the ghost data once in the time integration part of the code. The periodic condition is additionally imposed before the first derivative is computed. Note that the *_rk* is a compiled time assignment of real data type. There are three main data types for real numbers: float (single precision), double precision, and quadruple precision. In the current standard, they are unified under the real data type, but with different possible options. The *kind* option can be used with *real* to define precision; for example, *real(4)* stands for the old standard's *float*, and *real(8)* stands for *double precision*. Alternatively, one can define *integer, parameter:: rk = real32* and use the *real(rk)* declaration to define the real data. This allows the programmer to change the real data type by adjusting the value of *rk* once. In contrast, in the older standard, programmers must change it themselves at every declaration in the program. Note the equivalence of real(8) and real(real64).

```
!Stage 1----
call get_spatial_derivative_inplace(phi_pv, phi_cv, u, dx, n,
   ng, work%k1_pv, work%k1_cv, scheme)
work%tmp_pv = phi_pv + 0.5_rk * dt * work%k1_pv
work%tmp_cv = phi_cv + 0.5_rk * dt * work%k1_cv
!Stage 2----
call get_spatial_derivative_inplace(work%tmp_pv, work%tmp_cv,
   u, dx, n, ng, work%k2_pv, work%k2_cv, scheme)
work%tmp_pv = phi_pv + 0.5_rk * dt * work%k2_pv
work%tmp_cv = phi_cv + 0.5_rk * dt * work%k2_cv
!Stage 3----
call get_spatial_derivative_inplace(work%tmp_pv, work%tmp_cv,
   u, dx, n, ng, work%k3_pv, work%k3_cv, scheme)
work%tmp_pv = phi_pv + dt * work%k3_pv
work%tmp_cv = phi_cv + dt * work%k3_cv
!Stage 4----
call get_spatial_derivative_inplace(work%tmp_pv, work%tmp_cv,
   u, dx, n, ng, work%k4_pv, work%k4_cv, scheme)
!Final ensemble----
factor = dt / 6.0_rk
phi_pv = phi_pv + factor* ( work%k1_pv +2.0_rk * work%k2_pv &
                                       +2.0_rk * work%k3_pv +
   work%k4_pv )
phi_cv = phi_cv + factor* ( work%k1_cv +2.0_rk * work%k2_cv &
   &
                                       +2.0_rk * work%k3_cv +
   work%k4_cv )

```

```
22 call update_periodic_ghosts_pv(phi_pv, n, ng)
23 call update_periodic_ghosts_cv(phi_cv, n, ng)
```

Code 2.4: RK4 time integration (MMM_module_TimeINT.f90)

The final key part of the code is where we compute the spatial derivative depicted in Code.2.5. The code there is implemented true to the explicit scheme in Eq.(2.8). Thus, the negativity is pulled out of the coefficients. The outer negative sign represents the contribution of convection to the evolution of ϕ. We must move the convection term to the RHS before performing the time integration, so the sign is reversed. When multiple terms simultaneously contribute to the evolution, we should compute each term true to its value and apply the sign at the final ensemble. However, because the equation is simple, we can add the negative sign like this, or put it in the coefficients. The ensemble step is not required here.

```
1  ...
2  select case (trim(adjustl(scheme)))
3          case ('central4')
4                  c1coef = 2.0_rk * u / dx
5                  c2coef = 0.5_rk * u / dx
6                  do i = ng + 1, n + ng
7                  rhs_pv(i) = -(c1coef * ( phi_cv(i  ) - phi_cv(
   i-1) ) &
8                                - c2coef * ( phi_pv(i+1) - phi_pv(
   i-1) ) )
9                  end do
10         case ('central6')
11                 ...
```

Code 2.5: The code of the explicit Fourth-order multi-moment differentiation:MMEX4 (MMM_module_TimeINT.f90)

Now we have thoroughly reviewed the code. We are now able to use this code to validate the theoretical results from previous sections. Example 2.5 demonstrates the actual numerical error when applying both the multi-moment and finite-difference methods to solve the one-dimensional convection equation. The empirically observed behavior of these methods is then analyzed and compared directly against the theoretical dispersion limits established in this section.

Example 2.5 (Hands on with the one-dimensional convection equation)
Two FORTRAN source files, `ch01/FDM_1Dconvect.f90` and `ch01/MMM_1Dconvect.f90`, are provided to solve the one-dimensional convection equation using the finite-difference and multi-moment methods, respectively. Each is accompanied by the module file encapsulating the operation backend. The FDM program solves the standard discrete FDM equation, while the MMM program solves the same equation and couples it with the FVM discrete equation into a single, fully coupled system. The computational domain is set to $\Omega = [0, 2\pi]$ with periodic boundary conditions. The reader is encouraged to experiment with these scripts by adjusting parameters such as the number of cells, the time-step size, and the domain length. The spatial approximations and time-integration schemes can also be modified as desired.

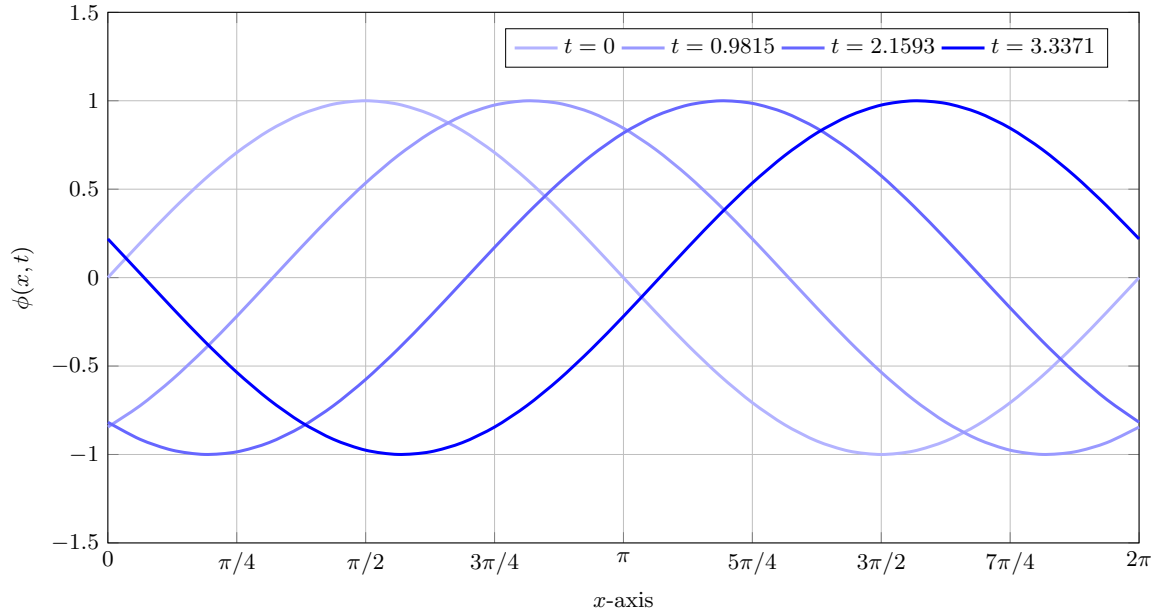


Fig. 2.11: Convection of a sine wave in a periodic domain.

Figure 2.11 shows the initial condition and the numerical solutions at three time instances. The simulation integrates up to a final time of $t = 20\pi$, meaning the wave effectively traverses the periodic domain ten times before returning to its original position. The maximum error (L_∞-norm) and the order of convergence for the fourth-order schemes are reported in Table 2.7. Both schemes clearly demonstrate fourth-order convergence. However, when comparing the methods at equivalent numbers of degrees of freedom (nDoFs), the multi-moment method proves to be significantly more accurate than its finite-difference counterpart. Specifically, at 32 degrees of freedom ($N_{cell} = 16$ for MMM versus $N_{cell} = 32$ for FDM), the multi-moment method produces an error that is only 44.3% of the FDM error. This empirical result closely matches the theoretical error ratio of 44.8% predicted by the Coupled Modified Wavenumber Analysis at this effective wavenumber ($kh_{eff} = \pi/16$), as shown previously in Fig. 2.4.

N_{Cell}	**Maximum Error (L_∞-Norm)**			**Order of Convergence**		
	MM-PV	**MM-CV**	**FD-PV**	**MM-PV**	**MM-CV**	**Rate FD-PV**
4	$3.635 \cdot 10^{-1}$	$2.705 \cdot 10^{-1}$	$1.997 \cdot 10^{0}$	-	-	-
6	$7.205 \cdot 10^{-2}$	$5.973 \cdot 10^{-2}$	$1.784 \cdot 10^{0}$	3.99	3.72	0.28
8	$2.240 \cdot 10^{-2}$	$2.027 \cdot 10^{-2}$	$6.747 \cdot 10^{-1}$	4.06	3.76	3.38
12	$4.383 \cdot 10^{-3}$	$4.215 \cdot 10^{-3}$	$1.518 \cdot 10^{-1}$	4.02	3.87	3.68
16	$1.373 \cdot 10^{-3}$	$1.361 \cdot 10^{-3}$	$4.889 \cdot 10^{-2}$	4.03	3.93	3.94
32	$8.619 \cdot 10^{-5}$	$8.618 \cdot 10^{-5}$	$3.099 \cdot 10^{-3}$	3.99	3.98	3.98

Table 2.7: Maximum error of the fourth-order finite-difference and multi-moment methods at the final time, and the order of convergence in one-dimensional convection at $t = 20\pi$.

2.10 Upwind Method

The convection equation describes the physical process of advection, whereby the bulk motion of a fluid transports a quantity. The transported quantity may be an intrinsic fluid property, such as internal energy, or an extrinsic substance carried by the flow, including dissolved salts in seawater, pollutants, or suspended particles such as smoke. From a continuum mechanics perspective, these quantities are advected at the local fluid velocity and therefore follow the fluid motion. Accurately resolving this transport while minimizing numerical artifacts remains one of the central challenges in computational fluid dynamics (CFD).

Unlike elliptic or parabolic partial differential equations (PDEs), the convection equation belongs to the class of *hyperbolic* PDEs. A defining characteristic of hyperbolic systems is that information propagates along specific trajectories known as characteristic curves. This property can be illustrated by considering a bonfire burning in an open field under a steady northerly wind. The generated smoke is transported exclusively downstream toward the south, influencing only the downwind region, commonly referred to as the domain of dependence. Neglecting molecular diffusion, it is physically impossible for the smoke to propagate upstream because the velocity field uniquely determines the direction of information transport. This directional propagation is a fundamental feature of convection-dominated flows and becomes particularly significant in high-Reynolds-number and inviscid flow regimes.

The development of numerical methods for solving convection-dominated problems has closely paralleled the evolution of modern CFD. Among the earliest approaches, the Finite Difference Method (FDM) emerged as a powerful framework based on Taylor-series approximations evaluated on structured grids. Its continued development has been driven by the pursuit of higher-order accuracy and improved shock-capturing capabilities. Advanced upwind-biased discretizations, including Essentially Non-Oscillatory (ENO) and Weighted Essentially Non-Oscillatory (WENO) schemes [26], were introduced to suppress spurious oscillations near discontinuities while maintaining numerical stability and high-order accuracy.

In parallel, the Finite Volume Method (FVM) was developed as a conservative formulation that enforces the exact balance of fluxes across discrete control volumes. This inherent conservation property makes FVM particularly attractive for complex geometries and unstructured meshes commonly encountered in industrial applications. The introduction of Godunov-type methods [8] marked a major milestone in the advancement of FVM, providing a rigorous framework for resolving hyperbolic conservation laws through local Riemann problems. Subsequent developments, including the Monotonic Upstream-Centered Scheme for Conservation Laws (MUSCL) [33] and the Quadratic Upstream Interpolation for Convective Kinematics (QUICK) scheme [25], further enhanced spatial accuracy while reducing numerical diffusion.

Today, both FDM and FVM constitute the foundation of high-fidelity CFD solvers and remain indispensable in large-scale simulations of compressible and convection-dominated flows. Their continued development has enabled increasingly accurate and efficient numerical predictions across a broad range of scientific and engineering applications.

For the one-dimensional convection equation with positive velocity ($u > 0$), information propagates exclusively from left to right. Consequently, numerical discretizations that utilize information solely from the *upwind* direction are more consistent with the underlying physics than symmetric central-difference approximations. This observation motivates the development of upwind-biased schemes, which form the basis of many modern discretizations for convection-dominated flows.

Consider the simplest upwind finite-difference approximation:

$$\left.\frac{\partial \phi}{\partial x}\right|_i \approx \frac{\phi_i - \phi_{i-1}}{h} \tag{2.115}$$

This expression corresponds to the first-order upwind approximation of the spatial derivative. As derived previously, the associated non-dimensional modified wavenumber is given by

$$k^* h = \sin(kh) - j(1 - \cos(kh)) \tag{2.116}$$

Despite its robustness and directional consistency, the first-order upwind scheme is well known for its excessive numerical dissipation. This behavior is directly reflected in the large imaginary component of the modified wavenumber shown in Fig. 2.12(b). For instance, at a cell wavenumber of approximately $kh \approx 1.5$, corresponding to roughly four computational cells per wavelength, the numerical solution experiences an artificial reduction in amplitude of nearly 50% over a single convective time unit.

While Fourier analysis quantifies the magnitude of this error in spectral space, additional physical insight can be obtained through *Modified Equation Analysis* (MEA). The central idea of MEA is to substitute the continuous Taylor-series expansion of the discrete nodal values into the semi-discrete governing equation, thereby revealing the differential operators implicitly introduced by the discretization. Expanding ϕ_{i-1} about node i gives

$$\frac{\partial \phi}{\partial t} + \frac{u}{h}\left[\phi_i - \left(\phi_i - h\frac{\partial \phi}{\partial x} + \frac{h^2}{2}\frac{\partial^2 \phi}{\partial x^2} - \frac{h^3}{6}\frac{\partial^3 \phi}{\partial x^3} + \dots\right)\right] = 0 \tag{2.117}$$

After simplification, the modified governing equation becomes

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = u\frac{h}{2}\frac{\partial^2 \phi}{\partial x^2} - u\frac{h^2}{6}\frac{\partial^3 \phi}{\partial x^3} + \text{H.O.T.} \tag{2.118}$$

The leading-order truncation error appears as a second-order derivative term, which is mathematically identical to a physical diffusion operator. The first-order upwind scheme therefore solves the original convection equation supplemented by an artificial diffusion process with an effective numerical diffusion coefficient.

$$K_{\text{num}} = \frac{uh}{2}.$$

This result explicitly explains the dissipative behavior observed in the modified-wavenumber analysis. As the mesh spacing increases, the magnitude of the artificial diffusion grows proportionally, leading to progressively stronger damping of the signal. In principle, one may attempt to compensate for this error by introducing an appropriate anti-diffusive correction.

More broadly, Modified Equation Analysis establishes a direct connection between truncation-error terms and their physical consequences: even-order spatial derivatives are associated with numerical dissipation and amplitude attenuation, whereas odd-order spatial derivatives introduce numerical dispersion, resulting in phase errors and spurious oscillatory behavior.

2.10.1 Upwind Multi-Moment Schemes

Following a similar methodology, we can derive the first-order upwind approximation for Multi-Moment schemes, arriving at:

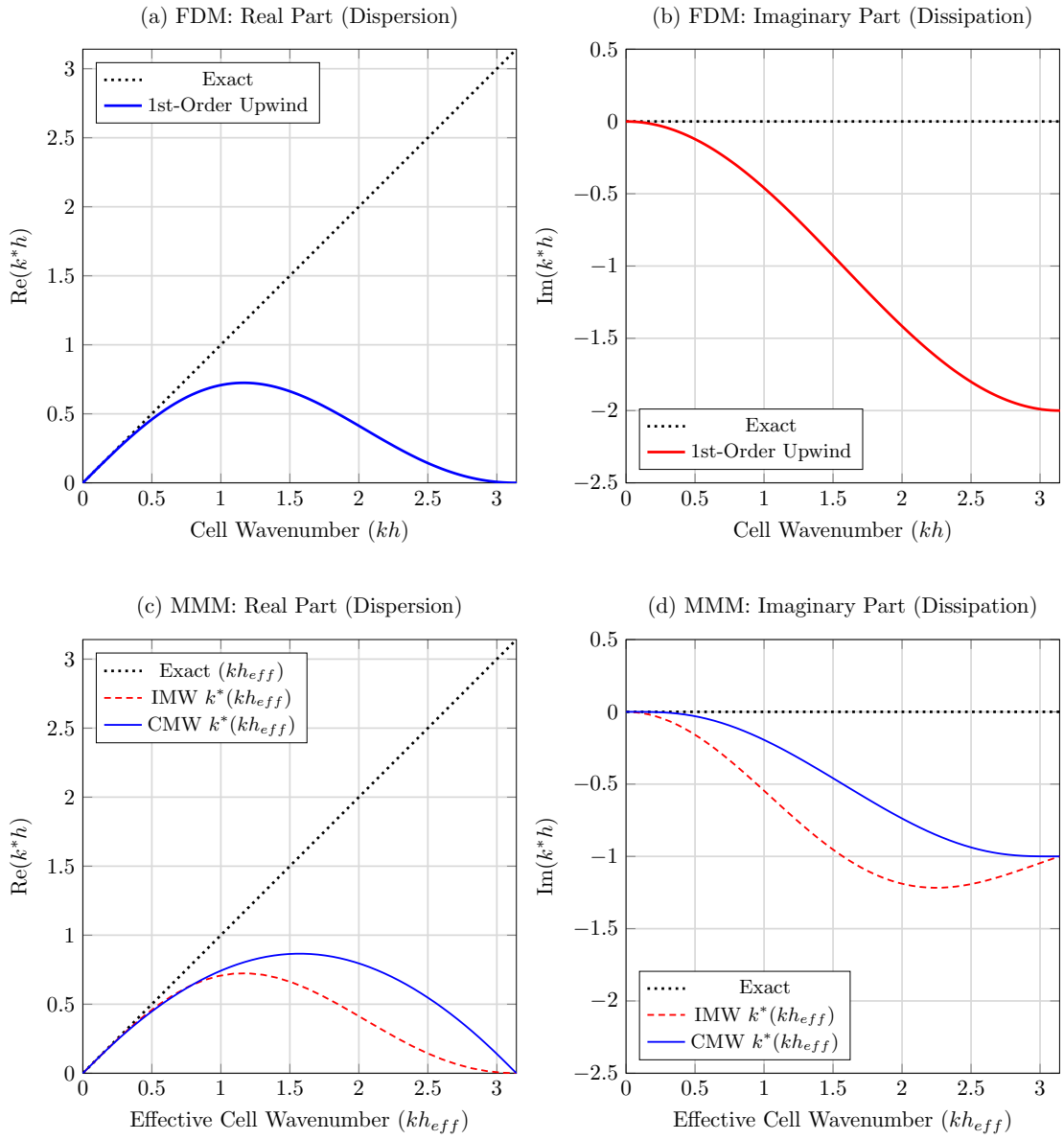


Fig. 2.12: Comparison of the non-dimensional modified wavenumber for the first-order upwind scheme. The numerical dispersion (real part) and dissipation (imaginary part) are evaluated for the standard Finite Difference Method (a, b) and the Multi-Moment Method (c, d).

$$\left.\frac{\partial \phi}{\partial x}\right|_i = 2\frac{\phi_i - \overline{\phi}_{i-1/2}}{h} \tag{2.119}$$

This formulation conceptually utilizes the midpoint to extract the nodal value $\phi_{i-1/2}$ from the cell average $\overline{\phi}_{i-1/2}$ before computing the gradient, which naturally yields the factor of two. While this assumption produces the correct leading-order coefficient for a first-order accurate method, relying on it rather than the exact cell average can yield an inaccurate representation of the modified wavenumber. The rigorous modified wavenumbers, denoted for the initial (k^*_{IMW}) and coupled (k^*_{CMW}) analyses, are given by:

$$k^*_{IMW} = \frac{2(1-\cos(kh))}{h^3 k} + j\left[\frac{2(\sin(kh)-hk)}{h^3 k}\right] \tag{2.120}$$

$$k^*_{CMW} = \frac{\sqrt{1-2\cos(2k)+2j\sin(2k)}-j}{2k} \tag{2.121}$$

These profiles are plotted in Fig. 2.12(c&d). At an equivalent number of degrees of freedom (nDoFs), the initial coupled modified wavenumber (CMW) shows a slight increase in accuracy in the high-wavenumber regime. Interestingly, the multi-moment is much less dissipative than the FDM. Regardless of these relative differences, both first-order methods introduce excessive numerical diffusion, rendering them overly dissipative and generally impractical for large-scale, high-fidelity computational frameworks.

2.10.2 Higher-order Upwind Multi-Moment Scheme

As demonstrated in the previous section, first-order upwind schemes respect the physics of characteristic information propagation but introduce substantial numerical diffusion through the second-derivative term appearing in the truncation error. In contrast, second-order central difference schemes eliminate this artificial diffusion. Still, they are susceptible to numerical dispersion associated with the leading third-derivative truncation term, which can manifest as oscillatory behavior near steep gradients. A natural extension is therefore to increase the stencil width and introduce an upwind bias while maintaining higher-order accuracy. For third-order upwind discretizations, the leading truncation error is proportional to the fourth spatial derivative. This error term provides scale-selective numerical dissipation, preferentially damping high-wavenumber components while exerting comparatively little influence on larger-scale flow structures.

The development of third-order upwind schemes represented an important step in the numerical treatment of convection-dominated flows, particularly in high-Reynolds-number applications where accurate discretization of the convective terms is essential. One of the early contributions in this area was made by Agarwal [2], who proposed a third-order upwind discretization for the three-dimensional Navier–Stokes equations. The study demonstrated that the asymmetric formulation retained favorable stability properties and preserved matrix diagonal dominance for challenging steady-flow configurations, including three-dimensional lid-driven cavity flows.

Subsequently, Kawamura and Kuwahara [18] introduced the four-point third-order upwind finite difference scheme that later became widely adopted in CFD applications. Their formulation was motivated by the need to reduce numerical instabilities associated with the discretization of convective terms. The resulting fourth-derivative dissipation acts primarily on the high-wavenumber portion of the solution spectrum, providing numerical damping while retaining substantially greater accuracy than first-order upwind approximations.

Using the method outlined previously, we can obtain an asymmetric stencil of the odd-order accurate scheme by foregoing the rightmost data of the central scheme in Eq.(2.8) and recalculating the approximation coefficients. For this section, we consider the following three high-order upwind multi-moment schemes:

$$\left.\frac{\partial \phi}{\partial x}\right|_{i,UPWEX3} = -\frac{7}{2h}\overline{\phi}_{i-1/2} + \frac{1}{2h}\overline{\phi}_{i+1/2} + \frac{1}{h}\phi_{i-1} + \frac{2}{h}\phi_i, \tag{2.122}$$

$$\left.\frac{\partial \phi}{\partial x}\right|_{i,UPWEX5} = -\frac{1}{18h}\overline{\phi}_{i-3/2} - \frac{55}{18h}\overline{\phi}_{i-1/2} + \frac{13}{9h}\overline{\phi}_{i+1/2} + \frac{1}{h}\phi_{i-1} + \frac{1}{h}\phi_i - \frac{1}{3h}\phi_{i+1}, \tag{2.123}$$

(2.124)

$$\left.\frac{\partial \phi}{\partial x}\right|_{i,UPWEX7} = -\frac{53}{216h}\overline{\phi}_{i-3/2} - \frac{725}{216h}\overline{\phi}_{i-1/2} + \frac{355}{216h}\overline{\phi}_{i+1/2} + \frac{1}{72h}\overline{\phi}_{i+3/2} + \frac{1}{18h}\phi_{i-2} + \frac{4}{3h}\phi_{i-1} + \frac{1}{h}\phi_i - \frac{4}{9h}\phi_{i+1}. \tag{2.125}$$

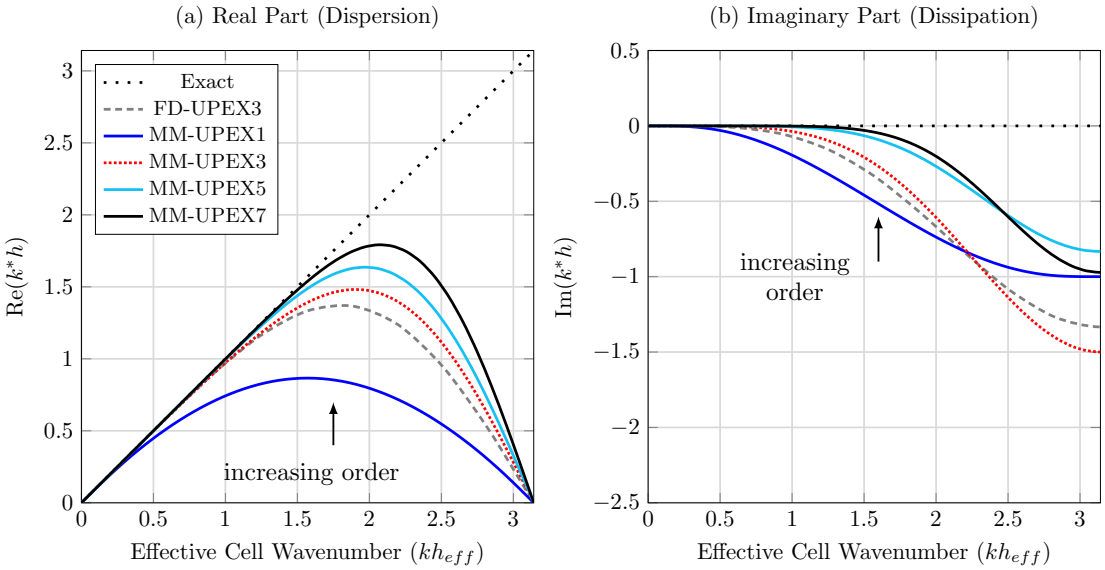


Fig. 2.13: Comparison of the modified wavenumber profiles for the first-, third-, fifth-, and seventh-order explicit upwind multi-moment schemes. The real (a) and imaginary (b) parts represent the numerical dispersion and dissipation, respectively. The classic third-order upwind (FD-UPEX3) is displayed for reference.

The coupled modified wavenumbers are plotted in Fig. 2.13, providing a direct comparison to the classic third-order scheme of Agarwal, Kawamura, and Kuwahara [2, 18]. The third-order multi-moment method (MM-UPEX3) yields slightly higher accuracy in wave-propagation speed and generates less numerical dissipation across the low-to-medium wavenumber range. Interestingly, it also introduces stronger damping at high frequencies. This combination of lower damping at resolved scales and stronger damping at high frequencies is a highly desirable property; it preserves the shape of the resolved signal more effectively while eliminating unphysical, spurious oscillations. As the formulation is extended to fifth- and seventh-order schemes, accuracy improves, albeit at the cost of higher computational expense. Because upwind multi-moment methods remain at the forefront of ongoing research, ample possibilities exist for further exploration, such as dynamically blending central and upwind schemes or directly optimizing the stencil coefficients.

2.11 Numerical Experiments

In this first test, we solve the linear convection-diffusion equation with different initial conditions and parameters. First, we solve a simple sine wave to complement the modified wave number analysis presented earlier. Next, we assess the robustness of the scheme by transporting the Runge function, an antithesis of high-order polynomials.

2.11.1 Convection of sine wave

A convergence study of the 1D convection equation ($u = 1.0,\ \Gamma = 0$) is carried out using the initial condition $\phi(x, 0) = sin(\pi x)$ on a periodic domain $x = [0, 40]$. This extra-large domain tests the resolving power beyond π. The solution is obtained up to $t = 6.5$, corresponding to 3.25 periods. Figure 2.14 shows an evolution of the same sine wave in a periodic domain using the EX12 scheme. Here we resolve the wave with two cells per wavelength, and the cell-wavenumber equals π—the normal Nyquist limit for a single grid. Initially, the PV data (black dots) treat the initial wave as zero, and the cell data are sampled at their maxima. The original wave is evolved to $t = 4$, where the waves are supposed to return to their original positions, and to $t = 6.5$, where the point values should be at their maxima.

The CV and PV data can recover the peak values. The L_2-norm can be transferred between the two data types, and it is a necessary mechanism to resolve the field beyond π. As we emphasized earlier, the method we used in this work is a reconstruction-type method. The EX12 scheme uses the 12th-order polynomial defined over six adjacent cells. We plot this polynomial for the two center cells. The P_{12} captures the sine wave very well. The phase error at the final time is 0.65%.

In the convergence table (Tab.2.8), the results are compared with the compact sixth-order FDM. At the three lowest resolutions (CPW=1.25, 1.4, and 2.0), the FDM cannot resolve the field. EX4, EX8, and EX12 are not accurate enough at the coarsest resolution, and the wave is out of phase, leading to large errors. For the same number of cells, the sixth-order multi-moment method is more accurate than the FDM, as expected. All the multi-moment schemes converge reasonably well at 2 CPW. In this test, we use the fifth-order Runge-Kutta method for time integration, which requires $4,000$ time steps to keep the time-integration error low. The accumulated round-off error eventually obscures the convergence-rate error of EX24 at 5 CPW and drowns out the spatial errors at 10 CPW. Note that EX6 is less accurate than the double-resolution FDCOM6. This is expected because the compact scheme is known for its high resolving power, which is normally comparable to the explicit scheme two orders higher. To compete with the double-resolution 6th-order implicit FDM, the explicit multi-moment method must use an eighth-order scheme, which introduces errors of 2.4E−2 and 9.8E−5 at 2.5 and 5 CPW, respectively.

2.11.2 Convection of Runge Function

In this test, we initialize the solution using a filtered Runge function:

$$\phi(x, 0) = f(x)\left(\frac{1}{1 + 25x^2}\right), \quad (-5 \le x \le 5) \tag{2.126}$$

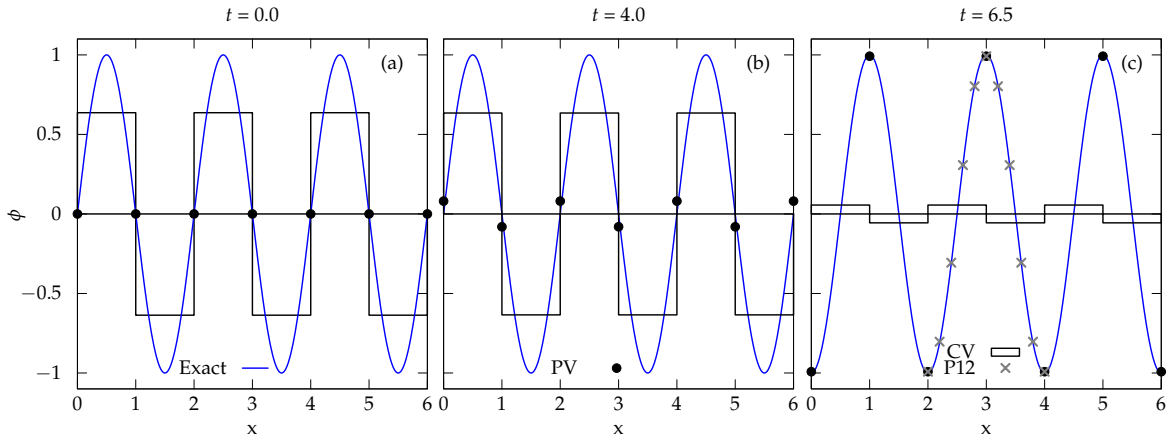


Fig. 2.14: The convection of $sin(\pi x)$ resolved by 2 cells per wave (cell wavenumber = π). The bars are the cell-average variable defined in FVM. The closed circles are the pointwise FDM variable. The three plots show the initial conditions and the numerical solutions at $t = 4$ and $t = 6.5$. P12 is part of the polynomial defined by the center-point stencil of EX12. The wave speed error at this resolution is 0.65%

CPW	Maximum Error					Convergence Rate				
	$FDCOM6$	$EX4$	$EX6$	$EX12$	$EX24$	$FDCOM6$	$EX4$	$EX6$	$EX12$	$EX24$
1.25	—	$1.68E+00$	$1.80E-00$	$1.53E+00$	$8.34E-01$	—	—	—	—	—
1.40	—	$1.86E+00$	$1.67E-00$	$8.10E-01$	$1.87E-01$	—	0.90	0.66	5.61	13.19
2.00	—	$5.28E-01$	$2.99E-01$	$2.69E-02$	$3.22E-04$	—	3.53	4.82	9.55	17.84
2.50	$1.44E+00$	$2.66E-01$	$9.69E-02$	$2.35E-03$	$2.92E-06$	—	3.07	5.05	10.92	21.08
5.00	$1.43E-02$	$1.48E-02$	$1.48E-03$	$6.20E-07$	$3.50E-13$	6.65	4.17	6.03	11.89	23.04
10.00	$1.93E-04$	$8.76E-04$	$2.55E-05$	$1.68E-10$	$4.16E-14$	6.21	4.08	5.86	11.85	—

Table 2.8: Maximum errors and convergence rate at $t = 4$ of the multi-moment methods solving the convection of $sin(\pi x)$.

with $f(x) = (erf\ (\rho(x+\beta)) - erf\ (\rho(x-\beta)))/2$, with $\beta = \sqrt{249}/5$, and $\rho = 10$. Using these filter parameters, Runge's function is undisturbed (up to the machine's accuracy) for $x \in [-2.5, 2.5]$ and $\phi(x, 0)$ drops to the machine's zero in the given domain such that the periodic conditions can be applied (see Fig.2.15).

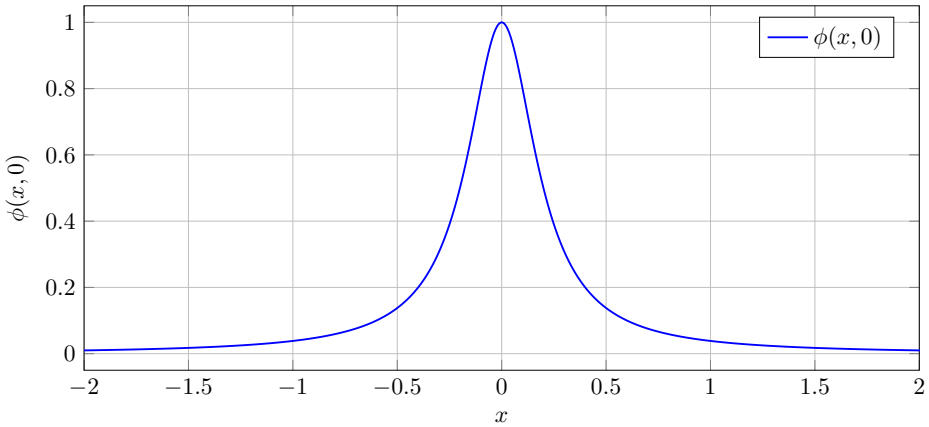


Fig. 2.15: Central domain of the initial condition of the convection problem of Runge's function.

The reconstruction method, which employs the information from the neighboring cells, can be criticized for lacking locality in contrast to the method using locally defined polynomial [36, 5] The explicit method we proposed for the multi-moment discretization here is a clear example of a wide stencil. Alternatively, one can use a locally defined polynomial for the numerical approximation. Nodal DGM can be considered the other extreme, because all DOF used to construct the polynomial are defined locally. Given the data locality, DGM should be more accurate than the reconstruction method. To investigate the effects of data locality, we compare the proposed scheme with the nodal DGM method [10]. Figure 2.16 displays the convergence curves of four explicit multi-moment schemes, In the FDM, the DoF is the point value (PV). In the multi-moment method, both CV and PV data are available. For the same number of cells, the explicit MM method has twice as many DOF as the compact FDM, but both define the polynomial over the same distance. The compact FDM uses the implicit derivative to complement the PV, while the explicit MM uses CV and PV. The number of DOF of DGM is the number of internal nodes multiplied by the number of cells. Note that DGM of order p can achieve a $p + 1$ convergence rate in a pure convection problem; thus, this is a fair comparison.

In the figure, EX4 starts off more accurate than the sixth-order compact FDM, but is eventually surpassed. With 8th-order accuracy, the explicit multi-moment method is more accurate than FDCOM6 everywhere. The twelfth-order scheme (EX12) is more accurate than the central flux DGM when the resolution is sufficient. Thus, it can be concluded that the explicit multi-moment method handles under-resolved fields better than the FDM but worse than the DGM. Data locality is highly beneficial for under-resolved simulations. To compete with DGM in that situation, the multi-moment method needs a compact scheme. It will be presented in another work.

2.11.3 Comparison of upwind schemes

To assess the influence of spatial discretization on solution quality, we consider the one-dimensional advection of a modified error-function profile,

$$\phi(x) = \frac{1}{2}\left[\mathrm{erf}(\rho(x + c_1)) - \mathrm{erf}(\rho(x - c_0))\right], \tag{2.127}$$

on the periodic domain $x \in [-20, 20]$. The solution is advanced from $t = 0$ to $t = 40$, corresponding to one complete revolution of the wave around the computational domain. The parameters $\rho = 30$ and $c_0 = c_1 = 2.5$ are chosen such that the profile closely resembles a sharp rectangular wave (Fig.2.17) while remaining infinitely differentiable.

This benchmark is particularly useful for evaluating numerical dissipation and dispersion. Although the initial condition is smooth, the steep gradients near the transition regions challenge spatial discretizations like discontinuous solutions. Consequently, insufficient numerical resolution often leads to either excessive diffusion or spurious oscillations. At the same time, the smoothness of the exact solution permits recovery of the formal convergence order under grid refinement. The performance of second- and fourth-order central difference schemes (EX2 and EX4) and first- and third-order upwind schemes (UPW1 and UPW3) is examined using both the conventional finite difference method (FDM) and the proposed multi-moment method (MMM). In addition, fifth- and seventh-order upwind schemes (UPW5 and UPW7) are evaluated within the MMM framework. To ensure a fair comparison, the total number of degrees of freedom is kept identical in all simulations. The FDM employs

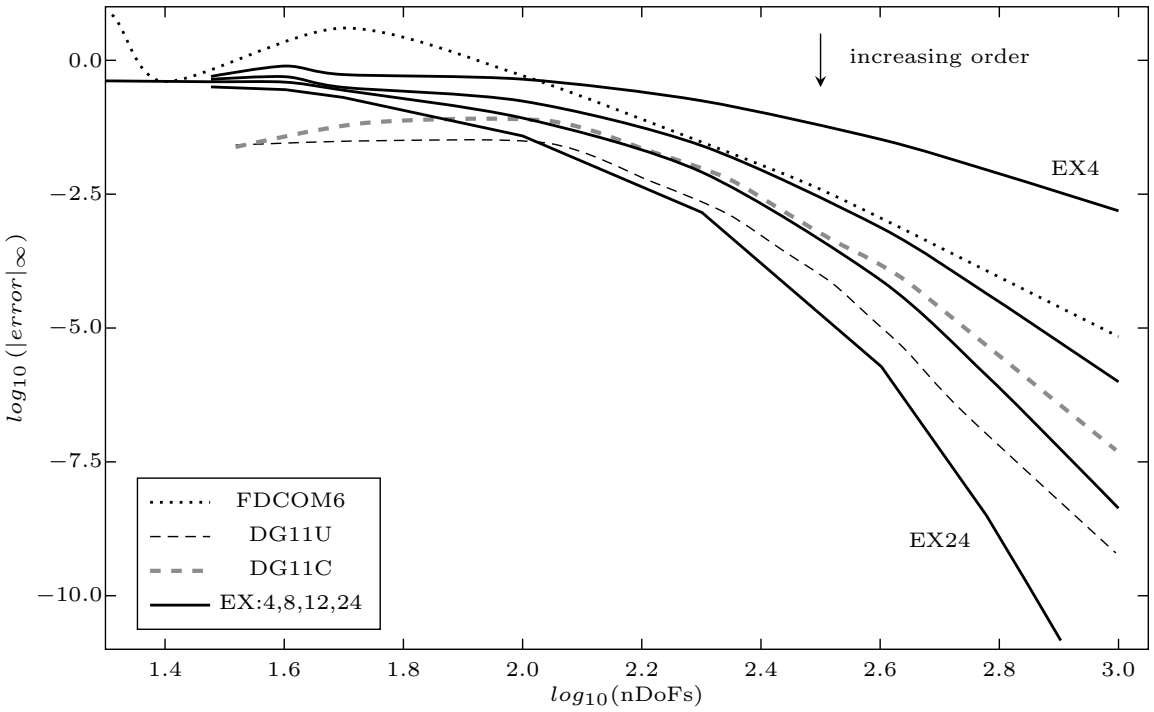


Fig. 2.16: Convergence plot of the fourth-, eighth-, twelfth-, and twenty-fourth-order multi-moment methods solving 1D convection of the filtered Runge's function. The sixth-order compact scheme(FDCOM6) and the eleventh-order nodal discontinuous Galerkin Method (DG11) with upwind flux (-U) and central flux (-C) are shown for comparison.

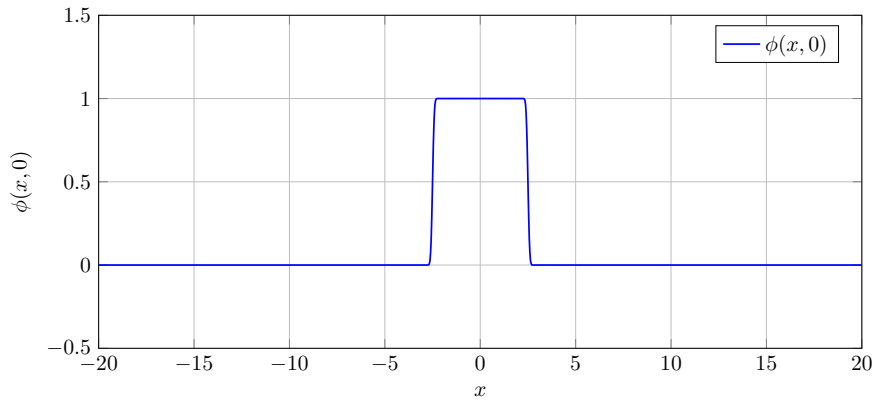


Fig. 2.17: Initial condition for the convection problem of the error function.

$N_{\text{cell}} = 1024$ grid points, whereas the MMM uses $N_{\text{cell}} = 512$ elements with two moments per element, yielding the same total number of unknowns. Time integration is performed using the classical fourth-order Runge–Kutta scheme at CFL=0.1.

Figure 2.18 compares the numerical solutions after one advection cycle, illustrating the impact of numerical dispersion and diffusion on scheme performance. Central-difference schemes, EX2 and EX4, preserve the profile amplitude but generate oscillations near steep gradients, revealing the influence of dispersive errors. Multi-moment methods produce sharper transitions and fewer oscillations but still exhibit some artifacts due to limited dissipation, which is important for evaluating their effectiveness. A magnified view of the upper-right corner in Figure 2.19 shows that the solutions from the second-order finite-difference method

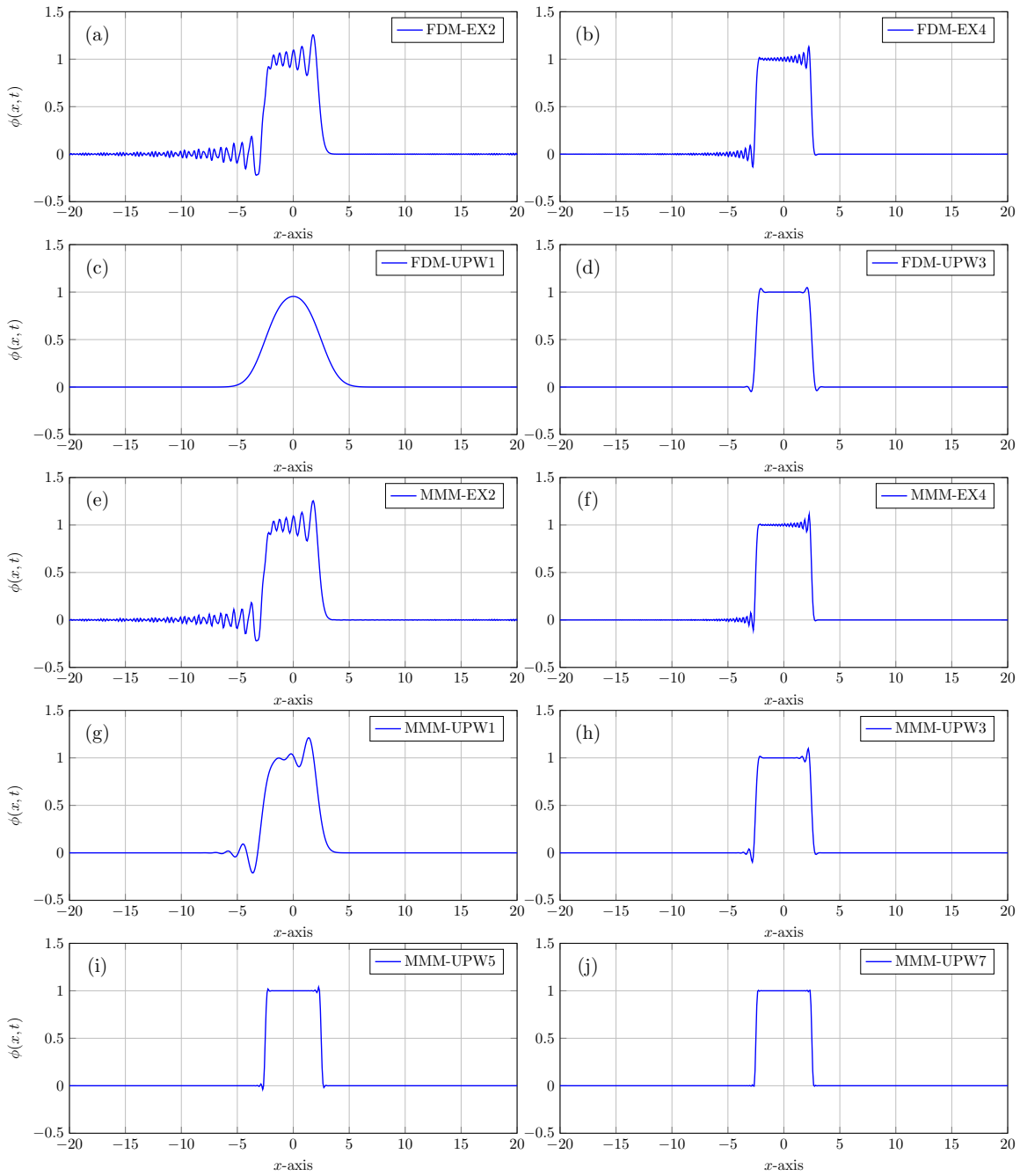


Fig. 2.18: Performance comparison of various spatial approximation schemes applied to the finite difference method (FDM) (a–d) and the multi-moment method (MMM) (e–j). The notations EX and UPW designate explicit central difference and upwind schemes, respectively. The appended numeral in each legend specifies the theoretical convergence order of the applied scheme. In all cases, the resolutions are matched with nDoFs=1024. The graphs share the same x and y axes.

and the second-order multi-moment method are nearly indistinguishable. This observation aligns with the coupled modified wavenumber analysis in Figure 2.3, which predicts similar dispersion and dissipation characteristics for the two schemes. For the fourth-order discretizations, however, a small but noticeable difference emerges: MMM-EX4 preserves the profile more accurately than FDM-EX4, resulting in improved resolution of the transported wave and a closer agreement with the exact solution.

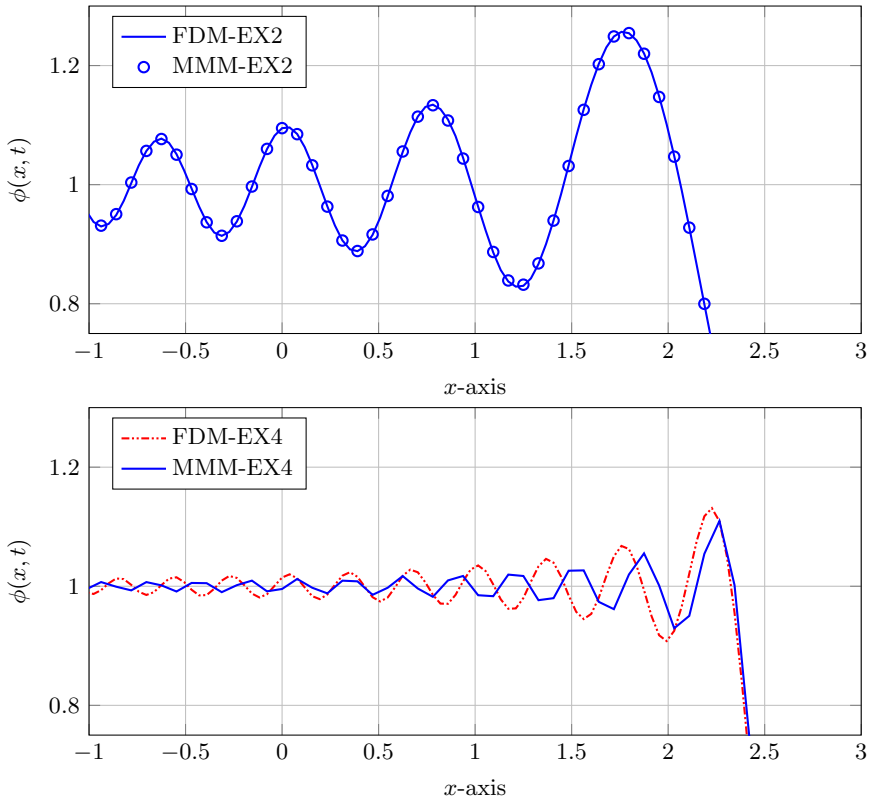


Fig. 2.19: Pointwise comparison of second-order (top) and fourth-order (bottom) central schemes using the finite difference method (FDM) and the multi-moment method (MMM), with a zoomed-in view near the top corner of the leading edge.

In contrast, the first-order upwind schemes exhibit strong numerical damping, which can undermine the accuracy of simulations. The FDM-UPW1 solution is substantially smeared, with the initial square-like profile evolving into a broad smooth hump. This behavior is consistent with the diffusive leading-order truncation error associated with first-order upwind discretizations. Recognizing this diffusive tendency emphasizes the need for higher-order methods to improve precision. A more favorable balance between stability and accuracy is obtained using the third-order upwind scheme. Both FDM-UPW3 and MMM-UPW3 preserve the overall shape of the advected profile while significantly reducing the dispersive oscillations observed in the central schemes. Small overshoots and undershoots remain near the corners, reflecting the well-known tendency of high-order linear schemes to generate local extrema in regions with steep gradients. While FDM-UPW3 shows slightly stronger numerical diffusion, MMM-UPW3 retains sharper interfaces at the cost of marginally larger oscillations near the bump's leading and trailing edges, indicating that most of the error is dispersion and that the diffusion error is less pronounced than in the FDM counterpart. Overall, the multi-moment formulation provides superior accuracy, achieving a lower L_∞

error (0.185 versus 0.200) while exhibiting improved conservation of the L_2 norm (0.9982 versus 0.9949).

The advantages of the multi-moment formulation become increasingly evident as the order of the upwind discretization is increased. The MMM-UPW5 and MMM-UPW7 solutions maintain sharp interfaces with negligible oscillations and minimal amplitude degradation. Notably, MMM-UPW7 reproduces the initial profile almost exactly after one complete advection cycle, demonstrating the method's robustness. This highlights how the additional local information in multi-moment representations enables high-order schemes to effectively reduce numerical diffusion and dispersion, ensuring accurate and stable solutions.

The computational cost of the multi-moment method remains competitive despite its higher-order reconstruction. For the present benchmark written in Python on an Intel 255H CPU, FDM-UPW3 requires 0.521 s, whereas MMM-UPW3, MMM-UPW5, and MMM-UPW7 require 0.374 s, 0.400 s, and 0.452 s, respectively. This demonstrates that multi-moment schemes can deliver superior solution quality efficiently, making them a compelling choice for practical applications. The source codes of both methods are provided with this book. The reader can independently evaluate and verify the performance under different conditions.

2.12 Effects of Time-Integration Error in Very High-Order Spatial Schemes

For low-order spatial discretizations, the error introduced by the Runge–Kutta time integration is usually negligible. In such cases, third- and fourth-order Runge–Kutta methods are sufficiently accurate, and the temporal error remains masked by the dominant spatial truncation error. As the spatial approximation is improved, however, the temporal error can become the limiting factor in the overall solution accuracy.

Figure 2.20 illustrates this effect for fourth-order spatial schemes. The multi-moment method (MM) operated at CFL = 0.6 uses the same physical time-step size as the finite-difference method (FDM) operated at CFL = 1.2 because, at an equal number of degrees of freedom (nDoFs), the MM cell size is twice that of the FDM. The resulting error levels are therefore directly comparable.

The results indicate that the temporal error associated with the classical fourth-order Runge–Kutta method (RK4) is already noticeable at this spatial order. In Fig. 2.20(b), RK4 reaches the accuracy limit of the fourth-order finite-difference scheme (FDEX4) using a time-step size between the two coarsest cases examined, corresponding approximately to $0.6\Delta x/u < \Delta t < 1.2\Delta x/u$. In contrast, Fig. 2.20(a) shows that CFL=0.6 and 0 .4 are insufficient for the fourth-order multi-moment scheme (MMEX4). A time-step size of approximately $0.25\Delta x/u$ is required to achieve the full accuracy. Referring to the stability limit reported in Table 3.3, this operating point corresponds to roughly 23% of the maximum stable time step. Although restrictive, it remains within the same order of magnitude as the stability constraint and is therefore still acceptable in practice.

The impact of temporal accuracy becomes substantially more severe as the spatial order increases. To investigate this effect, Fig. 2.21 compares several Runge–Kutta methods when coupled with the fourteenth-order multi-moment scheme (MMEX14). Figure 2.21(a) clearly demonstrates the limitation of RK4. To fully realize the spatial accuracy of MMEX14 across the entire resolution range, a CFL number of approximately 0.005 is required. This is nearly fifty times smaller than the time-step size needed for MMEX4. For engineering applications where an error level of 10^{-5} is acceptable, a CFL number of 0.05 still provides satisfactory results; however, this remains approximately five times smaller than that required by the fourth-order scheme.

Increasing the temporal order significantly alleviates this restriction. As shown in Fig. 2.21(b), the seventh-order Runge–Kutta method (RK7) enables MMEX14 to achieve its full spatial accuracy at CFL = 0.2, which is comparable to the CFL number required by RK4 for fourth-order spatial discretizations. The eighth-order Runge–Kutta method (RK8) offers only a modest additional improvement. In contrast, the ninth-order method (RK9) completely removes the temporal accuracy bottleneck and allows MMEX14 to operate at its largest stable CFL number without compromising accuracy.

The computational cost of increasing the spatial order grows approximately linearly with the order of the scheme. The resulting benefit, however, scales much more favorably because the reduction in spatial resolution requirements propagates through all spatial dimensions. In two dimensions, the savings scale with the square of the resolution reduction, while in three dimensions they scale cubically. When combined with the accompanying reduction in the number of time steps, the overall performance gain can scale approximately as $\mathcal{S}_\epsilon^4$.

The situation is markedly different for temporal integration. Unlike spatial discretizations, whose memory footprint is largely independent of the formal order of accuracy, higher-order Runge–Kutta methods require additional stages, resulting in increased memory usage

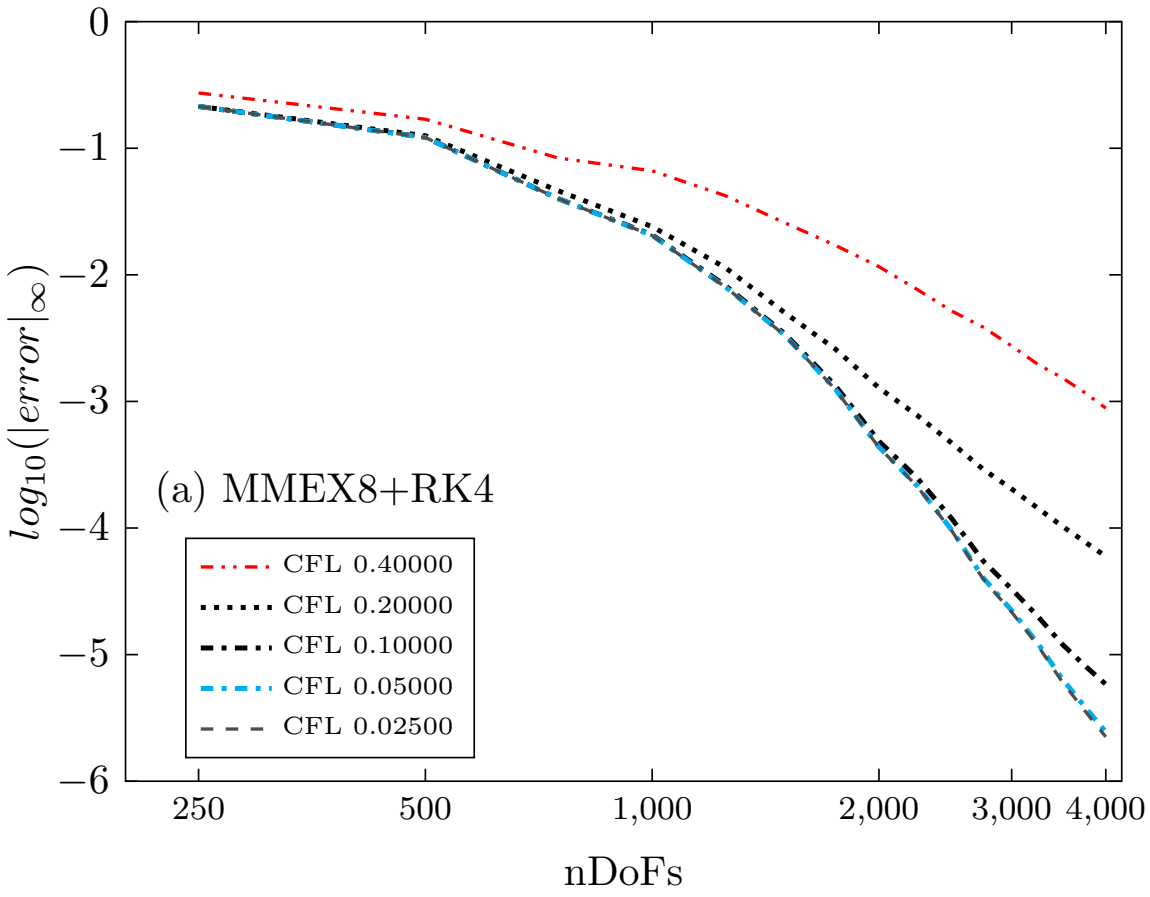


Fig. 2.20: Influence of RK4 temporal truncation error on fourth-order spatial discretizations: (a) fourth-order multi-moment scheme (MMEX4) and (b) fourth-order finite-difference scheme (FDEX4). The results demonstrate that temporal error becomes a limiting factor even for fourth-order spatial approximations.

and memory-access overhead. Consequently, the computational cost of temporal integration grows faster than linearly with temporal order. Achieving efficient high-order time integration is therefore a non-trivial task.

Tables 2.9 and 2.10 compare the computational cost of the fourth-, seventh-, and eighth-order Runge-Kutta methods (RK4, RK7, and RK9) for achieving nearly identical solution errors. The measurements were obtained from a Fortran implementation compiled with the Intel Fortran Compiler using the `-O3 -r8` optimization settings and executed on an Intel Xeon 8124 processor. Because RK4 possesses the lowest temporal order, it requires the smallest time-step size and therefore the largest number of time steps. In contrast, RK9 permits substantially larger CFL numbers while maintaining the same accuracy.

For MMEX14, upgrading from RK4 to RK7 reduces the total computational time by approximately a factor of 11.8. Replacing RK7 with RK9 yields an additional reduction of roughly 40%. Overall, RK9 produces solutions approximately 20.5 times faster than RK4 for the same error level. Naturally, this advantage diminishes if a larger numerical error can be tolerated.

These results demonstrate that very high-order spatial discretizations require correspondingly accurate time-integration methods. When the overall error is dominated by temporal discretization, both the multi-moment and finite-difference methods must operate with similar

time-step sizes, diminishing the advantage offered by the larger cell size of the multi-moment formulation. Conversely, when a sufficiently accurate time integrator is employed, allowing the time step to approach the stability limit, the overall performance approaches the values reported in Table 2.4.

To achieve an error level comparable to MMEX14, the fourteenth-order finite-difference method requires approximately 4,260 degrees of freedom, compared with 4,000 degrees of freedom for the multi-moment method. Even under this matched-accuracy comparison, the finite-difference formulation remains approximately 1.46–1.85 times slower than the corresponding multi-moment implementation. This range is close to the complexity ratio estimated in Ex.2.4, which is 1.86 for $p = 14$.

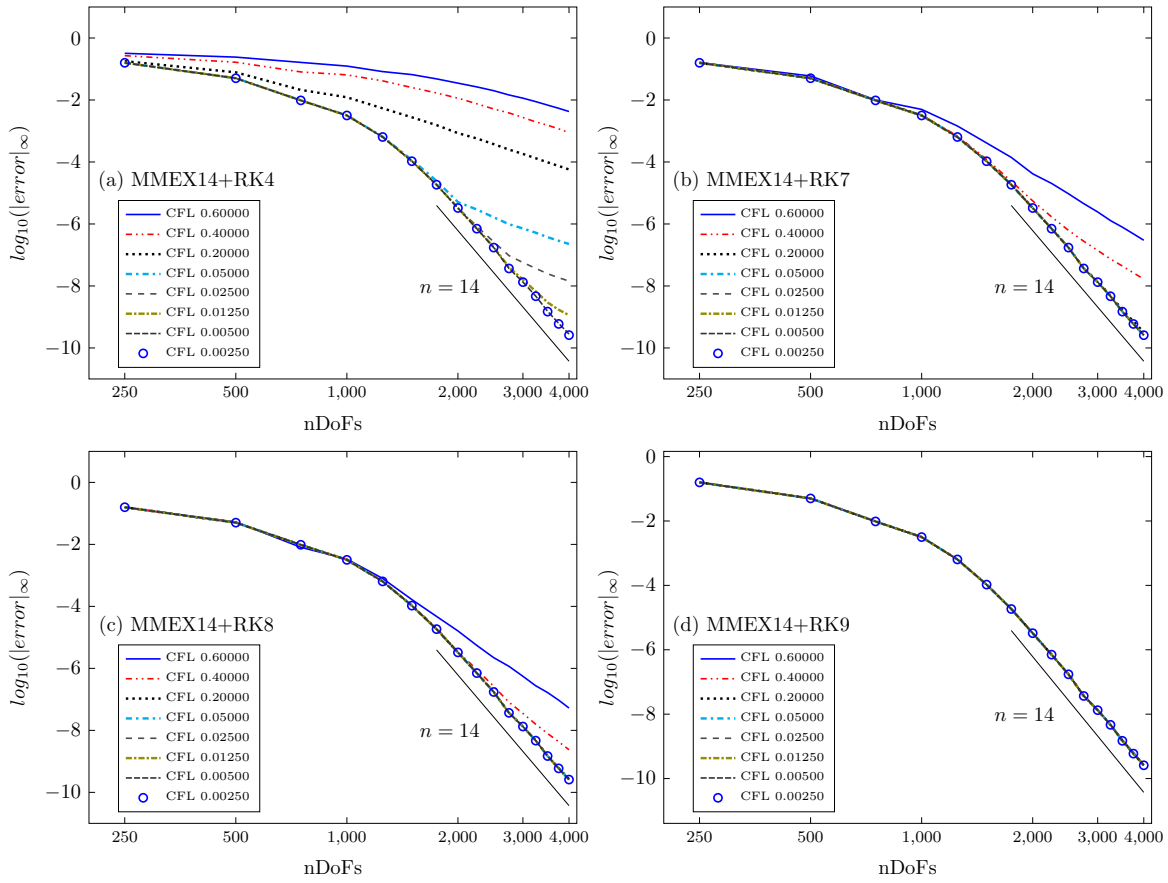


Fig. 2.21: Influence of temporal integration order on the fourteenth-order multi-moment scheme (MMEX14). Increasing the temporal order progressively removes the time-integration error as the dominant source of numerical error, allowing the full spatial accuracy of MMEX14 to be achieved at increasingly larger CFL numbers.

RK4			**RK7**			**RK9**		
CFL	Error	CPU-Time	CFL	Error	CPU-Time	CFL	Error	CPU-Time
0.00425	2.701×10^{-10}	13.7790	0.17	2.757×10^{-10}	1.0472	0.65	2.790×10^{-10}	0.6281
0.00420	2.695×10^{-10}	13.9650	0.16	2.650×10^{-10}	1.1810	0.60	2.667×10^{-10}	0.6799

Table 2.9: Performance comparison of RK4, RK7, and RK9 coupled with the fourteenth-order multi-moment scheme (MMEX14). Each configuration is selected to achieve approximately the same maximum error after one convection cycle using 4,000 degrees of freedom. The table reports the CFL number required to attain the target accuracy together with the corresponding CPU time.

RK4			**RK7**			**RK9**		
CFL	Error	CPU-Time	CFL	Error	CPU-Time	CFL	Error	CPU-Time
0.0070	2.719×10^{-10}	23.89	0.26	2.689×10^{-10}	1.6583	0.85	2.680×10^{-10}	0.9442
0.0065	2.708×10^{-10}	25.82	0.25	2.686×10^{-10}	1.7221	0.80	2.678×10^{-10}	1.0012

Table 2.10: Performance comparison of RK4, RK7, and RK9 coupled with the fourteenth-order finite-difference scheme (FDEX14). Each configuration is chosen to produce approximately the same maximum error after one convection cycle using 4,260 degrees of freedom. The table highlights the computational advantage of higher-order time integration when solutions are compared at matched accuracy.

Problems

Problem 2.1 Consider the one-dimensional linear convection equation

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = 0,$$

where $u > 0$ is constant. Using the method of characteristics, derive the exact solution for the initial condition $\phi(x, 0) = f(x)$. What two important physical properties of the initial profile are preserved by the exact solution?

Problem 2.2 Explain why the one-dimensional convection equation is classified as a hyperbolic partial differential equation. Compare its characteristic structure with that of the second-order wave equation

$$\frac{\partial^2 \phi}{\partial t^2} = c^2 \frac{\partial^2 \phi}{\partial x^2}.$$

In your answer, explain the meanings of the domain of dependence and the range of influence.

Problem 2.3 The one-dimensional multi-moment method simultaneously evolves point values ϕ_i and cell-averaged values $\overline{\phi}_{i+1/2}$. Write the corresponding semi-discrete finite-difference and finite-volume equations for the linear convection equation. Explain why the finite-volume equation requires no additional reconstruction of the convective flux when point values are available.

Problem 2.4 Consider the one-dimensional convection–diffusion equation

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = K\frac{\partial^2 \phi}{\partial x^2},$$

where u and K are constants. Define the cell-average moment over $\Omega_{i-1/2} = [x_{i-1}, x_i]$ by

$$\overline{\phi}_{i-1/2} = \frac{1}{h}\int_{x_{i-1}}^{x_i} \phi(x)\, dx.$$

Derive the evolution equations for the point value ϕ_i and the cell-average value $\overline{\phi}_{i-1/2}$. Show that the resulting coupled equations are

$$\frac{\partial \phi_i}{\partial t} + u\left(\frac{\partial \phi}{\partial x}\right)_i = K\left(\frac{\partial^2 \phi}{\partial x^2}\right)_i,$$

and

$$h\frac{\partial \overline{\phi}_{i-1/2}}{\partial t} + u(\phi_i - \phi_{i-1}) = K\left[\left(\frac{\partial \phi}{\partial x}\right)_i - \left(\frac{\partial \phi}{\partial x}\right)_{i-1}\right].$$

Explain why the point values can be used directly as numerical fluxes in the cell-average equation without a deconvolution operation.

Problem 2.5 Consider the fourth-order multi-moment approximation

$$\phi_i' = \frac{\beta_1}{h}\left(\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}\right) - \frac{\gamma_1}{h}\left(\phi_{i+1} - \phi_{i-1}\right).$$

Expand all point-value and cell-average quantities in Taylor series about x_i. Determine the coefficients β_1 and γ_1 required for fourth-order accuracy. Show that

$$\beta_1 = 2, \qquad \gamma_1 = \frac{1}{2},$$

and derive the leading truncation error

$$\text{LTE} = -\frac{h^4}{360}\phi_i^{(5)}.$$

Compare the physical stencil width with that of the conventional fourth-order explicit central finite-difference approximation.

Problem 2.6 Using Taylor-series expansions about x_i, derive the fourth-order explicit multi-moment approximation

$$\left.\frac{\partial \phi}{\partial x}\right|_i = \frac{2}{h}\left(\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}\right) - \frac{1}{2h}\left(\phi_{i+1} - \phi_{i-1}\right) + O(h^4).$$

Show explicitly how the leading second-order truncation-error terms cancel, and determine the leading truncation error of the resulting approximation.

Problem 2.7 Construct a sixth-order multi-moment approximation of the first derivative using the stencil

$$\phi_i' = \frac{\beta_1}{h}\left(\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}\right) + \frac{\beta_2}{h}\left(\overline{\phi}_{i+3/2} - \overline{\phi}_{i-3/2}\right) - \frac{\gamma_1}{h}\left(\phi_{i+1} - \phi_{i-1}\right).$$

Derive the linear system for β_1, β_2, and γ_1 using Taylor expansions, solve for the coefficients, and determine the leading truncation error. Verify the formal convergence rate numerically using $\phi(x) = e^x$ at $x = 0$ and the observed-order formula

$$p = \log_2\left(\frac{|E(h)|}{|E(h/2)|}\right).$$

Problem 2.8 Consider the multi-moment approximation of the second derivative

$$\phi_i'' = \frac{d}{h^2}\phi_i + \frac{c_1}{h^2}\left(\overline{\phi}_{i+1/2} + \overline{\phi}_{i-1/2}\right) + \frac{b_1}{h^2}\left(\phi_{i+1} + \phi_{i-1}\right).$$

Use Taylor expansions to determine d, c_1, and b_1 for the highest formal order attainable with this stencil. Derive the leading truncation error. Apply the resulting operator to $e^{\mathbf{i}kx}$ and obtain the squared modified wavenumber k^{2*}. Using

$$\epsilon_D = \frac{\left|k^{2*}h^2 - (kh)^2\right|}{(kh)^2},$$

determine the largest value of kh for which $\epsilon_D < 0.01$.

Problem 2.9 The Taylor-table system used to derive very-high-order explicit multi-moment schemes becomes increasingly Vandermonde-like as the approximation order increases. Explain why this creates a numerical difficulty. Describe the recommended procedure for computing high-order coefficients reliably, and explain why this difficulty does not necessarily increase the computational cost of the final flow solver.

Problem 2.10 Contrast an explicit multi-moment approximation with an implicit compact approximation of the first derivative. Why can the explicit multi-moment method achieve a compact physical footprint without requiring the solution of a global or semi-global linear system? Discuss the implications for parallel computing.

Problem 2.11 For the Fourier mode

$$\phi(x) = e^{\mathrm{j}kx},$$

derive the exact expression for the cell average $\overline{\phi}_{i+1/2}$ over the interval $[x_i, x_i + h]$. Show that the result can also be written in the form

$$\overline{\phi}_{i+1/2} = e^{\mathrm{j}k(x_i+h/2)} \frac{2\sin(kh/2)}{kh}.$$

What does the factor multiplying the pointwise Fourier mode indicate about the effect of cell averaging on high-wavenumber information?

Problem 2.12 Derive the modified wavenumbers of the second-order central finite-difference scheme and the second-order explicit multi-moment scheme:

$$k^*_{\mathrm{FDEX2}} = \frac{\sin(kh)}{h}, \qquad k^*_{\mathrm{MMEX2}} = \frac{4\sin^2(kh/2)}{kh^2}.$$

Expand both expressions for $kh \ll 1$ and compare their leading-order wavenumber errors.

Problem 2.13 Suppose a numerical differentiation scheme produces a complex modified wavenumber

$$k^* = k^*_R + \mathrm{j}k^*_I.$$

Show how the exact Fourier-mode solution of the convection equation is modified when k is replaced by k^*. Explain the physical meanings of separately k^*_R and k^*_I, and identify which component is responsible for dispersive error and which is responsible for dissipative error.

Problem 2.14 Let

$$\phi(x) = e^{\mathbf{i}kx}.$$

Show that its cell-average representation over a cell of width h is

$$\overline{\phi}(x) = \frac{2}{kh} \sin\left(\frac{kh}{2}\right) e^{\mathbf{i}kx}.$$

Apply the fourth-order multi-moment first-derivative approximation to this Fourier mode and derive its initial modified wavenumber k^*. Obtain the transfer function

$$T_D(kh) = \frac{k^*}{k},$$

and expand it about $kh = 0$ to verify the formal order of accuracy. Determine numerically the maximum wavenumber satisfying:

$$\left|\frac{k^* - k}{k}\right| < 0.01,$$

and compute the corresponding resolving efficiency

$$e(0.01) = \frac{(kh)_{\max}}{\pi}.$$

Problem 2.15 What is meant by the *Initial Modified Wavenumber Analysis* (IMWA) of a multi-moment scheme? Why does this analysis not necessarily predict the long-time propagation speed of the fully coupled point-value/cell-average system? Explain also why the effective nondimensional wavenumber:

$$kh_{\text{eff}} = \frac{kh}{\text{nDoFC}}$$

is introduced when comparing multi-moment and finite-difference methods.

Problem 2.16 For pure convection, consider the coupled Fourier system

$$\frac{\partial}{\partial t}\begin{bmatrix}\widehat{\widehat{\phi}}_k \\ \widehat{\phi}_k\end{bmatrix} = -\mathbf{i}uk\begin{bmatrix}\mathcal{B}_2 & \mathcal{B}_1 \\ 1 & 0\end{bmatrix}\begin{bmatrix}\widehat{\widehat{\phi}}_k \\ \widehat{\phi}_k\end{bmatrix}.$$

Derive the characteristic polynomial of the coupling matrix and show that the two eigenvalues are

$$T_{1,2} = \frac{1}{2}\left(\mathcal{B}_2 \pm \sqrt{\mathcal{B}_2^2 + 4\mathcal{B}_1}\right).$$

Define

$$\mathcal{F} = T_1 - T_2$$

and show that the modulation angular frequency is

$$\omega_m = uk\mathcal{F}.$$

For exact initialization, derive:

$$C_1 = \frac{1 - T_2}{\mathcal{F}}, \qquad C_2 = \frac{T_1 - 1}{\mathcal{F}}.$$

Explain the physical interpretation of the dominant and parasitic eigenmodes when $T_1 > 0$ and $T_2 < 0$.

Problem 2.17 Consider the linear test equation

$$\frac{dy}{dt} = \lambda y$$

and the degree-m Taylor stability polynomial

$$R_m(z) = \sum_{j=0}^{m} \frac{z^j}{j!}, \qquad z = \lambda\Delta t.$$

For $m = 4$, set $z = \mathbf{i}\eta$ and derive $|R_4(\mathbf{i}\eta)|^2$. Determine the largest positive value $\eta_{\max}$ satisfying

$$|R_4(\mathbf{i}\eta)| \leq 1.$$

For a multi-moment spatial approximation having

$$hk^*_{\max} = 3.58,$$

compute the estimated maximum convection CFL number from

$$\text{CFL}_{\max} = \frac{\eta_{\max}}{hk^*_{\max}}.$$

Repeat the calculation for a scheme having $hk^*_{\max} = 1.99$ and explain why the scheme with greater spectral resolving capability generally has a more restrictive stable time step.

Problem 2.18 The multi-moment Dirichlet boundary closure for the first derivative is

$$\phi_0' = \frac{23}{2h}\overline{\phi}_{1/2} + \frac{7}{2h}\overline{\phi}_{3/2} - \frac{6}{h}\phi_0 - \frac{8}{h}\phi_1 - \frac{1}{h}\phi_2.$$

The corresponding Neumann boundary approximation for the second derivative is

$$\phi_0'' = -\frac{97}{8h^2}\overline{\phi}_{1/2} - \frac{113}{8h^2}\overline{\phi}_{3/2} - \frac{13}{4h^2}\phi_0 + \frac{22}{h^2}\phi_1 + \frac{17}{4h^2}\phi_2.$$

Expand the pointwise and cell-average data about $x = 0$ and verify the formal order of each boundary approximation. Determine the leading truncation errors. Explain why a lower-order boundary closure can still be used together with a very-high-order interior approximation when the grid is sufficiently compressed near the boundary.

Problem 2.19 Suppose that evaluating a first-derivative operator requires M arithmetic operations. For the one-dimensional convection problem, assume that the coupled multi-moment formulation requires approximately:

$$C_{\mathrm{MM}} = M + 2$$

operations per cell, whereas a conventional finite-difference method on a once-refined grid requires approximately

$$C_{\mathrm{FD}} = 2M.$$

Derive the theoretical speedup:

$$S(M) = \frac{2M}{M+2}.$$

Evaluate S for

$$M = 3, \quad 7, \quad 15, \quad 31,$$

and determine

$$\lim_{M\to\infty} S(M).$$

Find the minimum value of M for which $S > 1.75$. Discuss why this operation-count estimate alone may not accurately predict measured wall-clock performance on modern processors.

Problem 2.20 Implement the coupled multi-moment method for the periodic one-dimensional advection equation

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = 0, \qquad u = 1,$$

with the initial condition

$$\phi(x, 0) = \sin(kx).$$

Initialize the point values exactly and obtain the corresponding cell-average values by analytical integration. Advance both moments using a sufficiently small time step and monitor:

$$G_{\mathrm{PV}}(t) = \frac{\|\phi_{\mathrm{PV}}(t)\|_2}{\|\phi_{\mathrm{PV}}(0)\|_2}, \qquad G_{\mathrm{CV}}(t) = \frac{\|\phi_{\mathrm{CV}}(t)\|_2}{\|\phi_{\mathrm{CV}}(0)\|_2}.$$

Measure the numerical modulation period and compare it with the CMWA prediction

$$T_m = \frac{2\pi}{uk|T_1 - T_2|}.$$

Repeat the experiment for at least three grid resolutions and determine how the amplitude of the modulation scales with grid spacing. Discuss whether the modulation grows, decays, or remains bounded and relate the result to the exchange of information between the PV and CV moments.

Solution Key

1. The characteristic equation associated with

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} = 0$$

is

$$\frac{dx}{dt} = u.$$

Hence,

$$x - ut = C,$$

where C is constant along a characteristic. Along this curve,

$$\frac{d\phi}{dt} = \frac{\partial \phi}{\partial t} + \frac{dx}{dt}\frac{\partial \phi}{\partial x} = 0.$$

Therefore ϕ remains constant along each characteristic. Using $\phi(x, 0) = f(x)$ gives

$$\boxed{\phi(x, t) = f(x - ut)}.$$

The exact solution therefore preserves:

a. the amplitude of the initial profile, and
b. the shape of the initial profile.

The profile is translated at velocity u, with neither physical dissipation nor dispersion.

2. The convection equation is hyperbolic because information propagates at a finite speed along characteristic curves. For constant u, there is one family of characteristics,

$$x - ut = \text{constant}.$$

For the second-order wave equation

$$\frac{\partial^2 \phi}{\partial t^2} = c^2\frac{\partial^2 \phi}{\partial x^2},$$

there are two families,

$$x - ct = \text{constant}, \qquad x + ct = \text{constant},$$

representing waves traveling in opposite directions.
The *domain of dependence* of a space–time point contains all earlier locations that can influence that point. The *range of influence* contains all future locations that can be affected by a disturbance introduced at that point. These regions are bounded by the characteristic curves.

3. For the point-value degree of freedom, the governing equation is

$$\boxed{\frac{\partial \phi_i}{\partial t} + u\left(\frac{\partial \phi}{\partial x}\right)_i = 0}.$$

Integrating the convection equation over $[x_i, x_{i+1}]$ gives

$$\frac{\partial}{\partial t}\int_{x_i}^{x_{i+1}} \phi \, dx + u(\phi_{i+1} - \phi_i) = 0.$$

Since

$$\overline{\phi}_{i+1/2} = \frac{1}{h}\int_{x_i}^{x_{i+1}} \phi \, dx,$$

the finite-volume equation becomes

$$\boxed{h\frac{\partial \overline{\phi}_{i+1/2}}{\partial t} + u(\phi_{i+1} - \phi_i) = 0}.$$

The interface values ϕ_i and ϕ_{i+1} are already dependent variables of the multi-moment formulation. Therefore the convective fluxes $u\phi_i$ and $u\phi_{i+1}$ are directly available and no deconvolution from the cell averages is required.

4. Applying the convection–diffusion equation directly at x_i gives

$$\boxed{\frac{\partial \phi_i}{\partial t} + u\left(\frac{\partial \phi}{\partial x}\right)_i = K\left(\frac{\partial^2 \phi}{\partial x^2}\right)_i}.$$

Integrating over $\Omega_{i-1/2} = [x_{i-1}, x_i]$ gives

$$\frac{\partial}{\partial t}\int_{x_{i-1}}^{x_i} \phi \, dx + u\int_{x_{i-1}}^{x_i} \phi_x \, dx = K\int_{x_{i-1}}^{x_i} \phi_{xx} \, dx.$$

Application of the fundamental theorem of calculus yields

$$\boxed{h\frac{\partial \overline{\phi}_{i-1/2}}{\partial t} + u(\phi_i - \phi_{i-1}) = K\left[\left(\frac{\partial \phi}{\partial x}\right)_i - \left(\frac{\partial \phi}{\partial x}\right)_{i-1}\right]}.$$

Because the point values occur exactly at the finite-volume interfaces, the convective fluxes are available directly. The diffusive flux still requires approximation of the first derivative.

5. The cell-average expansions are

$$\overline{\phi}_{i\pm1/2} = \phi_i \pm \frac{h}{2}\phi_i' + \frac{h^2}{6}\phi_i'' \pm \frac{h^3}{24}\phi_i''' + \frac{h^4}{120}\phi_i^{(4)} \pm \frac{h^5}{720}\phi_i^{(5)} + \cdots.$$

Hence,

$$\frac{\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}}{h} = \phi_i' + \frac{h^2}{12}\phi_i''' + \frac{h^4}{360}\phi_i^{(5)} + O(h^6).$$

Similarly,

$$\frac{\phi_{i+1} - \phi_{i-1}}{h} = 2\phi_i' + \frac{h^2}{3}\phi_i''' + \frac{h^4}{60}\phi_i^{(5)} + O(h^6).$$

Substitution into the proposed approximation gives the conditions

$$\beta_1 - 2\gamma_1 = 1,$$

$$\frac{\beta_1}{12} - \frac{\gamma_1}{3} = 0.$$

Therefore

$$\boxed{\beta_1 = 2, \qquad \gamma_1 = \frac{1}{2}}.$$

The resulting approximation is

$$\boxed{\phi_i' = \frac{2}{h}\left(\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}\right) - \frac{1}{2h}(\phi_{i+1} - \phi_{i-1})}.$$

Its expansion is

$$D_h\phi_i = \phi_i' - \frac{h^4}{360}\phi_i^{(5)} + O(h^6),$$

so

$$\boxed{\mathrm{LTE} = -\frac{h^4}{360}\phi_i^{(5)}}.$$

The MM stencil occupies only $[x_{i-1}, x_{i+1}]$, whereas the conventional fourth-order explicit FDM requires $[x_{i-2}, x_{i+2}]$.

6. From

$$\frac{\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}}{h} = \phi_i' + \frac{h^2}{12}\phi_i''' + \frac{h^4}{360}\phi_i^{(5)} + \cdots$$

and

$$\frac{\phi_{i+1} - \phi_{i-1}}{2h} = \phi_i' + \frac{h^2}{6}\phi_i''' + \frac{h^4}{120}\phi_i^{(5)} + \cdots,$$

taking twice the first expression minus the second gives

$$2\left[\phi_i' + \frac{h^2}{12}\phi_i''' + \frac{h^4}{360}\phi_i^{(5)}\right] - \left[\phi_i' + \frac{h^2}{6}\phi_i''' + \frac{h^4}{120}\phi_i^{(5)}\right].$$

The ϕ_i''' terms cancel exactly, yielding:

$$\boxed{\phi_i' = \frac{2}{h}\left(\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}\right) - \frac{\phi_{i+1} - \phi_{i-1}}{2h} + O(h^4)}.$$

The leading error of the discrete operator is

$$\boxed{-\frac{h^4}{360}\phi_i^{(5)}}.$$

7. The required Taylor expansions give

$$\frac{\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}}{h} = \phi' + \frac{h^2}{12}\phi''' + \frac{h^4}{360}\phi^{(5)} + \frac{h^6}{20160}\phi^{(7)} + \cdots,$$

$$\frac{\overline{\phi}_{i+3/2} - \overline{\phi}_{i-3/2}}{h} = 3\phi' + \frac{5}{4}h^2\phi''' + \frac{7}{40}h^4\phi^{(5)} + \frac{17}{1344}h^6\phi^{(7)} + \cdots,$$

and

$$\frac{\phi_{i+1} - \phi_{i-1}}{h} = 2\phi' + \frac{1}{3}h^2\phi''' + \frac{1}{60}h^4\phi^{(5)} + \frac{1}{2520}h^6\phi^{(7)} + \cdots.$$

Matching the coefficients of ϕ', ϕ''', and $\phi^{(5)}$ gives

$$\begin{bmatrix} 1 & 3 & -2 \\ 1/12 & 5/4 & -1/3 \\ 1/360 & 7/40 & -1/60 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \gamma_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Solving yields

$$\boxed{\beta_1 = \frac{9}{4}, \qquad \beta_2 = \frac{1}{36}, \qquad \gamma_1 = \frac{2}{3}}.$$

Therefore

$$\boxed{\phi_i' = \frac{9}{4h}\left(\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}\right) + \frac{1}{36h}\left(\overline{\phi}_{i+3/2} - \overline{\phi}_{i-3/2}\right) - \frac{2}{3h}(\phi_{i+1} - \phi_{i-1})}.$$

The leading truncation error is

$$\boxed{\text{LTE} = \frac{h^6}{5040}\phi_i^{(7)}}.$$

For $\phi = e^x$ at $x = 0$, $\phi^{(7)}(0) = 1$, so asymptotically

$$E(h) \simeq \frac{h^6}{5040}.$$

Thus,

$$\frac{E(h)}{E(h/2)} \to 2^6 = 64,$$

and

$$\boxed{p = \log_2(64) = 6}.$$

8. Using

$$\overline{\phi}_{i+1/2} + \overline{\phi}_{i-1/2} = 2\phi_i + \frac{h^2}{3}\phi_i'' + \frac{h^4}{60}\phi_i^{(4)} + \frac{h^6}{2520}\phi_i^{(6)} + \cdots$$

and

$$\phi_{i+1} + \phi_{i-1} = 2\phi_i + h^2\phi_i'' + \frac{h^4}{12}\phi_i^{(4)} + \frac{h^6}{360}\phi_i^{(6)} + \cdots,$$

the coefficient conditions are

$$d + 2c_1 + 2b_1 = 0,$$
$$\frac{c_1}{3} + b_1 = 1,$$
$$\frac{c_1}{60} + \frac{b_1}{12} = 0.$$

Solving gives

$$\boxed{d = -12, \qquad c_1 = \frac{15}{2}, \qquad b_1 = -\frac{3}{2}}.$$

Hence,

$$\boxed{\phi_i'' = -\frac{12}{h^2}\phi_i + \frac{15}{2h^2}\left(\overline{\phi}_{i+1/2} + \overline{\phi}_{i-1/2}\right) - \frac{3}{2h^2}(\phi_{i+1} + \phi_{i-1})}.$$

The leading truncation error is

$$\boxed{\text{LTE} = -\frac{h^4}{840}\phi_i^{(6)}}.$$

Let $\theta = kh$. For a Fourier mode,

$$h^2 D_2 = -12 + 15\frac{\sin\theta}{\theta} - 3\cos\theta.$$

Writing

$$D_2 e^{\mathbf{i}kx} = -k^{2*} e^{\mathbf{i}kx}$$

gives

$$\boxed{k^{2*}h^2 = 12 - 15\frac{\sin\theta}{\theta} + 3\cos\theta}.$$

Thus,

$$\epsilon_D = \frac{\left|12 - 15\frac{\sin\theta}{\theta} + 3\cos\theta - \theta^2\right|}{\theta^2}.$$

Numerically,

$$\boxed{(kh)_{\max} \approx 1.739}$$

for $\epsilon_D < 0.01$.

9. At high order, the Taylor-table matrix contains increasingly high powers of the stencil coordinates. Its structure therefore becomes Vandermonde-like and its condition number grows rapidly.
   Consequently, small floating-point errors can produce substantial errors in the calculated coefficients.
   A practical procedure is to:

   a. generate the coefficients during preprocessing using extended or quadruple precision,
   b. verify the Taylor constraints using the same high precision, and
   c. store the final coefficients in ordinary double precision for the actual solver.

   The high-precision calculation is performed only once during coefficient generation. It therefore does not significantly increase the computational cost of the time-dependent flow solver.
10. An explicit MM approximation evaluates a derivative directly from known PV and CV data,

$$\phi_i' = D(\phi, \overline{\phi}),$$

whereas a compact scheme typically has the form

$$a\phi_{i-1}' + \phi_i' + a\phi_{i+1}' = D(\phi).$$

Thus, compact schemes require the solution of a coupled linear system for the derivatives. The MM method obtains additional information by using multiple local data types rather than by increasing the spatial extent of the stencil. It can therefore achieve a compact physical footprint while remaining explicit.
This eliminates forward/backward substitution and reduces sequential dependencies, making explicit MM schemes particularly attractive for parallel architectures.

11. For

$$\phi(x) = e^{\mathrm{j}kx},$$

the cell average is

$$\overline{\phi}_{i+1/2} = \frac{1}{h}\int_{x_i}^{x_i+h} e^{\mathrm{j}kx}\,dx.$$

Therefore

$$\overline{\phi}_{i+1/2} = \frac{e^{\mathrm{j}k(x_i+h)} - e^{\mathrm{j}kx_i}}{\mathrm{j}kh}.$$

Using

$$e^{\mathrm{j}\theta} - 1 = 2\mathrm{j}e^{\mathrm{j}\theta/2}\sin\left(\frac{\theta}{2}\right),$$

we obtain

$$\boxed{\overline{\phi}_{i+1/2} = e^{\mathrm{j}k(x_i+h/2)}\frac{2\sin(kh/2)}{kh}}.$$

The factor

$$\boxed{\frac{2\sin(kh/2)}{kh}}$$

is a sinc-type averaging filter. It approaches unity for $kh \to 0$ and decreases for increasing wavenumber, demonstrating that cell averaging attenuates high-wavenumber content.

12. For the second-order central FDM,

$$D\phi_i = \frac{\phi_{i+1} - \phi_{i-1}}{2h}.$$

For a Fourier mode, this becomes

$$D\phi_i = \mathrm{j}\frac{\sin(kh)}{h}\phi_i,$$

so

$$\boxed{k^*_{\mathrm{FDEX2}} = \frac{\sin(kh)}{h}}.$$

For the second-order MM scheme,

$$D\phi_i = \frac{\overline{\phi}_{i+1/2} - \overline{\phi}_{i-1/2}}{h},$$

which gives

$$\boxed{k^*_{\mathrm{MMEX2}} = \frac{4\sin^2(kh/2)}{kh^2}}.$$

Let $\theta = kh$. Then

$$\frac{k^*_{\mathrm{FDEX2}}}{k} = \frac{\sin\theta}{\theta} = 1 - \frac{\theta^2}{6} + \frac{\theta^4}{120} + \cdots,$$

whereas

$$\frac{k^*_{\mathrm{MMEX2}}}{k} = \frac{4\sin^2(\theta/2)}{\theta^2} = 1 - \frac{\theta^2}{12} + \frac{\theta^4}{360} + \cdots.$$

Both are second-order accurate, but the leading wavenumber-error coefficient of MMEX2 is one-half that of FDEX2.

13. Let

$$k^* = k_R^* + \mathrm{j}k_I^*.$$

The Fourier solution becomes

$$\phi(x,t) = \widehat{\phi}_0 e^{-\mathrm{j}uk^*t} e^{\mathrm{j}kx}.$$

Therefore

$$\boxed{\phi(x,t) = \widehat{\phi}_0 e^{uk_I^*t} e^{\mathrm{j}(kx - uk_R^*t)}}.$$

The real part k_R^* determines the numerical phase speed. Consequently,

$$k_R^* \neq k$$

produces *dispersive error*.
The imaginary part controls the amplitude:

$$k_I^* < 0 \quad \Rightarrow \quad \text{numerical damping},$$

whereas

$$k_I^* > 0 \quad \Rightarrow \quad \text{numerical amplification}.$$

Thus k_I^* represents the dissipative or amplifying component of the spatial error.

14. Let

$$\theta = kh.$$

The fourth-order MM approximation gives

$$D_h\phi_i = \mathrm{j}k^*\phi_i,$$

with

$$\boxed{k^*h = \frac{4(1-\cos\theta)}{\theta} - \sin\theta}.$$

Therefore, the transfer function is

$$\boxed{T_D(\theta) = \frac{k^*}{k} = \frac{4(1-\cos\theta)}{\theta^2} - \frac{\sin\theta}{\theta}}.$$

Expanding about $\theta = 0$ gives

$$T_D(\theta) = 1 - \frac{\theta^4}{360} + \frac{\theta^6}{10080} + O(\theta^8).$$

Thus,

$$T_D - 1 = O(\theta^4),$$

confirming fourth-order accuracy.
Solving

$$|T_D(\theta) - 1| = 0.01$$

numerically gives approximately

$$\boxed{(kh)_{\max} \approx 1.402}.$$

Hence the 1% resolving efficiency is

$$\boxed{e(0.01) = \frac{1.402}{\pi} \approx 0.446}.$$

15. Initial Modified Wavenumber Analysis assumes that all moments are initialized exactly from the same continuous Fourier mode. The PV and CV quantities therefore initially possess exactly the continuous relationship implied by analytical averaging.
The resulting modified wavenumber describes the instantaneous response at $t = 0$. After evolution begins, however, the PV and CV variables satisfy different but coupled discrete equations. Their relative amplitudes and phases therefore evolve, and the initial modified wavenumber does not necessarily equal the propagation speed of the dominant eigenmode of the coupled system.
The MM method also possesses more degrees of freedom per geometric cell than conventional FDM. A normalized comparison therefore uses

$$\boxed{kh_{\text{eff}} = \frac{kh}{\text{nDoFC}}}.$$

This compares spectral resolution based on information density rather than physical grid spacing alone.

16. Let

$$\mathbf{M} = \begin{bmatrix} \mathcal{B}_2 & \mathcal{B}_1 \\ 1 & 0 \end{bmatrix}.$$

The characteristic equation is

$$\det(\mathbf{M} - T\mathbf{I}) = 0,$$

or

$$\begin{vmatrix} \mathcal{B}_2 - T & \mathcal{B}_1 \\ 1 & -T \end{vmatrix} = 0.$$

Hence

$$T^2 - \mathcal{B}_2 T - \mathcal{B}_1 = 0.$$

The eigenvalues are therefore

$$\boxed{T_{1,2} = \frac{1}{2}\left(\mathcal{B}_2 \pm \sqrt{\mathcal{B}_2^2 + 4\mathcal{B}_1}\right)}.$$

Their separation is

$$\mathcal{F} = T_1 - T_2.$$

Because the modal factors are

$$e^{-\mathrm{i}ukT_1 t}, \qquad e^{-\mathrm{i}ukT_2 t},$$

their relative phase evolves according to

$$e^{-\mathrm{i}uk(T_1 - T_2)t}.$$

Thus,

$$\boxed{\omega_m = uk\mathcal{F}}.$$

The eigenvector corresponding to T_j is proportional to

$$\begin{bmatrix} T_j \\ 1 \end{bmatrix}.$$

For exact initialization,

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = C_1 \begin{bmatrix} T_1 \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} T_2 \\ 1 \end{bmatrix}.$$

Therefore

$$C_1 + C_2 = 1, \qquad C_1 T_1 + C_2 T_2 = 1.$$

Solving gives

$$\boxed{C_1 = \frac{1 - T_2}{\mathcal{F}}, \qquad C_2 = \frac{T_1 - 1}{\mathcal{F}}}.$$

For $T_1 > 0$, the first mode propagates in the physical direction and is the dominant mode. If $T_2 < 0$, the second mode propagates in the opposite direction and represents a parasitic numerical mode. Its amplitude approaches zero as the grid is refined.

17. For RK4,

$$R_4(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}.$$

Setting $z = \mathrm{i}\eta$ gives

$$R_4(\mathrm{i}\eta) = 1 - \frac{\eta^2}{2} + \frac{\eta^4}{24} + \mathrm{i}\left(\eta - \frac{\eta^3}{6}\right).$$

Therefore

$$|R_4(\mathrm{i}\eta)|^2 = \left(1 - \frac{\eta^2}{2} + \frac{\eta^4}{24}\right)^2 + \left(\eta - \frac{\eta^3}{6}\right)^2.$$

Simplification gives

$$\boxed{|R_4(\mathrm{i}\eta)|^2 = 1 - \frac{\eta^6}{72} + \frac{\eta^8}{576} = 1 + \frac{\eta^6(\eta^2 - 8)}{576}}.$$

Thus stability along the imaginary axis requires

$$\eta^2 \le 8,$$

and

$$\boxed{\eta_{\max} = 2\sqrt{2} \approx 2.82843}.$$

For

$$hk^*_{\max} = 3.58,$$

$$\boxed{\mathrm{CFL}_{\max} = \frac{2.82843}{3.58} \approx 0.790}.$$

For

$$hk^*_{\max} = 1.99,$$

$$\boxed{\mathrm{CFL}_{\max} = \frac{2.82843}{1.99} \approx 1.421}.$$

A scheme capable of resolving larger numerical wavenumbers has a larger spectral radius. The same RK4 stability region must contain this enlarged spectrum, so a smaller time step is required.

18. For the Dirichlet first-derivative closure, expansion about $x = 0$ gives

$$D_h\phi_0 = \phi_0' - \frac{h^4}{90}\phi_0^{(5)} + O(h^5).$$

Hence the approximation is fourth-order:

$$\boxed{\mathrm{LTE}_D = -\frac{h^4}{90}\phi_0^{(5)}}.$$

For the stated Neumann second-derivative formula, however, a consistency problem occurs. Consider the constant function:

$$\phi(x) = C.$$

Then

$$\overline{\phi}_{1/2} = \overline{\phi}_{3/2} = \phi_0 = \phi_1 = \phi_2 = C,$$

while the exact second derivative is zero.
Substitution into the stated formula gives

$$D_h^{(2)}C = \frac{C}{h^2}\left(-\frac{97}{8} - \frac{113}{8} - \frac{13}{4} + 22 + \frac{17}{4}\right) = -\frac{13}{4h^2}C.$$

Therefore,

$$\boxed{D_h^{(2)}C \neq 0}$$

even for a constant function.
Consequently, the Neumann formula as written is not a consistent approximation to ϕ_0'' and no meaningful formal convergence order or conventional leading truncation error can be assigned to it. At least one coefficient in the stated closure must therefore be corrected before this portion of the problem can be completed.
In general, once a consistent boundary closure is used, a lower-order boundary approximation can remain acceptable if the grid spacing near the boundary is sufficiently small that its local error remains below the error generated by the higher-order interior discretization.

19. The theoretical speedup is

$$\boxed{S(M) = \frac{C_{\mathrm{FD}}}{C_{\mathrm{MM}}} = \frac{2M}{M+2}}.$$

The requested values are

$$S(3) = \frac{6}{5} = 1.200,$$

$$S(7) = \frac{14}{9} \approx 1.556,$$

$$S(15) = \frac{30}{17} \approx 1.765,$$

and

$$S(31) = \frac{62}{33} \approx 1.879.$$

Furthermore,

$$\boxed{\lim_{M\to\infty} S(M) = 2}.$$

To determine when $S > 1.75$,

$$\frac{2M}{M+2} > \frac{7}{4}.$$

Thus,

$$8M > 7M + 14,$$

or

$$M > 14.$$

For integer operation counts,

$$\boxed{M_{\min} = 15}.$$

Actual wall-clock performance also depends on memory bandwidth, cache behavior, vectorization, instruction scheduling, communication overhead, synchronization, and implementation details. Arithmetic-operation counts alone therefore provide only a theoretical estimate.

20. For the exact sinusoidal initial condition,

$$\phi_i(0) = \sin(kx_i).$$

The corresponding cell average is

$$\overline{\phi}_{i+1/2}(0) = \frac{1}{h}\int_{x_i}^{x_{i+1}} \sin(kx)\,dx,$$

giving

$$\boxed{\overline{\phi}_{i+1/2}(0) = \frac{\cos(kx_i) - \cos(kx_{i+1})}{kh}}.$$

For a single Fourier component, the coupled solution contains two eigenmodes. With

$$\mathcal{F} = T_1 - T_2,$$

the CMWA modulation factors may be written as

$$A_{\mathrm{PV}}(t) = C_1T_1 + C_2T_2e^{\mathrm{i}ukt\mathcal{F}},$$

and

$$A_{\mathrm{CV}}(t) = C_1 + C_2e^{\mathrm{i}ukt\mathcal{F}}.$$

Therefore,

$$G_{\mathrm{PV}}(t) = |A_{\mathrm{PV}}(t)|, \qquad G_{\mathrm{CV}}(t) = |A_{\mathrm{CV}}(t)|.$$

The angular frequency of the modulation is

$$\omega_m = uk|T_1 - T_2|,$$

so the predicted period is

$$\boxed{T_m = \frac{2\pi}{uk|T_1 - T_2|}}.$$

The measured period should approach this value when the time step is sufficiently small. For a spatial approximation of formal order p, the parasitic-mode contribution decreases asymptotically at the same rate as the leading spatial truncation error. Consequently, under grid refinement,

$$\boxed{A_{\mathrm{mod}} = O(h^p)}$$

for a fixed physical wavenumber.
The modulation does not represent an exponentially growing instability. For the periodic linear-convection problem, the semi-discrete coupled spectrum is purely imaginary, and the PV–CV interaction produces a

bounded periodic exchange of amplitude

between the two moments. The oscillation therefore becomes smaller as the grid is refined. A finite time step may introduce an additional slowly varying amplitude error, so the temporal step should be made sufficiently small when testing the CMWA prediction.

Chapter 3
Convection-Diffusion Equations

"The natural tendency of heat is to diffuse and equalize itself."
William Thomson (Lord Kelvin)

This chapter addresses the application of the multi-moment method to the combined effects of convection and diffusion. The convection-diffusion equation (CDE) is a fundamental tool for understanding heat and mass transfer phenomena in fluids. It can be applied to momentum conservation, chemical and aerosol transport, and heat transfer when combined with their respective physical principles. In situations where convection is dominant, the structures of the transported field can become even more complex than those of the base fluid flow. These complexities arise in various situations, such as scalar transport at high Schmidt numbers, chemical mixing at high Reynolds numbers, and heat transfer at low Prandtl numbers. Attempting to solve these coupled physical phenomena on a single grid can lead to either excessive resolution of the flow or an underresolved scalar field.

The first part of the chapter discusses the diffusion phenomena, their applications in various fields, governing equations, numerical techniques, and accuracy. The latter part addresses the combined effects.

3.1 Phenomena Governed by Diffusion

Diffusion is the macroscopic manifestation of countless microscopic interactions. To understand the origin of the diffusion equation, it is instructive to look at the physics before the mathematics, specifically observing how microscopic chaos reliably produces macroscopic order.

3.1.1 The Physical Mechanism: Brownian Motion and the Random Walk

At the microscopic level, particles within a fluid or gas are in a state of constant, chaotic thermal agitation, known as Brownian motion. These particles continuously collide with one another, resulting in highly erratic, unpredictable trajectories.

We can abstract this physical reality into a statistical model known as a one-dimensional random walk. Imagine a vast ensemble of independent particles suspended in a medium. In a small time interval Δt, each particle undergoes a collision that displaces it by a characteristic distance Δx. Because the thermal motion is entirely random and isotropic, a particle has an equal probability ($p = 1/2$) of jumping to the left ($-\Delta x$) or to the right ($+\Delta x$).

To bridge the gap between individual particle probability and macroscopic continuum mechanics, we must define how these discrete jumps manifest as a continuous field. Consider a small control volume around a point x. After each time step Δt, the random walk redistributes

particles into neighboring volumes. By aggregating the number of particles within a given volume and multiplying by the property of a single particle (such as mass or thermal energy), we obtain the local density. As the ensemble of particles grows vast ($N \to \infty$), this discrete statistical distribution smooths out into a continuous macroscopic concentration field, which we denote as $\phi(x,t)$ (as visualized in Fig. 3.1).

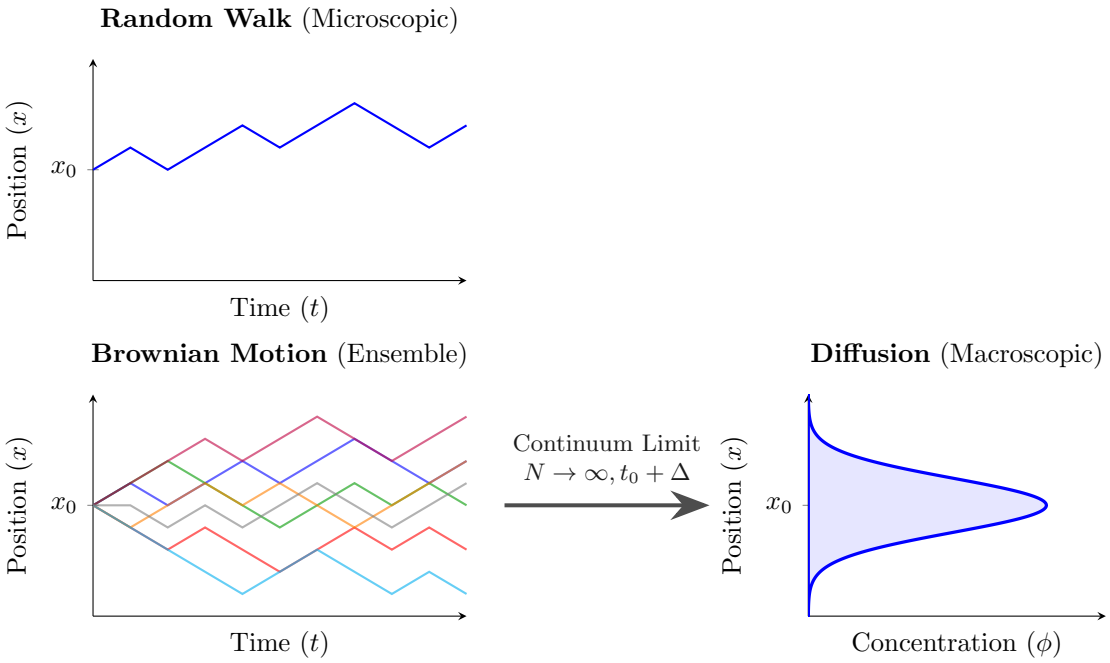


Fig. 3.1: Conceptual transition from a discrete, microscopic random walk of individual particles (left) to the macroscopic, continuous diffusion profile represented by a Gaussian distribution (right).

Having established $\phi(x,t)$ as the continuous concentration, we now seek to determine the net rate at which this quantity crosses an imaginary boundary located at x.

3.1.2 Emergence of Macroscopic Flux

The macroscopic flux, J_x, represents the net amount of the quantity ϕ passing through a unit area per unit time.

During the time interval Δt, half of the particles located at $x - \Delta x$ will randomly jump to the right, crossing the boundary at x. Simultaneously, half of the particles located at $x + \Delta x$ will randomly jump to the left, also crossing the boundary but in the opposite direction. The net flux moving in the positive x-direction is therefore the difference between these two streams:

$$J_x = \frac{\text{Jump Right} - \text{Jump Left}}{\Delta t} = \frac{1}{\Delta t}\left[\frac{1}{2}\phi(x-\Delta x,t)\Delta x - \frac{1}{2}\phi(x+\Delta x,t)\Delta x\right] \tag{3.1}$$

Factoring out the terms yields:

$$J_x = -\frac{(\Delta x)^2}{2\Delta t}\left[\frac{\phi(x+\Delta x,t) - \phi(x-\Delta x,t)}{\Delta x}\right] \tag{3.2}$$

To transition from discrete microscopic jumps to a continuous macroscopic model, we take the continuum limit ($\Delta x \to 0$ and $\Delta t \to 0$). The term in the brackets perfectly resolves into the spatial derivative, $\frac{\partial \phi}{\partial x}$. The coefficient in front must remain finite for the physics to hold, and we define this limit as the macroscopic diffusion coefficient, K:

$$K = \lim_{\Delta x, \Delta t \to 0} \frac{(\Delta x)^2}{2\Delta t} \tag{3.3}$$

Substituting K yields the constitutive relationship for diffusive flux:

$$J_x = -K \frac{\partial \phi}{\partial x} \tag{3.4}$$

This is one of the most important empirical relations in physics. When ϕ is temperature, this is *Fourier's Law of Heat Conduction*; when ϕ is mass concentration, this is *Fick's First Law of Diffusion*. The negative sign mathematically guarantees that a conserved quantity will always transport "down" the gradient—from regions of higher intensity to regions of lower intensity.

3.1.3 The Macroscopic Diffusion Equation

To track how the concentration ϕ changes over time at any given point, we must enforce the conservation of mass (or energy). The accumulation of ϕ within a tiny control volume is equal to the negative divergence of the flux (what flows in minus what flows out). In one dimension, this is expressed by the continuity equation:

$$\frac{\partial \phi}{\partial t} = -\frac{\partial J_x}{\partial x} \tag{3.5}$$

Substituting our macroscopic flux expression ($J_x = -K \frac{\partial \phi}{\partial x}$) into the continuity equation directly yields the classical one-dimensional diffusion equation:

$$\frac{\partial \phi}{\partial t} = \frac{\partial}{\partial x} \left(K \frac{\partial \phi}{\partial x} \right) = K \frac{\partial^2 \phi}{\partial x^2} \tag{3.6}$$

Thus, the second-order spatial derivative in the continuum diffusion equation is not merely a mathematical abstraction. It directly reflects the conservation of a quantity whose flux is driven by the microscopic, chaotic random walks of independent particles.

The general transient diffusion equation in three dimensions is expressed as:

$$\frac{\partial \phi}{\partial t} = \nabla \cdot (K \nabla \phi) + S, \tag{3.7}$$

where S represents any internal source terms.

3.1.4 Heat Conduction

Heat conduction describes the transfer of thermal energy through a material due to temperature gradients. The process is governed by Fourier's law, which states that the heat flux vector is proportional to the negative temperature gradient:

$$\mathbf{q} = -k\nabla T, \tag{3.8}$$

where $\mathbf{q}$ is the heat flux vector, k is the thermal conductivity, and T is the temperature.

Combining Fourier's law with the conservation of energy yields the transient heat diffusion equation:

$$\frac{\partial T}{\partial t} = \alpha\nabla^2 T + \frac{\dot{q}}{\rho c_p}, \tag{3.9}$$

where

$$\alpha = \frac{k}{\rho c_p} \tag{3.10}$$

is the thermal diffusivity, ρ is the density, c_p is the specific heat capacity, and $\dot{q}$ denotes volumetric heat generation.

Heat conduction problems are frequently employed in Method of Manufactured Solutions (MMS) studies because the governing equations possess a simple diffusive structure while remaining representative of many practical engineering applications.

3.1.5 Mass Diffusion

Mass diffusion refers to the transport of chemical species within a fluid or solid medium due to concentration gradients. Fick's first law describes this phenomenon:

$$\mathbf{J} = -D\nabla C, \tag{3.11}$$

where $\mathbf{J}$ is the diffusive mass flux, D is the mass diffusivity, and C is the species concentration.

Applying species conservation leads to Fick's second law:

$$\frac{\partial C}{\partial t} = D\nabla^2 C + R, \tag{3.12}$$

where R represents the rate of species generation or consumption due to chemical reactions.

Mass diffusion equations are widely used in pollutant transport, gas separation processes, moisture migration, chemical reactor analysis, and environmental modeling. Their mathematical form is identical to that of the heat diffusion equation, differing only in the physical interpretation of the transported quantity.

3.1.6 Momentum Diffusion

Due to the Brownian motion of fluids, there is always an exchange of mass with the adjacent fluid. This includes the momentum. Thus, when bulk momentum is imbalanced between adjacent fluid layers, Brownian motion creates a net exchange of momentum opposite to the momentum gradient. This phenomenon can be interpreted as the momentum diffusion or the viscous force whose constitutive relationship is Newton's law of viscosity:

$$\tau = \mu\frac{\partial u}{\partial y}, \tag{3.13}$$

where τ is the shear stress, μ is the dynamic viscosity, and u is the velocity component.

In fluid mechanics, momentum diffusion appears as the viscous term in the Navier–Stokes equations:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \tag{3.14}$$

where

$$\nu = \frac{\mu}{\rho} \tag{3.15}$$

is the kinematic viscosity, p is the pressure, and $\mathbf{f}$ denotes body forces. The term

$$\nu \nabla^2 \mathbf{u} \tag{3.16}$$

represents the diffusion of momentum and is mathematically analogous to the diffusion terms encountered in heat and mass transport equations.

3.1.7 Common Mathematical Structure

Although heat, mass, and momentum diffusion involve different physical quantities, they all share the same underlying mathematical structure. Assuming a constant diffusion coefficient and neglecting source terms, Equation (3.7) reduces to

$$\frac{\partial \phi}{\partial t} = K \nabla^2 \phi. \tag{3.17}$$

Equation (3.17) is commonly referred to as the classical diffusion equation and forms the basis of numerous numerical verification studies.

Regardless of the transported variable, diffusion acts as a natural smoothing mechanism that reduces spatial gradients, dissipates fluctuations, and promotes equilibrium throughout the domain. This behavior contrasts with convection-dominated transport processes, where steep gradients and propagating fronts may develop.

The universality of Equation (3.17) makes diffusion problems particularly suitable for verification using the Method of Manufactured Solutions (MMS). By constructing an analytical solution and introducing a corresponding source term, MMS enables rigorous assessment of numerical accuracy, code correctness, and convergence properties across a broad range of engineering applications.

3.2 Mathematical Regimes: Parabolic and Elliptic Behavior

Depending on the temporal state of the system, the diffusion equation exhibits two distinct mathematical behaviors, classifying it as either a parabolic or an elliptic partial differential equation (PDE). Understanding these regimes is critical for selecting appropriate spatial and temporal discretization schemes in the multi-moment method.

3.2.1 Classification of Second-Order Partial Differential Equations

The most general linear second-order partial differential equation in two independent variables can be written as

$$A\,\phi_{xx} + B\,\phi_{xy} + C\,\phi_{yy} + D\,\phi_x + E\,\phi_y + F\,u = G, \tag{3.18}$$

where A, B, C, D, E, F, and G may be constants or functions of the independent variables.

The classification of Equation (3.18) is determined by the discriminant

$$\Delta = B^2 - 4AC. \tag{3.19}$$

Depending on the sign of Δ, the PDE is classified as

$$\begin{cases} \Delta < 0, & \text{Elliptic,} \\ \Delta = 0, & \text{Parabolic,} \\ \Delta > 0, & \text{Hyperbolic.} \end{cases} \tag{3.20}$$

This classification is analogous to that of quadratic equations and provides valuable insight into the behavior of the solution. Elliptic equations are typically associated with equilibrium or steady-state phenomena, parabolic equations describe diffusive transport processes, and hyperbolic equations govern wave propagation and information transport along characteristic curves.

3.2.2 The Hyperbolic Regime (Thermal Wave Propagation)

When the heat input to a material is extremely intense, or occurs over vanishingly short time scales, the classical Fourier heat conduction theory ceases to provide an adequate description of thermal transport. Fourier's law fundamentally implies that thermal disturbances propagate instantaneously throughout the entire domain, corresponding to an infinite speed of heat propagation. While this mathematical continuum assumption is perfectly acceptable for the vast majority of engineering applications, it breaks down in regimes characterized by ultrafast heating, micro- and nanoscale dimensions, or ultra-low temperatures.

Notable examples where Fourier's law fails include short-pulse laser heating of metallic surfaces [3, 39], nanoscale quantum heat transport [22], and highly transient processes in materials with a nonhomogeneous inner structure (such as porous-capillary bodies or suspensions) [17]. Under these conditions, the interaction of structural elements causes thermal transport to exhibit wave-like characteristics, propagating at a finite speed. In this regime, the temperature field is modeled by the hyperbolic heat conduction equation (derived via the Cattaneo-Vernotte formulation):

$$\frac{K}{c^2}\frac{\partial^2 T}{\partial t^2} + \rho C_p \frac{\partial T}{\partial t} = K\frac{\partial^2 T}{\partial x^2}, \tag{3.21}$$

where ρ is the density (kg/m^3), C_p is the specific heat capacity ($J/(kg \cdot K)$), K is the thermal conductivity ($W/(m \cdot K)$), and c is the finite thermal-wave propagation speed (m/s). The term $K/(\rho C_p c^2)$ is commonly defined as the thermal relaxation time, τ.

Unlike the parabolic diffusion equation, this formulation introduces a second-order time derivative, fundamentally altering the equation's character. Consequently, thermal disturbances propagate as damped thermal waves rather than purely diffusive fronts. Information is transported strictly along characteristic directions bounded by $x \pm ct$. This guarantees that a thermal disturbance can only influence a limited spatial region of the domain during a finite time interval, maintaining causality.

From the perspective of partial differential equation (PDE) classification, this formulation satisfies the discriminant condition $\Delta > 0$, placing it squarely in the hyperbolic class. Analogous to acoustic and mechanical wave equations, its solutions exhibit sharp wavefronts and distinct characteristic curves. The domain outside these characteristics remains entirely unaffected until the physical thermal wave arrives, as illustrated by the restricted domain of dependence in Fig. 3.2.

3.2.3 The Parabolic Regime (Transient Diffusion)

For systems operating under standard macroscopic conditions where thermal relaxation times are negligible, the domain is governed by the classical transient diffusion equation. This parabolic PDE takes the general form:

$$\frac{\partial \phi}{\partial t} = \alpha \nabla^2 \phi + S, \tag{3.22}$$

where ϕ is the transported scalar (e.g., temperature), α is the thermal diffusivity, and S represents internal source terms.

In the parabolic regime, the system possesses a "fading memory" of its initial state. Information advances strictly forward in time, and local disturbances smoothly dissipate outward, progressively influencing the surrounding continuum. A defining mathematical feature of parabolic equations is their horizontally unbounded domain of dependence. The speed of signal propagation is theoretically infinite; a macroscopic perturbation introduced anywhere in the domain is mathematically felt everywhere else instantaneously, albeit with an exponentially decaying magnitude. While this violates the finite speed of physical molecular collisions, it remains an incredibly robust and standard approximation for general transport phenomena.

3.2.4 The Elliptic Regime (Steady-State Diffusion)

As time approaches infinity ($t \to \infty$), the transient effects fully decay, and the system reaches an equilibrium state. The time derivative vanishes ($\partial \phi / \partial t = 0$), reducing the governing equation to Poisson's equation (or Laplace's equation if $S = 0$):

$$\alpha \nabla^2 \phi + S = 0$$

This represents the elliptic regime. In this state, the system no longer retains a memory of its initial conditions; the boundary conditions and internal sources entirely dictate the internal field. Elliptic PDEs represent a state of global interconnectedness. A change made to any single boundary condition requires the entire internal domain to adjust to a new equilibrium simultaneously. Consequently, solving equations in the elliptic regime requires

numerical methods that can couple the entire domain simultaneously, often necessitating robust implicit formulations or iterative global matrix solvers.

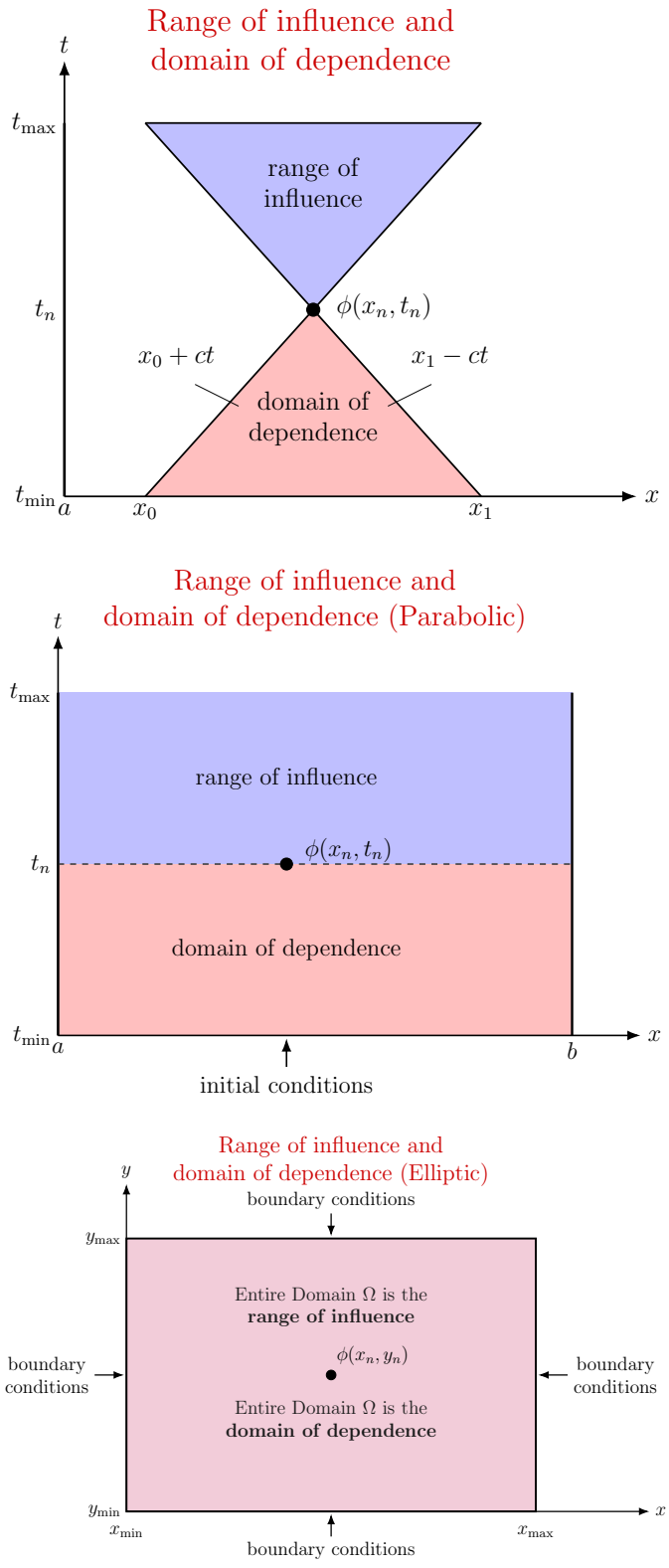


Fig. 3.2: Domain of dependence and domain of influence in hyperbolic (top), parabolic (middle), and elliptic (bottom) PDEs

3.3 Multi-Moment Approximation of the Second Derivative

Similar to the convection equation, we introduce two basic approximations for the second-order derivative: ii.) the explicit scheme and ii) the compact scheme.

3.3.1 Explicit Multi-Moment Schemes

A general form of the multi-moment approximation for the second-order derivative is given by:

$$\phi_i'' = \frac{d}{h^2}\phi_i + \sum_{\tau=1}^{m_1} \frac{b_\tau}{h^2}\left(\overline{\phi}^x_{i+\tau-\frac{1}{2}} + \overline{\phi}^x_{i-\tau+\frac{1}{2}}\right) + \sum_{\tau=1}^{m_2} \frac{c_\tau}{h^2}\left(\phi_{i+\tau} + \phi_{i-\tau}\right), \tag{3.23}$$

where the corresponding structural coefficients are listed in Tab. 3.1.

The squared modified wavenumbers $(k^{2*}h)$ for the explicit multi-moment schemes are displayed in Fig. 3.3(a) for the initial modified wavenumber square. We limit ourselves to this type of modified wavenumber analysis. The effects of coupled Fourier Analysis will be discussed in chapter 4.

According to Eq.3.23 and Tab. 3.1, the fourth-order approximation for the second-derivative in multi-moment discretization reads

$$\phi_i'' = -\frac{12}{h^2}\phi_i + \frac{15}{2h^2}\left(\overline{\phi}_{i+\frac{1}{2}} + \overline{\phi}_{i-\frac{1}{2}}\right) - \frac{3}{2h^2}(\phi_{i+1} + \phi_{i-1}). \tag{3.24}$$

Likewise,

The spectral resolution properties are evaluated using the resolving efficiency, defined as $e = (kh)_{\max}/\pi$, where $(kh)_{\max}$ represents the maximum wavenumber up to which the relative error satisfies $|k^{2*}h - k^2h|/k^2h < \epsilon$.

As expected, numerical approximations of the second-order derivative exhibit higher resolving efficiencies compared to their first-derivative counterparts within the same class of methods. All explicit schemes presented deliver an error of less than 10% ($\epsilon = 0.1$) up to the single-grid limit ($kh \leq \pi$). For a stricter 1% relative error tolerance ($\epsilon = 0.01$), the eighth-order scheme (EX8) utilizes a four-cell stencil to accurately capture wavenumbers up to $kh = 0.972\pi$, spanning nearly the entire Nyquist limit. Extending the stencil by two additional cells (MMEX12) reduces the relative error by a decade ($\epsilon = 0.001$) while maintaining an exceptionally wide resolution window ($kh \leq 0.934\pi$). Notably, standard compact finite-difference schemes cannot resolve wavenumbers beyond the grid limit of π; hence, the benchmark FDCOM6 curve terminates at this threshold.

Example 3.1 (Modified wavenumber square of the fourth-order explicit scheme for the second-order derivatives.)
Let us compute the initial modified wavenumber square by using the Fourier analysis similar to section 2.3 for each k-th mode of Eq.(3.24):

$$\begin{aligned}
\mathrm{FFT}\left(\phi_i''\right) &= -k^2 e^{jkx}, & \mathrm{FFT}\left(\phi_i\right) &= 1 \cdot e^{jkx} \\
\mathrm{FFT}\left(\overline{\phi}_{i-\frac{1}{2}}\right) &= -\frac{-j\cos(kh) + j - \sin(kh)}{hk} e^{jkx}, & \mathrm{FFT}\left(\phi_{i-1}\right) &= (\cos(kh) - j\sin(kh)) e^{jkx} \\
\mathrm{FFT}\left(\overline{\phi}_{i+\frac{1}{2}}\right) &= \frac{-j\cos(kh) + j + \sin(kh)}{hk} e^{jkx}, & \mathrm{FFT}\left(\phi_{i+1}\right) &= (\cos(kh) + j\sin(kh)) e^{jkx}
\end{aligned}$$

It follows that:

$$-k^2 = -\frac{3(\cos(kh)hk + 4kh - 5\sin(kh))}{h^3 k},$$

which is consistent as $h \to 0$, and instead of the exact $-k^2$, we got the modified wavenumber square in its place.

$$k^{2*} = \frac{3(\cos(kh)hk + 4kh - 5\sin(kh))}{h^3 k},$$

There are two widely adopted conventions for visualizing the modified wavenumber squared, k^{2*}. One may plot k^{2*} directly, as illustrated in Fig. 3.3, or plot $-k^{2*}$, which appears as the eigenvalue associated with the Fourier mode e^{jkx} in the discrete approximation of the second-derivative operator. Care must be taken when evaluating these quantities numerically. The expressions for both k^* and k^{2*} typically involve ratios whose numerator and denominator vanish simultaneously at $k\Delta x = 0$ and at integer multiples of π. Although the resulting singularities are removable, direct numerical evaluation can suffer from catastrophic cancellation, leading to substantial round-off errors and inaccurate graphical representations. While arbitrary-precision arithmetic can mitigate these issues, the required precision increases rapidly with the order of the scheme. In practice, high-order methods may require quadruple precision to produce acceptable results. Therefore, it is generally preferable to evaluate the limiting values analytically using L'Hôpital's Rule or equivalent asymptotic expansions. This approach removes the singular behavior explicitly and yields numerically stable evaluations throughout the entire wavenumber range.

3.3.2 Compact Multi-Moment Schemes

To further optimize spectral resolution while preserving a compact spatial stencil, an implicit multi-moment framework is introduced. The general formulation for the compact multi-moment approximation of the second-order derivative is expressed as:

$$\begin{aligned}
\phi_i'' + \sum_{\tau=1}^{m_3} a_\tau \left(\phi_{i+\tau}'' + \phi_{i-\tau}''\right) = \frac{d}{h^2}\phi_i + \sum_{\tau=1}^{m_1} \frac{b_\tau}{h^2} \left(\overline{\phi}^x_{i-\tau+\frac{1}{2}} + \overline{\phi}^x_{i+\tau-\frac{1}{2}}\right) \\
+ \sum_{\tau=1}^{m_2} \frac{c_\tau}{h^2} \left(\phi_{i-\tau} + \phi_{i+\tau}\right). \qquad (3.25)
\end{aligned}$$

The sets of unique coefficients for these implicit formulations are detailed in Tab. 3.2. For example, the sixth-order compact approximation for the second derivative in multi-moment

discretization reads

$$\phi_i'' - \frac{1}{32}\left(\phi_{i+1}'' + \phi_{i-1}''\right) = -\frac{15}{h^2}\phi_i + \frac{315}{32h^2}\left(\overline{\phi}_{i+\frac{1}{2}} + \overline{\phi}_{i-\frac{1}{2}}\right) - \frac{75}{32h^2}(\phi_{i+1} + \phi_{i-1}), \tag{3.26}$$

and the eighth-order compact scheme is

$$\begin{aligned}\phi_i'' - \frac{1}{20}\left(\phi_{i+1}'' + \phi_{i-1}''\right) = &-\frac{159}{10h^2}\phi_i + \frac{421}{40h^2}\left(\bar{\phi}^x_{i-\frac{1}{2}} + \bar{\phi}^x_{i+\frac{1}{2}}\right) - \frac{1}{40h^2}\left(\bar{\phi}^x_{i-\frac{3}{2}} + \bar{\phi}^x_{i+\frac{3}{2}}\right) \\ &- \frac{51}{20h^2}\left(\phi_{i-1} + \phi_{i+1}\right).\end{aligned} \tag{3.27}$$

The squared modified wavenumbers ($k^{2*}h$) obtained from the compact multi-moment schemes are illustrated in Fig. 3.3(b). Mirroring the trends observed in the explicit variants, the resolving efficiency increases monotonically with the order of the scheme. However, the implicit formulation shifts the spectral curves upward, demonstrating noticeably superior high-wavenumber resolution compared with explicit schemes of equivalent stencil sizes. Under a relaxed error tolerance of $\epsilon = 0.1$, the lowest-order compact scheme (MMCOM6) significantly improves the resolving efficiency ($e = 1.409$) over the explicit EX4 scheme ($e = 1.038$). As the accuracy threshold tightens, the benefits of the implicit coupling become more pronounced. At a 1% error tolerance ($\epsilon = 0.01$), the mid-order compact scheme MMCOM8 resolves up to $kh = 1.0972\pi$, effectively surpassing the standard single-grid limit. For high-fidelity simulations requiring extreme precision ($\epsilon = 0.0001$), the MMCOM14 scheme sustains a remarkable resolving efficiency of $e = 1.0201$. This indicates that MMCOM14 can resolve small-scale fluctuations without suffering from the premature uncoupling or severe underestimation of wave dispersion typically found in traditional localized finite-difference methods.

Example 3.2 (Modified wavenumber square of the compact sixth-order scheme for the second-order derivatives.)
Similar to the previous example, we derive the Fourier transform of each mode k for each term in Eq.(3.26):

$$\begin{aligned}
\mathrm{FFT}\left(\phi_i''\right) &= -k^2 e^{jkx}, & \mathrm{FFT}\left(\phi_i\right) &= 1 \cdot e^{jkx} \\
\mathrm{FFT}\left(\phi_{i\pm1}''\right) &= -k^2(\cos(kh) \pm j\sin(kh))e^{jkx}, & & \\
\mathrm{FFT}\left(\overline{\phi}_{i-\frac{1}{2}}\right) &= -\frac{-j\cos(kh) + j - \sin(kh)}{hk}e^{jkx}, & \mathrm{FFT}\left(\phi_{i-1}\right) &= (\cos(kh) - j\sin(kh))e^{jkx} \\
\mathrm{FFT}\left(\overline{\phi}_{i+\frac{1}{2}}\right) &= \frac{-j\cos(kh) + j + \sin(kh)}{hk}e^{jkx}, & \mathrm{FFT}\left(\phi_{i+1}\right) &= (\cos(kh) + j\sin(kh))e^{jkx}
\end{aligned}$$

It follows that by equating the left- and right-hand sides, we have

$$-k^2\left(1 - \frac{1}{16}\cos(kh)\right) = \frac{315\sin(kh) - 75kh\cos(kh) - 240kh}{16h^3k}.$$

Then we can isolate the k^2 on the right-hand side:

$$k^2 = \frac{15(16kh - 21\sin(kh) + 5kh\cos(kh))}{h^3k(16 - \cos(kh))}.$$

The left-hand side of the equation is the analytical k^2, and the whole RHS is the *numerical approximation* of it. The above equation is consistent because they agree as $h \to 0$. However,

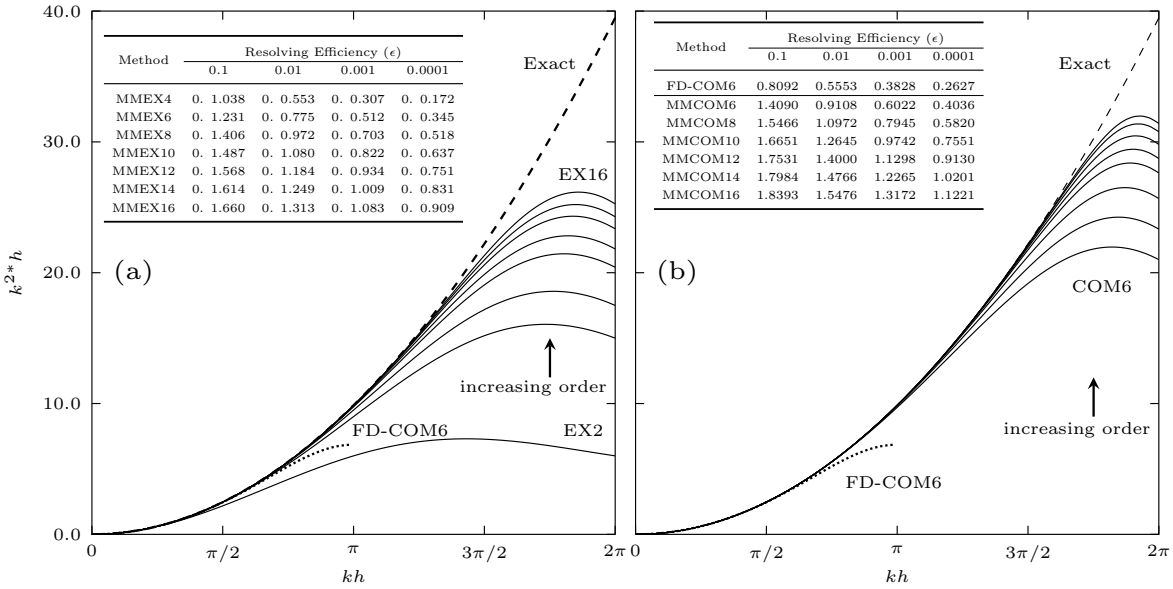
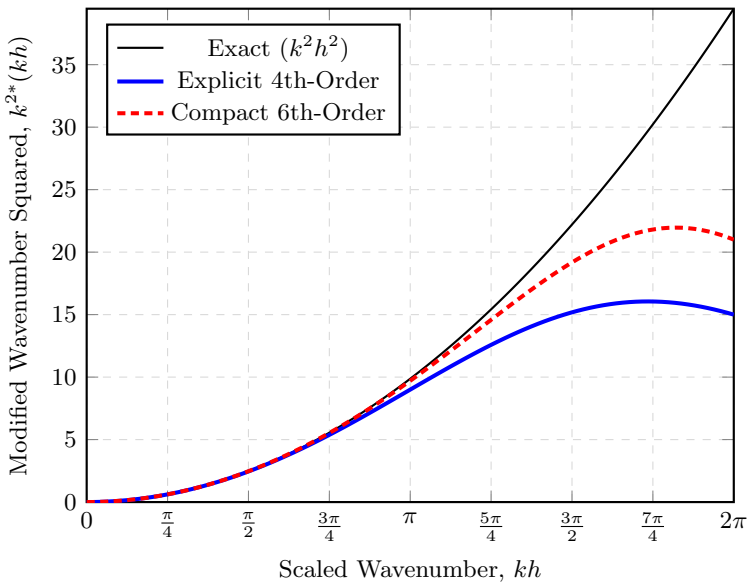


Fig. 3.3: Comparison of the squared modified wavenumber ($k^{2*}h$) as a function of the grid-scaled wavenumber (kh) for second-derivative approximations: (a) explicit multi-moment schemes (EX2–EX16) and (b) compact multi-moment schemes (COM4, MMCOM6–MMCOM20). The standard sixth-order compact finite-difference scheme (FDCOM6) [24] is plotted in both panels for benchmark comparison. Embedded tables provide the resolving efficiencies (e) across four relative error tolerances (ϵ). Both panels share the identical vertical axis scale shown in (a).

the numerical approximation introduces approximation error, and we have the modified wavenumber square

$$k^{2*} = \frac{15(16kh - 21\sin(kh) + 5kh\cos(kh))}{h^3k(16 - \cos(kh))}.$$

instead. The comparison of this approximation and the explicit fourth-order scheme can be visualized in Fig.3.4 below.



$Order$	d_1	b_1	b_2	b_3	b_4	b_5	b_6
2	-6	3	—	—	—	—	—
4	-12	$\frac{15}{2}$	—	—	—	—	—
6	$-\frac{27}{2}$	$\frac{209}{24}$	$\frac{1}{24}$	—	—	—	—
8	-15	$\frac{1439}{144}$	$\frac{31}{144}$	—	—	—	—
10	$-\frac{47}{3}$	$\frac{38083}{3600}$	$\frac{1183}{3600}$	$\frac{1}{900}$	—	—	—
12	$-\frac{49}{3}$	$\frac{201209}{18000}$	$\frac{4417}{9000}$	$\frac{167}{18000}$	—	—	—
14	$-\frac{401}{24}$	$\frac{40642487}{3528000}$	$\frac{2116727}{3528000}$	$\frac{63431}{3528000}$	$\frac{3}{78400}$	—	—
16	$-\frac{205}{12}$	$\frac{586085413}{49392000}$	$\frac{36056101}{49392000}$	$\frac{2493}{5488000}$	$\frac{96}{2}$	—	—
24	$\frac{-5369}{300}$	$\frac{4483308727523}{355000060800}$	$\frac{384765751523}{355000060800}$	$\frac{19002573199}{177500030400}$	$\frac{1172497199}{177500030400}$	$\frac{64252693}{355000060800}$	$\frac{90641}{71000012160}$
$Order$	—	c_1	c_2	c_3	c_4	c_5	c_6
2	—	—	—	—	—	—	—
4	—	$-\frac{3}{2}$	—	—	—	—	—
6	—	-2	—	—	—	—	—
8	—	$-\frac{8}{3}$	$-\frac{1}{24}$	—	—	—	—
10	—	-3	$-\frac{3}{40}$	—	—	—	—
12	—	$-\frac{27}{8}$	$-\frac{27}{200}$	$-\frac{1}{600}$	—	—	—
14	—	$-\frac{18}{5}$	$-\frac{9}{5}$	$-\frac{2}{525}$	—	—	—
16	—	$-\frac{6}{25}$	$-\frac{32}{3675}$	$-\frac{3}{3675}$	$-\frac{3}{39200}$	—	—
24	—	$-\frac{216}{49}$	$-\frac{675}{1568}$	$-\frac{50}{1323}$	$-\frac{3}{1568}$	$-\frac{6}{148225}$	$-\frac{1}{5122656}$

Table 3.1: Coefficients of the explicit multi-moment approximation of the second derivative in Eq.(3.23).

Fig. 3.4: Comparison of the implicit sixth-order and the explicit fourth-order approximations of the second-order derivative.

3.3.3 Boundary Closure

Within the interior of the computational domain, the governing equation can be discretized using information from both sides of each solution point. At a physical boundary, this is no longer possible because part of the stencil lies outside the domain. The numerical treatment must therefore satisfy both the prescribed physical boundary condition and the requirements of the discrete governing equation. This introduces two boundary-closure problems. The first is to determine the appropriate boundary quantity, such as the solution value, gradient, or flux, from the physical condition of the problem. The second is to construct a discrete approximation using only the information available within the computational domain. In particular, centered interior stencils must generally be replaced by asymmetric closures near the boundary.

The difficulty becomes more pronounced when a thin boundary layer is present. In convection–diffusion problems, the imposed boundary condition can generate strong local gradients as the near-wall solution adjusts from the interior state to the boundary state. If the grid is too coarse to resolve this variation, polynomial reconstruction may produce large errors

Order	a_1	a_2	a_3	a_4	b_1	b_2	b_3	b_4
6	$-\frac{1}{32}$	—	—	—	$\frac{315}{32}$	—	—	—
8	$-\frac{1}{20}$	—	—	—	$\frac{421}{40}$	$-\frac{1}{40}$	—	—
10	$-\frac{1}{12}$	—	—	—	$\frac{1175}{108}$	$-\frac{5}{27}$	—	—
12	$-\frac{34}{261}$	$\frac{7}{8352}$	—	—	$\frac{6486715}{601344}$	$-\frac{330565}{601344}$	—	—
14	$-\frac{146}{899}$	$\frac{53}{28768}$	—	—	$\frac{109664981}{10356480}$	$-\frac{2892473}{3452160}$	$\frac{13}{40455}$	—
16	$-\frac{549}{2696}$	$\frac{45}{10784}$	—	—	$\frac{990197537}{97056000}$	$-\frac{121201213}{97056000}$	$\frac{201103}{48528000}$	—
18	$-\frac{880389}{3506560}$	$\frac{191241}{21916000}$	$-\frac{1781}{87664000}$	—	$\frac{1269639222971}{131496000000}$	$-\frac{459625307183}{262992000000}$	$\frac{5217629221}{262992000000}$	—
20	$-\frac{136593}{478553}$	$\frac{626643}{47855300}$	$-\frac{673}{11963825}$	—	$\frac{18539293107361}{2009922600000}$	$-\frac{4250265215639}{2009922600000}$	$\frac{78511405243}{2009922600000}$	$-\frac{24747}{5359793600}$
22	$-\frac{141399}{435695}$	$\frac{172017}{8713900}$	$-\frac{347}{2178475}$	—	$\frac{156044132981411}{17933206200000}$	$-\frac{45370805292229}{17933206200000}$	$\frac{1350289295393}{17933206200000}$	$-\frac{103199049}{1195547080000}$
24	$-\frac{19878976}{54322025}$	$\frac{17341906}{597542275}$	$-\frac{1732928}{4182795925}$	$\frac{60791}{133849469600}$	$\frac{119368112575631753}{14691317783296000}$	$-\frac{10859766529150345 27}{367282944582400000}$	$\frac{5947042234855067}{44073953349888000}$	$-\frac{637842268943069}{1101848833747200000}$

Order	c_1	c_2	c_3	c_4	d	LTE
6	$-\frac{75}{32}$	—	—	—	-15	$-\frac{h^6}{40320}$
8	$-\frac{51}{20}$	—	—	—	$-\frac{159}{10}$	$\frac{h^8}{1039500}$
10	$-\frac{89}{36}$	$\frac{1}{36}$	—	—	$-\frac{33}{2}$	$-\frac{h^{10}}{37065600}$
12	$-\frac{1570}{783}$	$\frac{10855}{100224}$	—	—	$-\frac{1935}{116}$	$\frac{h^{12}}{1519333200}$
14	$-\frac{4310}{2697}$	$\frac{60437}{345216}$	—	—	$-\frac{179695}{10788}$	$-\frac{79h^{14}}{3364509657600}$
16	$-\frac{21255}{21568}$	$\frac{290679}{1078400}$	$-\frac{869}{1617600}$	—	$-\frac{66655}{4044}$	$\frac{137h^{16}}{202765901184000}$
18	$-\frac{2955771}{14026240}$	$\frac{817086123}{2191600000}$	$-\frac{91999649}{26299200000}$	—	$-\frac{5316451}{328740}$	$-\frac{8249h^{18}}{476175629529600000}$
20	$\frac{3580101}{9571060}$	$\frac{2105322849}{4785530000}$	$-\frac{26470957}{3589147500}$	—	$-\frac{182728721}{11485272}$	$\frac{71h^{20}}{120891995451727500}$
22	$\frac{461331990101}{43569500}$	$\frac{434253771}{871390000}$	$-\frac{481310447}{32023582500}$	$\frac{10863}{1067452750}$	$-\frac{32577979}{2091336}$	$-\frac{60791h^{22}}{3592528557882264000000}$
24	$\frac{461331990101}{43569500}$	$\frac{434253771}{871390000}$	$-\frac{481310447}{32023582500}$	$\frac{10863}{1067452750}$	$-\frac{17439603755}{1147281168}$	$\frac{11162299h^{24}}{252202976859623572282500000}$

Table 3.2: Coefficients and leading truncation error (LTE) for the compact multi-moment approximation of the second derivative (Eq.(3.25)).

or spurious oscillations. One-sided boundary stencils also differ from the centered schemes used in the interior and may have less favorable stability or dispersion properties. Adequate near-boundary grid resolution is therefore important. When the steep gradient is sufficiently resolved, a simpler or lower-order closure can often be employed near the boundary while the higher-order interior discretization is recovered farther inside the domain.

We now turn to the construction of numerical closures for the two most fundamental classes of boundary conditions: *Dirichlet* and *Neumann* conditions. In the Dirichlet case, the solution value is prescribed at the boundary, and the numerical closure must provide the derivatives required by the governing equation. In the Neumann case, the boundary gradient or flux is prescribed, while the boundary solution value remains an unknown that must be coupled

consistently to the interior discretization. These two cases provide the basic framework from which more specialized boundary treatments can be developed.

Dirichlet Boundary Conditions

At a physical boundary where the value of the solution is prescribed, the corresponding pointwise degree of freedom can be set directly to the known boundary value. Such a condition is referred to as a *Dirichlet boundary condition*. For example, at the left boundary,

$$\phi_0 = \phi_{\mathrm{b}}, \tag{3.28}$$

where ϕ_{b} denotes the prescribed boundary value.

Although the solution value is known at the boundary, the discretization of the diffusion term also requires the first derivative, or equivalently the diffusive flux, at the boundary. Since a centered stencil cannot be applied at the domain boundary, an asymmetric closure must be used. For the present multi-moment formulation, we employ the following fourth-order approximation:

$$\phi_0' = \frac{23}{2h}\overline{\phi}_{1/2} + \frac{7}{2h}\overline{\phi}_{3/2} - \frac{6}{h}\phi_0 - \frac{8}{h}\phi_1 - \frac{1}{h}\phi_2. \tag{3.29}$$

The stencil combines the prescribed boundary point value with neighboring pointwise and cell-averaged moments, yielding a one-sided, fourth-order approximation of the boundary derivative.

As the stencil moves into the interior, additional neighboring data become available, allowing the use of the fourth-order or higher-order centered schemes listed in Table 2.1. The approximation order may therefore be increased gradually until the same order as the interior discretization is recovered.

For a non-uniform grid, the coefficients in Eq. (3.29) are no longer constant. They can be obtained systematically by expanding the neighboring pointwise values and cell averages in Taylor series about the boundary location and solving the resulting coefficient-matching system to satisfy the desired order of accuracy.

Neumann Boundary Conditions

Under a Neumann boundary condition, the boundary gradient is known,

$$\phi_0' = g_{\mathrm{N}}, \tag{3.30}$$

whereas the boundary value ϕ_0 is not prescribed. Since the diffusion operator involves the second derivative of the solution, a closure for ϕ_0'' is still required at the boundary.

Using the same reconstruction polynomial introduced for the Dirichlet boundary treatment, the boundary second derivative is approximated by

$$\phi_0'' = -\frac{97}{8h^2}\overline{\phi}_{1/2} - \frac{113}{8h^2}\overline{\phi}_{3/2} - \frac{13}{4h^2}\phi_0 + \frac{22}{h^2}\phi_1 + \frac{17}{4h^2}\phi_2. \tag{3.31}$$

Only the boundary location requires this asymmetric treatment. At the next interior location, the available neighboring data are sufficient to apply the centered fourth-order scheme.

An important feature of this closure is that the unknown boundary value ϕ_0 enters explicitly into the approximation of the diffusion operator. The boundary degree of freedom is therefore coupled directly to the interior diffusive dynamics, in addition to satisfying the prescribed Neumann condition.

3.4 Stability Limit of Runge-Kutta Time Integration in Diffusion Problems

We learned from Chapter 2 that the modified wavenumber from the initial Fourier analysis and the coupled Fourier analysis are close approximations of each other. Thus, for the purpose of estimating the stability limit, it is sufficient to use the modified wavenumber from the initial Fourier analysis together with the von Neumann analysis. Thus, in this section, let us consider the one-dimensional linear diffusion equation:

$$\frac{\partial \phi}{\partial t} = K \frac{\partial^2 \phi}{\partial x^2}, \tag{3.32}$$

where $K > 0$ represents the constant diffusion coefficient. Substituting a continuous Fourier mode $\phi(x,t) = \hat{\phi}(t) e^{jkx}$ into the exact spatial second-derivative operator yields $-Kk^2\hat{\phi}$. Thus, the exact continuous eigenvalue associated with the continuous spatial operator is purely real and negative:

$$\lambda_{\text{num}} = -Kk^2. \tag{3.33}$$

This stands in sharp contrast to linear convection problems, where the exact eigenvalues are purely imaginary and lie entirely along the imaginary axis of the complex plane.

When the spatial second derivative is approximated using a discrete multi-moment framework, the local reconstruction of cell averages and higher-order boundary moments alters the wavenumber characteristics. We characterize this discrete dispersion and dissipation behavior by introducing the non-dimensional modified wavenumber squared, h^2k^{2*}. The semi-discrete system in spectral space becomes:

$$\frac{d\hat{\phi}}{dt} = -Kk^{2*}\hat{\phi}, \tag{3.34}$$

yielding the discrete numerical eigenvalue $\lambda = -Kk^{2*}$. Because the symmetric multi-moment formulations considered here employ central stencils for structural symmetry, h^2k^{2*} remains strictly real and positive across the entire resolvable spectrum $kh \in [0, \pi]$. Consequently, the discrete spatial eigenvalues λ lie entirely on the negative real axis of the complex plane.

We can now map the entire spatial spectrum directly into the complex stability parameter z used in temporal Ordinary Differential Equation (ODE) analysis:

$$z = \lambda \Delta t = -Kk^{2*}\Delta t = -\left(h^2k^{2*}\right)\sigma, \tag{3.35}$$

where h is the spatial grid spacing, Δt is the temporal step size, and $\sigma = K\Delta t/h^2$ denotes the non-dimensional grid Fourier number (or cell diffusion number).

For an explicit time integration scheme to remain numerically stable, the complex parameter z must lie entirely within the scheme's geometric stability region $\mathcal{R}_z$. Given that z is purely real for symmetric spatial operators, the stability criterion is determined exclusively by the intersection of $\mathcal{R}_z$ with the negative real axis, defined by the interval $[-z_{\text{real, max}}, 0]$. Table 3.3 outlines the real-axis stability thresholds $z_{\text{real, max}}$ for standard and high-order explicit multi-stage Runge-Kutta (RK) methods.

To guarantee that the solution does not experience non-physical growth, the maximum magnitude of the discrete parameter $|z|$ must satisfy the stability boundary across all resolvable wavenumbers:

$$\max(h^2k^{2*})\sigma \leq z_{\text{real,max}}. \tag{3.36}$$

We can rearrange this inequality to establish the maximum permissible grid Fourier number that maintains numerical stability:

$$\sigma \leq \frac{z_{\text{real,max}}}{\max(h^2 k^{2*})}. \tag{3.37}$$

Equation 3.37 reveals a fundamental coupling between the spatial scheme's spectral resolution and temporal step constraints. High-wavenumber modes near the Nyquist limit ($kh = \pi$) typically dictate the maximum value of the modified wavenumber squared. Table 3.3 presents a comprehensive breakdown of the resulting maximum stable grid Fourier numbers ($\sigma_{\max}$) across various explicit (EXp) and compact (COMp) multi-moment variations.

Method / Scheme	$h^2 k^{2*}_{\max}$	$\lvert h\rho(\boldsymbol{C})\rvert$	$z_{\text{real,max}}/h^2 k^{2*}_{\max}$						
			RK3	RK4	RK5	RK7	RK8	RK9	RK10
$\lambda dt_{\max}$	—	—	2.51	2.78	3.31	3.95	5.18	5.65	6.22
EX4	15.58	TBA	0.16	0.18	0.21	0.25	0.33	0.36	0.40
EX6	18.16	TBA	0.14	0.15	0.18	0.22	0.29	0.31	0.34
EX8	21.14	TBA	0.12	0.13	0.16	0.19	0.25	0.27	0.29
EX10	22.57	TBA	0.11	0.12	0.15	0.18	0.23	0.25	0.28
COM6	20.47	TBA	0.12	0.14	0.16	0.19	0.25	0.28	0.30
COM8	22.01	TBA	0.11	0.13	0.15	0.18	0.24	0.26	0.28
COM10	23.25	TBA	0.11	0.12	0.14	0.17	0.22	0.24	0.27
SPM (on $h/2$-grid)	39.48	39.48	0.06	0.07	0.08	0.10	0.13	0.14	0.16

Table 3.3: Stability limit $\sigma_{\max} = \lambda \Delta t_{\max}/h^2 k^{2*}_{\max}$ from the simplified von Neumann analysis. The limits shown are for the h-grid, except for the spectral method (SPM).

Evaluating these parameters reveals several important trends within the multi-moment framework.

For the explicit fourth-order multi-moment scheme (EX4), the localized moment constraints produce a maximum spatial eigenvalue of $h^2 k^{2*}_{\max} = 15.58$. When combined with the classical fourth-order Runge–Kutta (RK4) time integration scheme, whose stability region intersects the negative real axis at $z_{\text{real,max}} = 2.78$, the maximum allowable diffusion cell number is $\sigma \leq 0.18$.

For the compact sixth-order multi-moment scheme (COM6), the implicit coupling of moments across neighboring cell interfaces enhances the spectral response near the Nyquist wavenumber, increasing the maximum spatial eigenvalue to $[h^2 k^{2*}]_{max} = 20.47$.

Although the compact formulation provides superior spectral resolution and significantly reduces dissipation and dispersion errors over a broad range of wavenumbers, the larger spectral radius increases the stiffness of the semi-discrete system. Consequently, when coupled with the same RK4 integrator, the stability limit becomes more restrictive, $\sigma \leq 0.14$.

These results illustrate a fundamental trade-off in the design of high-order numerical methods. Increasing the spatial accuracy improves the representation of small-scale features and extends the range of accurately resolved wavenumbers. However, these benefits are accompanied by a larger spectral radius, which reduces the maximum stable time step available to

explicit time integration methods. In other words, improved spatial fidelity is achieved at the expense of explicit temporal efficiency.

Recalling Equation 3.37, the stability constraint on the physical time step is

$$\Delta t \leq \frac{z_{\text{real,max}}}{\max(h^2 k^{2*})} \frac{h^2}{K}.$$

This expression shows that the influence of the spatial discretization is contained in the factor

$$\frac{z_{\text{real,max}}}{\max(h^2 k^{2*})},$$

whereas the dominant dependence arises from the quadratic scaling with the grid spacing, h^2. Consequently, mesh refinement rapidly decreases the maximum stable time step, regardless of the particular scheme employed. For very fine grids, the diffusion term therefore becomes the primary stability constraint. In addition, the stronger the diffusion coefficient, the stricter the limit on the time step size. This behavior is especially important in convection–diffusion simulations. The explicit stability limit associated with convection typically scales as $O(h)$, whereas the diffusion stability limit scales as $O(h^2)$. As the mesh is refined, the diffusive restriction becomes increasingly severe and eventually dominates the allowable time step. As a result, even though diffusion is physically a smoothing process, it often dictates the computational cost of explicit simulations. For this reason, many computational fluid dynamics (CFD) and magnetohydrodynamics (MHD) solvers adopt mixed temporal integration strategies. Convective terms are advanced explicitly to preserve efficiency and accurately capture wave propagation, while diffusive terms are treated implicitly to circumvent the severe $O(h^2)$ stability restriction. Such implicit–explicit (IMEX) approaches combine the advantages of both formulations and are particularly effective for convection–diffusion systems on highly refined meshes [19].

3.5 Diffusion of a Gaussian Profile

The diffusion of a Gaussian profile is a fundamental benchmark problem in the numerical analysis of partial differential equations. Physically, this process represents the gradual smoothing of a concentrated scalar quantity, such as heat or mass, over time within a continuum. Mathematically, the smoothness of the Gaussian function (denoted as C_∞) and its strictly positive nature make it an ideal test case for evaluating the dissipative properties of a numerical scheme. It also assesses the scheme's ability to accurately preserve mass and amplitude, without introducing non-physical oscillations or artifacts.

The evolution of the scalar field $\phi(x,t)$ the one-dimensional parabolic diffusion equation governs :

$$\frac{\partial \phi}{\partial t} = K \frac{\partial^2 \phi}{\partial x^2}, \tag{3.38}$$

where $\alpha > 0$ is the constant diffusion coefficient (or thermal diffusivity).

To formulate the initial value problem, we prescribe an initial condition consisting of a Gaussian pulse centered at x_c with an initial standard deviation (or characteristic width) σ_0, and an initial peak amplitude A:

$$\phi(x,0) = A \exp\left(-\frac{(x-x_c)^2}{\sigma_0^2}\right). \tag{3.39}$$

If we assume the domain is unbounded ($x \in \mathbb{R}$), the exact analytical solution can be derived using the Fourier transform or by convolving the initial condition with the fundamental solution (the heat kernel). The exact solution at any time $t \geq 0$ is given by:

$$\phi(x,t) = A\frac{\sigma_0}{\sqrt{\sigma_0^2 + 4Kt}} \exp\left(-\frac{(x-x_c)^2}{\sigma_0^2 + 4Kt}\right). \tag{3.40}$$

This analytical solution elegantly demonstrates the core properties of parabolic diffusion. As time progresses, the profile variance increases linearly ($\sigma^2(t) = \sigma_0^2 + 4\alpha t$), broadening the pulse (Fig.3.5). Simultaneously, the peak amplitude, given by $A\sigma_0/\sqrt{\sigma_0^2 + 4alphat}$, decays inversely with the square root of time. Despite these changes in shape, the total area under the curve (representing the system's total mass or energy) remains perfectly conserved for all t, reinforcing the reliability of the mathematical model and its physical relevance.

3.5.1 Truncation of the Computational Domain

In many practical simulations, the physical domain is either very large or theoretically unbounded. Since such a domain cannot be represented directly in a finite computation, it must be truncated to a computational domain of manageable size. The resulting artificial boundaries should be placed sufficiently far from the region of interest so that their influence on the interior solution remains negligible over the time interval considered. This procedure is commonly referred to as *domain truncation* or *computational domain selection*.

For the diffusion of a Gaussian profile, the required domain size can be estimated relatively easily because the analytical solution retains its Gaussian form as it evolves. Under linear diffusion, the profile broadens continuously, while its peak amplitude decreases and its variance increases in a predictable manner. The computational domain can therefore be chosen so that the exponentially decaying tails of the Gaussian remain sufficiently small at the boundaries throughout the simulation. In practice, the initial profile should first be located far enough from the boundaries that its tails are already negligible there. The domain must then be extended further to account for the subsequent diffusive spreading of the solution. The required extent therefore depends not only on the initial profile, but also on the diffusion coefficient and the total simulation time.

Once the computational domain has been selected, an appropriate treatment must be imposed at its artificial boundaries. The boundary conditions complete the truncated mathematical problem and strongly influence how faithfully the finite computational domain reproduces the behavior of the original open or unbounded domain. In the context of localized scalar transport, three boundary formulations are of particular interest:

- **Homogeneous Dirichlet conditions:** One simple way to truncate an unbounded Gaussian-diffusion problem is to place the artificial boundaries sufficiently far from the center of the profile and set

$$\phi = 0. \tag{3.41}$$

  This is a homogeneous Dirichlet boundary condition. Since a Gaussian decays exponentially, its exact value at a distant boundary can be extremely small, although it is never exactly zero at a finite location. The imposed boundary condition therefore introduces a truncation error of approximately the same order as the neglected Gaussian tail.
  If the boundary position is chosen such that

$$|\phi(x_\mathrm{b}, t)| < \varepsilon_\mathrm{tol} \tag{3.42}$$

for the entire simulation interval, where ε_tol denotes an acceptable numerical tolerance, the influence of the artificial boundary is negligible. In particular, choosing

$$|\phi(x_\mathrm{b}, t)| \lesssim \epsilon_\mathrm{mach} \tag{3.43}$$

makes the truncation error smaller than the resolution of floating-point arithmetic. In practice, it is usually sufficient for the boundary error to remain well below the discretization error of the numerical method.

- **Periodic conditions:** A periodic boundary condition does not approximate an isolated open domain. Instead, it changes the physical interpretation of the computational domain: the finite domain is treated as a representative cell that repeats indefinitely in space. Mathematically, the two opposite boundaries are identified so that

$$\phi(x_\mathrm{L}, t) = \phi(x_\mathrm{R}, t), \tag{3.44}$$

together with the corresponding periodicity requirements for any additional moments or derivatives used by the discretization.

This approximation is useful for systems containing many similar repeating structures. Consider, for example, a heat sink composed of a large array of fins, as illustrated in Figure 3.6. The outermost fins are affected by the finite extent of the heat sink and therefore experience different thermal conditions from the fins in the interior. The influence of these edge conditions decreases with distance from the outer boundary, and sufficiently far inside the array the thermal behavior of neighboring fins becomes nearly repetitive.

If the number of fins is large enough that a substantial interior region exists, this region may be represented by a periodic unit cell. The heat transfer obtained from the periodic calculation then characterizes the bulk behavior of the interior fins. The total heat-transfer rate of a large heat sink may consequently be estimated from this representative behavior, with the understanding that the approximation neglects the distinct contribution of the edge fins.

In practice, this boundary condition is highly effective for modeling transport phenomena within symmetric, repetitive physical architectures or for analyzing pure spectral properties without the influence of physical walls. For a diffusing Gaussian, periodicity implies that the scalar mass flux exiting one boundary simultaneously re-enters through the opposite boundary, eventually leading to a uniform scalar field across the domain.

- **Neumann conditions:** Whereas a Dirichlet condition specifies the value of the solution, a Neumann condition specifies its normal derivative. For example,

$$\frac{\partial \phi}{\partial x} = 0 \tag{3.45}$$

defines a homogeneous Neumann boundary condition. For a diffusion process, the diffusive flux is proportional to the gradient,

$$q_\mathrm{d} = -K \frac{\partial \phi}{\partial x}. \tag{3.46}$$

Hence, a zero-gradient condition implies

$$q_\mathrm{d} = 0, \tag{3.47}$$

and the boundary behaves as an insulating or impermeable wall. In a purely diffusive closed system, this prevents scalar mass or thermal energy from crossing the boundary.
The prescribed gradient need not be zero. Setting

$$\frac{\partial \phi}{\partial x} = g_{\mathrm{N}} \tag{3.48}$$

instead imposes a specified diffusive exchange with the exterior.
Care is required when interpreting the same condition for convection–diffusion problems. A homogeneous Neumann condition removes the diffusive flux at the outlet, but it does not remove the convective flux. It can therefore provide a useful approximation when the flow is directed out of the domain and diffusion is weak near the boundary.
For a purely convective equation, however, the downstream solution is carried out of the domain by the characteristics. If all characteristics point outward, no independent boundary information is required. A Neumann condition imposed at such an outlet can therefore over-constrain the problem rather than improve its physical representation.

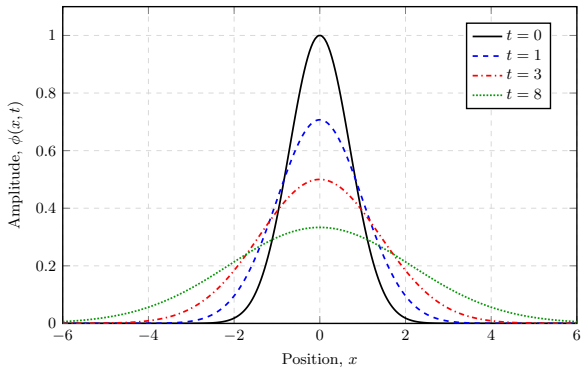


Fig. 3.5: Diffusion of Gaussian Profile over time with $A = 0$, $x_c = 0$, $K = 0.25$, $\sigma_0 = 1$.

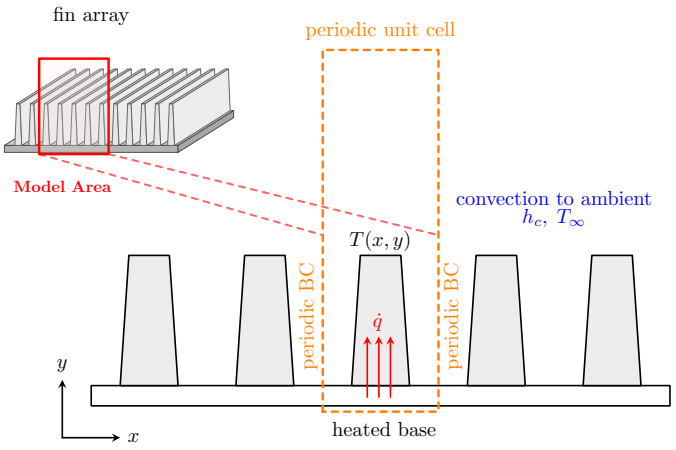


Fig. 3.6: Periodic domain assignment to the representative unit of the fin.

Example 3.3 (Effect of domain truncation with Periodic Boundary conditions)
The analytical solution given in Eq. (3.40) is strictly valid only within an infinite domain. In practical computations, however, truncated domains are necessary due to finite computing resources. To investigate the impact of domain truncation, we use the same physical conditions as in Fig. 3.5 and evaluate boundary effects when we terminate the domain at $x = \pm 4$.
For low-resolution grids, a domain spanning $x \in [-4, 4]$ proves reasonable, keeping the maximum absolute error below 2% (Fig.3.7). Doubling the resolution to 8 cells reduces the near-center errors to about 1-2 per mille. The errors at the boundary are now much larger than those in the inner domain. On the 64-cell solution, the boundary-induced errors become dominant and begin to pollute the interior solution. A logarithmic plot would reveal that the boundary error is four orders of magnitude larger than that at the center. These boundary errors propagate inward and decay as a power function of x. Consequently, to safeguard the region of interest from these edge effects in this specific problem, the boundaries should be positioned sufficiently away from the primary dynamics.

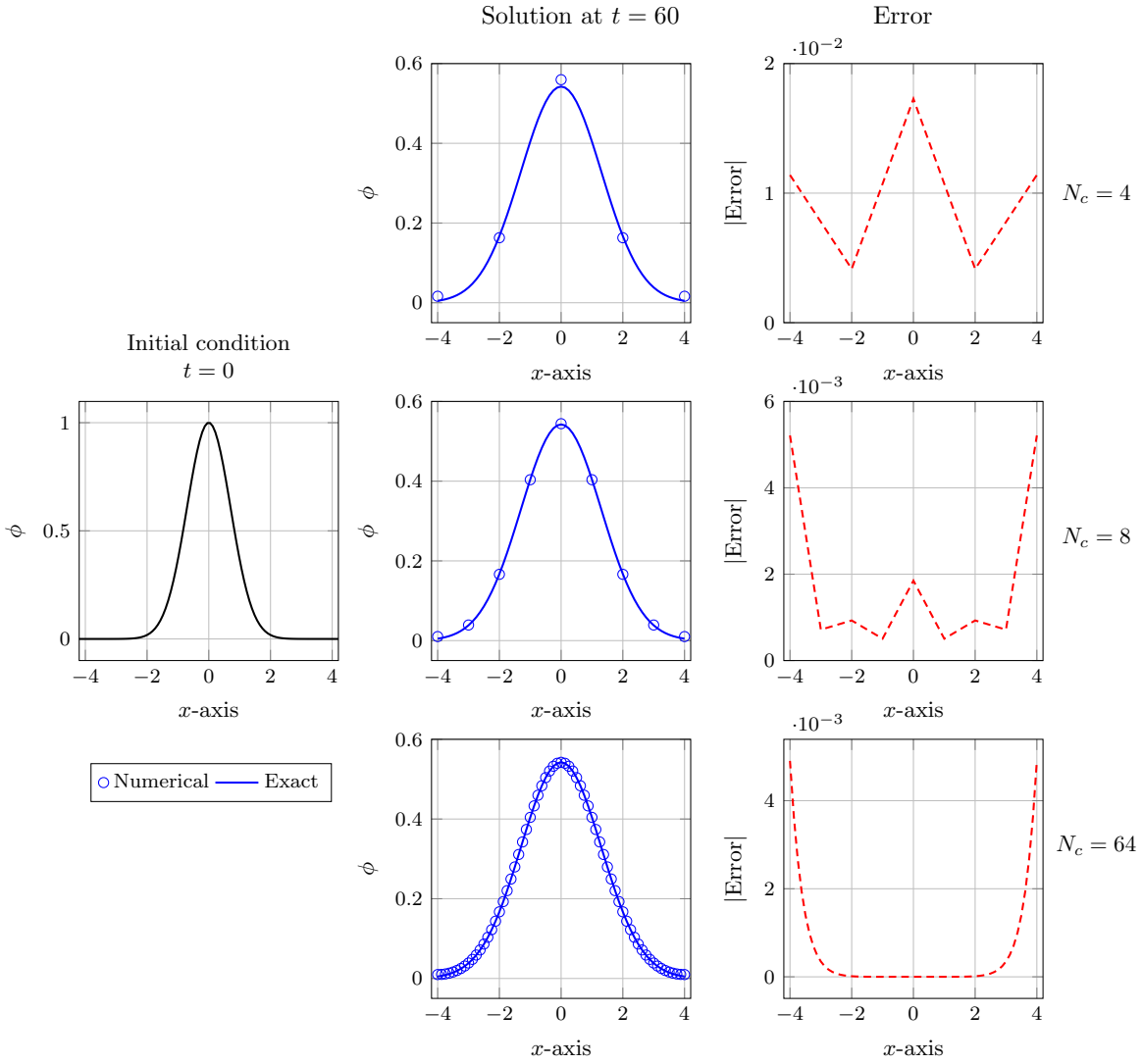


Fig. 3.7: Numerical solution and absolute error for the 1D advection equation at $t = 60$ using central fourth-order spatial discretization. The left panel shows the initial Gaussian profile at $t = 0$. The middle and right columns illustrate the convergence of the numerical solution (ϕ) and the spatial distribution of the absolute error across successive mesh refinements ($N_c = 4, 8$, and 64).

Additionally, it is worth noting the behavior of the periodic boundary conditions used here. Because periodic boundaries transfer flux from one end of the domain to the other, and given the symmetric nature of the profile, the incoming and outgoing fluxes balance exactly. As a result, the initial total quantity of ϕ is conserved. Over a long period, the asymptotic solution does not decay to zero; rather, it approaches a uniform state equal to the spatial average of the initial ϕ distribution over the domain.

Problems

Problem 3.1 Consider a symmetric one-dimensional random walk in which a particle moves either $+\Delta x$ or $-\Delta x$ with equal probability during each time interval Δt. Show that the continuum limit leads to a diffusion coefficient of the form

$$K = \lim_{\Delta x, \Delta t \to 0} \frac{(\Delta x)^2}{2\Delta t}.$$

Explain physically why the resulting macroscopic flux must act opposite to the concentration gradient and state the corresponding constitutive law.

Problem 3.2 The diffusive flux of a scalar quantity is given by

$$J_x = -K \frac{\partial \phi}{\partial x}.$$

Starting from the conservation equation

$$\frac{\partial \phi}{\partial t} = -\frac{\partial J_x}{\partial x},$$

derive the one-dimensional diffusion equation for constant K. Generalize the result to three dimensions and include an arbitrary source term S.

Problem 3.3 Heat conduction, mass diffusion, and momentum diffusion have the same mathematical structure despite representing different physical processes. For each of these three phenomena, identify:

1. the transported variable,
2. the corresponding diffusion coefficient, and
3. the physical constitutive law responsible for the diffusive flux.

Show how a generic diffusion equation can represent all three.

Problem 3.4 Consider the one-dimensional diffusion equation

$$\frac{\partial \phi}{\partial t} = K \frac{\partial^2 \phi}{\partial x^2}, \qquad K > 0.$$

Assume a Fourier mode

$$\phi(x, t) = \widehat{\phi}(t) e^{\mathbf{i} k x}.$$

Derive the exact temporal evolution of $\widehat{\phi}(t)$. Explain why diffusion preferentially removes high-wavenumber fluctuations and therefore acts as a smoothing mechanism.

Problem 3.5 A general second-order PDE in two independent variables is written as

$$A\phi_{xx} + B\phi_{xy} + C\phi_{yy} + D\phi_x + E\phi_y + F\phi = G.$$

State the discriminant used to classify the equation and classify each of the following:

$$\phi_{xx} + \phi_{yy} = 0,$$

$$\phi_t - \alpha\phi_{xx} = 0,$$

and

$$\phi_{tt} - c^2\phi_{xx} = 0.$$

Explain the physical meaning of each classification.

Problem 3.6 The hyperbolic heat-conduction equation may be written as

$$\frac{K}{c^2}\frac{\partial^2 T}{\partial t^2} + \rho C_p \frac{\partial T}{\partial t} = K\frac{\partial^2 T}{\partial x^2}.$$

Introduce the thermal diffusivity

$$\alpha = \frac{K}{\rho C_p}$$

and derive the thermal relaxation time τ. Rewrite the equation in terms of α and τ. What happens in the limit $\tau \to 0$? Contrast the domains of dependence of the hyperbolic, parabolic, and elliptic regimes.

Problem 3.7 Consider the explicit multi-moment approximation

$$\phi_i'' = \frac{d}{h^2}\phi_i + \frac{b_1}{h^2}\left(\overline{\phi}_{i+1/2} + \overline{\phi}_{i-1/2}\right) + \frac{c_1}{h^2}\left(\phi_{i+1} + \phi_{i-1}\right).$$

Using Taylor expansions about x_i, determine d, b_1, and c_1 so that the approximation has the highest possible formal order. Show that the resulting scheme is

$$\phi_i'' = -\frac{12}{h^2}\phi_i + \frac{15}{2h^2}\left(\overline{\phi}_{i+1/2} + \overline{\phi}_{i-1/2}\right) - \frac{3}{2h^2}\left(\phi_{i+1} + \phi_{i-1}\right),$$

and determine its leading truncation error.

Problem 3.8 Apply the fourth-order explicit multi-moment second-derivative approximation to the Fourier mode

$$\phi(x) = e^{\mathbf{i}kx}.$$

Let $\theta = kh$ and define the squared modified wavenumber by

$$D_2\phi = -k^{2*}\phi.$$

Show that

$$h^2k^{2*} = 12 - 15\frac{\sin\theta}{\theta} + 3\cos\theta.$$

Expand this expression about $\theta = 0$ and verify the formal order of the scheme.

Problem 3.9 For the fourth-order explicit multi-moment scheme, define the relative second-derivative spectral error by

$$\epsilon_D(\theta) = \frac{\left| h^2 k^{2*} - \theta^2 \right|}{\theta^2}.$$

Determine numerically the largest $\theta = kh$ satisfying

$$\epsilon_D < 0.01.$$

Compute the corresponding resolving efficiency

$$e(0.01) = \frac{\theta_{\max}}{\pi}.$$

Problem 3.10 Consider a compact multi-moment approximation of the form

$$\phi_i'' + a\left(\phi_{i+1}'' + \phi_{i-1}''\right) = \frac{d}{h^2}\phi_i + \frac{b}{h^2}\left(\overline{\phi}_{i+1/2} + \overline{\phi}_{i-1/2}\right) + \frac{c}{h^2}\left(\phi_{i+1} + \phi_{i-1}\right).$$

Use Taylor expansions to determine the coefficients required for sixth-order consistency. Show that the resulting scheme is

$$\phi_i'' - \frac{1}{32}\left(\phi_{i+1}'' + \phi_{i-1}''\right) = -\frac{15}{h^2}\phi_i + \frac{315}{32h^2}\left(\overline{\phi}_{i+1/2} + \overline{\phi}_{i-1/2}\right) - \frac{75}{32h^2}\left(\phi_{i+1} + \phi_{i-1}\right).$$

Determine the leading residual of the compact relation.

Problem 3.11 Apply the sixth-order compact scheme from the previous problem to $e^{\mathbf{i}kx}$. Show that its squared modified wavenumber satisfies

$$h^2 k^{2*} = \frac{240 - 315\dfrac{\sin\theta}{\theta} + 75\cos\theta}{16 - \cos\theta}, \qquad \theta = kh.$$

Expand this expression for $\theta \ll 1$ and verify that it is sixth-order accurate. Explain why compact schemes generally possess better high-wavenumber resolution than explicit schemes of comparable stencil width.

Problem 3.12 At a left Dirichlet boundary, the multi-moment fourth-order closure is

$$\phi_0' = \frac{23}{2h}\overline{\phi}_{1/2} + \frac{7}{2h}\overline{\phi}_{3/2} - \frac{6}{h}\phi_0 - \frac{8}{h}\phi_1 - \frac{1}{h}\phi_2.$$

Expand all quantities about $x = 0$ and verify that the approximation is fourth-order accurate. Determine its leading truncation error. Why must centered interior stencils generally be replaced by asymmetric approximations near a physical boundary?

Problem 3.13 For the diffusion equation

$$\frac{\partial \phi}{\partial t} = K\frac{\partial^2 \phi}{\partial x^2},$$

let the discrete second-derivative operator have modified wavenumber k^{2*}. Show that its Fourier-space evolution equation is

$$\frac{d\widehat{\phi}}{dt} = -Kk^{2*}\widehat{\phi}.$$

Define the grid Fourier number

$$\sigma = \frac{K\Delta t}{h^2}$$

and show that the stability parameter can be written as

$$z = -\left(h^2k^{2*}\right)\sigma.$$

If an explicit time integrator is stable on the negative real axis for $-z_{\text{real,max}} \le z \le 0$, derive the maximum allowable value of σ.

Problem 3.14 The classical fourth-order Runge–Kutta method has stability polynomial

$$R_4(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}.$$

On the negative real axis, set $z = -r$ and determine numerically the nonzero value $r_{\max}$ for which

$$|R_4(-r_{\max})| = 1.$$

For the fourth-order explicit multi-moment diffusion operator,

$$h^2k^{2*} = 12 - 15\frac{\sin\theta}{\theta} + 3\cos\theta,$$

show that

$$\max_{0\le\theta\le\pi} h^2k^{2*} = 9.$$

Estimate the maximum stable grid Fourier number for RK4.

Problem 3.15 For an explicit convection–diffusion calculation, suppose the convective time-step restriction scales as

$$\Delta t_c = O(h),$$

while the diffusive restriction scales as

$$\Delta t_d = O(h^2).$$

Explain which restriction dominates as the grid is refined. If the grid spacing is reduced from h to $h/2$, by what factors do the two time-step limits change? Explain why this behavior motivates implicit–explicit (IMEX) time integration for convection–diffusion systems.

Problem 3.16 Consider the diffusion of the Gaussian initial condition

$$\phi(x,0) = A\exp\left[-\frac{(x-x_c)^2}{\sigma_0^2}\right]$$

on an infinite domain. Show that

$$\phi(x,t) = A\frac{\sigma_0}{\sqrt{\sigma_0^2 + 4Kt}}\exp\left[-\frac{(x-x_c)^2}{\sigma_0^2 + 4Kt}\right].$$

Using this expression, show that the width of the profile increases with time, the peak amplitude decreases, and the total integral

$$M(t) = \int_{-\infty}^{\infty}\phi(x,t)\,dx$$

remains constant.

Problem 3.17 For the Gaussian diffusion problem, let

$$A = 1, \qquad x_c = 0, \qquad \sigma_0 = 1, \qquad K = 0.25.$$

At $t = 8$, determine:

1. the width parameter $\sqrt{\sigma_0^2 + 4Kt}$,
2. the peak value $\phi(0, 8)$,
3. the value $\phi(3, 8)$, and
4. the total scalar mass on the infinite domain.

Problem 3.18 A Gaussian diffusion problem is truncated to the symmetric finite domain

$$x \in [x_c - L, x_c + L].$$

Using the analytical solution, derive a condition on L such that

$$|\phi(x_c \pm L, T)| \leq \epsilon_{\text{tol}}$$

at the final simulation time T. Show that

$$L \geq \sqrt{\left(\sigma_0^2 + 4KT\right) \ln \left[\frac{A\sigma_0}{\epsilon_{\text{tol}} \sqrt{\sigma_0^2 + 4KT}} \right]}.$$

Estimate L for

$$A = 1, \quad \sigma_0 = 1, \quad K = 0.25, \quad T = 8, \quad \epsilon_{\text{tol}} = 10^{-6}.$$

Problem 3.19 Compare homogeneous Dirichlet, periodic, and homogeneous Neumann boundary conditions for a localized diffusing Gaussian profile. For each condition, describe its physical interpretation and whether the total scalar mass is expected to remain conserved. What is the long-time solution of a purely diffusive problem on a periodic domain?

Problem 3.20 Use the Method of Manufactured Solutions to verify a numerical diffusion solver. Consider

$$\phi(x, t) = e^{-t} \sin x$$

on the periodic domain $x \in [0, 2\pi]$, governed by

$$\frac{\partial \phi}{\partial t} = K \frac{\partial^2 \phi}{\partial x^2} + S(x, t).$$

Determine the source term $S(x, t)$ required to make the prescribed $\phi(x, t)$ an exact solution. Describe a convergence experiment that would verify the fourth-order spatial accuracy of the MMEX4 scheme while ensuring that the temporal integration error does not contaminate the measurement.

Solution Keys

Here h is the uniform cell width, $\overline{\phi}_{i+1/2} = h^{-1}\int_{x_i}^{x_{i+1}} \phi(x)\,dx$, and $K > 0$ denotes diffusivity unless explicitly identified as thermal conductivity. For an explicit stencil, the derivative error means $D_h\phi - \phi''$. For a compact relation, the residual means its left side minus its right side after inserting the exact function and derivatives.

1. **Random walk and the diffusion coefficient**
   Let $p(x,t)$ denote particle probability density. The unbiased walk obeys

   $$p(x, t+\Delta t) = \tfrac{1}{2}p(x-\Delta x, t) + \tfrac{1}{2}p(x+\Delta x, t).$$

   Taylor expansion gives

   $$\Delta t\, p_t + O(\Delta t^2) = \frac{(\Delta x)^2}{2} p_{xx} + O((\Delta x)^4).$$

   Taking the diffusive limit with $(\Delta x)^2/\Delta t$ finite yields

   $$\boxed{p_t = K p_{xx}, \qquad K = \lim \frac{(\Delta x)^2}{2\Delta t}.}$$

   Equivalently, after $n = t/\Delta t$ independent steps, $\langle X^2\rangle = n(\Delta x)^2 = 2Kt$. More particles cross an interface from its more concentrated side, so the net flux points down the concentration gradient. The constitutive law is Fick's law, $\boxed{J_x = -K\phi_x}$.
2. **Conservation and the diffusion equation**
   Substitution of the constitutive law into conservation gives

   $$\phi_t = -\partial_x(-K\phi_x) = K\phi_{xx}$$

   for constant K. In three dimensions,

   $$\boxed{\phi_t = K(\phi_{xx} + \phi_{yy} + \phi_{zz}) + S = K\nabla^2\phi + S.}$$

   For a spatially variable isotropic diffusivity, the appropriate expression is $\phi_t = \nabla\cdot(K\nabla\phi)+S$; K cannot generally be taken outside the divergence.
3. **Heat, mass, and momentum diffusion**
   For constant material properties and the corresponding purely diffusive setting, the three examples are:

Process	Variable	Diffusivity	Constitutive law
Heat	Temperature T	$\alpha = k/(\rho C_p)$	Fourier: $\boldsymbol{q} = -k\nabla T$
Mass	Concentration c	Molecular D	Fick: $\boldsymbol{j} = -D\nabla c$
Momentum	Velocity u	$\nu = \mu/\rho$	Newtonian shear: $\tau_{xy} = \mu\,\partial_y u$

   For shear flow $u = u(y,t)$, momentum balance gives $\rho u_t = \partial_y\tau_{xy} = \mu u_{yy}$. Written as a conservative diffusive momentum flux, the flux is $J_{u,y} = -\mu u_y$; this distinguishes the

flux sign from the stress convention. Heat balance similarly gives $\rho C_p T_t = k\nabla^2 T$. After division by the storage coefficient, all three reduce to

$$\boxed{\phi_t = \kappa\nabla^2\phi + s,}$$

where κ is respectively α, D, or ν and s is the source per unit storage. General fluid momentum also contains convection, pressure, and possibly other forces, which are absent from this reduction.

4. **Decay of a Fourier mode**
For $\phi = \widehat{\phi}(t)e^{\mathrm{i}kx}$,

$$\frac{d\widehat{\phi}}{dt} = -Kk^2\widehat{\phi}, \qquad \boxed{\widehat{\phi}(t) = \widehat{\phi}(0)e^{-Kk^2t}.}$$

The decay time is $1/(Kk^2)$ for $k \neq 0$. Thus, short waves decay faster than long waves, while the constant mode $k = 0$ is unchanged. This is the spectral explanation for smoothing and preserving the mean.

5. **PDE classification**
With the mixed term written as $B\phi_{xy}$, the discriminant is $\Delta = B^2 - 4AC$. Negative, zero, and positive values correspond to elliptic, parabolic, and hyperbolic equations, respectively.

- $\phi_{xx} + \phi_{yy} = 0$: $A = C = 1$, $B = 0$, so $\Delta = -4$. It is elliptic and describes equilibrium controlled by boundary data throughout the domain.
- $\phi_t - \alpha\phi_{xx} = 0$: in variables (x, t), $A = -\alpha$, $B = C = 0$, so $\Delta = 0$. It is parabolic: initial data spread diffusively, and disturbances have no finite propagation front.
- $\phi_{tt} - c^2\phi_{xx} = 0$: $A = -c^2$, $C = 1$, $B = 0$, so $\Delta = 4c^2 > 0$. It is hyperbolic, with characteristics $x \pm ct$ and finite propagation speed c.

Lower-order terms do not change this principal-part classification.

6. **Thermal relaxation and domains of dependence**
In this problem, K is the thermal conductivity. Divide the stated equation by ρC_p to obtain

$$\frac{K}{\rho C_p c^2}T_{tt} + T_t = \frac{K}{\rho C_p}T_{xx}.$$

Consequently,

$$\boxed{\alpha = \frac{K}{\rho C_p}, \qquad \tau = \frac{\alpha}{c^2}, \qquad \tau T_{tt} + T_t = \alpha T_{xx}.}$$

For $\tau \to 0$ with α fixed, the equation reduces to $T_t = \alpha T_{xx}$, apart from a possible initial relaxation layer. For $\tau > 0$ the characteristic speed is $\sqrt{\alpha/\tau} = c$: only initial data within $[x - ct, x + ct]$ can influence (x, t) on the infinite line. Parabolic diffusion instead has a nonzero heat kernel over the entire initial line for every $t > 0$. An elliptic equilibrium problem has no causal time cone; its solution is determined globally by the boundary data and interior sources.

7. **Fourth-order explicit second derivative**
Taylor expansion of the symmetric data pairs gives

$$\overline{\phi}_{i+1/2} + \overline{\phi}_{i-1/2} = 2\phi_i + \frac{h^2}{3}\phi_i'' + \frac{h^4}{60}\phi_i^{(4)} + \frac{h^6}{2520}\phi_i^{(6)} + O(h^8),$$

$$\phi_{i+1} + \phi_{i-1} = 2\phi_i + h^2\phi_i'' + \frac{h^4}{12}\phi_i^{(4)} + \frac{h^6}{360}\phi_i^{(6)} + O(h^8).$$

Cancellation of the constant term, normalization of ϕ'', and cancellation of $\phi^{(4)}$ require

$$d + 2b_1 + 2c_1 = 0, \qquad b_1/3 + c_1 = 1, \qquad b_1/60 + c_1/12 = 0.$$

Their unique solution is

$$\boxed{d = -12, \qquad b_1 = 15/2, \qquad c_1 = -3/2.}$$

These are the three available independent consistency conditions. The next coefficient does not vanish:

$$\frac{b_1}{2520} + \frac{c_1}{360} = -\frac{1}{840}.$$

Therefore, the stencil in the question satisfies

$$\boxed{D_h\phi = \phi'' - \frac{h^4}{840}\phi^{(6)} + O(h^6).}$$

It has the highest possible order, four, for the stated three-parameter family. If truncation error is defined as exact derivative minus stencil, its leading sign is reversed.

8. **Explicit modified wavenumber**
The exact cell averages of a Fourier mode satisfy

$$\overline{\phi}_{i\pm1/2} = e^{\mathrm{i}kx_i} e^{\pm\mathrm{i}\theta/2} \frac{\sin(\theta/2)}{\theta/2}.$$

Their sum is $2(\sin\theta/\theta)e^{\mathrm{i}kx_i}$, and the nodal sum is $2\cos\theta\, e^{\mathrm{i}kx_i}$. Hence

$$D_h\phi = \frac{-12 + 15\sin\theta/\theta - 3\cos\theta}{h^2}\phi,$$

so

$$\boxed{h^2k^{2*} = 12 - 15\frac{\sin\theta}{\theta} + 3\cos\theta.}$$

Taking $\sin\theta/\theta = 1$ at zero by continuity, the series is

$$h^2k^{2*} = \theta^2 - \frac{\theta^6}{840} + \frac{\theta^8}{30240} + O(\theta^{10}).$$

Dividing by h^2 shows an $O(h^4k^6)$ error at fixed k, in agreement with Problem 7.

9. **One-percent resolving efficiency**
On the interval connected to $\theta = 0$, the threshold is obtained from

$$\frac{\left|12 - 15\sin\theta/\theta + 3\cos\theta - \theta^2\right|}{\theta^2} = 0.01.$$

Bisection on $(0, \pi)$ gives

$$\boxed{\theta_{\text{threshold}} = 1.7385708717, \qquad e(0.01) = \frac{\theta_{\text{threshold}}}{\pi} = 0.5534042963.}$$

The small-wavenumber expansion $\epsilon_D = \theta^4/840 - \theta^6/30240 + O(\theta^8)$ also checks the limiting behavior. Strictly, the question's inequality $\epsilon_D < 0.01$ excludes the threshold itself: the reported endpoint is the supremum of the resolved interval. This calculation uses exact cell averages of a Fourier mode, as stipulated, rather than the full coupled PV–CV evolution spectrum.

10. **Sixth-order compact coefficients and residual**
For the exact function, the left side expands as

$$(1+2a)\phi'' + ah^2\phi^{(4)} + \frac{ah^4}{12}\phi^{(6)} + \frac{ah^6}{360}\phi^{(8)} + O(h^8).$$

Using the data-pair expansions in Problem 7, sixth-order consistency requires

$$\begin{aligned} d + 2b + 2c &= 0, & b/3 + c &= 1 + 2a, \\ b/60 + c/12 &= a, & b/2520 + c/360 &= a/12. \end{aligned}$$

Solving gives

$$\boxed{a = -\frac{1}{32}, \quad d = -15, \quad b = \frac{315}{32}, \quad c = -\frac{75}{32}.}$$

At order $h^6\phi^{(8)}$, the exact left side minus right side is

$$\frac{a}{360} - \frac{b}{181440} - \frac{c}{20160} = -\frac{1}{40320}.$$

The leading residual of the relation, with its h^{-2} factors retained, is therefore

$$\boxed{\mathcal{R} = -\frac{h^6}{40320}\phi^{(8)} + O(h^8).}$$

If the entire relation is multiplied by h^2, its residual is instead $-h^8\phi^{(8)}/40320 + O(h^{10})$. The residual differs from the error in the solved derivative. Since the compact left operator has leading factor $1 + 2a = 15/16$, inversion gives

$$D_h\phi - \phi'' = \frac{h^6}{37800}\phi^{(8)} + O(h^8).$$

11. **Compact modified wavenumber**
The derivative coupling has Fourier multiplier $1 + 2a\cos\theta = 1 - \cos\theta/16$. The right side has multiplier

$$\frac{1}{h^2}\left[-15 + \frac{315}{16}\frac{\sin\theta}{\theta} - \frac{75}{16}\cos\theta\right].$$

Solving the scalar Fourier relation yields

$$\boxed{h^2k^{2*} = \frac{240 - 315\sin\theta/\theta + 75\cos\theta}{16 - \cos\theta}.}$$

Its expansion is

$$h^2k^{2*} = \theta^2 - \frac{\theta^8}{37800} + \frac{\theta^{10}}{594000} + O(\theta^{12}).$$

Thus the error in k^{2*} is $O(h^6k^8)$, confirming sixth order. The compact denominator produces a rational spectral approximation, adding flexibility without widening the right-side data stencil. This often reduces intermediate- and high-wavenumber errors compared with explicit formulas of comparable width. It does not make the approximation exact at all wavenumbers, and requires a coupled derivative solve.

12. **Dirichlet boundary derivative closure**
At $x = 0$, write $\phi_0^{(n)} = \phi^{(n)}(0)$. The required series are

$$\overline{\phi}_{1/2} = \sum_{n\geq 0} \frac{h^n}{(n+1)!}\phi_0^{(n)},$$
$$\overline{\phi}_{3/2} = \sum_{n\geq 0} \frac{(2^{n+1}-1)h^n}{(n+1)!}\phi_0^{(n)},$$
$$\phi_j = \sum_{n\geq 0} \frac{j^n h^n}{n!}\phi_0^{(n)}.$$

Inserting these into the stated closure, the coefficients of $\phi_0/h, \phi_0', h\phi_0'', h^2\phi_0^{(3)}, h^3\phi_0^{(4)}$ are respectively $0, 1, 0, 0, 0$. For $h^4\phi_0^{(5)}$, the coefficient is

$$\frac{1}{5!}\left[\frac{23}{12} + \frac{7(2^6-1)}{12} - 8 - 2^5\right] = -\frac{1}{90}.$$

Consequently,

$$\boxed{D_{b,h}\phi = \phi_0' - \frac{h^4}{90}\phi_0^{(5)} + O(h^5).}$$

This is fourth-order accurate. A centered stencil at the left boundary would require exterior values or averages that the physical domain does not supply. A one-sided reconstruction, or a consistent ghost-data construction using the boundary condition, closes the system.

13. **Diffusive Fourier evolution and time-step bound**
For the discrete spatial operator, $D_h\phi = -k^{2*}\phi$. Therefore

$$\boxed{\frac{d\widehat{\phi}}{dt} = -Kk^{2*}\widehat{\phi}, \qquad z = -K\Delta t\, k^{2*} = -\sigma(h^2k^{2*}), \quad \sigma = \frac{K\Delta t}{h^2}.}$$

Let $Q_{\max} = \max_\theta(h^2k^{2*})$, with nonnegative real k^{2*} for all relevant modes. If the integrator's stable real-axis interval is $[-z_{\text{real,max}}, 0]$, then

$$\boxed{\sigma \leq \frac{z_{\text{real,max}}}{Q_{\max}}, \qquad \Delta t \leq \frac{z_{\text{real,max}}h^2}{KQ_{\max}}.}$$

For a fully coupled PV–CV discretization, apply the stability polynomial to every eigenvalue branch of its actual semidiscrete operator; a single reconstructed-field symbol is not automatically the full spectrum.

14. **RK4 stability for the explicit diffusion symbol**
On the negative real axis,

$$R_4(-r) = 1 - r + \frac{r^2}{2} - \frac{r^3}{6} + \frac{r^4}{24}.$$

This expression is positive: it equals $\left[(r^2-2r)^2 + 8(r-3/2)^2 + 6\right]/24$. Thus the nonzero stability endpoint occurs at $R_4(-r) = 1$. Factoring out r gives

$$r^3 - 4r^2 + 12r - 24 = 0, \qquad \boxed{r_{\max} = 2.7852935634.}$$

To find the spatial maximum, set $Q(\theta) = 12 - 15\sin\theta/\theta + 3\cos\theta$. Its derivative is

$$Q'(\theta) = 15\frac{\sin\theta - \theta\cos\theta}{\theta^2} - 3\sin\theta.$$

For $0 < t \le \theta \le \pi$, $\sin t/t \ge \sin\theta/\theta$. It follows that

$$\sin\theta - \theta\cos\theta = \int_0^\theta t\sin t\,dt \ge \frac{\theta^2\sin\theta}{3},$$

so $Q'(\theta) \ge 2\sin\theta > 0$ on $(0,\pi)$. Hence $Q_{\max} = Q(\pi) = 12 - 3 = 9$, and

$$\boxed{\sigma_{\max} = \frac{2.7852935634}{9} = 0.3094770626, \qquad \Delta t_{\max} \approx 0.3094770626\frac{h^2}{K}.}$$

This is the estimate for the symbol and wavenumber interval specified in the question. Boundary closures or additional coupled modes can change the actual solver restriction.

15. **Convection versus diffusion under refinement**
Writing $\Delta t_c = C_c h$ and $\Delta t_d = C_d h^2$ gives $\Delta t_d/\Delta t_c = (C_d/C_c)h \to 0$. Thus diffusion eventually imposes the tighter explicit time-step bound. On changing h to $h/2$,

$$\boxed{\Delta t_c \mapsto \Delta t_c/2, \qquad \Delta t_d \mapsto \Delta t_d/4.}$$

An IMEX method can treat diffusion implicitly and convection explicitly, removing the explicit diffusive stability restriction when the implicit part has suitable stability. Accuracy and the explicit convective restriction still constrain the time step.

16. **Gaussian diffusion on the infinite line**
Let $y = x - x_c$, $s(t) = \sigma_0^2 + 4Kt$, and try $\phi = B(t)e^{-y^2/s(t)}$. Then

$$\frac{\phi_t}{\phi} = \frac{B'}{B} + \frac{y^2 s'}{s^2}, \qquad \frac{\phi_{xx}}{\phi} = -\frac{2}{s} + \frac{4y^2}{s^2}.$$

Matching terms in $\phi_t = K\phi_{xx}$ gives $s' = 4K$ and $B'/B = -2K/s$, so $B = A\sigma_0/\sqrt{s}$ from the initial data. Therefore

$$\boxed{\phi(x,t) = \frac{A\sigma_0}{\sqrt{\sigma_0^2 + 4Kt}}\exp\left[-\frac{(x-x_c)^2}{\sigma_0^2 + 4Kt}\right].}$$

The width parameter is $w = \sqrt{s}$, which increases with time, while the peak $A\sigma_0/\sqrt{s}$ decreases for $A > 0$. Direct integration gives

$$\boxed{M(t) = \frac{A\sigma_0}{\sqrt{s}}\sqrt{\pi s} = A\sigma_0\sqrt{\pi}.}$$

For this exponent convention, the normalized Gaussian's actual variance is $s/2$, not s. Its standard deviation is $w/\sqrt{2}$.

17. **Numerical Gaussian values**
With the stated parameters, $s(8) = 1 + 4(0.25)(8) = 9$. Thus

Width parameter: $\boxed{w = 3}$,

$$\text{Peak:} \quad \boxed{\phi(0,8) = 1/3 \approx 0.3333333333},$$
$$\text{At } x = 3: \quad \boxed{\phi(3,8) = e^{-1}/3 \approx 0.1226264804},$$
$$\text{Total mass:} \quad \boxed{M = \sqrt{\pi} \approx 1.7724538509}.$$

The standard deviation is $3/\sqrt{2} \approx 2.1213203436$.

18. **Domain size for a boundary-value tolerance**
Let $s_T = \sigma_0^2 + 4KT$. The exact boundary magnitude is

$$|\phi(x_c \pm L, T)| = \frac{|A|\sigma_0}{\sqrt{s_T}} e^{-L^2/s_T}.$$

If $|A|\sigma_0/\sqrt{s_T} > \epsilon_{\text{tol}} > 0$, taking logarithms gives

$$\boxed{L \geq \sqrt{s_T \ln\left(\frac{|A|\sigma_0}{\epsilon_{\text{tol}}\sqrt{s_T}}\right)}}.$$

For $A > 0$, this is the expression in the question. If the final peak is already below the tolerance, every $L \geq 0$ satisfies the final-time boundary-value condition; the square-root formula then needs no use. For the numerical parameters,

$$\boxed{L \geq \sqrt{9 \ln\left(\frac{1}{3 \times 10^{-6}}\right)} = 10.6982280974.}$$

For example, choosing $L = 10.70$ meets the stated final-time condition. This is a bound on the exact boundary value, not a bound on the numerical solution error. A tolerance required at all times must maximize the boundary magnitude over $0 \leq t \leq T$. For a fixed $L > 0$, its logarithmic derivative with respect to s is $(L^2 - s/2)/s^2$; the maximum occurs at $s = 2L^2$ if this lies in the time interval's range of s, otherwise at the appropriate endpoint. Here $2L^2 > s_T = 9$, so the boundary magnitude increases up to T and the final-time check also bounds earlier times.

19. **Boundary conditions, mass, and long-time behavior**
On $[a, b]$, integration of the diffusion equation gives

$$\frac{dM}{dt} = K\left[\phi_x(b,t) - \phi_x(a,t)\right] = J(a,t) - J(b,t).$$

- Homogeneous Dirichlet, $\phi(a,t) = \phi(b,t) = 0$: the endpoints are reservoirs fixed at zero scalar value. Boundary gradients generally allow flux, so mass is not conserved. A nonnegative localized profile typically loses mass and tends to zero.
- Periodic: endpoint values and fluxes match, so mass is conserved. The profile wraps around a repeated domain; it eventually approaches the constant $\boxed{\phi_\infty = M(0)/(b-a)}$ as all nonzero Fourier modes decay.
- Homogeneous Neumann, $\phi_x(a,t) = \phi_x(b,t) = 0$: the boundaries are insulating or impermeable. Both fluxes vanish, mass is conserved, and the purely diffusive long-time state is also the initial mean.

These finite-domain solutions differ from the isolated infinite-line Gaussian once boundary effects become significant.

20. **Manufactured solution and fourth-order verification**
For $\phi = e^{-t}\sin x$, $\phi_t = -e^{-t}\sin x$ and $\phi_{xx} = -e^{-t}\sin x$. The required source is therefore

$$\boxed{S = \phi_t - K\phi_{xx} = (K-1)e^{-t}\sin x.}$$

Use periodic grids with $N, 2N, 4N, \ldots$ cells, $h = 2\pi/N$, and a fixed final time T. Initialize both moment types exactly:

$$\phi_i(0) = \sin x_i, \qquad \overline{\phi}_{i+1/2}(0) = \frac{\cos x_i - \cos x_{i+1}}{h}.$$

For CV evolution, also use the exact cell-averaged source

$$\overline{S}_{i+1/2}(t) = (K-1)e^{-t}\frac{\cos x_i - \cos x_{i+1}}{h}.$$

Evaluate time-dependent forcing at each integrator stage time. Compare nodal values with $e^{-T}\sin x_i$ and cell averages with $e^{-T}(\cos x_i - \cos x_{i+1})/h$. Use a complete fourth-order spatial discretization for both PV and CV equations, so another lower-order component does not set the result.
With RK4, choose $\Delta t = Ch^2$ for a sufficiently small stable C (and adjust the final step to land exactly on T). Then the global temporal error is $O(\Delta t^4) = O(h^8)$, below the expected $O(h^4)$ spatial error. Check this separation by repeating with $\Delta t/2$ on representative grids. For instance, compute $E_N = \max_i |\phi_i^{\mathrm{num}}(T) - \phi_i(T)|$ and an analogous CV error. In the asymptotic range,

$$\boxed{E_N/E_{2N} \longrightarrow 16, \qquad p_{\mathrm{obs}} = \log_2(E_N/E_{2N}) \longrightarrow 4.}$$

Avoid grids so fine that roundoff controls the error. Exact source averaging and exact initialization prevent unrelated data errors from masking the spatial convergence rate.

Chapter 4
Coupled Fourier Analysis of Multi-moment method

"We are only as strong as we are united, as weak as we are divided."
Albus Dumbledore, in J. K. Rowling's Harry Potter and the Goblet of Fire

In Chapter 2, we found that coupling the point-value (PV) and cell-value (CV) representations enhances the modified-wavenumber transfer function of the multi-moment discretization. This result illustrates an important feature of the method: the two moments are not merely redundant descriptions of the same solution. Each carries information that complements the other, and their combination enables a more accurate representation of the underlying spatial variation. The resulting gain in accuracy is therefore a collective property of the multi-moment formulation. The PV approximation benefits from information contained in the cell-integrated representation, while the CV evolution is driven by flux information supplied by the point values. As the solution advances in time, this mutual dependence creates a continuous exchange of information between the two representations, shaping both the accuracy and the spectral behavior of the numerical solution.

In this chapter, we study this interaction through a coupled Fourier analysis. The objective is to move beyond the initial modified-wavenumber description and examine how the coupled moments evolve together. Particular attention is given to their influence on convection and diffusion, the resulting amplitude and wave-speed undulations, and the interaction between the spatial discretization and time integration. This coupled perspective provides a more complete explanation of why combining different moments can yield a numerical method whose accuracy is greater than that suggested by either representation considered independently.

4.1 Coupled Fourier analysis of the convection term

Recall the one-dimensional equation of the convection-diffusion equation in the multi-moment dual discretization:

$$\frac{\partial \phi_i}{\partial t} + u\left(\frac{\partial \phi}{\partial x}\right)_i = K\left(\frac{\partial^2 \phi}{\partial x^2}\right)_i, \text{ and} \tag{4.1}$$

$$|\Omega_{i-\frac{1}{2}}|\frac{\partial \overline{\phi}_{i-\frac{1}{2}}}{\partial t} + u\left(\phi_i - \phi_{i-1}\right) = K\left[\left(\frac{\partial \phi}{\partial x}\right)_i - \left(\frac{\partial \phi}{\partial x}\right)_{i-1}\right], \tag{4.2}$$

While the initial modified-wavenumber analysis (IMWA) correctly describes the spectral response at the initial condition, it does not capture the subsequent interaction between the finite-difference (FD) and finite-volume (FV) representations. In particular, errors generated in one equation can feed back into the other through the coupling between the point-value (PV) and cell-value (CV) degrees of freedom. The pointwise value, the first derivative, and the cell average can be expressed in Fourier space as:

$$\phi_i = \int_{-\infty}^{\infty} \widehat{\phi}_k e^{\mathbf{i}kx_i}\,\mathrm{d}k, \tag{4.3}$$

$$\phi_i' = \mathbf{i}k \int_{-\infty}^{\infty} \widehat{\phi}_k e^{\mathbf{i}kx_i}\,\mathrm{d}k, \tag{4.4}$$

$$\overline{\phi}_i = \frac{2}{kh}\sin\left(\frac{kh}{2}\right) \int_{-\infty}^{\infty} \widehat{\phi}_k e^{\mathbf{i}kx_i}\,\mathrm{d}k, \tag{4.5}$$

respectively.

The analysis in this section is restricted to the convective subsystem, i.e. $K = 0$, to isolate this coupling mechanism. To describe the dynamic interaction between the two representations, we regard the function represented by the pointwise data and the function underlying the cell-average data as distinct numerical fields. We denote the former by $\widetilde{\phi}$ and the latter by $\widehat{\phi}$. Following the procedure used in the preceding analysis, Eqs. (4.1) and (4.2) are written in Fourier space for the k-th mode as

$$\frac{\partial \widehat{\widetilde{\phi}_k}}{\partial t} + u\mathbf{i}k\left(\mathcal{B}_2\widehat{\widetilde{\phi}_k} + \mathcal{B}_1'\widehat{\widehat{\phi}_k}\right) = 0, \text{ and} \tag{4.6}$$

$$\frac{\partial \widehat{\widehat{\phi}_k}}{\partial t} + u\mathbf{i}\left[\frac{2}{h}\sin\left(\frac{kh}{2}\right)\right]\widehat{\widetilde{\phi}_k} = 0. \tag{4.7}$$

These equations form a coupled system of ordinary differential equations for the Fourier components associated with the PV and CV representations. The operators $\mathcal{B}_2$ and $\mathcal{B}_1'$ represent the contributions of the PV and CV data, respectively, to the approximation of the first derivative. Here, $\mathcal{B}_1'$ is an intermediate operator used in the construction of the final coupling operator $\mathcal{B}_1$.

Using the relation in Eq. (4.5), the discrete cell-average representation can be expressed in terms of the corresponding continuous Fourier mode. Combining the resulting averaging factor with $\mathcal{B}_1'$ gives $\mathcal{B}_1$. The coupled system can then be written as

$$\frac{\partial}{\partial t}\begin{bmatrix} \widehat{\widetilde{\phi}_k} \\ \widehat{\widehat{\phi}_k} \end{bmatrix} = -uk\mathbf{i}\begin{bmatrix} \mathcal{B}_2 & \mathcal{B}_1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} \widehat{\widetilde{\phi}_k} \\ \widehat{\widehat{\phi}_k} \end{bmatrix}. \tag{4.8}$$

Unlike the IMWA, each equation now contains the Fourier component of the other numerical representation. Owing to this explicit coupling, we refer to the resulting analysis as the Coupled Modified Wavenumber Analysis (CMWA).

For an exactly initialized solution, the two underlying representations coincide at the initial time. The operators $\mathcal{B}_1$ and $\mathcal{B}_2$ then recover the transfer function of the initial modified-wavenumber analysis,

$$T_D(kh) = \mathcal{B}_1(kh) + \mathcal{B}_2(kh), \qquad k^*(kh) = k\,T_D(kh).$$

The eigenvalues of the coupled Fourier matrix are obtained from its characteristic equation. Let $\mathcal{F}$ denote the square root of the discriminant of the characteristic polynomial,

$$\mathcal{F} = \sqrt{\mathcal{B}_2^2 + 4\mathcal{B}_1},$$

the two eigenvalues are

$$T_1 = \frac{1}{2}(\mathcal{B}_2 + \mathcal{F}), \qquad T_2 = \frac{1}{2}(\mathcal{B}_2 - \mathcal{F}).$$

Furthermore,

$$\mathcal{F} = T_1 - T_2,$$

so that $\mathcal{F}$ also measures the separation between the two coupled eigenmodes. Next, we define

$$C_1 = \frac{1 - T_2}{\mathcal{F}}, \qquad C_2 = \frac{T_1 - 1}{\mathcal{F}},$$

The first form of the solution can be written as :

$$\widehat{\widetilde{\phi}}_k(t) = \widehat{\widetilde{\phi}}_k(0)\left[C_1 T_1 e^{-\mathbf{i}uktT_1} + C_2 T_2 e^{-\mathbf{i}uktT_2}\right], \tag{4.9}$$

$$\widehat{\widehat{\phi}}_k(t) = \widehat{\widehat{\phi}}_k(0)\left[C_1 e^{-\mathbf{i}uktT_1} + C_2 e^{-\mathbf{i}uktT_2}\right]. \tag{4.10}$$

This representation follows directly from the eigendecomposition of the coupled system and can be verified by direct substitution. It shows that a single initial Fourier mode of wavenumber k excites two eigenmodes. The first is the dominant physical mode, which propagates in the original direction. For the schemes considered here, $T_2 < 0$ over the relevant wavenumber range, so the second eigenmode propagates in the opposite direction and may therefore be interpreted as a parasitic mode. As $kh \to 0$, the amplitude of the dominant mode approaches unity, whereas the amplitude of the parasitic mode vanishes at the same asymptotic rate as the leading truncation error.

An equivalent interpretation can be obtained by defining the time-dependent modulation factors

$$\widetilde{\mathcal{A}} = C_1 T_1 + C_2 T_2 e^{ukt\mathbf{i}\mathcal{F}}, \qquad \widehat{\mathcal{A}} = C_1 + C_2 e^{ukt\mathbf{i}\mathcal{F}}.$$

The solutions can then be rearranged as

$$\widehat{\widetilde{\phi}}_k(t) = \widehat{\widetilde{\phi}}_k(0)\widetilde{\mathcal{A}} e^{-\mathbf{i}uktT_1}, \tag{4.11}$$

$$\widehat{\widehat{\phi}}_k(t) = \widehat{\widehat{\phi}}_k(0)\widehat{\mathcal{A}} e^{-\mathbf{i}uktT_1}. \tag{4.12}$$

In this form, the solution can be interpreted as a dominant wave propagating with the modified phase-speed transfer function T_1, while its complex amplitude varies periodically in time. This interpretation differs substantially from that provided by the IMW analysis. First, the effective phase-speed transfer function is T_1 rather than T_D. Second, the PV and CV amplitudes are multiplied by the modulation factors $\widetilde{\mathcal{A}}$ and $\widehat{\mathcal{A}}$, respectively. Third, each modulation factor contains a mean component and an oscillatory component whose relative phase is governed by $\mathcal{F} = T_1 - T_2$, with a corresponding modulation angular frequency of $uk\mathcal{F}$. The oscillatory component periodically drives the amplitudes back toward unity. For the schemes considered here, T_1, C_1T_1, and C_1 approach unity at the same asymptotic rate as T_D, whereas C_2 and C_2T_2 approach zero. The quantities C_1T_1 and C_1 may therefore be interpreted as the *mean modified amplitudes* of the PV and CV moments, respectively. The analysis thus reveals that the interaction between different moments produces a coupled exchange that affects both the amplitude and the effective propagation speed of the numerical solution.

Furthermore, if we compare the wave speeds predicted by both analyses and define the ratio

$$R = \frac{k^*_{\mathrm{CMW}}(kh)}{k^*_{\mathrm{IMW}}(kh)} = \frac{\mathcal{B}_2 + \sqrt{\mathcal{B}_2^2 + 4\mathcal{B}_1}}{2\,(\mathcal{B}_2 + \mathcal{B}_1)}. \tag{4.13}$$

It follows directly that

$$R \geq 1 \qquad \text{if} \qquad \mathcal{B}_2 + \mathcal{B}_1 \leq 1.$$

Hence, in the absence of overshoot by the multi-moment (MM) scheme, the initial modified wavenumber provides a lower bound for the modified wavenumber for the point value.

At first sight, the resulting amplitude undulation may appear to indicate a deficiency in the numerical method or its implementation. Indeed, this was our initial interpretation when the behavior was first observed in the numerical experiments. The coupled modified wave number analysis (CMWA) instead shows that, at the semidiscrete level considered here, the modulation is bounded and oscillatory and is an intrinsic consequence of the coupling between the PV and CV representations. When one representation is used in the evolution equation of the other, information is transferred between their distinct numerical representation spaces. This coupling provides the mechanism through which the two sets of degrees of freedom exchange information and reinforce one another.

To the best of our knowledge, comparable amplitude undulations have not previously been reported for multi-moment methods. The time-evolution conversion (TEC) used in other formulations may modify or suppress this behavior. Imai [14] performed a related coupled Fourier analysis for the IDO scheme using a different set of degrees of freedom (ϕ and ϕ'), but did not report a corresponding oscillatory modulation. We were likewise unable to identify a comparable analysis for the IDO-CF scheme in the literature. A likely reason why these amplitude undulations have not been reported previously is their small magnitude. The oscillatory component scales with the truncation error and can therefore be difficult to distinguish without a sensitive L_2-norm analysis. We already plot a comparison of the modified wavenumbers predicted by the IMWA and CMWA in Fig. 2.2 back in chapter 2. The value predicted by the CMWA, represented by T_1, is consistently higher than that predicted by the IMWA through T_D.

The present CMWA deliberately considers only the convective subsystem ($K = 0$) to isolate the mechanism responsible for the PV-CV amplitude exchange. Extending the analysis to the pure diffusion equation or to the complete convection–diffusion equation introduces a second coupled operator associated with the multi-moment approximation of the second derivative. The resulting eigenvalues generally contain both imaginary convective and real dissipative contributions. A complete characterization must therefore account simultaneously for moment coupling, numerical diffusion, and the stability region of the selected time-integration scheme. Such an analysis is beyond the scope of the present work and will be addressed separately.

4.2 Long-time integration of a sine wave

Let us return to the periodic convection of the sine wave considered in Fig. 2.14. The purpose of the present test is not only to examine the conventional phase and amplitude errors over a few periods, but also to investigate how the numerical solution behaves after a very large number of time steps.

For the one-dimensional constant-coefficient advection equation,

$$\frac{\partial \phi}{\partial t} + a\frac{\partial \phi}{\partial x} = 0, \tag{4.14}$$

subject to periodic boundary conditions, the continuous L_2-norm is conserved. Multiplying Eq. (4.14) by ϕ and integrating over the periodic domain gives

$$\frac{1}{2}\frac{d}{dt}\int_\Omega \phi^2\,dx = -\frac{a}{2}\int_\Omega \frac{\partial \phi^2}{\partial x}\,dx = 0. \tag{4.15}$$

Hence,

$$L_2(t) = L_2(0) \tag{4.16}$$

for the exact solution. Therefore, the normalized quantity

$$\mathcal{N}(t) = \frac{L_2(t)}{L_2(0)} \tag{4.17}$$

should remain identically equal to unity in the absence of numerical dissipation or other discrete changes of amplitude.

The long-time evolution of $\mathcal{N}(t)$ is shown in Fig. 4.1. The numerical solution is advanced to $t = 1200$ with a fourth-order Runge–Kutta method using

$$\Delta t \in \{0.2,\ 0.1,\ 0.05\}. \tag{4.18}$$

The spatial resolution is fixed at four cells per wavelength,

$$\text{4CPW}, \qquad \Delta x = 0.5, \tag{4.19}$$

so that changes produced by varying Δt can be attributed primarily to the temporal discretization.

Panel (a) compares the relative norms obtained with the EX8 PV and CV schemes and the EX4 CV scheme. The EX4 PV result is omitted for visual clarity. Two distinct features can be identified in the numerical solution:

1. a slowly varying decay of the mean amplitude over long times, and
2. a much faster oscillatory variation, or *undulation*, superimposed on this slowly varying envelope.

These two effects have different numerical origins and should therefore be distinguished.

4.2.1 Long-time decay produced by the time integrator

Consider first the slowly varying envelope. For a single Fourier mode, the semi-discrete spatial approximation may be written in the form

$$\frac{d\widehat{\boldsymbol{\phi}}}{dt} = \mathbf{A}(k)\widehat{\boldsymbol{\phi}}, \tag{4.20}$$

where $\mathbf{A}(k)$ is the Fourier symbol of the multi-moment spatial operator.

If $\lambda_m(k)$ denotes an eigenvalue of $\mathbf{A}(k)$, application of the classical fourth-order Runge–Kutta method gives the amplification factor

$$G_m = R_4\left(\lambda_m \Delta t\right), \tag{4.21}$$

where

$$R_4(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}. \tag{4.22}$$

For the exact time evolution of a nondissipative Fourier mode, the corresponding amplification factor has unit magnitude. The RK4 polynomial, however, reproduces the exact exponential only to fourth order. Consequently,

$$|G_m| \neq 1 \tag{4.23}$$

in general, even when the underlying semi-discrete spatial mode is nondissipative.

A very small deviation of $|G_m|$ from unity may appear insignificant over one time step, but the effect accumulates after many thousands of steps. If a mode is advanced through N time steps, its amplitude contains the factor

$$|G_m|^N. \tag{4.24}$$

Thus, a small dissipative error in the time integrator can produce a substantial reduction of the norm when the integration time is very long. This explains the strong dependence of the envelope in Fig. 4.1(a) on Δt. At $\Delta t = 0.2$ the accumulated decay is substantial, whereas reducing the time step to $\Delta t = 0.1$ and then to $\Delta t = 0.05$ progressively moves the long-time norm closer to unity.

To quantify this temporal convergence, define the final norm defect as

$$D(\Delta t) = 1 - \frac{L_2(T;\Delta t)}{L_2(0)}, \qquad T = 1200. \tag{4.25}$$

For two calculations whose time steps differ by a factor of two, an observed convergence order may be estimated from

$$p = \frac{\log\left[D(\Delta t)/D(\Delta t/2)\right]}{\log 2}. \tag{4.26}$$

Using the two smallest time steps gives an observed order of approximately

$$p = 4.26, \tag{4.27}$$

which is consistent with the fourth-order accuracy of the Runge–Kutta time integration.

The long-time envelope should therefore not be interpreted primarily as spatial dissipation of the MM scheme. Its strong convergence with Δt indicates that, for this experiment, the accumulated decay is dominated by the temporal discretization.

In contrast to the slowly varying decay, the frequency and amplitude of the rapid norm undulations change little under time-step refinement, indicating that these oscillations are principally associated with the spatial multi-moment discretization.

4.2.2 Amplitude and phase-speed undulations

The coupled modified-wavenumber analysis developed above predicts two closely related consequences of the interaction between the point-value (PV) and cell-value (CV) representations. First, the amplitudes of the two moments undergo a bounded periodic modulation. Second, the phase of each moment contains a time-dependent contribution in addition to the

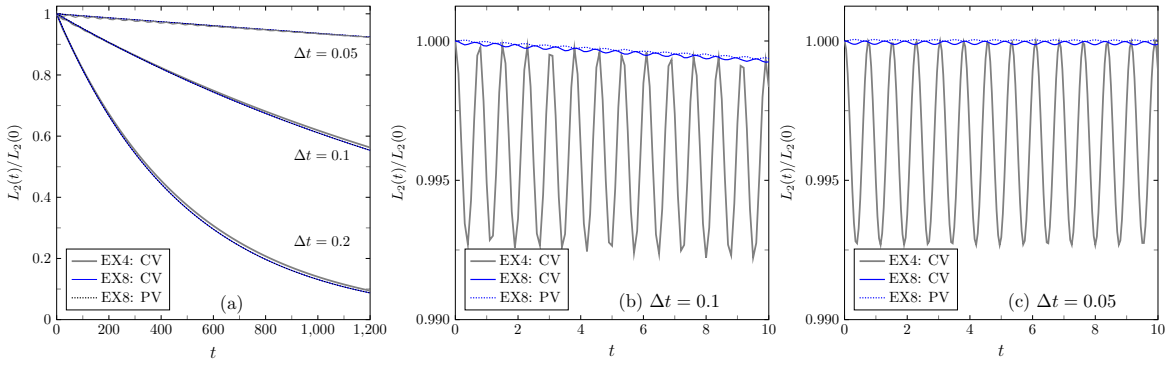


Fig. 4.1: Evolution of the relative discrete norm $L_2(t)/L_2(0)$ for the periodic advection of $\sin(\pi x)$ using fourth-order Runge–Kutta time integration at four cells per wavelength (4CPW, $\Delta x = 0.5$). Panel (a) shows the long-time evolution to $t = 1200$ for $\Delta t \in \{0.05, 0.1, 0.2\}$. The progressive reduction of the long-time norm defect under time-step refinement is consistent with the fourth-order temporal discretization. Panels (b) and (c) show the short-time behavior over $0 \le t \le 10$ for $\Delta t = 0.1$ and $\Delta t = 0.05$, respectively.

dominant phase propagation governed by T_1. The present section examines these predictions directly in the numerical solution.

To resolve the oscillations more clearly, the computational domain is extended to 8π, the initial condition is chosen as

$$\phi(x, 0) = \sin(x), \tag{4.28}$$

and a sufficiently small time step is employed so that temporal errors are small compared with the spatial effects of interest.

4.2.2.1 Measurement of the apparent amplitude

We first define an effective amplification factor from the discrete L_2-norm,

$$G_{\mathrm{ap}}(t) = \frac{\|\boldsymbol{\phi}_{\mathrm{sol}}(t)\|_2}{\|\boldsymbol{\phi}_{\mathrm{sol}}(0)\|_2}. \tag{4.29}$$

For a pure sinusoidal wave whose shape is preserved, this quantity is identical to the amplitude ratio. In a numerical calculation, small phase and shape errors may also be present, and G_{ap} should therefore be interpreted more generally as an *effective numerical amplitude*.

The CMWA provides a direct prediction for this behavior. Recall that the PV and CV Fourier components may be written as

$$\widetilde{\widehat{\phi}}_k(t) = \widetilde{\widehat{\phi}}_k(0)\widetilde{\mathcal{A}}(t)e^{-\mathrm{i}uktT_1}, \tag{4.30}$$

$$\widehat{\widehat{\phi}}_k(t) = \widehat{\widehat{\phi}}_k(0)\widehat{\mathcal{A}}(t)e^{-\mathrm{i}uktT_1}, \tag{4.31}$$

where

$$\widetilde{\mathcal{A}}(t) = C_1T_1 + C_2T_2e^{\mathrm{i}uktF}, \tag{4.32}$$

$$\widehat{\mathcal{A}}(t) = C_1 + C_2e^{\mathrm{i}uktF}, \tag{4.33}$$

and

$$\mathcal{F} = T_1 - T_2. \tag{4.34}$$

For the nondissipative convection problem considered here, the corresponding CMWA amplitude predictions are therefore

$$G_{\mathrm{PV}}^{\mathrm{CMWA}}(t) = \left|\widetilde{\mathcal{A}}(t)\right|, \tag{4.35}$$

$$G_{\mathrm{CV}}^{\mathrm{CMWA}}(t) = \left|\widehat{\mathcal{A}}(t)\right|. \tag{4.36}$$

When T_1, T_2, C_1, and C_2 are real, these relations may be written explicitly as

$$\left(G_{\mathrm{PV}}^{\mathrm{CMWA}}\right)^2 = (C_1T_1)^2 + (C_2T_2)^2 + 2C_1C_2T_1T_2\cos(ukt\mathcal{F}), \tag{4.37}$$

$$\left(G_{\mathrm{CV}}^{\mathrm{CMWA}}\right)^2 = C_1^2 + C_2^2 + 2C_1C_2\cos(ukt\mathcal{F}). \tag{4.38}$$

Equations (4.37) and (4.38) make the origin of the amplitude undulation particularly clear. The oscillation is generated by the PV–CV coupling and has angular frequency

$$\boxed{\omega_{\mathrm{und}} = uk\mathcal{F} = uk(T_1 - T_2)}. \tag{4.39}$$

The corresponding period is

$$\boxed{\tau_{\mathrm{und}} = \frac{2\pi}{uk\mathcal{F}}}. \tag{4.40}$$

Thus, both the magnitude and the period of the amplitude modulation are determined by the spatial MM discretization through the quantities C_1, C_2, T_1, and T_2.

4.2.2.2 Measurement of the apparent phase speed

To quantify the propagation error, the numerical solution is compared with a translated and amplitude-scaled exact wave. At time t_n, define the search interval

$$\mathcal{I}_n = \left[\overline{u}_{\mathrm{ap}}(t_{n-1}) - \delta,\ \overline{u}_{\mathrm{ap}}(t_{n-1}) + \delta\right], \tag{4.41}$$

where δ is chosen sufficiently small to follow continuously the same phase branch. The apparent phase speed is then defined by

$$\boxed{\overline{U}_{\mathrm{ap}}(t_n) = \underset{c\in\mathcal{I}_n}{\arg\min}\left\|G_{\mathrm{ap}}(t_n)\phi_{\mathrm{exact}}\left(\boldsymbol{x} - ct_n\right) - \boldsymbol{\phi}_{\mathrm{sol}}(t_n)\right\|_2.} \tag{4.42}$$

This definition measures the velocity associated with the best-fit translation accumulated between the initial time and t_n. It should therefore be interpreted as an *apparent* or *cumulative phase speed*, rather than a strictly instantaneous characteristic velocity.

The optimization-based definition is convenient because the same physical-space measure can be applied to both the PV and CV solutions, while allowing for the small amplitude and waveform errors produced by the numerical discretization.

4.2.2.3 CMWA interpretation of the phase-speed undulation

The phase-speed undulation can be derived directly from the complex phase of the CMWA modulation factor. Consider a numerical Fourier component written in the generic form

$$\widehat{\phi}_k(t) = \widehat{\phi}_k(0)\mathcal{A}(t)e^{-\mathbf{i}ukT_1}, \tag{4.43}$$

where T_1 represents the dominant propagation branch of the coupled system, while $\mathcal{A}(t)$ describes the additional time-dependent modulation generated by the interaction between the coupled moments.

In general, $\mathcal{A}(t)$ is a complex quantity. It may therefore be written in polar form as

$$\mathcal{A}(t) = |\mathcal{A}(t)|e^{\mathbf{i}\theta_{\mathcal{A}}(t)}, \qquad \theta_{\mathcal{A}}(t) = \arg\left[\mathcal{A}(t)\right]. \tag{4.44}$$

Here, $\theta_{\mathcal{A}}(t)$ denotes the phase of the modulation factor, with the argument followed continuously in time. In other words, if the conventional principal value of $\arg[\mathcal{A}(t)]$ undergoes an artificial jump of 2π, the phase branch is continued so that $\theta_{\mathcal{A}}(t)$ remains continuous. For the exactly initialized state, the phase is chosen such that

$$\theta_{\mathcal{A}}(0) = 0, \tag{4.45}$$

which is consistent with $\mathcal{A}(0) = 1$.

Substituting the polar representation of $\mathcal{A}(t)$ into Eq. (4.43) gives

$$\widehat{\phi}_k(t) = \widehat{\phi}_k(0)|\mathcal{A}(t)| \exp\left[-\mathbf{i}ukT_1 + \mathbf{i}\theta_{\mathcal{A}}(t)\right]. \tag{4.46}$$

This expression separates the effect of the modulation factor into an amplitude contribution and a phase contribution. The magnitude $|\mathcal{A}(t)|$ controls the amplitude of the numerical Fourier component, whereas $\theta_{\mathcal{A}}(t)$ provides an additional phase correction.

It is important to distinguish $\theta_{\mathcal{A}}(t)$ from ukT_1. The quantity ukT_1 represents the dominant propagation phase associated with the principal coupled eigenmode, while $\theta_{\mathcal{A}}(t)$ arises solely from the phase of the modulation factor.

To determine how this additional phase affects the propagation velocity, the spatial dependence of the Fourier mode must also be included. The complete numerical mode is proportional to

$$\phi_k(x,t) \propto |\mathcal{A}(t)| \exp\left(\mathbf{i}\left[kx - ukT_1 + \theta_{\mathcal{A}}(t)\right]\right). \tag{4.47}$$

The total phase of the numerical wave is therefore

$$\Psi(x,t) = kx - ukT_1 + \theta_{\mathcal{A}}(t). \tag{4.48}$$

The instantaneous phase speed is obtained by following a point of constant phase. Along such a trajectory,

$$\frac{d\Psi}{dt} = 0. \tag{4.49}$$

Differentiating Eq. (4.48) with respect to time gives

$$k\frac{dx}{dt} - ukT_1 + \frac{d\theta_{\mathcal{A}}}{dt} = 0. \tag{4.50}$$

Identifying the velocity of the constant-phase point as

$$u_{\mathrm{ph}}(t) = \frac{dx}{dt}, \tag{4.51}$$

we obtain

$$u_{\mathrm{ph}}(t) = uT_1 - \frac{1}{k}\frac{d\theta_{\mathcal{A}}}{dt}. \tag{4.52}$$

After normalization by the exact advection velocity u, the instantaneous phase-speed transfer function becomes

$$\boxed{\frac{u_{\mathrm{ph}}(t)}{u} = T_1 - \frac{1}{uk}\frac{d\theta_{\mathcal{A}}}{dt}}. \tag{4.53}$$

Equation (4.53) shows that the numerical phase speed consists of two contributions. The first contribution, T_1, represents the dominant propagation speed of the coupled eigenmode. The second contribution,

$$-\frac{1}{uk}\frac{d\theta_{\mathcal{A}}}{dt}, \tag{4.54}$$

is a time-dependent correction generated by the changing phase of the modulation factor. Consequently, the same PV–CV coupling that produces the amplitude undulation also produces an oscillatory correction to the instantaneous phase speed.

4.2.2.4 Amplitude behavior

Panels (a)–(c) of Fig. 4.2 compare the apparent amplitudes of the PV and CV moments with the predictions of the CMWA. The numerical oscillations closely follow the modulation described by Eqs. (4.35) and (4.36).

The quantities

$$C_1T_1 \qquad \text{and} \qquad C_1 \tag{4.55}$$

are the nonoscillatory components of the PV and CV modulation factors, respectively. Since the oscillatory terms C_2T_2 and C_2 become small as the grid is refined, these quantities also provide the leading approximations to the center levels of the observed amplitude undulations.

The numerical results show approximately fourth-order convergence of these center levels. The measured convergence rates are approximately 4.6 for the PV field and 3.8 for the CV field, consistent with the expected fourth-order behavior of EX4. At four cells per wavelength, the amplitude error remains below approximately 1% in the present test.

As the grid is coarsened, the oscillatory contribution becomes more pronounced and the undulation period changes. This behavior is also consistent with the CMWA because both the oscillation amplitude and the frequency

$$\omega_{\mathrm{und}} = uk\mathcal{F}(kh) \tag{4.56}$$

depend explicitly on the nondimensional wavenumber kh.

4.2.2.5 Evolution of the phase speed

Panels (d)–(f) of Fig. 4.2 show the corresponding evolution of the apparent phase speed.

At the initial time, the CV equation receives its flux directly from the exactly initialized PV field. Its initial phase speed is therefore the exact value,

$$\frac{u_{\mathrm{CV}}(0)}{u} = 1. \tag{4.57}$$

The PV derivative approximation, in contrast, initially sees the two representations in their exactly consistent state. The initial modified-wavenumber therefore describes its initial phase speed transfer function,

$$\frac{u_{\mathrm{PV}}(0)}{u} = T_D = \mathcal{B}_1 + \mathcal{B}_2. \tag{4.58}$$

Once the calculation begins, however, the PV and CV representations evolve through the coupled system and no longer behave as independent approximations. The error generated in one representation enters the evolution equation of the other. Consequently, the initial description provided by T_D is no longer sufficient.

The CMWA predicts that the dominant coupled propagation branch is

$$T_1 = \frac{1}{2}\left(\mathcal{B}_2 + \mathcal{F}\right). \tag{4.59}$$

The numerical results show that the apparent phase speeds of both moments evolve toward this value while undergoing small oscillatory departures from it. The PV and CV variations are approximately out of phase, reflecting the reciprocal nature of their coupled evolution.

The two modified-wavenumber analyses therefore describe different stages of the numerical dynamics:

$$\boxed{\begin{array}{ll} T_D & : \quad \text{initial response of the exactly consistent PV–CV state,} \\ T_1 & : \quad \text{dominant phase-speed branch of the coupled evolution.} \end{array}} \tag{4.60}$$

In this sense, the IMWA and CMWA are complementary rather than contradictory. The IMWA describes the initial spatial response before the two numerical representations have developed different errors, whereas the CMWA describes their subsequent coupled evolution.

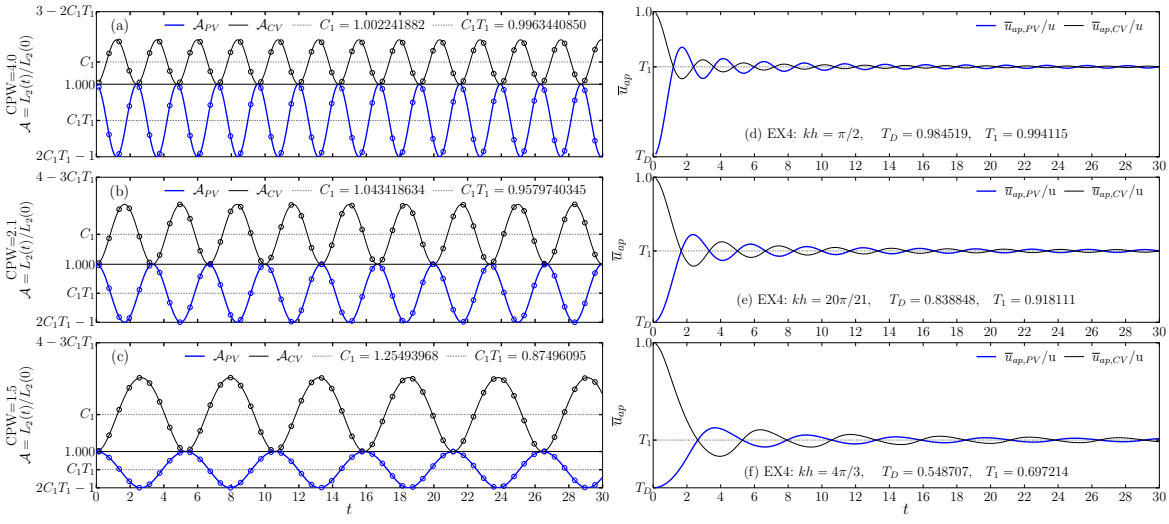


Fig. 4.2: Amplitude and apparent phase-speed undulations for the fourth-order multi-moment method applied to one-dimensional periodic advection. Panels (a)–(c) show the amplitude behavior, while panels (d)–(f) show the corresponding apparent phase speed at progressively different spatial resolutions. The numerical results illustrate the bounded modulation predicted by the coupled modified-wavenumber analysis and the approach of the phase speed toward the dominant transfer function T_1.

4.2.2.6 Boundedness of the undulation

No instability is observed over the time interval considered. Consistent with the CMWA, the amplitude modulation remains bounded, and its magnitude decreases as the spatial resolution is increased.

For the periodic linear-convection problem considered here, the semidiscrete coupled operator has a purely imaginary spectrum. Hence, the CMWA contains no exponentially growing eigenmode. The observed amplitude variation should therefore not be interpreted as a numerical instability. Instead, it is an intrinsic consequence of the coupling between the PV and CV representations.

This distinction is important. A numerical instability is characterized by an error component whose amplitude grows without bound. The present behavior is qualitatively different:

$$\boxed{\text{PV–CV coupling} \quad \Longrightarrow \quad \text{bounded periodic modulation,}} \tag{4.61}$$

with an oscillatory contribution that decreases with spatial refinement.

The modulation is of the same asymptotic order as the spatial truncation error and can therefore be difficult to detect from conventional solution plots. The L_2-based amplitude diagnostic used here provides a more sensitive measure of this effect.

4.2.2.7 Relation to discrete conservation

The bounded amplitude modulation also raises a broader question concerning the algebraic structure of the multi-moment spatial operator. Conservative flux differencing guarantees preservation of the scalar integral under periodic boundary conditions. Exact preservation of a discrete quadratic quantity such as the L_2 norm, however, is commonly associated with a spatial operator that is skew-symmetric, or skew-adjoint with respect to an appropriate discrete inner product [28, 34].

The present MM operator is not strictly skew-symmetric in the conventional Euclidean inner product, even though its spectrum is purely imaginary for the periodic linear-convection problem. The coexistence of conservation, bounded PV–CV amplitude exchange, and a purely imaginary spectrum therefore suggests a more general question: whether an alternative weighted inner product exists under which the coupled MM operator becomes skew-adjoint.

This question is particularly important on non-uniform grids, where the point-value and cell-value moments naturally carry different geometric weights. Identifying the appropriate algebraic structure would provide a more complete explanation of how a multi-moment discretization can simultaneously preserve the conservative integral and control the quadratic norm.

4.3 Coupled Fourier Analysis of the Diffusion Equation

We now extend the coupled Fourier analysis to the diffusion operator. In contrast to the convection problem, for which the eigenvalues of the semidiscrete system produce modified propagation speeds, diffusion acts through real decay rates. The coupling between the point-value (PV) and cell-value (CV) representations nevertheless remains present because the second derivative appearing in the PV equation and the diffusive flux appearing in the CV equation are both reconstructed from the coupled multi-moment representation.

To isolate this mechanism, we consider the pure diffusion subsystem by setting $u = 0$ in Eqs. (4.1) and (4.2). The governing equations then become

$$\frac{\partial \phi_i}{\partial t} = K \left(\frac{\partial^2 \phi}{\partial x^2} \right)_i , \tag{4.62}$$

$$|\Omega_{i-\frac{1}{2}}| \frac{\partial \overline{\phi}_{i-\frac{1}{2}}}{\partial t} = K \left[\left(\frac{\partial \phi}{\partial x} \right)_i - \left(\frac{\partial \phi}{\partial x} \right)_{i-1} \right] . \tag{4.63}$$

For the uniform grid considered here, $|\Omega_{i-\frac{1}{2}}| = h$.

As in the convection analysis, the functions represented by the PV and CV degrees of freedom are regarded as distinct numerical fields and are denoted by $\widetilde{\phi}$ and $\widehat{\phi}$, respectively. The pointwise approximation of the second derivative can be represented in Fourier space in the general form

$$\widehat{\left(\frac{\partial^2 \phi}{\partial x^2} \right)}_k = -k^2 \left(\mathcal{D}_2 \widetilde{\widehat{\phi}}_k + \mathcal{D}_1 \widehat{\widehat{\phi}}_k \right) , \tag{4.64}$$

where $\mathcal{D}_2$ and $\mathcal{D}_1$ denote the contributions of the PV and CV representations, respectively, to the approximation of the second derivative. As with $\mathcal{B}_1$ in the convection analysis, the cell-averaging factor in Eq. (4.5) is understood to have been incorporated into $\mathcal{D}_1$.

The PV diffusion equation therefore becomes

$$\frac{\partial \widetilde{\widehat{\phi}}_k}{\partial t} = -Kk^2 \left(\mathcal{D}_2 \widetilde{\widehat{\phi}}_k + \mathcal{D}_1 \widehat{\widehat{\phi}}_k \right) . \tag{4.65}$$

The Fourier representation of the CV equation can be obtained from the first-derivative operator already introduced for the convection problem. At a point x_i, the numerical gradient may be written as

$$\widehat{\phi'_k} = \mathbf{i}k \left(\mathcal{B}_2 \widetilde{\widehat{\phi}}_k + \mathcal{B}_1 \widehat{\widehat{\phi}}_k \right) . \tag{4.66}$$

Consequently, the difference of the two gradients bounding the control volume is

$$\begin{aligned} (\phi_x)_i - (\phi_x)_{i-1} &= \mathbf{i}k \left(\mathcal{B}_2 \widetilde{\widehat{\phi}}_k + \mathcal{B}_1 \widehat{\widehat{\phi}}_k \right) \times \left[e^{\mathbf{i}kh/2} - e^{-\mathbf{i}kh/2} \right] e^{\mathbf{i}kx_{i-\frac{1}{2}}} \\ &= -2k \sin \left(\frac{kh}{2} \right) \left(\mathcal{B}_2 \widetilde{\widehat{\phi}}_k + \mathcal{B}_1 \widehat{\widehat{\phi}}_k \right) e^{\mathbf{i}kx_{i-\frac{1}{2}}} . \end{aligned}$$

Using the cell-averaging relation

$$\widetilde{\widehat{\phi}}_k = \frac{2}{kh} \sin \left(\frac{kh}{2} \right) \widehat{\widehat{\phi}}_k , \tag{4.67}$$

the geometric factor associated with the finite-volume difference cancels the Fourier cell-averaging factor. Equation (4.63) therefore reduces to

$$\frac{\partial \widehat{\widehat{\phi}}_k}{\partial t} = -Kk^2 \left(\mathcal{B}_2 \widehat{\widetilde{\phi}}_k + \mathcal{B}_1 \widehat{\widehat{\phi}}_k \right) . \tag{4.68}$$

Equations (4.65) and (4.68) form the coupled Fourier system for the pure diffusion equation:

$$\frac{\partial}{\partial t} \begin{bmatrix} \widetilde{\widehat{\phi}}_k \\ \widehat{\widehat{\phi}}_k \end{bmatrix} = -Kk^2 \begin{bmatrix} \mathcal{D}_2 & \mathcal{D}_1 \\ \mathcal{B}_2 & \mathcal{B}_1 \end{bmatrix} \begin{bmatrix} \widetilde{\widehat{\phi}}_k \\ \widehat{\widehat{\phi}}_k \end{bmatrix} . \tag{4.69}$$

For convenience, define the coupled diffusion matrix as

$$\mathcal{M}_K = \begin{bmatrix} \mathcal{D}_2 & \mathcal{D}_1 \\ \mathcal{B}_2 & \mathcal{B}_1 \end{bmatrix}. \tag{4.70}$$

The structure of Eq. (4.69) is analogous to that obtained for convection in Eq. (4.8). The important difference is that the convection matrix is multiplied by $-\mathbf{i}uk$, whereas the diffusion matrix is multiplied by the real factor $-Kk^2$. This distinction changes the physical interpretation of the coupled eigenmodes.

4.3.1 Eigenmodes of the coupled diffusion operator

Let λ denote an eigenvalue of $\mathcal{M}_K$. The characteristic equation is

$$\lambda^2 - (\mathcal{D}_2 + \mathcal{B}_1)\,\lambda + (\mathcal{D}_2\mathcal{B}_1 - \mathcal{D}_1\mathcal{B}_2) = 0. \tag{4.71}$$

Defining

$$\mathcal{G} = \sqrt{(\mathcal{D}_2 - \mathcal{B}_1)^2 + 4\mathcal{D}_1\mathcal{B}_2}, \tag{4.72}$$

the two eigenvalues can be rewritten as

$$\lambda_1 = \frac{1}{2}\left(\mathcal{D}_2 + \mathcal{B}_1 + \mathcal{G}\right), \tag{4.73}$$

$$\lambda_2 = \frac{1}{2}\left(\mathcal{D}_2 + \mathcal{B}_1 - \mathcal{G}\right). \tag{4.74}$$

Thus,

$$\mathcal{G} = \lambda_1 - \lambda_2, \tag{4.75}$$

in direct analogy with $\mathcal{F} = T_1 - T_2$ in the convection analysis.

The corresponding eigenvalues of the complete semidiscrete evolution operator are

$$\lambda_j = -Kk^2\lambda_j, \qquad j = 1, 2. \tag{4.76}$$

Hence, when λ_j is real and positive, the corresponding eigenmode decays exponentially. A negative value of λ_j, in contrast, would generate an exponentially growing mode and therefore indicate a semidiscrete diffusion instability.

The physical diffusion branch is identified as the eigenvalue λ_p that approaches the exact diffusion transfer function in the long-wave limit,

$$\lambda_\mathrm{p} \longrightarrow 1, \qquad kh \longrightarrow 0, \tag{4.77}$$

while the remaining eigenvalue is denoted by λ_s and represents the secondary coupled mode. This definition is preferable to identifying the physical branch solely with the positive or negative sign in Eq. (4.74), since the appropriate branch is determined by its asymptotic behavior.

For either eigenvalue, the ratio between the PV and CV components of its eigenvector can be written as

$$P_j = \frac{\lambda_j - \mathcal{B}_1}{\mathcal{B}_2} = \frac{\mathcal{D}_1}{\lambda_j - \mathcal{D}_2}, \qquad j \in \mathrm{p}, \mathrm{s}, \tag{4.78}$$

whenever the corresponding denominators are nonzero. An eigenvector can therefore be represented as

$$\boldsymbol{v}_j = \begin{bmatrix} P_j & 1 \end{bmatrix}. \tag{4.79}$$

4.3.2 Evolution from an exactly initialized state

As in the convection analysis, the two underlying numerical representations coincide when the solution is initialized exactly:

$$\widehat{\widetilde{\phi}}_k(0) = \widehat{\widehat{\phi}}_k(0) = \widehat{\phi}_k(0). \tag{4.80}$$

The initial state can consequently be decomposed into the two coupled eigenvectors according to

$$\begin{bmatrix} 1 & 1 \end{bmatrix} = A_\mathrm{p} \begin{bmatrix} P_\mathrm{p} & 1 \end{bmatrix} + A_\mathrm{s} \begin{bmatrix} P_\mathrm{s} & 1 \end{bmatrix}, \tag{4.81}$$

where

$$A_\mathrm{p} = \frac{1 - P_\mathrm{s}}{P_\mathrm{p} - P_\mathrm{s}}, \qquad A_\mathrm{s} = \frac{P_\mathrm{p} - 1}{P_\mathrm{p} - P_\mathrm{s}}. \tag{4.82}$$

These coefficients satisfy

$$A_\mathrm{p} + A_\mathrm{s} = 1, \qquad A_\mathrm{p} P_\mathrm{p} + A_\mathrm{s} P_\mathrm{s} = 1, \tag{4.83}$$

and therefore recover the exactly consistent initial PV–CV state.

The Fourier components subsequently evolve as

$$\widehat{\widetilde{\phi}}_k(t) = \widehat{\phi}_k(0) \left[A_\mathrm{p} P_\mathrm{p} e^{-Kk^2 t \lambda_\mathrm{p}} + A_\mathrm{s} P_\mathrm{s} e^{-Kk^2 t \lambda_\mathrm{s}} \right], \tag{4.84}$$

$$\widehat{\widehat{\phi}}_k(t) = \widehat{\phi}_k(0) \left[A - \mathrm{p} e^{-Kk^2 t \lambda_\mathrm{p}} + A_\mathrm{s} e^{-Kk^2 t \lambda_\mathrm{s}} \right]. \tag{4.85}$$

Thus, a single Fourier mode in the physical initial condition generally excites two numerical diffusion modes. Both modes contain the same spatial wavenumber k, but they decay at different rates determined by λ_p and λ_s.

An alternative form, directly analogous to Eqs. (4.11) and (4.12), is obtained by factoring out the physical diffusion branch:

$$\widehat{\widetilde{\phi}}_k(t) = \widehat{\phi}_k(0) \widetilde{\mathcal{A}}_K(t) e^{-Kk^2 t \lambda_\mathrm{p}}, \tag{4.86}$$

$$\widehat{\widehat{\phi}}_k(t) = \widehat{\phi}_k(0) \widehat{\mathcal{A}}_K(t) e^{-Kk^2 t \lambda_\mathrm{p}}, \tag{4.87}$$

where

$$\widetilde{\mathcal{A}}_K(t) = A_\mathrm{p} P_\mathrm{p} + A_\mathrm{s} P_\mathrm{s} e^{-Kk^2 t (\lambda_\mathrm{s} - \lambda_\mathrm{p})}, \tag{4.88}$$

$$\widehat{\mathcal{A}}_K(t) = A_\mathrm{p} + A_\mathrm{s} e^{-Kk^2 t (\lambda_\mathrm{s} - \lambda_\mathrm{p})}. \tag{4.89}$$

At the initial time,

$$\widetilde{\mathcal{A}}_K(0) = \widehat{\mathcal{A}}_K(0) = 1, \tag{4.90}$$

as required.

Equations (4.88) and (4.89) reveal an important distinction between the coupled convection and diffusion operators. In the convection problem, the relative contribution of the second eigenmode contains the factor

$$e^{\mathrm{i} u k t (T_1 - T_2)}, \tag{4.91}$$

which produces the persistent amplitude and phase-speed undulations described previously. For diffusion, the corresponding factor is

$$e^{-Kk^2t(\lambda_s-\lambda_p)}. \tag{4.92}$$

When the two diffusion branches are real, and the secondary branch is more strongly damped,

$$\lambda_s > \lambda_p > 0. \tag{4.93}$$

This contribution decreases exponentially rather than oscillating. Consequently, the PV–CV coupling produces a transient redistribution between the two moments rather than a persistent periodic undulation.

The characteristic decay rate of this coupling transient is

$$\boxed{\sigma_{cpl} = Kk^2\left(\lambda_s - \lambda_p\right)}, \tag{4.94}$$

with the associated relaxation time

$$\boxed{\tau_{cpl} = \frac{1}{Kk^2\left(\lambda_s - \lambda_p\right)}} \tag{4.95}$$

when the eigenvalues are real and $\lambda_s > \lambda_p$. Thus, the separation between the two coupled eigenvalues again controls the exchange between the moments. In convection this separation determines an undulation frequency; in diffusion it determines a relaxation rate.

In the long-time limit, the more strongly damped secondary mode disappears and

$$\widetilde{\mathcal{A}}_K \longrightarrow A_p P_p, \tag{4.96}$$

$$\widehat{\mathcal{A}}_K \longrightarrow A_p. \tag{4.97}$$

The numerical solution is then governed almost entirely by the physical coupled diffusion mode. Moreover, because $P_p \to 1$ as $kh \to 0$, the PV and CV representations recover the same underlying solution in the well-resolved limit.

4.3.3 Initial and coupled diffusion transfer functions

The exact solution of the continuous diffusion equation for a Fourier mode is

$$\widehat{\phi}_k(t) = \widehat{\phi}_k(0)e^{-Kk^2t}. \tag{4.98}$$

Hence, the exact nondimensional diffusion transfer function is unity.

At the initial time, when the PV and CV representations are exactly consistent, Eq. (4.65) gives

$$\left.\frac{\partial\widetilde{\widehat{\phi}}_k}{\partial t}\right|_{t=0} = -Kk^2\left(\mathcal{D}_1 + \mathcal{D}_2\right)\widehat{\phi}_k(0). \tag{4.99}$$

The initial diffusion transfer function seen by the PV equation is therefore

$$T^{IMW}_{K,PV} = \mathcal{D}_1 + \mathcal{D}_2. \tag{4.100}$$

Similarly, the CV equation initially gives

$$T^{IMW}_{K,CV} = \mathcal{B}_1 + \mathcal{B}_2 = T_D. \tag{4.101}$$

The appearance of T_D in Eq. (4.101) follows from the conservative form of the CV diffusion equation: the discrete divergence contributes one wavenumber factor exactly through the flux difference, whereas the remaining approximation is contained in the reconstructed first derivative.

After the two moments begin to evolve, however, neither initial transfer function by itself describes the complete numerical dynamics. The dominant coupled diffusion branch is instead governed by λ_{p}. We may therefore define the CMWA diffusion transfer function as

$$\boxed{T_K^{\mathrm{CMW}} = \lambda_{\mathrm{p}}} \tag{4.102}$$

or, equivalently, introduce a coupled modified diffusion wavenumber through

$$\boxed{\left(k^*_{K,\mathrm{CMW}}\right)^2 = k^2\lambda_{\mathrm{p}}.} \tag{4.103}$$

The exact diffusion rate is recovered when $\lambda_{\mathrm{p}} = 1$. Thus,

$$\lambda_{\mathrm{p}} < 1 \qquad \Longrightarrow \qquad \text{under-diffusion,} \tag{4.104}$$

whereas

$$\lambda_{\mathrm{p}} > 1 \qquad \Longrightarrow \qquad \text{over-diffusion.} \tag{4.105}$$

The ratio between the dominant numerical amplitude and the exact diffusive amplitude contains the factor

$$\exp\left[-Kk^2t\left(\lambda_{\mathrm{p}} - 1\right)\right]. \tag{4.106}$$

The total PV and CV errors additionally contain the modulation factors $\widetilde{\mathcal{A}}_K$ and $\widehat{\mathcal{A}}_K$. The CMWA therefore separates two effects that the initial modified-wavenumber cannot distinguish analysis alone: the error in the dominant numerical diffusion rate and the transient error associated with excitation of the secondary coupled mode.

This distinction is analogous to that obtained for convection. The IMWA describes the instantaneous response of an exactly consistent PV–CV state, whereas the CMWA describes the subsequent evolution after the two numerical representations have begun to exchange information. For diffusion, this distinction may be summarized as

$$\boxed{\begin{aligned} \mathcal{D}_1 + \mathcal{D}_2, ; \mathcal{B}_1 + \mathcal{B}_2 &: \quad \text{initial diffusion responses of the PV and CV equations,} \\ \lambda_{\mathrm{p}} &: \quad \text{dominant diffusion branch of the coupled evolution.} \end{aligned}}$$

The secondary eigenvalue is also important even when its modal amplitude is small. Because the semidiscrete eigenvalues are $-Kk^2\lambda_j$, the more strongly damped branch may determine the largest negative eigenvalue of the complete spatial operator and therefore the admissible time step of an explicit time integrator. For the fourth-order Runge–Kutta method used in the preceding numerical experiments, the amplification factor of each coupled diffusion mode is

$$G_j = R_4\left(-Kk^2\lambda_j\Delta t\right), \qquad j \in \mathrm{p,s}, \tag{4.107}$$

where R_4 is given by Eq. (4.22). Stability must therefore be satisfied for both coupled branches. This result illustrates an additional role of the secondary mode: although its contribution to an accurately initialized solution may rapidly disappear, it can still control the stability restriction of the fully discrete method.

The coupled Fourier analysis consequently provides a different interpretation of diffusion from that obtained for convection. In the convective subsystem, the two coupled eigenmodes possess different propagation speeds, and their interference produces the bounded amplitude and phase-speed undulations observed in the numerical solution. In the diffusive subsystem, the two modes instead possess different decay rates. Their interaction produces a transient exchange between the PV and CV representations, after which the more weakly damped physical branch dominates. In both cases, however, the central mechanism is the same: the PV and CV moments do not evolve as independent approximations, and the spectral behavior of the multi-moment method is determined by the eigenstructure of their coupled evolution rather than by either moment considered in isolation.

Problems

Problem 4.1 For the pure convection problem, consider the coupled Fourier system

$$\frac{\partial}{\partial t}\begin{bmatrix}\widehat{\phi}_{\mathrm{PV}}\\ \widehat{\phi}_{\mathrm{CV}}\end{bmatrix} = -\mathbf{i}uk\begin{bmatrix}\mathcal{B}_2 & \mathcal{B}_1\\ 1 & 0\end{bmatrix}\begin{bmatrix}\widehat{\phi}_{\mathrm{PV}}\\ \widehat{\phi}_{\mathrm{CV}}\end{bmatrix}.$$

Derive the characteristic polynomial of the coupling matrix and show that its two eigenvalues are

$$T_{1,2} = \frac{1}{2}\left(\mathcal{B}_2 \pm \sqrt{\mathcal{B}_2^2 + 4\mathcal{B}_1}\right).$$

Define

$$\mathcal{F} = T_1 - T_2.$$

Explain why the coupled modified-wavenumber analysis (CMWA) provides more information about the subsequent numerical evolution than the initial modified-wavenumber analysis (IMWA).

Problem 4.2 Assume that the point-value and cell-value representations are initialized exactly from the same Fourier mode. Decompose the initial state into the two coupled convection eigenmodes and show that

$$C_1 = \frac{1 - T_2}{T_1 - T_2}, \qquad C_2 = \frac{T_1 - 1}{T_1 - T_2}.$$

Show that the corresponding Fourier components may be written as

$$\widehat{\phi}_{\mathrm{PV}}(t) = \widehat{\phi}_0\left[C_1T_1e^{-\mathbf{i}uktT_1} + C_2T_2e^{-\mathbf{i}uktT_2}\right],$$

and

$$\widehat{\phi}_{\mathrm{CV}}(t) = \widehat{\phi}_0\left[C_1e^{-\mathbf{i}uktT_1} + C_2e^{-\mathbf{i}uktT_2}\right].$$

Explain the physical meaning of the two branches when $T_1 > 0$ and $T_2 < 0$.

Problem 4.3 Define the modulation factors

$$\mathcal{A}_{\mathrm{PV}}(t) = C_1T_1 + C_2T_2e^{\mathbf{i}ukt(T_1-T_2)},$$

and

$$\mathcal{A}_{\mathrm{CV}}(t) = C_1 + C_2e^{\mathbf{i}ukt(T_1-T_2)}.$$

Assuming that C_1, C_2, T_1, and T_2 are real, derive explicit expressions for

$$G_{\mathrm{PV}}^2(t) = |\mathcal{A}_{\mathrm{PV}}(t)|^2$$

and

$$G_{\mathrm{CV}}^2(t) = |\mathcal{A}_{\mathrm{CV}}(t)|^2.$$

Show that the angular frequency and period of the amplitude undulation are

$$\omega_{\mathrm{und}} = uk(T_1 - T_2), \qquad \tau_{\mathrm{und}} = \frac{2\pi}{uk(T_1 - T_2)}.$$

Why should this bounded oscillation not be interpreted as numerical instability?

Problem 4.4 A numerical Fourier component governed by the dominant coupled convection branch can be expressed as

$$\widehat{\phi}_k(t) = \widehat{\phi}_k(0)\mathcal{A}(t)e^{-\mathbf{i}uktT_1},$$

where

$$\mathcal{A}(t) = |\mathcal{A}(t)|e^{\mathbf{i}\theta_A(t)}.$$

Derive the instantaneous numerical phase speed and show that

$$\frac{u_{\mathrm{ph}}(t)}{u} = T_1 - \frac{1}{uk}\frac{d\theta_A}{dt}.$$

Explain why both the PV and CV phase speeds can oscillate even though the dominant eigenvalue T_1 is constant.

Problem 4.5 For periodic linear convection, the exact continuous L_2 norm is conserved:

$$L_2(t) = L_2(0).$$

Suppose the semi-discrete MM operator has purely imaginary eigenvalues, but the solution is advanced using the classical fourth-order Runge–Kutta method with

$$R_4(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}.$$

Explain why the numerical L_2 norm may nevertheless decay slowly over a very long simulation. If the final norm defect is defined by

$$D(\Delta t) = 1 - \frac{L_2(T;\Delta t)}{L_2(0)},$$

derive the observed-order estimate

$$p = \frac{\log[D(\Delta t)/D(\Delta t/2)]}{\log 2}.$$

How can time-step refinement be used to distinguish temporal dissipation from the rapid PV–CV amplitude undulation?

Problem 4.6 The periodic multi-moment convection discretization conserves the scalar integral through conservative flux differencing, yet the separate PV and CV L_2 norms may undergo bounded oscillations. Explain why conservation of the scalar integral does not automatically imply conservation of the Euclidean discrete L_2 norm. What algebraic property of a spatial operator is commonly associated with exact preservation of a quadratic norm? Explain why a weighted inner product may be relevant for a multi-moment discretization.

Problem 4.7 For pure diffusion, consider the coupled Fourier matrix

$$\mathbf{M}_K = \begin{bmatrix} D_2 & D_1 \\ B_2 & B_1 \end{bmatrix},$$

so that

$$\frac{\partial \widehat{\boldsymbol{\phi}}}{\partial t} = -Kk^2\mathbf{M}_K\widehat{\boldsymbol{\phi}}.$$

Derive the characteristic polynomial

$$\lambda^2 - (D_2 + B_1)\lambda + (D_2 B_1 - D_1 B_2) = 0.$$

Defining

$$\mathcal{G} = \sqrt{(D_2 - B_1)^2 + 4 D_1 B_2},$$

show that

$$\lambda_{1,2} = \frac{1}{2}\left(D_2 + B_1 \pm \mathcal{G}\right).$$

How is the physical diffusion branch identified, and what would a negative value of one of these eigenvalues imply?

Problem 4.8 Let the physical and secondary diffusion eigenvalues be denoted by λ_p and λ_s, with corresponding eigenvectors

$$\mathbf{v}_j = \begin{bmatrix} P_j \\ 1 \end{bmatrix}, \qquad P_j = \frac{\lambda_j - B_1}{B_2}.$$

For the exactly initialized state

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

show that

$$A_p = \frac{1 - P_s}{P_p - P_s}, \qquad A_s = \frac{P_p - 1}{P_p - P_s}.$$

Derive the subsequent PV and CV Fourier-mode evolution and explain why a single physical Fourier mode generally excites two numerical diffusion modes.

Problem 4.9 Assume that the two diffusion branches are real and satisfy

$$\lambda_s > \lambda_p > 0.$$

Show that the relative contribution of the secondary branch contains the factor

$$\exp\left[-K k^2 t(\lambda_s - \lambda_p)\right].$$

Hence, derive the coupling relaxation rate and relaxation time,

$$\sigma_{\mathrm{cpl}} = K k^2(\lambda_s - \lambda_p), \qquad \tau_{\mathrm{cpl}} = \frac{1}{K k^2(\lambda_s - \lambda_p)}.$$

Contrast this behavior with the convection problem, where the corresponding eigenvalue separation generates an oscillation rather than exponential relaxation.

Problem 4.10 For the coupled diffusion problem, the initial transfer functions of the PV and CV equations are

$$T^{\mathrm{IMW}}_{K,\mathrm{PV}} = D_1 + D_2, \qquad T^{\mathrm{IMW}}_{K,\mathrm{CV}} = B_1 + B_2,$$

whereas the dominant coupled diffusion transfer function is

$$T^{\mathrm{CMW}}_{K} = \lambda_p.$$

Explain the distinction between these quantities. Show that

$$\lambda_p < 1$$

corresponds to under-diffusion and

$$\lambda_p > 1$$

to over-diffusion.

Finally, for RK4, show that each coupled diffusion mode has amplification factor

$$G_j = R_4\left(-Kk^2\lambda_j\Delta t\right), \qquad j \in \{p, s\}.$$

Explain why the secondary mode may control the maximum stable time step even when its amplitude in an accurately initialized solution is very small.

Solution Key

1. Let

$$\mathbf{M}_c = \begin{bmatrix} \mathcal{B}_2 & \mathcal{B}_1 \\ 1 & 0 \end{bmatrix}.$$

The eigenvalues satisfy

$$\det(\mathbf{M}_c - T\mathbf{I}) = 0.$$

Hence

$$\begin{vmatrix} \mathcal{B}_2 - T & \mathcal{B}_1 \\ 1 & -T \end{vmatrix} = 0,$$

which gives

$$T^2 - \mathcal{B}_2 T - \mathcal{B}_1 = 0.$$

Therefore

$$\boxed{T_{1,2} = \frac{1}{2}\left(\mathcal{B}_2 \pm \sqrt{\mathcal{B}_2^2 + 4\mathcal{B}_1}\right)}.$$

Defining

$$\mathcal{F} = \sqrt{\mathcal{B}_2^2 + 4\mathcal{B}_1},$$

we have

$$\boxed{\mathcal{F} = T_1 - T_2}.$$

The IMWA assumes that the PV and CV representations retain the exact relationship inherited from the continuous Fourier mode. It therefore describes only the instantaneous response of the exactly initialized state.

The CMWA instead treats PV and CV as two dynamically coupled numerical fields. It therefore captures the exchange of error and information between the two moments and predicts the two coupled propagation branches that govern the subsequent evolution.

2. For an eigenvalue T_j, an eigenvector may be written in the form

$$\mathbf{v}_j = \begin{bmatrix} T_j \\ 1 \end{bmatrix}.$$

Exact initialization requires

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = C_1 \begin{bmatrix} T_1 \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} T_2 \\ 1 \end{bmatrix}.$$

Thus,

$$C_1 + C_2 = 1,$$

and

$$C_1 T_1 + C_2 T_2 = 1.$$

Solving gives

$$\boxed{C_1 = \frac{1 - T_2}{T_1 - T_2}},$$

and

$$\boxed{C_2 = \frac{T_1 - 1}{T_1 - T_2}}.$$

Each eigenmode evolves according to

$$e^{-\mathbf{i}ukT_j t}.$$

Hence

$$\boxed{\widehat{\phi}_{\mathrm{PV}}(t) = \widehat{\phi}_0 \left[C_1 T_1 e^{-\mathbf{i}ukT_1 t} + C_2 T_2 e^{-\mathbf{i}ukT_2 t} \right]}$$

and

$$\boxed{\widehat{\phi}_{\mathrm{CV}}(t) = \widehat{\phi}_0 \left[C_1 e^{-\mathbf{i}ukT_1 t} + C_2 e^{-\mathbf{i}ukT_2 t} \right]}.$$

When $T_1 > 0$, the first eigenmode propagates in the physical downstream direction and represents the dominant branch.
When $T_2 < 0$, the second mode propagates in the opposite direction and is therefore interpreted as a parasitic coupled mode. Its amplitude vanishes asymptotically as the grid is refined.

3. For the PV modulation,

$$\mathcal{A}_{\mathrm{PV}} = C_1 T_1 + C_2 T_2 e^{\mathbf{i}\omega t},$$

where

$$\omega = uk(T_1 - T_2).$$

Therefore

$$|\mathcal{A}_{\mathrm{PV}}|^2 = \left(C_1 T_1 + C_2 T_2 e^{\mathbf{i}\omega t} \right) \left(C_1 T_1 + C_2 T_2 e^{-\mathbf{i}\omega t} \right).$$

Hence

$$\boxed{G_{\mathrm{PV}}^2 = (C_1 T_1)^2 + (C_2 T_2)^2 + 2C_1 C_2 T_1 T_2 \cos(\omega t)}.$$

Similarly,

$$\boxed{G_{\mathrm{CV}}^2 = C_1^2 + C_2^2 + 2C_1 C_2 \cos(\omega t)}.$$

Thus

$$\boxed{\omega_{\mathrm{und}} = uk(T_1 - T_2)}$$

and

$$\boxed{\tau_{\mathrm{und}} = \frac{2\pi}{uk(T_1 - T_2)}}.$$

The oscillation is bounded because both contributions are purely oscillatory. There is no exponential factor of the form

$$e^{\sigma t}, \qquad \sigma > 0.$$

Thus the variation represents a periodic exchange between the PV and CV representations rather than an unstable growing mode.

4. The complete Fourier mode is

$$\phi_k(x,t) \propto |\mathcal{A}(t)| \exp\left\{ \mathbf{i} \left[kx - ukT_1 t + \theta_A(t) \right] \right\}.$$

The total phase is

$$\Psi(x,t) = kx - ukT_1 t + \theta_A(t).$$

Along a point of constant phase,

$$\frac{d\Psi}{dt} = 0.$$

Therefore

$$k\frac{dx}{dt} - ukT_1 + \frac{d\theta_A}{dt} = 0.$$

Since

$$u_{\mathrm{ph}} = \frac{dx}{dt},$$

we obtain

$$u_{\mathrm{ph}} = uT_1 - \frac{1}{k}\frac{d\theta_A}{dt}.$$

Hence

$$\boxed{\frac{u_{\mathrm{ph}}}{u} = T_1 - \frac{1}{uk}\frac{d\theta_A}{dt}}.$$

Although T_1 is constant for a specified wavenumber, the complex modulation factor varies periodically because the dominant and secondary eigenmodes interfere. Its phase $\theta_A(t)$ therefore changes in time, producing the oscillatory correction

$$-\frac{1}{uk}\frac{d\theta_A}{dt}.$$

Consequently, the apparent phase speeds of both moments can oscillate around the dominant coupled value T_1.

5. For a semidiscrete nondissipative mode,

$$\lambda = \mathbf{i}\omega$$

is purely imaginary. The exact time evolution therefore satisfies

$$|e^{\lambda\Delta t}| = 1.$$

RK4, however, replaces the exponential by

$$R_4(\lambda\Delta t).$$

Although RK4 is fourth-order accurate,

$$R_4(z) \neq e^z$$

for finite z, and therefore

$$|R_4(\mathbf{i}\omega\Delta t)| \neq 1$$

in general.
After N time steps, the amplitude contains the factor

$$|R_4|^N.$$

A very small per-step amplitude error can therefore accumulate into a large long-time norm defect.
For

$$D(\Delta t) \approx C(\Delta t)^p,$$

we have

$$D(\Delta t/2) \approx C \left(\frac{\Delta t}{2}\right)^p.$$

Thus

$$\frac{D(\Delta t)}{D(\Delta t/2)} \approx 2^p,$$

so

$$\boxed{p = \frac{\log[D(\Delta t)/D(\Delta t/2)]}{\log 2}}.$$

If the slowly varying norm decay decreases strongly when Δt is reduced, it is primarily temporal.

By contrast, if the rapid PV–CV undulation retains approximately the same frequency and amplitude under time-step refinement, it is primarily a property of the spatially coupled discretization.

6. Conservation of the scalar integral concerns a linear quantity, for example

$$\sum_i \overline{\phi}_{i+1/2} h.$$

Conservative flux differencing ensures that internal interface fluxes cancel pairwise, so the total scalar quantity remains constant under periodic conditions.

The discrete L_2 norm,

$$\|\boldsymbol{\phi}\|_2^2 = \boldsymbol{\phi}^T \boldsymbol{\phi},$$

is instead a quadratic quantity. Its conservation requires stronger algebraic structure than simple flux conservation.

In the conventional Euclidean inner product, exact quadratic-norm preservation is commonly associated with a skew-symmetric or skew-adjoint spatial operator:

$$\boxed{\mathbf{L}^T = -\mathbf{L}}.$$

Then

$$\frac{d}{dt}\left(\boldsymbol{\phi}^T \boldsymbol{\phi}\right) = \boldsymbol{\phi}^T (\mathbf{L} + \mathbf{L}^T)\boldsymbol{\phi} = 0.$$

PV and CV variables represent geometrically different quantities and may naturally possess different weights. Therefore, the appropriate quadratic invariant may have the form

$$\|\boldsymbol{\phi}\|_W^2 = \boldsymbol{\phi}^T \mathbf{W} \boldsymbol{\phi},$$

rather than the ordinary Euclidean norm.

A corresponding weighted skew-adjoint condition is

$$\boxed{\mathbf{W}\mathbf{L} + \mathbf{L}^T \mathbf{W} = 0}.$$

7. Let

$$\mathbf{M}_K = \begin{bmatrix} D_2 & D_1 \\ B_2 & B_1 \end{bmatrix}.$$

Then

$$\det(\mathbf{M}_K - \lambda \mathbf{I}) = 0,$$

or

$$\begin{vmatrix} D_2 - \lambda & D_1 \\ B_2 & B_1 - \lambda \end{vmatrix} = 0.$$

Therefore

$$(D_2 - \lambda)(B_1 - \lambda) - D_1 B_2 = 0,$$

which expands to

$$\boxed{\lambda^2 - (D_2 + B_1)\lambda + (D_2 B_1 - D_1 B_2) = 0}.$$

The discriminant is

$$(D_2 + B_1)^2 - 4(D_2 B_1 - D_1 B_2),$$

which simplifies to

$$(D_2 - B_1)^2 + 4D_1 B_2.$$

Thus, defining

$$\mathcal{G} = \sqrt{(D_2 - B_1)^2 + 4D_1 B_2},$$

we obtain

$$\boxed{\lambda_{1,2} = \frac{1}{2}(D_2 + B_1 \pm \mathcal{G})}.$$

The physical branch is not identified merely by choosing the plus or minus sign. The long-wave condition defines

$$\boxed{\lambda_p \to 1 \qquad \text{as} \qquad kh \to 0}.$$

Because the complete semidiscrete eigenvalue is

$$-Kk^2\lambda_j,$$

a real positive λ_j gives exponential decay.
If

$$\lambda_j < 0,$$

then

$$-Kk^2\lambda_j > 0,$$

and the corresponding mode grows exponentially. This indicates a semidiscrete diffusion instability.

8. The exactly initialized state satisfies

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = A_p \begin{bmatrix} P_p \\ 1 \end{bmatrix} + A_s \begin{bmatrix} P_s \\ 1 \end{bmatrix}.$$

Hence,

$$A_p + A_s = 1,$$

and

$$A_p P_p + A_s P_s = 1.$$

Using

$$A_s = 1 - A_p,$$

we obtain

$$A_p(P_p - P_s) = 1 - P_s,$$

so

$$\boxed{A_p = \frac{1 - P_s}{P_p - P_s}}.$$

Similarly,

$$\boxed{A_s = \frac{P_p - 1}{P_p - P_s}}.$$

The PV component evolves as

$$\boxed{\widehat{\phi}_{\mathrm{PV}}(t) = \widehat{\phi}_0 \left[A_p P_p e^{-Kk^2\lambda_p t} + A_s P_s e^{-Kk^2\lambda_s t} \right]}.$$

The CV component evolves as

$$\boxed{\widehat{\phi}_{\mathrm{CV}}(t) = \widehat{\phi}_0 \left[A_p e^{-Kk^2\lambda_p t} + A_s e^{-Kk^2\lambda_s t} \right]}.$$

Although the physical initial condition contains only one Fourier wavenumber k, the MM discretization has two coupled degrees of freedom. The same spatial Fourier mode therefore projects onto two eigenvectors of the coupled numerical operator, producing two numerical decay rates.

9. Factor the physical-mode decay from the two-mode solution:

$$e^{-Kk^2\lambda_s t} = e^{-Kk^2\lambda_p t} e^{-Kk^2 t(\lambda_s - \lambda_p)}.$$

Thus the relative secondary contribution contains

$$\boxed{e^{-Kk^2 t(\lambda_s - \lambda_p)}}.$$

If

$$\lambda_s > \lambda_p > 0,$$

this factor decays exponentially.
The characteristic relaxation rate is therefore

$$\boxed{\sigma_{\mathrm{cpl}} = Kk^2(\lambda_s - \lambda_p)}.$$

The corresponding relaxation time is

$$\boxed{\tau_{\mathrm{cpl}} = \frac{1}{Kk^2(\lambda_s - \lambda_p)}}.$$

For convection, the corresponding relative factor is

$$e^{\mathbf{i}uk t(T_1 - T_2)},$$

whose magnitude is unity. The eigenvalue separation therefore produces a periodic modulation with angular frequency

$$uk(T_1 - T_2).$$

For diffusion, the relative factor is real and exponentially decaying, so the eigenvalue separation determines a relaxation rate rather than an oscillation frequency.

10. At $t = 0$, the PV and CV representations are exactly consistent. Therefore, each equation sees a particular instantaneous combination of its coupled operators:

$$\boxed{T^{\mathrm{IMW}}_{K,\mathrm{PV}} = D_1 + D_2}$$

and

$$\boxed{T^{\mathrm{IMW}}_{K,\mathrm{CV}} = B_1 + B_2}.$$

These quantities describe the initial response only.
After the two moments evolve and exchange information, the long-time the physical eigenvalue governs behavior

$$\boxed{T^{\mathrm{CMW}}_{K} = \lambda_p}.$$

The exact Fourier diffusion rate is

$$-Kk^2.$$

The numerical physical branch has rate

$$-Kk^2\lambda_p.$$

Therefore, if

$$\lambda_p < 1,$$

the magnitude of the numerical decay rate is smaller than the exact one, giving

$$\boxed{\text{under-diffusion}}.$$

If

$$\lambda_p > 1,$$

the numerical decay is too strong, giving

$$\boxed{\text{over-diffusion}}.$$

For RK4, a semidiscrete eigenmode satisfies

$$\frac{d\widehat{\phi}_j}{dt} = -Kk^2\lambda_j\widehat{\phi}_j.$$

Therefore, its fully discrete amplification factor is

$$\boxed{G_j = R_4\left(-Kk^2\lambda_j\Delta t\right), \qquad j \in \{p, s\}}.$$

Stability requires

$$|G_p| \le 1 \qquad \text{and} \qquad |G_s| \le 1.$$

Even if the secondary eigenmode has a very small coefficient in an accurately initialized solution, it remains an eigenmode of the complete spatial operator. If

$$\lambda_s > \lambda_p,$$

it generates the more negative semidiscrete eigenvalue,

$$-Kk^2\lambda_s.$$

Consequently, this secondary branch may reach the boundary of the RK4 stability region first and therefore determine the allowable time step.

Chapter 5
Make it work in multi-dimension

"No one can whistle a symphony. It takes a whole orchestra to play it."
Halford E. Luccock

5.1 Multidimensional Discretization

The core concept of our multi-moment method is to combine multiple discretization schemes such that their approximations mutually reinforce one another's resolving power. Consider a cell-averaged finite-volume discretization where fluxes are exchanged through the enclosing surfaces (Fig. 1.4). Instead of interpolating these fluxes solely from cell-averaged values, as in standard Finite-Volume Methods (FVM), we define the surface-averaged values as independent, evolving degrees of freedom.

The cell-averaged variables are evolved via FVM using numerical fluxes derived from these surface-averaged quantities. In turn, the surface averages are advanced using the Finite-Surface Method [13]. In three dimensions, FSM operates as a multi-discretization scheme combining two-dimensional FVM-like updates along the surface with a finite-difference-like update normal to it. The flux exchanges across surface boundaries occur through the enclosing 1D finite lines. While traditional high-order methods treat these line-averaged values (LVs) as auxiliary reconstructed quantities, the present Multi-Moment Method treats LVs as inherent degrees of freedom, evolving them directly via their own governing equations.

5.1.1 Data Structure and Multidimensional Indexing

To extend this framework to multiple dimensions, we distinguish among variable types, their physical dimensions, and their spatial locations. We introduce superscripts to denote the direction(s) of spatial integration. For instance, averaging a scalar field ϕ over the x-direction is denoted by $\overline{\phi}^x$, while $\overline{\phi}^y$ and $\overline{\phi}^{xy}$ denote y-directional and two-dimensional area integration, respectively.

To facilitate seamless code implementation and eliminate fractional indices, we adopt a dual-indexing system mapping node-based grid points x_i ($i = 1, \ldots, n_x$) to cell-centered indices ic ($ic = 1, \ldots, n_x - 1$). A cell index ic corresponds to the half-integer index $i + 1/2$ defining the interval $[x_i, x_{i+1}]$:

$$xc_{ic} = x_{i+1/2} = \frac{1}{2}\left(x_i + x_{i+1}\right). \tag{5.1}$$

Under this notation, $\overline{\phi}^x_{ic} = \overline{\phi}^x_{i+1/2}$ represents the 1D line average over $[x_i, x_{i+1}]$. This dual-indexing structure directly translates into array structures in programming languages, avoiding half-integer subscript handling.

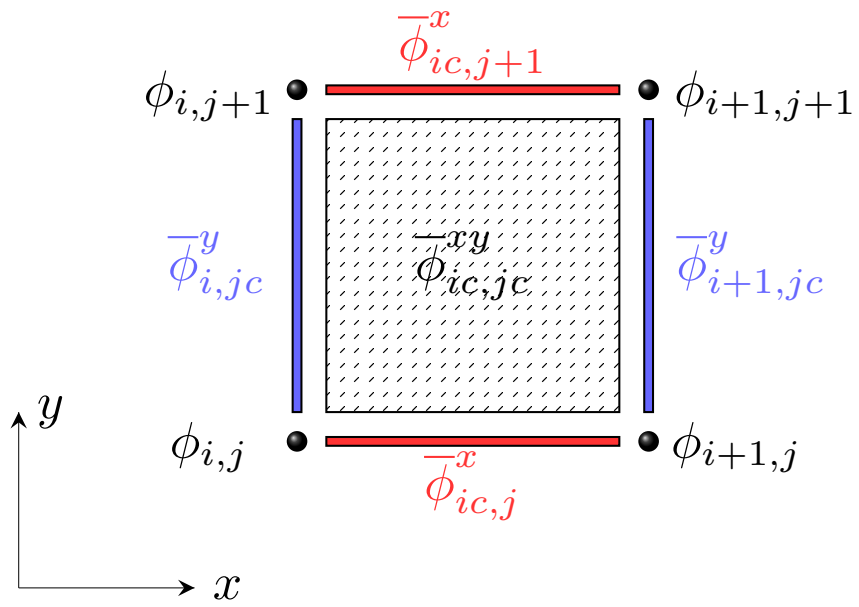


Fig. 5.1: Exploded view of the two-dimensional control volume and its associated edges and nodes. The multi-moment discretization uses four data types: (i) volume-average $\overline{\phi}^{xy}_{ic,jc}$, (ii) vertical-edge average $\overline{\phi}^{y}_{i,jc}$, (iii) horizontal-edge average $\overline{\phi}^{x}_{ic,j}$, and (iv) pointwise $\phi_{i,j}$.

5.1.2 Two-Dimensional Multi-Moment Degrees of Freedom

The arrangement of the multi-moment degrees of freedom for a two-dimensional Cartesian cell is illustrated in Fig. 5.1. The control surface $\Omega_{ic,jc}$ is defined by:

$$\Omega_{ic,jc} = [x_i, x_{i+1}] \times \left[y_j, y_{j+1}\right], \tag{5.2}$$

bounded by the four boundary edges $\partial\Omega_{ic,jc} = \Gamma_{\text{East}} \cup \Gamma_{\text{West}} \cup \Gamma_{\text{North}} \cup \Gamma_{\text{South}}$, where:

$$\begin{aligned} \Gamma_{\text{East}} &= \{x_{i+1}\} &&\times[y_j, y_{j+1}], \\ \Gamma_{\text{West}} &= \{x_i\} &&\times[y_j, y_{j+1}], \\ \Gamma_{\text{North}} &= [x_i, x_{i+1}] &&\times\{y_{j+1}\}, \\ \Gamma_{\text{South}} &= [x_i, x_{i+1}] &&\times\{y_j\}. \end{aligned} \tag{5.3}$$

The nodal points are defined at the grid vertices $X_{i,j} = (x_i, y_j)$.

In two dimensions, the multi-moment method simultaneously evolves four distinct data types:

1. 2D Area Averages (2D-CV): $\overline{\phi}^{xy}_{ic,jc}$, defined on control surfaces $\Omega_{ic,jc}$,
2. 1D Horizontal Edge Averages (1D-LV): $\overline{\phi}^{x}_{ic,j}$, defined on horizontal edges $\Lambda_{\text{South}}/\Lambda_{\text{North}}$,
3. 1D Vertical Edge Averages (1D-LV): $\overline{\phi}^{y}_{i,jc}$, defined on vertical edges $\Lambda_{\text{West}}/\Lambda_{\text{East}}$,
4. 0D Pointwise Values (0D-PV): $\phi_{i,j}$, defined at grid nodes $X_{i,j}$.

5.1.3 Two-Dimensional Multi-Moment Method for Convection–Diffusion Equation

Consider the integral form of the convection–diffusion equation:

$$\frac{\partial}{\partial t}\int_{\Omega}\phi\ d\Omega + \oint_{A}(\mathbf{u}\cdot\mathbf{n})\phi\ dA = K\oint_{A}\boldsymbol{n}\cdot(\nabla\phi)\ dA. \tag{5.4}$$

We can apply this equation to the corresponding geometrical entity and obtain the governing equation for the variable ϕ defined there.

5.1.3.1 Finite-Volume Discretization for $\overline{\boldsymbol{\phi}}^{\boldsymbol{xy}}_{\boldsymbol{ic,jc}}$

We begin with the standard finite-volume discretization for $\overline{\phi}^{xy}_{ic,jc}$:

$$|\Omega_{ic,jc}|\frac{\partial\overline{\phi}^{xy}_{ic,jc}}{\partial t} = -C^{xy}_{ic,jc} + \mathcal{D}^{xy}_{ic,jc}, \tag{5.5}$$

with:

$$C^{xy}_{ic,jc} = \left(\overline{u\phi}^{y}_{i+1,jc} - \overline{u\phi}^{y}_{i,jc}\right)\triangle y_{jc} + \left(\overline{v\phi}^{x}_{ic,j+1} - \overline{v\phi}^{x}_{ic,j}\right)\triangle x_{ic}, \tag{5.6}$$

and:

$$\mathcal{D}^{xy}_{ic,jc} = K\left(\left(\overline{\frac{\partial\phi}{\partial x}}^{y}\right)_{i+1,jc} - \left(\overline{\frac{\partial\phi}{\partial x}}^{y}\right)_{i,jc}\right)\triangle y_{jc} + \left(\left(\overline{\frac{\partial\phi}{\partial y}}^{x}\right)_{ic,j+1} - \left(\overline{\frac{\partial\phi}{\partial y}}^{x}\right)_{ic,j}\right)\triangle x_{ic}. \tag{5.7}$$

Note that the nonlinear flux, e.g., $\overline{u\phi}^{y}_{i,jc}$, requires special attention, which will be addressed later.

5.1.3.2 Finite Line Discretization for $\overline{\boldsymbol{\phi}}^{\boldsymbol{x}}_{\boldsymbol{ic,j}}$

The horizontal and vertical lines bounding the cell edges require a finite-line method. Let us consider the horizontal line on Λ_{South}, and taking the limiting form of the area-integrated equation as the transverse width tends to zero yields the edge evolution equation for $\overline{\phi}^{x}_{ic,j}$:

$$\triangle x_{ic}\frac{\partial\overline{\phi}^{x}_{ic,j}}{\partial t} = -C^{x}_{ic,j} + \mathcal{D}^{x}_{ic,j}, \tag{5.8}$$

with:

$$C^{x}_{ic,j} = \left(u\phi_{i+1,j} - u\phi_{i,j}\right) + \triangle x_{ic}\left(\overline{\frac{\partial v\phi}{\partial y}}^{x}\right)_{ic,j}, \tag{5.9}$$

and:

$$\mathcal{D}^{x}_{ic,j} = K\left(\left(\frac{\partial\phi}{\partial x}\right)_{i+1,j} - \left(\frac{\partial\phi}{\partial x}\right)_{i,j}\right) + \triangle x_{ic}K\left(\overline{\frac{\partial^2\phi}{\partial y^2}}^{x}\right)_{ic,j}. \tag{5.10}$$

Note that the above equations are obtained for an infinitely small $\triangle y$. The above equations show that the evolution requires fluxes in the x-direction. We call this direction FVM-like. In the other direction, the exchange is FDM-like. The evolution of $\overline{\phi}^{y}_{i,jc}$ is constructed analogously and is therefore omitted.

To compute the first and second derivatives in Eqs. (5.9) and (5.10), the formulas from Section 2.1 are applied by substituting 2D averages for 1D averages and using line-averaged data instead of point data. The only term left to resolve is the derivative in Eq. (5.9). We calculate this term by:

$$\left(\overline{\frac{\partial v\phi}{\partial y}}^{x}\right)_{ic,j} = \left(\overline{v\frac{\partial \phi}{\partial y}}^{x}\right)_{ic,j} + \left(\overline{\phi\frac{\partial v}{\partial y}}^{x}\right)_{ic,j}. \tag{5.11}$$

Both terms on the right-hand side of the above equation require special attention similar to $\overline{u\phi}^{y}_{i,jc}$ mentioned previously. The approximation of this convolution is addressed later in Section 5.1.4.

5.1.3.3 Finite-Difference Discretization for $\phi_{i,j}$

Even though the FDM is well known, we are obliged to describe it here for completeness:

$$\frac{\partial \phi_{i,j}}{\partial t} = -\left(\frac{\partial u\phi}{\partial x} + \frac{\partial v\phi}{\partial y}\right)_{i,j} + K\left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}\right)_{i,j}, \tag{5.12}$$

which could be expanded as:

$$\frac{\partial \phi_{i,j}}{\partial t} = -\left(u\frac{\partial \phi}{\partial x} + v\frac{\partial \phi}{\partial y}\right)_{i,j} - \phi\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)_{i,j} + K\left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}\right)_{i,j}. \tag{5.13}$$

It has been widely demonstrated that retaining the continuity residual (the second term on the right-hand side of Eq. (5.13)) facilitates skew-symmetric convective formulations and effectively suppresses aliasing errors [4, 20, 27]. Thus, we explicitly retain this term in the finite-difference nodal updates of our multi-moment scheme.

From a computational perspective, the availability of point values $\phi_{i,j}$ and their spatial derivatives ($\partial\phi/\partial x$ and $\partial\phi/\partial y$) directly at grid nodes eliminates the need for explicit deconvolution and differentiation in the FVM-like directions (Eqs. (5.9) and (5.10)). Similarly, the line-average values $\overline{\phi}^{x}_{ic,j}$ and $\overline{\phi}^{y}_{i,jc}$, along with their FDM-like convective transport, directly supply the required numerical fluxes for the finite-volume discretization (Eqs. (5.6) and (5.7)).

Instead of evaluating 16 derivative terms across both FVM- and FDM-like directions (including those for $\overline{\phi}^{y}_{i,jc}$), the proposed multi-moment formulation reduces the number of evaluations to only 8 terms in the FDM-like directions:

$$\frac{\partial \phi}{\partial x},\quad \frac{\partial \phi}{\partial y},\quad \frac{\partial^2 \phi}{\partial x^2},\quad \frac{\partial^2 \phi}{\partial y^2},\quad \overline{\frac{\partial \phi}{\partial y}}^{x},\quad \overline{\frac{\partial \phi}{\partial x}}^{y},\quad \overline{\frac{\partial^2 \phi}{\partial y^2}}^{x},\quad \text{and}\quad \overline{\frac{\partial^2 \phi}{\partial x^2}}^{y}.$$

When these velocity derivatives are already available from the flow solver, no additional derivative evaluation is required for scalar transport.

Thus, the multi-moment method achieves a computational saving per degree of freedom in two dimensions—matching the efficiency gains demonstrated in 1D. The same reuse mechanism extends naturally to the three-dimensional formulation.

Recalling the geometric entity layout in Fig. 1.4, the discretization procedure described in this section can be directly extended to the remaining faces. The final variable to be evolved in three dimensions is the cell-average value, which is solved using a standard Finite Volume Method where the numerical fluxes are obtained directly from the other moments.

5.1.4 Treatment of Nonlinear Flux

The treatment of the nonlinear flux e.g., $\overline{u\phi}^y$, requires careful consideration because the simple product of averages, $\overline{u}^y\overline{\phi}^y$, yields only second-order accuracy, i.e., $\overline{u\phi}^y = \overline{u}^y\overline{\phi}^y + O(h^2)$. To maintain the formal accuracy of the overall scheme, the discretization of the nonlinear terms must match the precision of the linear terms. One approach to improving this approximation is to apply nonlinear corrections that augment the second-order flux with fourth-order [12, 30] and sixth-order [13] correction terms. Previous investigations into the instabilities of plane channel flow have demonstrated that nonlinearity plays a crucial role in energy transfer; specifically, it has been shown that a fourth-order FVM requires a fourth-order accurate nonlinear flux to remain effective [12]. Conversely, the observed benefits of computing high-order nonlinear fluxes in fully turbulent flows were relatively marginal for fourth- and sixth-order methods [12, 13]. This phenomenon is often attributed to discretization errors in the high-frequency components of the linear terms dominating, thereby masking errors originating from the nonlinear terms. It is therefore of significant interest to investigate how these fourth- and sixth-order nonlinear corrections perform within the multi-moment framework.

5.1.4.1 Nonlinear Correction

The concept of nonlinear correction was first proposed by Pereira et al. [30] for fourth-order FVM, and a sixth-order formulation was subsequently developed for FSM [13]. The present multi-moment discretization differs from FSM in that pointwise values are evolved directly together with integral moments, providing a different set of discrete information from which the nonlinear correction can be constructed. The correction compensates for the discrepancy between the average of a product and the product of the corresponding averages. Introducing the fourth- and sixth-order correction terms $\mathcal{N}_4$ and $\mathcal{N}_6$, respectively, gives

$$\overline{u\phi}^y_{i,jc} = \overline{u}^y_{i,jc}\,\overline{\phi}^y_{i,jc} + \mathcal{N}_4 + \mathcal{N}_6 + O(h^6). \tag{5.14}$$

Hereafter, NC4 denotes the fourth-order nonlinear-flux correction obtained by adding $\mathcal{N}_4$, while NC6 denotes the sixth-order correction obtained by including both $\mathcal{N}_4$ and $\mathcal{N}_6$. Exploiting the independently evolved pointwise values, we derive below a new low-cost NC6 formulation specifically for the multi-moment discretization.

The nonlinear correction terms can be computed from the imbalance between the Taylor expansions of $\overline{u\phi}$ and $\overline{u}\,\overline{\phi}$:

$$\mathcal{N}_4 = \frac{h_y^2}{12}\left(\frac{\partial u}{\partial y}\frac{\partial \phi}{\partial y}\right)_{i,jc} \tag{5.15}$$

$$\mathcal{N}_6 = \frac{h_y^4}{480}\left(\frac{\partial u}{\partial y}\frac{\partial^3 \phi}{\partial y^3} + \frac{\partial \phi}{\partial y}\frac{\partial^3 u}{\partial y^3}\right)_{i,jc} + \frac{h_y^4}{720}\left(\frac{\partial^2 u}{\partial y^2}\frac{\partial^2 \phi}{\partial y^2}\right)_{i,jc} \tag{5.16}$$

Let us define the differentiation operators needed to evaluate each term in the correction above:

$$\left.\frac{\partial \phi}{\partial y}\right|_{i,jc} \approx D_1(\phi_{i,jc}) = \frac{1}{h_y}\left(\phi_{i,jc+\frac{1}{2}} - \phi_{i,jc-\frac{1}{2}}\right) + O(h_y^2), \tag{5.17}$$

$$\left.\frac{\partial^2 \phi}{\partial y^2}\right|_{i,jc} \approx D_2(\phi_{i,jc}) = \frac{6}{h_y^2}\left(\phi_{i,jc+\frac{1}{2}} - 2\overline{\phi}^y_{i,jc} + \phi_{i,jc-\frac{1}{2}}\right) + O(h_y^2), \tag{5.18}$$

$$\left.\frac{\partial^3 \phi}{\partial y^3}\right|_{i,jc} \approx D_3(\phi_{i,jc}) = \frac{3}{h_y^3}\left(\overline{\phi}^y_{i,jc+1} - \overline{\phi}^y_{i,jc-1}\right) - \frac{6}{h_y^3}\left(\phi_{i,jc+\frac{1}{2}} - \phi_{i,jc-\frac{1}{2}}\right) + O(h_y^2). \tag{5.19}$$

The D_1 operator can be directly substituted into the expression for NC4 to yield a fourth-order-accurate nonlinear flux. In fact, the first derivative in Eq. (5.14) can be obtained by evaluating $\partial\phi/\partial y$ used in the convection of $\overline{\phi}^y_{i,jc}$. Both approaches are formally fourth-order accurate, but their truncation errors differ. In practical CFD implementations, it is common to compute the temporal term on a variable-by-variable basis and to treat convection and diffusion terms separately. This approach is preferred over computing every term at once, because some temporary variables can be reused and it reduces the total memory footprint. Motivated by this, we propose computing the correction using central differences of the variables under consideration. This helps reduce memory access overhead. The associated floating-point operation count is also minimal. However, a straightforward implementation might instead evaluate the correction from $\overline{u}^y_{i,jc}$ and $\overline{\phi}^y_{i,jc}$ as done previously in FVM [12, 30]. However, such a computation only resolves $kh < \pi$. The form proposed above is denser and resolves $kh < 2\pi$.

The sixth-order nonlinear correction term corresponding to the operator defined above is given by:

$$\mathcal{N}_6 = \frac{h_y^4}{720}\left(D_2(\overline{u}^y_{i,jc})D_2(\overline{\phi}^y_{i,jc})\right) - \frac{h_y^4}{720}\left(D_1(\overline{u}^y_{i,jc})D_3(\overline{\phi}^y_{i,jc}) + D_1(\overline{\phi}^y_{i,jc})D_3(\overline{u}^y_{i,jc})\right). \tag{5.20}$$

This $\mathcal{N}_6$ term is intrinsically coupled to the evaluation of $\mathcal{N}_4$ described above. The nonlinear corrections for other variables, such as those appearing in Eq. (5.11), can be obtained analogously by replacing u and ϕ with the corresponding variables involved in each line integral. It should be emphasized that both $\mathcal{N}_4$ and $\mathcal{N}_6$ must be included to achieve sixth-order accuracy of the nonlinear flux.

We next examine the computational cost associated with the nonlinear correction. The fourth-order correction requires a total of 5 floating-point operations (FLOP), comprising four operations for evaluating the correction term and one operation for adding it to the linear convection term. Throughout this work, we use the notation *FLOP* instead of *FLOPS*, since the latter is commonly used to denote floating-point operations per second. In contrast, the fourth-order correction of IDO-CF [15] can be implemented in 21 FLOP and our $\mathcal{N}_6$ requires 33 FLOP.

In principle, the nonlinear flux can be evaluated by interpolating the solution to Gauss quadrature points and integrating the resulting products directly. However, the cost of this procedure scales approximately as p^2 with polynomial order, and each interpolation to a quadrature point incurs a cost comparable to that of a differentiation operation. Consequently, quadrature-based flux evaluation becomes expensive even at fourth order and increasingly prohibitive at higher orders. If relatively inexpensive fourth- and sixth-order flux corrections can be shown to preserve the accuracy of substantially higher-order schemes, as previously reported [12, 13], the multi-moment method would offer a particularly efficient alternative for high-fidelity simulations.

5.1.5 The Advantage of the Multi-Moment Method in Linear Problems

Let M be the cost of the first derivative approximation and N be the number of cells in one direction. If we take the cost of the second derivative, which is $M + 2$ for an explicit scheme, we arrive at the complexity of the multi-moment method for the one-dimensional convection–diffusion equation (Eq.1.25 and Eq.1.26):

$$C_{MMM} = (2M + 11)N \tag{5.21}$$

The multi-moment achieves the $kh < 2\pi$ resolution range because it has twice as many degrees of freedom per cell. Thus, for a fair comparison, we compare it with the complexity of the double-resolution FDM, $C_{FDM} = (2M + 5)(2N)$. When the same scheme is used for differentiation, each differentiation in the FD and MM methods costs the same amount (M), and the computing time should be linearly proportional to the complexity. Thus, we can compute the ratio between the two complexities and define it as the theoretical equal-order per-time-step speedup of the multi-moment method relative to the finite-difference method of the same formal order of convergence:

$$S_{MM/FD} = \frac{C_{FDM}}{C_{MMM}} = 2 \times \frac{2M + 5}{2M + 11}. \tag{5.22}$$

For an explicit scheme, the differentiation cost is given by $M = 2P - 1$, with P denoting the formal order of accuracy. Thus, the $S_{MM/FD}$ approach approaches 2 as the order increases. However, this asymptotic complexity ratio does not tell us useful information at practical accuracy, e.g., at 1% or 0.1%. Alternatively, by leveraging the spectral response of the methods, we can estimate the potential savings in computational time by comparing the number of degrees of freedom and the number of time-integration steps required to solve a problem.

To this end, let us define the normalized wavenumber:

$$k^*_{\text{eff}} = \frac{k^*}{nDoFC}, \tag{5.23}$$

where $nDoFC$ denotes the number of degrees of freedom per cell, which is two for the multi-moment method and one for finite-difference and finite-volume methods. Then, we can determine the required number of degrees of freedom needed for each relative error from the resolving efficiency in 1D ($N1D$), and then compute the *degree-of-freedom saving potential* for scheme X as:

$$\Lambda_X(\epsilon) = \left[\frac{N1D_{FDEX2}(\epsilon)}{N1D_X(\epsilon)}\right]^d \left(\frac{N1D_{FDEX2}(\epsilon)}{N1D_X(\epsilon)}\right), \tag{5.24}$$

relative to the second-order FDM as a baseline. The first term represents the advantage gained from reducing the *number of degrees of freedom needed to resolve the field*. Crucially, having fewer degrees of freedom allows for a larger time-step size, an effect quantified by the second term, under the assumption that the CFL restriction of the time-step size is a linear function of the number of degrees of freedom per cell. Thus, the product of two terms indicates the total number of degrees of freedom that the CPU must retrieve throughout the simulation. Note that internal methods such as DGM cannot directly be compared using this formula. Further correction of the temporal evolution is required.

The theoretical computational speedup can then be estimated as the product of the saving potential and the complexity ratio between the baseline second-order FDM (FDEX2) and the scheme X being compared:

$$\mathcal{S}_X(\epsilon) = \frac{W_{FDEX2}}{W_X}\Lambda_X(\epsilon). \tag{5.25}$$

Assuming the computation time is a linear function of the total complexity, we can also interpret this ratio as the theoretical speedup of the scheme relative to FDEX2. If the performance of the two schemes being compared is compute-bound, we would expect the relative speedup to scale with $\mathcal{S}_X(\epsilon)$. However, if they are memory-bound, the computing cost will have less impact, and $\mathcal{S}_X(\epsilon)$ should scale with $\Lambda_X(\epsilon)$.

Table 5.1 reports the degree-of-freedom-saving potential of several finite-difference and multi-moment methods, based on the resolving efficiency of the coupled modified wavenumber. The primary advantage of higher-order schemes arises from this reduction in the required computational complexity. However, the actual performance gain ultimately depends on the additional computational cost of the higher-order scheme relative to the second-order FDM. As a reference, our previous work on a compact fourth-order FVM [12] achieved a tenfold speedup over the second-order FVM for turbulent flow simulations. Multiplying the flux-computation cost ratio of the two schemes (3/8) by the advantage ratio at $\epsilon = 0.1$ yields 11.9, which is close to the implementation's measured performance. This agreement suggests that Eq. (5.25) provides a reliable a priori estimate of the achievable speedup of a numerical method. In addition, for the same order of accuracy in the error range above, EX4 gains the most advantage ($\Lambda_{EX4}/\Lambda_{FDEX4}$) over the FDM counterpart. This is due to EX4's resolving efficiency. The advantage of the multi-moment method gradually increases from EX6 as the order of accuracy is increased.

Regarding the degree-of-freedom saving from reducing degrees of freedom (N1D), the proposed fourth-order multi-moment method requires fewer degrees of freedom than the explicit fourth-order finite-difference scheme (FDEX4). A comparable pattern is evident for the sixth-order methods, where EX6 demonstrates superior performance relative to FDEX6. Additionally, the explicit multi-moment schemes achieve performance levels intermediate to those of explicit finite-difference schemes of adjacent orders: EX4 is positioned approximately halfway between FDEX4 and FDEX6, while EX6 is situated between FDEX6 and FDEX8. Although the explicit multi-moment schemes offer less complexity reduction than compact finite-difference schemes of equivalent formal order, their normalized resolution efficiency remains competitive. Specifically, EX6 and EX8 are marginally less accurate than FDCOM4 and FDCOM6, respectively, at relatively loose error tolerances. As the error tolerance becomes more stringent, the performance gap narrows. EX6 exceeds FDCOM4 at an error threshold of 10^{-3}, and EX8 similarly surpasses FDCOM6 at 10^{-4}.

In summary, Eq. (5.22) describes the asymptotic one-dimensional advantage of the proposed explicit multi-moment schemes over explicit finite-difference methods, whereas Eq. (5.25) predicts the theoretical speedup at a specified relative-error level. The realized speedup also depends on resolving efficiency and implementation-related costs relative to the second-order FDM, including memory traffic and bandwidth effects, which may favor schemes with fewer data accesses. For linear problems, such as the Poisson, Helmholtz, Schrödinger, and acoustic wave equations, the predicted advantage can be largely retained. In strongly nonlinear problems, however, the nonlinear correction introduces additional computational overhead and reduces the attainable speedup. This effect is investigated in the final test case of this paper.

Method	Norm. Resolv. Eff. ($e^*(\epsilon)$)				$N1D(\epsilon)$			2D $\Lambda(\epsilon)$			3D $\Lambda(\epsilon)$		
	0.1	0.01	0.001	0.0001	0.1	0.01	0.001	0.1	0.01	0.001	0.1	0.01	0.001
FDEX2	0.501	0.156	0.049	0.016	7.98	25.64	81.63	1.00	1.00	1.00	1.00	1.00	1.00
FDEX4	0.888	0.479	0.266	0.149	4.50	8.35	15.04	5.57	28.95	160.0	9.87	88.89	868.4
FDEX6	1.087	0.700	0.467	0.315	3.68	5.71	8.57	10.21	90.35	865.7	22.16	405.4	8,251
FDEX8	1.210	0.852	0.621	0.459	3.31	4.70	6.44	14.07	162.9	2,033	33.97	889.5	25,753
FDCOM4	1.188	0.711	0.410	0.232	3.37	5.63	9.76	13.33	94.67	585.8	31.62	431.5	4,902
FDCOM6	1.403	1.005	0.703	0.485	2.85	3.98	5.69	21.96	267.4	2953	61.50	1,723	42,368
EX4	1.001	0.569	0.323	0.182	4.00	7.03	12.38	7.98	48.52	286.4	15.94	177.0	1,888
EX6	1.133	0.752	0.510	0.347	3.53	5.32	7.84	11.57	112.0	1,128	26.16	540.0	11,735
EX8	1.259	0.913	0.679	0.509	3.18	4.38	5.89	15.87	200.5	2,661	39.88	1,173	36,872

Table 5.1: DoF-saving potential (Λ) in two and three dimensions of finite-difference and multi-moment methods compared to the second-order FDM. The table lists the number of degrees of freedom in one-dimensional (N1D) required to meet the spatial resolution required by the normalized resolving efficiency ($e^*(\epsilon)$).

5.2 Numerical Examples

This section evaluates the accuracy, resolving behavior, and computational performance of the multi-moment method in two-dimensional convection and convection–diffusion problems.

5.2.1 Two-dimensional convection–diffusion equation

This test demonstrates the applicability of the multi-moment method in two dimensions. To this end, we solve the two-dimensional convection–diffusion equation (Eqs. (5.5) and (5.8)) using the method described in Section 5.1. The point values are initialized exactly. The edge- and cell-averages are computed numerically using an adaptive integral, ensuring that the integration error remains below 1×10^{-13}. The time integration is performed using the standard fourth-order Runge–Kutta scheme with a sufficiently small time step to ensure the temporal error is negligible.

$$\phi(x, y, 0) = \frac{1}{\mathrm{erf}(5/8)} \cos\left(\frac{xy\pi}{2}\right) \cos\left(\frac{\pi r^2}{2}\right) f(r^2), \tag{5.26}$$

on $(x, y) \in [-7.5, 7.5]^2$ with $r^2 = x^2 + y^2$ and the radial filter:

$$f(\xi) = \frac{1}{2}\left[\mathrm{erf}\left(\rho(\xi + \beta)\right) - \mathrm{erf}\left(\rho(\xi - \beta)\right)\right].$$

The filter parameters are set to $\beta = 1/8$ and $\rho = 5$. The convection velocities are $u = v = 1.0$, and the diffusion coefficient is $K = 1 \times 10^{-6}$. A surface plot of the initial condition is shown in Fig. 5.2. The profile exhibits structures across multiple spatial scales, becoming increasingly complex away from the center while simultaneously decreasing in amplitude. In the outer (highly damped) region, successive cosine lobes are approximately one order of magnitude smaller than those in the preceding layer.

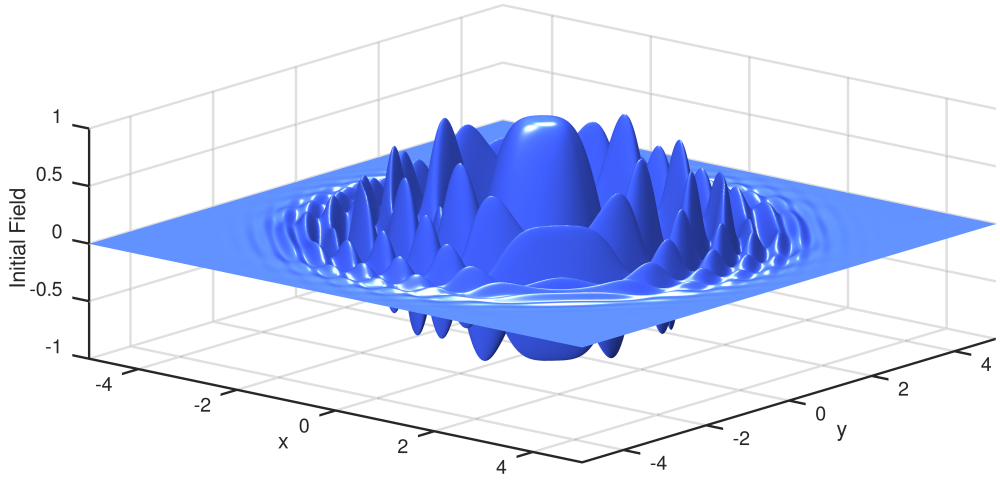


Fig. 5.2: Initial condition of the 2D test case zoomed to the center area.

In Fig. 5.3, we compare contours in the central region at levels $[-0.7 : 0.2 : 0.7]$ and ± 0.005. On coarse grids, standard contouring algorithms may not represent underresolved high-order data reliably; therefore, the numerical fields are first resampled onto a grid that is eight times finer in each spatial direction, resulting in 64 times as many grid points. The interpolation is performed using a reconstruction of the same formal order as the underlying numerical scheme. For the multi-moment method, all available data types are included in the reconstruction. No low-pass filtering is applied.

Despite using only 50^2 cells, EX24 produces contours that are more accurate than those obtained with FDCOM8 on 100^2 cells. Increasing the EX24 resolution to 60^2 cells further improves the representation of the outer lobes. The dashed and solid contours denote the negative and positive levels, respectively, with the outermost pair corresponding to $\phi = -0.005$ and $\phi = 0.005$. At the finest resolution shown, the EX24 solution with nDoF $= 4 \times 75^2$ and the FDCOM8 solution with nDoF $= 200^2$ are visually indistinguishable from the reference solution. At this resolution, the maximum error of EX24 is 2.2×10^{-4}.

Table 5.2(a) shows that, at matched total numbers of degrees of freedom, FDCOM4 is more accurate than EX4 but less accurate than EX6. At coarse resolutions, a double-resolution compact finite-difference scheme of order n provides accuracy comparable to that of explicit multi-moment schemes of orders $n + 2$ to $n + 4$. This behavior is consistent with the high resolving efficiency of compact schemes; in particular, FDCOM4 typically provides resolution comparable to that of a sixth-order explicit finite-difference scheme. At an error tolerance of 10^{-2}, EX4 requires approximately 27% more degrees of freedom per spatial direction than FDCOM4, corresponding to approximately 61% more degrees of freedom in two dimensions. By contrast, EX6 requires only 53% of the two-dimensional degrees of freedom used by FDCOM4. This improvement is greater than that predicted by a monochromatic modified-wavenumber analysis, likely because the error in this test is determined by a broadband distribution of spatial scales weighted by the energy spectrum.

Overall, EX4 is slightly less accurate per degree of freedom than the double-resolution FDCOM4 scheme, whereas EX6 provides greater accuracy per degree of freedom.

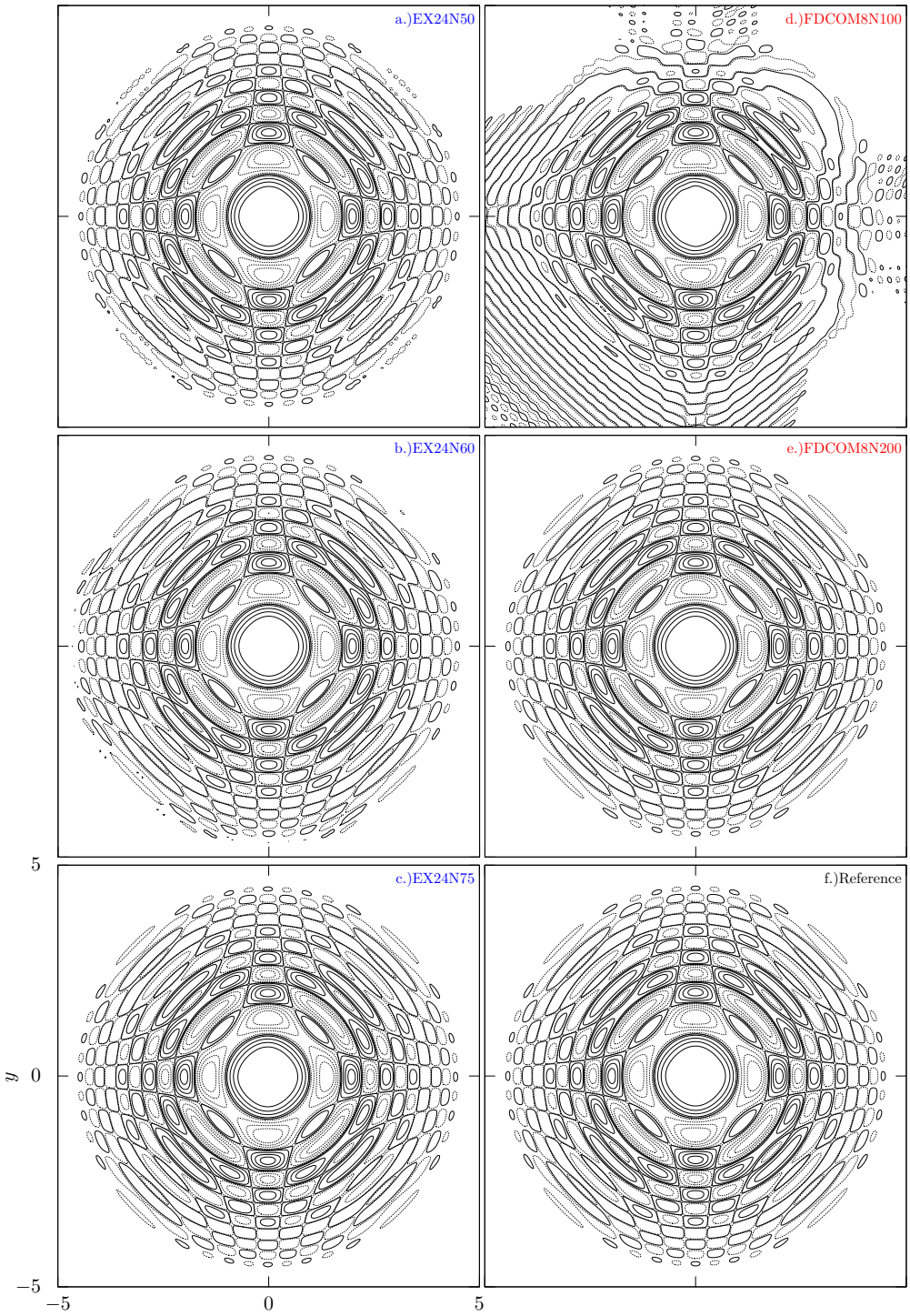


Fig. 5.3: Contours at $t = 15$ obtained with the twenty-fourth-order multi-moment method, EX24 [panels (a)–(c)], and the eighth-order compact finite-difference method, FDCOM8 [panels (d)–(f)]. The reference contours are computed from the exact solution on a 1500^2-point grid. All panels use identical axes and contour levels; axis labels and scales are shown in panel (c).

5.2.2 Two-dimensional convection with a spatially varying velocity field

To demonstrate the applicability of the proposed method to a broader class of problems, we consider the two-dimensional form of Eq. (5.4) with a spatially varying velocity field:

a.) Maximum Error

N_x, N_y	FDCOM4	FDCOM8	EX4	EX6	EX8	EX12	EX24
50	1.06×10^{0}	8.09×10^{-1}	9.542×10^{-1}	6.304×10^{-1}	3.28×10^{-1}	1.59×10^{-1}	4.19×10^{-2}
60	8.58×10^{-1}	4.50×10^{-1}	7.442×10^{-1}	3.693×10^{-1}	1.79×10^{-1}	5.01×10^{-2}	6.45×10^{-3}
120	4.90×10^{-1}	3.04×10^{-2}	1.537×10^{-1}	1.1947×10^{-2}	1.89×10^{-3}	5.15×10^{-5}	1.69×10^{-8}
160	2.84×10^{-1}	2.61×10^{-3}	4.581×10^{-2}	3.309×10^{-3}	1.85×10^{-4}	1.69×10^{-6}	2.48×10^{-11}
200	1.33×10^{-1}	3.72×10^{-4}	1.850×10^{-2}	8.797×10^{-4}	3.08×10^{-5}	1.173×10^{-7}	4.79×10^{-13}
400	6.97×10^{-3}	1.44×10^{-6}	1.149×10^{-3}	1.372×10^{-5}	1.19×10^{-7}	1.45×10^{-11}	—
End Rate	**4.25**	**8.23**	**4.00**	**6.00**	**8.02**	**11.36**	**23.1**

b.) Grid designed for 10^{-2} error

Quantity	FDCOM4	FDCOM8	EX4	EX6	EX8	EX12	EX24
N_x, N_y	366	138	233	134	98	76	57
$\|\text{Error}\|_\infty$	1.00×10^{-2}	1.11×10^{-2}	1.00×10^{-2}	1.05×10^{-2}	1.00×10^{-2}	1.06×10^{-2}	1.12×10^{-2}

c.) Grid designed for 10^{-4} error

Quantity	FDCOM4	FDCOM8	EX4	EX6	EX8	EX12	EX24
N_x, N_y	1146	230	730	286	170	113	85
$\|\text{Error}\|_\infty$	1.02×10^{-4}	1.13×10^{-4}	1.02×10^{-4}	1.02×10^{-4}	1.10×10^{-4}	1.06×10^{-4}	1.05×10^{-4}

Table 5.2: Numerical errors of the multi-moment scheme in 2D convection-diffusion problem compared to fourth- and sixth-order compact FDM.

$$\frac{\partial \phi}{\partial t} + \frac{\partial u\phi}{\partial x} + \frac{\partial v\phi}{\partial y} = K\left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}\right). \tag{5.27}$$

Two test cases are considered in this section: the rotating Gaussian pulse and the idealized kinematic frontogenesis problem.

In both cases, the rotating velocity is given by:

$$u(x, y) = -y \cdot \omega(x, y) \tag{5.28}$$

$$v(x, y) = x \cdot \omega(x, y) \tag{5.29}$$

with a distinct $\omega(x, y)$ in each case.

5.2.2.1 Rotating Gaussian Pulse Problem

This problem is a scalar transport problem for a Gaussian signal in a fluid undergoing a solid-body rotation. The angular velocity is constant: $\omega(x, y) = 2\pi$. The general exact solution is given by:

$$\phi(x, y, t) = A_0 \frac{\sigma^2}{\sigma^2 + 2Kt} \exp\left[-\frac{(x - x_c(t))^2 + (y - y_c(t))^2}{2(\sigma^2 + 2Kt)}\right], \tag{5.30}$$

with:

$$x_c(t) = x_0 \cos(2\pi t) - y_0 \sin(2\pi t), \tag{5.31}$$

$$y_c(t) = x_0 \sin(2\pi t) + y_0 \cos(2\pi t). \tag{5.32}$$

In this test case, we set $A_0 = 1$, $\sigma^2 = 0.5$, and $K = 0$. The domain is set to $(x, y) \in [-11, 11]^2$ with periodic boundary conditions. This size is selected to ensure that the scalar at the boundary is comparable to machine epsilon. This problem represents the transport of a localized temperature or concentration distribution by a prescribed incompressible velocity field and provides an analytical benchmark for assessing variable-coefficient convective transport. It therefore serves as a model for the advection of thermal or concentration structures in nonuniform flows.

Figure 5.4(a) shows the convergence of the MM method using the sixth-order nonlinear correction (NC6), which combines $\mathcal{N}_4$ and $\mathcal{N}_6$ from Eq. (5.14). The errors are plotted against $\sqrt{\text{nDoF}}$, representing the effective resolution in each spatial direction. For comparison, an FDM grid with 120^2 cells has the same total number of degrees of freedom as a multi-moment method on 60^2 cells. With NC6, EX10 and EX12 achieve convergence rates approaching tenth order. EX12 retains a lower error level, by approximately a factor of 3–5 in the asymptotic regime. However, slight curvature at the finest resolutions indicates that the nonlinear correction imposes a limitation. This limitation is more apparent in panel (b), where we use the fourth-order nonlinear correction (NC4). Once the error of the linear discretization becomes sufficiently small, the nonlinear-correction error begins to dominate, and the convergence curves depart from their formal linear rates. Beyond $\sqrt{\text{nDoF}} \approx 80$, EX10 and EX12 exhibit nearly identical behavior, with a terminal convergence rate of approximately 3.56. EX8 remains unaffected over the range considered and achieves an average rate of about 7.5, although with a higher error level than EX10 and EX12. Except for EX4, all MM schemes outperform the fourth-order compact FDM over the tested range.

Panel (c) demonstrates the necessity of the nonlinear correction. Without it, the higher-order multi-moment schemes exhibit a substantial loss of accuracy, whereas EX4 is only weakly affected. Previous work by the first and last authors [13] showed that the nonlinear correction can be two to three orders of magnitude smaller than the linear convective contribution, $\overline{u}\overline{\phi}$. Thus, despite its lower formal order, the nonlinear-correction error may remain negligible until sufficiently fine resolutions are reached because of its small coefficient. A similar behavior is observed here for both NC4 and NC6. Panel (d) compares the MM schemes using the inexpensive NC4 treatment with their explicit FDM counterparts. Over the range considered, NC4 is sufficiently accurate for schemes up to eighth order. EX10 and EX12, however, eventually become limited by the NC4 error and fall behind their FDM counterparts at the finest resolutions, although they retain superior accuracy over a substantial range. At higher resolutions, EX8 is likewise expected to encounter the NC4 accuracy barrier.

The present observations are consistent with earlier FVM studies [12, 13], in which the nonlinear correction was found to have negligible influence on turbulent-flow statistics. Formally, achieving the theoretical convergence rate requires the nonlinear-flux discretization to have an accuracy compatible with that of the linear scheme. In practical computations, however, the solution typically contains a broad spectrum of spatial scales. The dominant numerical error is then often governed by the smaller-scale structures, whose accurate advection is limited by dispersive and under-resolution errors. The range $30 < \sqrt{\text{nDoF}} < 60$ in the present results is representative of this practically relevant regime.

To distinguish the nature of the errors produced by a relatively lower-order scheme with a consistent nonlinear correction from those of a higher-order scheme without a nonlinear

correction, we consider EX4+NC4 on 60^2 cells and EX10+NC0 on 30^2 cells. Both solutions are advanced to $t = 4$, corresponding to four complete revolutions of the Gaussian pulse. Figure 5.5 shows the resulting error fields. At $t = 0.5$, the two schemes have comparable maximum errors: $|\varepsilon|_\infty = 8.8 \times 10^{-3}$ for EX4+NC4 and 9.4×10^{-3} for EX10+NC0. After four revolutions, however, the error of EX4+NC4 increases to 6.7×10^{-2}, whereas that of EX10+NC0 reaches only 3.2×10^{-2}. This comparison reveals two different error mechanisms. For EX4+NC4, the lower-order linear discretization produces appreciable dispersive error in the resolved pulse, causing the error to accumulate over a broad region surrounding the large-scale structure. For EX10+NC0, the high-order linear discretization preserves the pulse itself much more accurately.

At the same time, the dominant error is localized near the pulse base, where the absence of a nonlinear-flux correction creates a localized error. The resulting under-resolution produces small-amplitude numerical oscillations. Thus, although EX10+NC0 exhibits more pronounced small-scale wiggles, it maintains the large-scale solution substantially more accurately than EX4+NC4 over long integration times.

These results should be interpreted over the practically relevant error range rather than as a statement about the ultimate asymptotic order. NC4 is generally sufficient to retain the high-order accuracy advantage of EX8 and EX10 over the tolerances considered. At sufficiently small errors, however, the nonlinear-flux correction becomes the order-limiting component; accordingly, NC6 is required for EX10 at the 10^{-5} target.

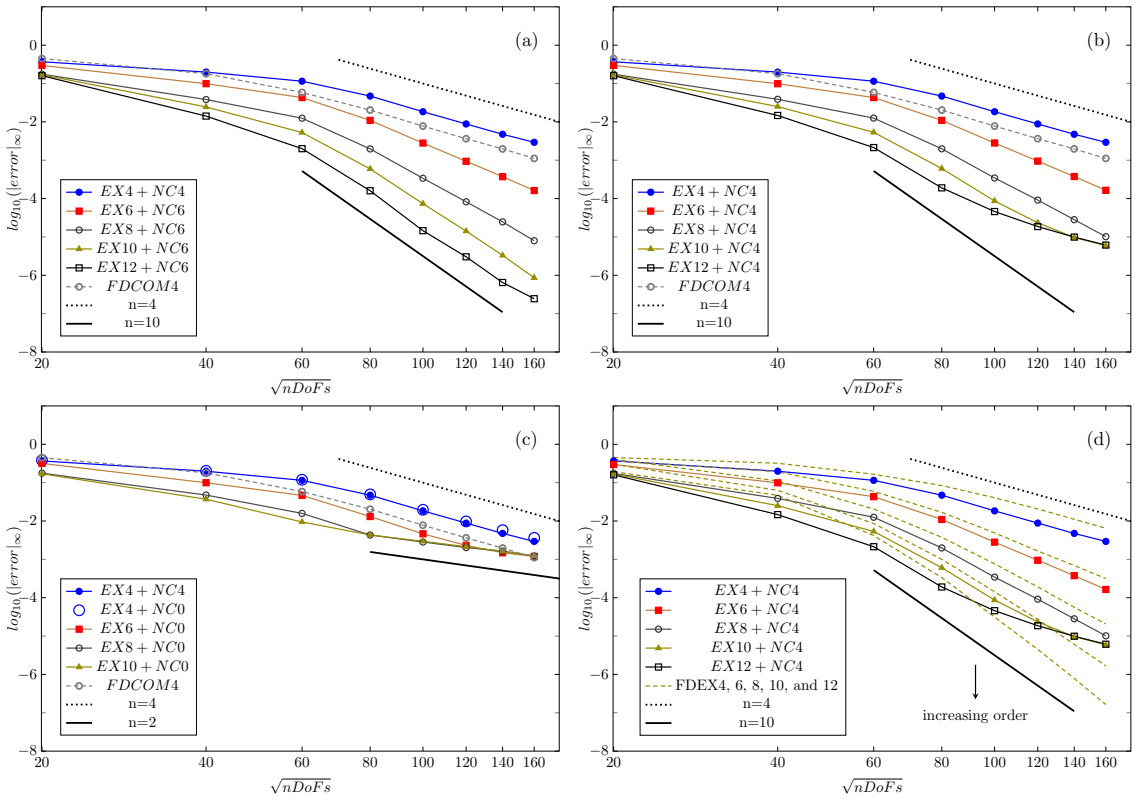


Fig. 5.4: Convergence analysis of the rotating Gaussian pulse at $t = 1.0$. Panels (a) and (b) illustrate the multi-moment scheme with sixth-order (NC6) and fourth-order (NC4) nonlinear corrections, respectively, while (c) shows the results without corrections. Panel (d) provides a comparative accuracy analysis between the proposed method (using NC4) and its corresponding FDM counterparts.

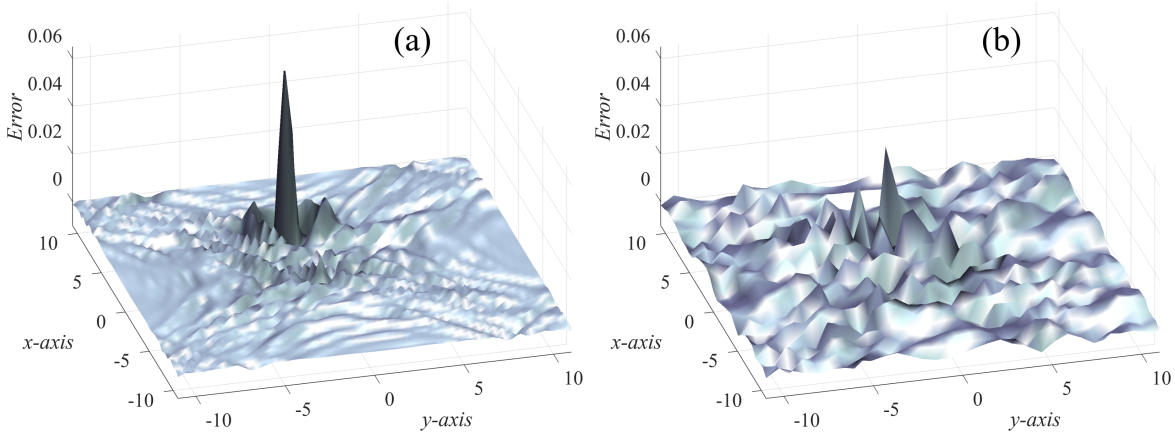


Fig. 5.5: Error fields for the rotating Gaussian pulse at $t = 4$: (a) EX4+NC4 on a 60^2-cell grid and (b) EX10+NC0 on a 30^2-cell grid.

5.2.2.2 Idealized kinematic frontogenesis

In our final test case, we evaluate the performance of the proposed method by simulating the idealized kinematic frontogenesis problem introduced by Doswell [6]. Compared to the previous examples, this case features a significantly more complex and rapidly varying velocity field:

$$u(x, y) = -y \cdot \omega(x, y) \tag{5.33}$$

$$v(x, y) = x \cdot \omega(x, y) \tag{5.34}$$

$$\omega(x, y) = V_T(x, y)/r, \tag{5.35}$$

$$V_T(x, y) = \frac{3\sqrt{3}}{2} \operatorname{sech}^2(r) \tanh(r), \tag{5.36}$$

with $r = \sqrt{x^2 + y^2}$. The exact analytical solution for the inviscid case is given by:

$$\phi(x, y, t) = -\tanh\left(\frac{y\cos(\omega t)}{2} - \frac{x\sin(\omega t)}{2}\right). \tag{5.37}$$

The computational domain is restricted to a compact region, $(x, y) \in [-5, 5]^2$. In this test case, $K = 10^{-10}$ is retained to represent a convection-dominated case and minimize regularization from diffusion. A comparison with the inviscid calculation confirmed that its contribution to the maximum error at $t = 10$ is well below the smallest error threshold considered.

The frontogenesis velocity field provides a demanding scalar-transport benchmark because it continuously stretches the transported field into progressively finer structures, analogous to the thin thermal and concentration layers encountered in high-Prandtl- and high-Schmidt-number transport. This strong deformation also provides a stringent test of the nonlinear-flux correction. To isolate interior-discretization errors from boundary effects, ghost data outside the computational domain are prescribed from the time-dependent inviscid analytical solution. The pointwise values are imposed exactly, whereas the corresponding averaged quantities are evaluated numerically. The exact solution at $t = 8$ is shown in Fig. 5.6. Over this interval, the vortex progressively deforms the initial hyperbolic-tangent profile, producing increasingly sharp alternating layers of high and low scalar concentration.

Temporal integration is performed using the classical fourth-order Runge–Kutta method. Compared with the preceding advection problem, this test is less sensitive to temporal-discretization error, and a normalized CFL number $nDoFC\,|\mathbf{u}|\,\Delta t/\Delta x = 0.8$, is sufficient over the error range considered. The same normalized CFL number is therefore used for all cases. Since the FDM and MM formulations employ one and two degrees of freedom per cell, respectively, this corresponds to $\Delta t = 0.8\,\Delta x/|\boldsymbol{u}|_\infty$ for the FDM and $\Delta t = 0.4\,\Delta x/|\boldsymbol{u}|_\infty$ for the MM method.

Figure 5.7 illustrates the temporal evolution of the scalar field morphology in the two-dimensional frontogenesis benchmark at four representative times ($t = 0.0, 3.3, 6.6$, and 10.0). At the initial state ($t = 0.0$), the field presents a horizontal boundary layer separating two distinct scalar regions. As time progresses under the influence of the nonlinear velocity field, the interface undergoes localized spiraling and roll-up ($t = 3.3$). By $t = 6.6$ and $t = 10.0$, intense deformation and shearing generate tightly wound, multi-turn spiral filaments around the core. No visually apparent excessive dissipation or spurious oscillations are observed at the resolution shown.

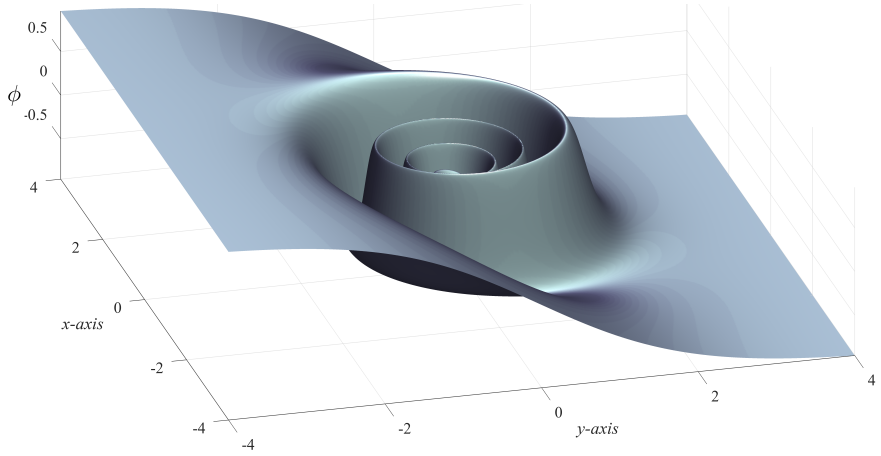


Fig. 5.6: Surface plot of the inviscid analytical solution in the central region of the idealized kinematic frontogenesis solution at $t = 8$.

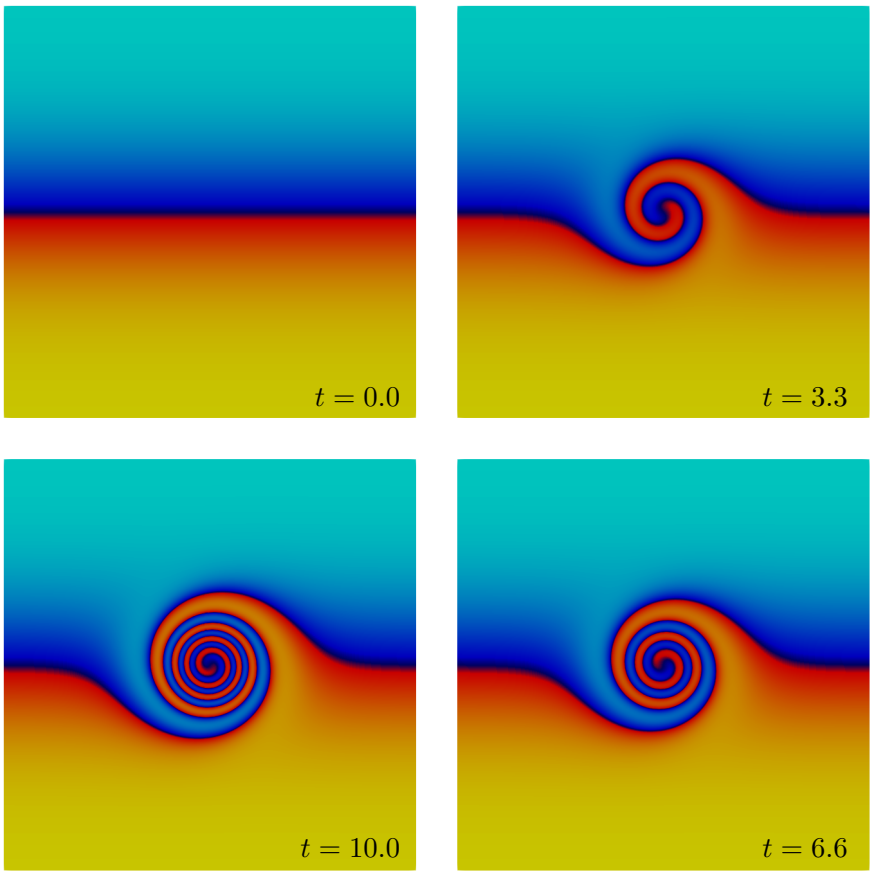


Fig. 5.7: Time evolution of the scalar field morphology in the frontogenesis problem from $t = 0.0$ to $t = 10.0$, computed using the EX8+NC4 scheme on a 272^2 grid. The top and right boundaries of each panel correspond to $y = 5$ and $x = 5$, respectively.

Threshold (ϵ)	**Multi-Moment Method**				**Finite Difference Method**			
	EX4	EX6	EX8	EX10	FDEX4	FDEX6	FDEX8	FDEX10
Number of Cells per direction (N_x, N_y)								
10^{-2}	160	122	114	111	380	268	244	236
10^{-3}	302	220	186	174	728	478	398	362
10^{-4}	550	342	272	250	1338	750	584	514
10^{-5}	970	514	378	328	2394	1118	816	684
Average CPU-Time (second)								
10^{-2}	0.760	0.345	0.347	0.271	0.976	0.385	0.448	0.428
10^{-3}	4.915	1.983	1.278	1.164	10.86	2.493	1.400	1.136
10^{-4}	55.16	8.00	3.89	3.24	125.36	12.81	5.52	4.91
10^{-5}	381.68	44.05	12.36	9.71	767.07	67.49	18.69	10.84
Actual Maximum Error								
10^{-2}	$1.1e{-2}$	$1.0e{-2}$	$1.0e{-2}$	$1.0e{-2}$	$1.0e{-2}$	$1.0e{-2}$	$1.0e{-2}$	$1.1e{-2}$
10^{-3}	$1.0e{-3}$	$1.1e{-3}$	$1.1e{-3}$	$1.1e{-3}$	$1.0e{-3}$	$1.0e{-3}$	$1.0e{-3}$	$1.0e{-3}$
10^{-4}	$1.0e{-4}$	$1.1e{-4}$	$1.0e{-4}$	$1.1e{-4}$	$1.0e{-4}$	$1.0e{-4}$	$1.1e{-4}$	$1.0e{-4}$
10^{-5}	$1.0e{-5}$	$1.0e{-5}$	$1.1e{-5}$	$1.0e{-5}$	$1.0e{-5}$	$1.1e{-5}$	$1.1e{-5}$	$1.1e{-5}$

Table 5.3: Grid resolutions required to achieve four target relative error thresholds, along with the corresponding mean CPU times and actual maximum errors for the frontogenesis problem at the final time ($t = 10$).

The computational performance of the proposed schemes was evaluated by determining the grid resolutions required to satisfy target error tolerances ranging from $\varepsilon = 10^{-2}$ to 10^{-5}, together with the corresponding maximum errors and mean CPU times reported in Table 5.3. All benchmark codes were compiled and executed on the *LANTA* supercomputer at the Thailand National Science and Technology Development Agency (NSTDA) Supercomputer Center using `gfortran 12.2.0` with the target-specific optimization flags `-O3 -march=znver3`. Benchmarks were scheduled through Slurm using the automated benchmarking script. Sixteen independent benchmark instances of each scheme were executed concurrently on an exclusively allocated dual-socket node equipped with two 64-core AMD EPYC 7713 processors. Each benchmark was repeated eight times, yielding a total of 128 timing samples for each tested configuration. Every instance was pinned using the `taskset` command to a designated physical core. This fixed processor affinity prevents process migration during execution, preserves cache locality, and improves the reproducibility of the timing measurements. The reported CPU time encompasses the evaluation of the governing right-hand side operators—Eqs. (5.5), (5.8), and (5.13)—and all core routines within the time-stepping loop. To compare the core spatial operators independently of the test-specific analytical boundary integration, the latter was excluded from timing for all schemes.

Table 5.3 summarizes the grid resolutions, maximum errors, and mean CPU times for the MM and explicit finite-difference (FDM) schemes in the frontogenesis benchmark at $t = 10$. For each target error level between 10^{-2} and 10^{-5}, the grid resolution was chosen such that the resulting maximum error closely matched the prescribed value. Each reported CPU time is averaged over 128 independent executions, with the standard deviation ranging from 0.63% to 12.30% of the mean. The MM calculations employ NC4, except for the EX10 scheme at the strictest error tolerance, which utilizes NC6. Across the tested accuracy range, the MM schemes require fewer cells and fewer degrees of freedom per direction than their corresponding FDM counterparts. This reduction in spatial resolution generally translates to a corresponding decrease in computational runtime. At the 10^{-5} target error, the matched-order speedups are approximately 2.01×, 1.53×, 1.51×, and 1.12× for EX4, EX6, EX8, and EX10, respectively. Across all target error levels, the mean matched-order speedups are 1.94×, 1.38×, 1.33×, and 1.30×, yielding an overall average speedup of approximately 1.49×. Only one of the sixteen comparisons—EX10 at the 10^{-3} target—exhibited a marginally longer MM runtime, with the difference being approximately 2.5%. Overall, the measured speedup trends are consistent with the theoretical efficiency estimates in Table 5.1.

The results clearly demonstrate the advantage of increasing the approximation order when stringent error tolerances are required. At an error level of approximately $\epsilon \approx 10^{-5}$, EX8 and EX10 require only 12.36 s and 9.71 s, respectively, compared with 767.07 s for the fourth-order FDM. These values correspond to speedups of approximately 62× and 79×, respectively. The principal source of this improvement is the ability of the higher-order schemes to attain the prescribed accuracy on substantially coarser grids.

To provide a direct reference for the DoF-saving potential reported in Tab. 5.1, the classical second-order finite-difference scheme (FDEX2) was also benchmarked. This additional test is particularly relevant because FDEX2 serves as the reference scheme in the definition of $\Lambda(\epsilon)$. The corresponding FDEX2 timing results are reported separately and are not included in the timing table. For this direct comparison, the error levels $\epsilon \approx 10^{-2}$ and 10^{-3} were selected deliberately. These tolerances represent a practically relevant accuracy range over which the second-order scheme can still be evaluated at computationally manageable grid resolutions. Although FDEX2 can, in principle, attain smaller errors through further grid

refinement, the required number of degrees of freedom increases rapidly. Extending the benchmark to substantially smaller tolerances would therefore move the comparison into an increasingly impractical large-grid regime while providing limited additional insight into the relative computational performance. The selected tolerances consequently provide a practical and representative basis for comparison.

At $\epsilon \approx 10^{-2}$ and 10^{-3}, FDEX2 requires grids of 1290^2 and 4080^2, with mean CPU times of 106.49 s and 3961.9 s, respectively. In comparison, the corresponding EX8 calculations require only 0.347 s and 1.278 s, yielding measured speedups of approximately 307× and 3100×, respectively. It is important to distinguish these measured speedups from the complexity-saving potential Λ reported in Tab. 5.1. The values of Λ quantify the combined reduction in spatial degrees of freedom and time-step requirements. They do not, however, account for the higher computational cost per degree of freedom of the higher-order schemes. Thus, Λ represents a theoretical work-reduction potential rather than a direct prediction of wall-clock speedup. For EX8, Tab. 5.1 gives two-dimensional complexity-saving potentials of approximately 200 and 2661 at $\epsilon \approx 10^{-2}$ and 10^{-3}, respectively. Since the arithmetic cost of EX8 is approximately four times that of FDEX2 per computational unit, the corresponding operation-count-based speedup estimates are approximately 50× and 665×, respectively. These cost-adjusted estimates are substantially smaller than the measured speedups of 307× and 3100×. The discrepancy therefore cannot be explained by resolving efficiency and arithmetic operation counts alone.

The measured timings indicate that the cost of data movement also plays an important role. This is evident in cases where schemes with substantially different arithmetic complexity exhibit nearly identical runtimes. For example, FDEX4 on a 728^2 grid requires 10.86 s, whereas FDEX10 on a 684^2 grid requires 10.84 s, despite the latter having approximately 2.7 times greater arithmetic complexity. The runtimes are therefore nearly identical even though the higher-order scheme performs considerably more floating-point operations. Similar behavior is observed for EX4 on a 302^2 grid and EX8 on a 272^2 grid, for which the runtime does not increase in proportion to the nominal arithmetic cost. This behavior is consistent with a memory-bandwidth-limited execution regime. The AMD EPYC 7713 processor has a nominal FP64 peak throughput of approximately 3.76 TFLOP s^{-1}, whereas its eight-channel DDR4-3200 memory subsystem provides a theoretical peak memory bandwidth of approximately 204.8 GB s^{-1} [1]. Consequently, for kernels with relatively low arithmetic intensity, attainable performance is constrained more strongly by data movement than by peak floating-point throughput. A higher-order scheme may therefore perform additional arithmetic at relatively small incremental cost when the required data are already resident in the cache hierarchy.

In addition to the memory-bound behavior of the code, the memory hierarchy and, in particular, cache utilization are likely to contribute to the observed performance difference. The estimated data movement of FDEX2 is approximately 32 and 120 times larger than that of EX8 at the two target error levels considered above. At $\epsilon \approx 10^{-3}$, the principal EX8 working set on the 186^2 grid is sufficiently small that a substantial fraction may reside within the L3 cache associated with the assigned CCD (32 MB). By contrast, the corresponding FDEX2 arrays on the 4080^2 grid occupy approximately 399 MB, exceeding the available local L3-cache capacity by more than an order of magnitude and therefore generating substantially greater main-memory traffic.

Although hardware performance counters were not used to measure cache misses, memory bandwidth, or arithmetic intensity directly, the large difference in working-set size and expected cache residency provides a plausible explanation for the measured performance

beyond that predicted from arithmetic complexity alone. The observed acceleration therefore results from three related effects: the higher resolving efficiency reduces the required grid size, the smaller problem correspondingly reduces total data movement, and the reduced working set makes more effective use of the cache hierarchy. Consequently, the practical benefit of the higher-order schemes can substantially exceed the speedup predicted by a cost-adjusted operation-count model alone.

Problems

Problem 5.1 In the multidimensional multi-moment formulation, superscripts are used to identify the direction of spatial integration. For example, ϕ^x, ϕ^y, and ϕ^{xy} represent quantities integrated in the x-direction, y-direction, and over an xy-area, respectively.

On a one-dimensional grid with nodal coordinates

$$x_i, \qquad i = 1, \ldots, n_x,$$

the cell-centered index i_c is introduced through

$$x_{c,i_c} = x_{i+1/2} = \frac{x_i + x_{i+1}}{2}.$$

Explain why the dual-indexing system (i, i_c) is preferable to using half-integer indices directly in a computer program. For a two-dimensional grid containing $n_x \times n_y$ nodes, state the array dimensions required to store the point values, the x-averaged line values, the y-averaged line values, and the cell-averaged values.

Problem 5.2 Describe the hierarchy of degrees of freedom used by the multi-moment method on a two-dimensional Cartesian grid. Distinguish among

$$\phi_{i,j}, \qquad \phi^x_{i_c,j}, \qquad \phi^y_{i,j_c}, \qquad \phi^{xy}_{i_c,j_c}.$$

Identify the geometric entity associated with each quantity and explain which numerical viewpoint—finite difference, finite line, finite surface, or finite volume—is naturally associated with its evolution.

How does this geometric hierarchy extend to three spatial dimensions?

Problem 5.3 Consider the two-dimensional convection–diffusion equation

$$\frac{\partial \phi}{\partial t} + \frac{\partial (u\phi)}{\partial x} + \frac{\partial (v\phi)}{\partial y} = K \left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} \right).$$

Define the y-averaged line moment

$$\phi^y_{i,j_c} = \frac{1}{h_y} \int_{y_j}^{y_{j+1}} \phi(x_i, y, t)\, dy.$$

Integrate the governing equation over $[y_j, y_{j+1}]$ and derive the evolution equation for ϕ^y_{i,j_c}. Show explicitly which terms remain as derivatives in the x-direction and which terms become flux differences at y_j and y_{j+1}.

Derive the analogous equation for the x-averaged moment $\phi^x_{i_c,j}$.

Problem 5.4 One computational advantage of the multidimensional multi-moment method is that information evolved by one discretization can be reused directly by another. In two dimensions, the formulation reduces the required derivative evaluations to the following eight FDM-like quantities:

$$\frac{\partial \phi}{\partial x}, \qquad \frac{\partial \phi}{\partial y}, \qquad \frac{\partial^2 \phi}{\partial x^2}, \qquad \frac{\partial^2 \phi}{\partial y^2},$$

$$\frac{\partial \phi^x}{\partial y}, \qquad \frac{\partial \phi^y}{\partial x}, \qquad \frac{\partial^2 \phi^x}{\partial y^2}, \qquad \frac{\partial^2 \phi^y}{\partial x^2}.$$

Explain why other fluxes and derivatives required by the integral equations can be obtained directly from the independently evolved moments. Why does this eliminate deconvolution operations that would otherwise be required in a conventional high-order finite-volume formulation?

Discuss how the same reuse mechanism extends naturally to three dimensions.

Problem 5.5 For a nonlinear convective flux, the average of a product is not equal to the product of the averages. Consider two smooth functions $u(y)$ and $\phi(y)$ over a cell of width h_y centered at y_c. Define

$$\overline{u\phi} = \frac{1}{h_y}\int_{y_c-h_y/2}^{y_c+h_y/2} u(y)\phi(y)\,dy.$$

Using Taylor expansions about y_c, show that

$$\overline{u\phi} = \overline{u}\,\overline{\phi} + \frac{h_y^2}{12}\frac{\partial u}{\partial y}\frac{\partial \phi}{\partial y} + O(h_y^4).$$

Hence identify the fourth-order nonlinear correction

$$N_4 = \frac{h_y^2}{12}\frac{\partial u}{\partial y}\frac{\partial \phi}{\partial y}.$$

Explain why simply using $\overline{u}\,\overline{\phi}$ limits the nonlinear flux to second-order accuracy even when the underlying spatial discretization is of much higher order.

Problem 5.6 The nonlinear-flux approximation in the chapter is written schematically as

$$\overline{u\phi} = \overline{u}\,\overline{\phi} + N_4 + N_6 + O(h^6),$$

where N_4 and N_6 denote fourth- and sixth-order corrections.

Explain why both N_4 and N_6 must be retained to obtain a sixth-order nonlinear-flux approximation. Discuss why a high-order linear scheme such as EX8 or EX10 may still exhibit good practical accuracy when only NC4 is used, even though NC4 eventually becomes the order-limiting approximation at sufficiently small errors.

Compare this correction-based strategy with direct Gauss-quadrature evaluation of the nonlinear flux, noting that the quadrature-based cost scales approximately as p^2 with polynomial order p. Why is the correction approach attractive for very-high-order multi-moment methods?

Problem 5.7 Let M denote the computational cost of a first-derivative approximation. The chapter estimates the one-dimensional convection–diffusion costs as

$$C_{\mathrm{MM}} = (2M + 11)N$$

for the multi-moment method and

$$C_{\mathrm{FD}} = (2M + 5)(2N)$$

for a finite-difference calculation at twice the spatial resolution.

Show that the theoretical equal-order speedup is

$$S_{\mathrm{MM/FD}} = 2\frac{2M + 5}{2M + 11}.$$

For an explicit approximation, take

$$M = 2P - 1,$$

where P is the formal order of accuracy. Evaluate $S_{\mathrm{MM/FD}}$ for

$$P = 4, \qquad 6, \qquad 8, \qquad 10,$$

and determine

$$\lim_{P \to \infty} S_{\mathrm{MM/FD}}.$$

Explain why this operation-count ratio alone does not completely describe the multidimensional computational advantage of a high-resolution method.

Problem 5.8 Suppose method A requires N_A degrees of freedom in each spatial direction to achieve a specified error tolerance, whereas a reference method requires N_R. Define the one-dimensional resolution ratio

$$r = \frac{N_R}{N_A}.$$

Show that the spatial degree-of-freedom saving scales as

$$r^d$$

in d spatial dimensions. If the stable explicit time step is proportional to the grid spacing, show that the total work-saving potential scales approximately as

$$r^{d+1}.$$

For

$$r = 2,$$

compute the theoretical work-saving factors in one, two, and three spatial dimensions. Explain why even a moderate reduction in the number of required grid points per direction can produce a large computational advantage in three-dimensional simulations.

Problem 5.9 Consider the rotating Gaussian-pulse problem with velocity field

$$u(x, y) = -2\pi y, \qquad v(x, y) = 2\pi x,$$

and $K = 0$.

First show that the velocity field is incompressible:

$$\nabla \cdot \mathbf{u} = 0.$$

The center of an initially displaced Gaussian pulse, (x_0, y_0), evolves according to

$$x_c(t) = x_0 \cos(2\pi t) - y_0 \sin(2\pi t),$$

$$y_c(t) = x_0 \sin(2\pi t) + y_0 \cos(2\pi t).$$

Determine the location of the pulse center at

$$t = \frac{1}{4}, \qquad t = \frac{1}{2}, \qquad t = \frac{3}{4}, \qquad t = 1.$$

What should happen to the shape, peak amplitude, and total scalar quantity after one complete revolution when $K = 0$? Explain why this problem is a useful multidimensional verification case for variable-coefficient convection and nonlinear-flux treatment.

Problem 5.10 In the idealized kinematic frontogenesis test, the numerical results at the error threshold

$$\epsilon = 10^{-4}$$

require the following numbers of cells per spatial direction:

$$N_{\mathrm{EX8}} = 272, \qquad N_{\mathrm{FDEX8}} = 584.$$

The corresponding measured CPU times are

$$t_{\mathrm{EX8}} = 3.89\ \mathrm{s}, \qquad t_{\mathrm{FDEX8}} = 5.52\ \mathrm{s}.$$

Compute:

1. the ratio of grid resolutions,
$$\frac{N_{\mathrm{FDEX8}}}{N_{\mathrm{EX8}}},$$
2. the ratio of the total number of two-dimensional cells,
$$\left(\frac{N_{\mathrm{FDEX8}}}{N_{\mathrm{EX8}}}\right)^2,$$
and
3. the measured CPU-time speedup,
$$\frac{t_{\mathrm{FDEX8}}}{t_{\mathrm{EX8}}}.$$

Explain why the cell-count saving can be substantially larger than the measured wall-clock speedup. Discuss the roles of higher local arithmetic cost, memory access, nonlinear-flux corrections, and time integration.

Finally, explain why the frontogenesis problem is more demanding than the solid-body rotating Gaussian test and why the formation of tightly wound scalar filaments provides a stringent assessment of multidimensional resolving power.

Solution Key

1. The dual-indexing convention distinguishes nodal locations from cell-centered locations using integer array indices. Thus,

$$i = 1, \ldots, n_x$$

denotes nodal positions, whereas

$$i_c = 1, \ldots, n_x - 1$$

denotes intervals between consecutive nodes.
The correspondence is

$$i_c \longleftrightarrow i + \frac{1}{2},$$

so that

$$x_{c,i_c} = x_{i+1/2} = \frac{x_i + x_{i+1}}{2}.$$

This convention is preferable in computer programs because programming languages require integer array indices. It avoids storing or interpreting half-integer indices such as

$$i + \frac{1}{2},$$

and makes the data layout directly compatible with ordinary arrays.
For an $n_x \times n_y$ nodal grid, the required array dimensions are

$$\boxed{\phi_{i,j} : \qquad n_x \times n_y}$$

for point values,

$$\boxed{\phi^x_{i_c,j} : \qquad (n_x - 1) \times n_y}$$

for line averages in the x-direction,

$$\boxed{\phi^y_{i,j_c} : \qquad n_x \times (n_y - 1)}$$

for line averages in the y-direction, and

$$\boxed{\phi^{xy}_{i_c,j_c} : \qquad (n_x - 1) \times (n_y - 1)}$$

for cell-averaged quantities.
Thus each geometric type of degree of freedom has a natural integer array representation.
2. The four two-dimensional moments are

$$\phi_{i,j}, \qquad \phi^x_{i_c,j}, \qquad \phi^y_{i,j_c}, \qquad \phi^{xy}_{i_c,j_c}.$$

Their geometric interpretations are:

$$\boxed{\phi_{i,j} : \text{point value at the node } (x_i, y_j)}$$

and its evolution is naturally associated with the *Finite Difference Method*.
The quantity

$$\boxed{\phi^x_{i_c,j} = \frac{1}{h_x}\int_{x_i}^{x_{i+1}} \phi(x, y_j)\, dx}$$

is a line average along an x-directed edge.
Likewise,

$$\boxed{\phi^y_{i,j_c} = \frac{1}{h_y}\int_{y_j}^{y_{j+1}} \phi(x_i, y)\, dy}$$

is a line average along a y-directed edge.
These quantities are naturally associated with the *Finite Line Method*.
Finally,

$$\boxed{\phi^{xy}_{i_c,j_c} = \frac{1}{h_x h_y}\int_{x_i}^{x_{i+1}}\int_{y_j}^{y_{j+1}} \phi(x, y)\, dy\, dx}$$

is an area- or cell-averaged quantity.
Geometrically it is a two-dimensional surface moment. Its conservative evolution is equivalent to a two-dimensional finite-volume update over the control cell. Thus it may be viewed as the two-dimensional counterpart of either the finite-surface or finite-volume level, depending on whether the emphasis is geometric or conservative.
In three dimensions, the hierarchy becomes

$$\boxed{\text{point} \longrightarrow \text{line} \longrightarrow \text{surface} \longrightarrow \text{volume}}$$

with the corresponding discretizations

$$\boxed{\text{FDM} \longrightarrow \text{FLM} \longrightarrow \text{FSM} \longrightarrow \text{FVM.}}$$

For example, ϕ^{xy}, ϕ^{xz}, and ϕ^{yz} become surface-averaged degrees of freedom, whereas

$$\phi^{xyz}$$

is the three-dimensional cell-volume average.

3. Define

$$\phi^y_{i,j_c} = \frac{1}{h_y}\int_{y_j}^{y_{j+1}} \phi(x_i, y, t)\, dy.$$

Integrating

$$\phi_t + (u\phi)_x + (v\phi)_y = K(\phi_{xx} + \phi_{yy})$$

over the interval $[y_j, y_{j+1}]$ gives

$$\frac{1}{h_y}\int_{y_j}^{y_{j+1}} \phi_t\, dy + \frac{1}{h_y}\int_{y_j}^{y_{j+1}} (u\phi)_x\, dy + \frac{1}{h_y}\int_{y_j}^{y_{j+1}} (v\phi)_y\, dy$$

$$= \frac{K}{h_y}\int_{y_j}^{y_{j+1}} \phi_{xx}\, dy + \frac{K}{h_y}\int_{y_j}^{y_{j+1}} \phi_{yy}\, dy.$$

Define the averaged flux

$$(u\phi)^y_{i,j_c} = \frac{1}{h_y}\int_{y_j}^{y_{j+1}} u\phi\, dy.$$

Then

$$\boxed{\frac{\partial \phi^{y}_{i,j_c}}{\partial t} + \frac{\partial (u\phi)^{y}_{i,j_c}}{\partial x} + \frac{(v\phi)_{i,j+1} - (v\phi)_{i,j}}{h_y}}$$

$$\boxed{= K \left[\frac{\partial^2 \phi^{y}_{i,j_c}}{\partial x^2} + \frac{\left(\phi_y\right)_{i,j+1} - \left(\phi_y\right)_{i,j}}{h_y} \right].}$$

Thus the x-direction remains in differential form because no integration has been performed in x, whereas the y-direction becomes a difference of boundary fluxes.
Similarly, defining

$$\phi^{x}_{i_c,j} = \frac{1}{h_x} \int_{x_i}^{x_{i+1}} \phi(x, y_j, t)\, dx,$$

we obtain

$$\boxed{\frac{\partial \phi^{x}_{i_c,j}}{\partial t} + \frac{(u\phi)_{i+1,j} - (u\phi)_{i,j}}{h_x} + \frac{\partial (v\phi)^{x}_{i_c,j}}{\partial y}}$$

$$\boxed{= K \left[\frac{(\phi_x)_{i+1,j} - (\phi_x)_{i,j}}{h_x} + \frac{\partial^2 \phi^{x}_{i_c,j}}{\partial y^2} \right].}$$

The two equations are therefore complementary: integration in one coordinate converts that coordinate's derivatives into boundary flux differences while leaving derivatives in the transverse direction.

4. The eight FDM-like derivative evaluations are

$$\phi_x, \qquad \phi_y, \qquad \phi_{xx}, \qquad \phi_{yy},$$

and

$$(\phi^x)_y, \qquad (\phi^y)_x, \qquad (\phi^x)_{yy}, \qquad (\phi^y)_{xx}.$$

Other quantities required by the integral equations occur naturally at the boundaries of the corresponding geometric entities.
For example, a cell-average update requires fluxes through its edges. In the multi-moment method, the edge or line moments are already independent evolving degrees of freedom. Likewise, an edge update requires information at its endpoints, which is supplied directly by the independently evolved point values.
Thus the hierarchy is

$$\boxed{\text{point values} \longrightarrow \text{line fluxes} \longrightarrow \text{cell fluxes.}}$$

In a conventional high-order finite-volume method, the cell average alone does not provide the pointwise interface state. A reconstruction or deconvolution must therefore be performed:

$$\overline{\phi} \longrightarrow \phi_{\text{interface}}.$$

In the multi-moment method, the required interface quantity is already stored:

$$\boxed{\phi_{\text{interface}} \text{ is an evolved degree of freedom.}}$$

Therefore, the reconstruction step is eliminated.
In three dimensions, the same principle becomes

$$\boxed{\text{point} \longrightarrow \text{line} \longrightarrow \text{surface} \longrightarrow \text{volume.}}$$

A volume update obtains its surface fluxes from independently evolved surface moments; a surface update obtains its edge fluxes from line moments; and the line equations obtain endpoint information from point values. This reuse of information is one of the principal structural advantages of the multidimensional formulation.

5. Let

$$\xi = y - y_c.$$

Expand about the cell center:

$$u = u_0 + u_0'\xi + \frac{1}{2}u_0''\xi^2 + O(\xi^3),$$

and

$$\phi = \phi_0 + \phi_0'\xi + \frac{1}{2}\phi_0''\xi^2 + O(\xi^3).$$

Their product is

$$u\phi = u_0\phi_0 + (u_0\phi_0' + u_0'\phi_0)\xi + \left(\frac{1}{2}u_0\phi_0'' + u_0'\phi_0' + \frac{1}{2}u_0''\phi_0\right)\xi^2 + O(\xi^3).$$

Over the symmetric interval,

$$\overline{\xi} = 0, \qquad \overline{\xi^2} = \frac{h_y^2}{12}.$$

Therefore

$$\overline{u\phi} = u_0\phi_0 + \frac{h_y^2}{24}\left(u_0\phi_0'' + 2u_0'\phi_0' + u_0''\phi_0\right) + O(h_y^4).$$

Separately,

$$\overline{u} = u_0 + \frac{h_y^2}{24}u_0'' + O(h_y^4),$$

and

$$\overline{\phi} = \phi_0 + \frac{h_y^2}{24}\phi_0'' + O(h_y^4).$$

Hence

$$\overline{u}\,\overline{\phi} = u_0\phi_0 + \frac{h_y^2}{24}\left(u_0\phi_0'' + u_0''\phi_0\right) + O(h_y^4).$$

Subtracting gives

$$\boxed{\overline{u\phi} - \overline{u}\,\overline{\phi} = \frac{h_y^2}{12}u_0'\phi_0' + O(h_y^4).}$$

Therefore

$$\boxed{\overline{u\phi} = \overline{u}\,\overline{\phi} + \frac{h_y^2}{12}\frac{\partial u}{\partial y}\frac{\partial \phi}{\partial y} + O(h_y^4)}$$

and

$$\boxed{N_4 = \frac{h_y^2}{12}\frac{\partial u}{\partial y}\frac{\partial \phi}{\partial y}.}$$

Without N_4, the leading error is proportional to h_y^2. Consequently,

$$\boxed{\overline{u}\,\overline{\phi} = \overline{u\phi} + O(h_y^2),}$$

so the nonlinear flux remains only second-order accurate regardless of the formal order of the linear differentiation scheme.

6. The uncorrected product contains an error expansion of the form

$$\overline{u\phi} - \overline{u}\,\overline{\phi} = C_2 h^2 + C_4 h^4 + C_6 h^6 + \cdots .$$

The correction N_4 removes the leading h^2 contribution:

$$\overline{u\phi} = \overline{u}\,\overline{\phi} + N_4 + O(h^4).$$

To obtain sixth-order accuracy, the next h^4 contribution must also be removed. Hence

$$\boxed{\overline{u\phi} = \overline{u}\,\overline{\phi} + N_4 + N_6 + O(h^6).}$$

Thus NC6 necessarily contains both corrections:

$$\boxed{\text{NC6} = N_4 + N_6.}$$

A high-order linear discretization such as EX8 or EX10 may still perform very well with NC4 over a practically relevant range because the coefficient multiplying the nonlinear-correction error may be small. The linear spatial error can therefore remain dominant over a wide range of grid resolutions.
As the grid is refined further, however, the high-order linear error decreases more rapidly than the fourth-order nonlinear-flux error. Eventually,

$$\boxed{\text{NC4 becomes the order-limiting contribution.}}$$

The manuscript observes this behavior precisely: NC4 remains effective over a substantial practical range, while NC6 becomes necessary when very small target errors are sought.
A direct Gauss-quadrature treatment requires interpolation to multiple quadrature points and evaluation of the nonlinear product there. If the polynomial order is p, the computational work scales approximately as

$$\boxed{O(p^2).}$$

The correction approach instead evaluates a relatively small number of derivative-based terms. Consequently, its additional cost grows much more slowly with formal order, making it particularly attractive for very-high-order multi-moment schemes.

7. The costs are

$$C_{\text{MM}} = (2M + 11)N$$

and

$$C_{\text{FD}} = (2M + 5)(2N).$$

Therefore

$$S_{\text{MM/FD}} = \frac{C_{\text{FD}}}{C_{\text{MM}}} = \frac{2N(2M + 5)}{N(2M + 11)}.$$

Hence

$$\boxed{S_{\text{MM/FD}} = 2\frac{2M + 5}{2M + 11}.}$$

For an explicit approximation,

$$M = 2P - 1.$$

Thus

$$S(P) = 2\frac{4P+3}{4P+9}.$$

For $P = 4$,

$$S = 2\frac{19}{25} = \boxed{1.520}.$$

For $P = 6$,

$$S = 2\frac{27}{33} = \boxed{1.636}.$$

For $P = 8$,

$$S = 2\frac{35}{41} = \boxed{1.707}.$$

For $P = 10$,

$$S = 2\frac{43}{49} = \boxed{1.755}.$$

Finally,

$$\boxed{\lim_{P\to\infty} S_{\mathrm{MM/FD}} = 2.}$$

This result represents an equal-order, per-time-step arithmetic-cost comparison. It does not include the potentially much larger advantage arising from higher resolving efficiency. In multiple dimensions, a more accurate method may require far fewer grid points in each direction. This reduces not only arithmetic work but also:

total memory,	memory traffic,	communication,	and number of time steps.

Consequently, multidimensional wall-clock performance cannot be predicted from operation counts alone.

8. For a fixed physical domain, the number of spatial degrees of freedom scales approximately as

$$N^d.$$

If

$$r = \frac{N_R}{N_A},$$

then the ratio of spatial degrees of freedom is

$$\frac{N_R^d}{N_A^d} = \boxed{r^d}.$$

Since

$$h \propto \frac{1}{N},$$

method A uses a grid spacing approximately r times larger:

$$h_A \approx r h_R.$$

If the explicit stable time step obeys

$$\Delta t \propto h,$$

then

$$\frac{\Delta t_A}{\Delta t_R} \approx r.$$

Hence, method A requires approximately r times fewer time steps. Combining spatial and temporal savings gives

$$\boxed{\text{work saving} \sim r^{d+1}.}$$

For $r = 2$:
In one dimension,

$$\boxed{r^{1+1} = 2^2 = 4.}$$

In two dimensions,

$$\boxed{r^{2+1} = 2^3 = 8.}$$

In three dimensions,

$$\boxed{r^{3+1} = 2^4 = 16.}$$

The corresponding spatial savings alone are

$$2, \qquad 4, \qquad 8,$$

respectively.
Thus a factor of only two reduction in the required resolution per direction can, in principle, reduce total explicit three-dimensional work by a factor of

$$\boxed{16}.$$

This dimensional amplification is one of the principal motivations for high-resolution numerical methods.

9. The velocity field is

$$u = -2\pi y, \qquad v = 2\pi x.$$

Therefore

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}.$$

Since

$$\frac{\partial u}{\partial x} = 0, \qquad \frac{\partial v}{\partial y} = 0,$$

we obtain

$$\boxed{\nabla \cdot \mathbf{u} = 0.}$$

The angular position is

$$\theta = 2\pi t.$$

At

$$t = \frac{1}{4}, \qquad \theta = \frac{\pi}{2},$$

so

$$\boxed{(x_c, y_c) = (-y_0, x_0).}$$

At

$$t = \frac{1}{2}, \qquad \theta = \pi,$$

$$\boxed{(x_c, y_c) = (-x_0, -y_0).}$$

At

$$t = \frac{3}{4}, \qquad \theta = \frac{3\pi}{2},$$

$$\boxed{(x_c, y_c) = (y_0, -x_0).}$$

At

$$t = 1, \qquad \theta = 2\pi,$$

$$\boxed{(x_c, y_c) = (x_0, y_0).}$$

Thus one unit of time corresponds to one complete revolution.
Because $K = 0$ and the velocity field represents incompressible solid-body rotation, the exact solution undergoes no deformation. Consequently, after one revolution,

$$\boxed{\text{shape is unchanged,}}$$

$$\boxed{\text{peak amplitude is unchanged,}}$$

and

$$\boxed{\int_\Omega \phi \, d\Omega \text{ is conserved.}}$$

Any change in width, amplitude, or shape is therefore purely numerical.
The problem is particularly useful because the velocity varies spatially even though the motion is exactly known. Thus the numerical method must correctly evaluate fluxes such as

$$u(x, y)\phi \qquad \text{and} \qquad v(x, y)\phi,$$

making it an effective verification case for multidimensional variable-coefficient transport and the nonlinear-flux correction.

10. The ratio of grid resolutions is

$$r_N = \frac{N_{\mathrm{FDEX8}}}{N_{\mathrm{EX8}}} = \frac{584}{272}.$$

Therefore

$$\boxed{r_N \approx 2.147.}$$

The corresponding ratio of two-dimensional cell counts is

$$r_{\mathrm{cell}} = \left(\frac{584}{272}\right)^2.$$

Hence

$$\boxed{r_{\mathrm{cell}} \approx 4.61.}$$

Thus, FDEX8 requires approximately 4.6 times as many geometric cells as EX8 to reach the specified error threshold.
The measured CPU-time speedup is

$$S_t = \frac{t_{\mathrm{FDEX8}}}{t_{\mathrm{EX8}}} = \frac{5.52}{3.89},$$

so

$$\boxed{S_t \approx 1.42.}$$

The reduction in geometric cell count does not translate directly into the same wall-clock speedup because a multi-moment cell contains more degrees of freedom and requires more local work than a single-moment finite-difference cell.
The total cost is influenced by:

a. the additional PV, line-average, and cell-average variables;
b. the higher arithmetic cost of the high-order derivative operators;
c. nonlinear-flux corrections;
d. the number of Runge–Kutta stages;
e. the stable time-step restriction;
f. memory access and cache behavior; and
g. implementation-specific vectorization and data movement.

For the normalized CFL used in the chapter, the MM formulation also uses a smaller time step per geometric cell because it contains a higher information density. Consequently, fewer cells alone do not determine the final runtime.
The frontogenesis problem is substantially more demanding than solid-body Gaussian rotation. In solid-body rotation, the velocity field transports the Gaussian without changing its shape. The exact solution therefore retains approximately the same range of spatial scales throughout the calculation.
Frontogenesis, in contrast, continuously stretches and folds the scalar field. The solution develops progressively thinner layers and tightly wound spiral filaments. Thus its characteristic wavenumbers increase continuously with time.
The test consequently probes the ability of the numerical method to retain

$$\boxed{\text{high-wavenumber information,}}$$

while controlling

$$\boxed{\text{dispersion,} \quad \text{numerical diffusion,} \quad \text{and nonlinear-flux error.}}$$

The formation of multiple thin scalar filaments therefore provides a much more stringent multidimensional test of resolving power than the rigid rotation of an undeformed Gaussian profile.

Chapter 6
Outlook and Future Research Directions

"The limits of today's methods are the starting points of tomorrow's discoveries."
Arpiruk Hokpunna

The objective of this book has been to introduce the multi-moment method from a fundamental point of view and to demonstrate how multiple types of discrete information can be evolved simultaneously to construct accurate and compact numerical approximations. Beginning with point values and cell averages in one dimension, the formulation has been extended to convection, diffusion, convection–diffusion systems, coupled Fourier analysis, boundary treatment, and multidimensional discretization.

The results presented in the preceding chapters suggest that the multi-moment framework possesses several attractive properties. It provides high-order spatial resolution using relatively compact physical stencils, permits different discretizations to share information directly, and can reduce the computational cost associated with repeated reconstruction or deconvolution. At the same time, the framework introduces new questions that do not arise, or arise differently, in conventional single-moment discretizations.

In particular, the simultaneous evolution of several numerical moments creates a coupled dynamical system whose behavior must be understood not only through the truncation error of each individual approximation, but also through the interaction among the moments. Likewise, extension to nonlinear systems, irregular geometries, complex boundary conditions, and large-scale three-dimensional simulations requires additional developments beyond the foundational formulations described in this book.

This chapter discusses several possible directions for future research. These directions should not be interpreted as a complete research program. Rather, they identify problems that arise naturally from the numerical structure developed in the preceding chapters and indicate where the multi-moment concept may be extended, improved, or better understood.

6.1 Extension to General Systems of Conservation Laws

Most of the fundamental development in this book has been presented using the scalar convection–diffusion equation,

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\mathbf{u}\phi) = \nabla \cdot (K \nabla \phi),$$

because this equation contains the principal numerical mechanisms of convection and diffusion while remaining sufficiently simple for detailed analysis.

A natural next step is the systematic extension of the multi-moment framework to systems of conservation laws,

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}(\mathbf{U}) = \nabla \cdot \mathbf{F}_v(\mathbf{U}, \nabla \mathbf{U}) + \mathbf{S},$$

where $\mathbf{U}$ denotes a vector of conserved variables.

The compressible Euler and Navier–Stokes equations are particularly important targets. In such systems, the pointwise and integral moments would no longer describe a single scalar field but several coupled physical variables. The interaction among density, momentum, total energy, and possibly additional species introduces both mathematical and numerical complications.

Several questions arise immediately:

1. Which set of moments provides the most efficient representation for each conserved variable?
2. Should all governing variables employ identical moment sets, or should different physical equations use different combinations of moments?
3. How should characteristic decomposition be incorporated into multi-moment reconstruction?
4. How should positivity of density, pressure, temperature, or species concentration be guaranteed?
5. Can conservation, high-order accuracy, and compactness be maintained simultaneously in strongly nonlinear regimes?

The multi-discretization interpretation developed in this book provides one possible framework for addressing these questions. Each geometric moment can be associated with its own integral or differential evolution equation, while information required by one discretization may be obtained directly from the other evolved moments.

An important research objective is therefore to determine whether the efficiency advantage observed for scalar transport remains equally significant for realistic systems of conservation laws.

6.2 High-Order Treatment of Nonlinear Fluxes

One of the central challenges in extending the method to nonlinear problems is the evaluation of averaged nonlinear fluxes. In general,

$$\overline{u\phi} \neq \overline{u}\,\overline{\phi}.$$

As discussed previously, the difference can be corrected systematically. For example,

$$\overline{u\phi} = \overline{u}\,\overline{\phi} + N_4 + N_6 + O(h^6),$$

where N_4 and N_6 denote fourth- and sixth-order nonlinear corrections.

The numerical experiments presented in this book show an important practical feature. A nonlinear correction having lower formal order than the linear spatial approximation may remain sufficiently accurate over a substantial range of practically useful resolutions because the coefficient of the nonlinear error can be small. Eventually, however, the nonlinear-flux error becomes dominant and prevents the complete scheme from reaching the formal accuracy of the underlying linear discretization.

This observation motivates several research directions. *First*, inexpensive correction formulas beyond sixth order should be derived. A general correction of order $2p$ could potentially be constructed in the form

$$\overline{fg} = \overline{f}\,\overline{g} + \sum_{m=1}^{p-1} N_{2m+2} + O(h^{2p}),$$

where the required derivative information is obtained as far as possible from moments that are already available in the numerical solution. *Second*, it is important to establish when high-order nonlinear correction is actually necessary. In practical turbulent simulations, truncation and dispersion errors associated with unresolved high-frequency components may dominate the error introduced by the nonlinear flux. Thus, computing a formally tenth- or twelfth-order nonlinear flux may provide little benefit if the solution itself is not sufficiently resolved.

A useful goal would therefore be to establish quantitative criteria of the form

$$E_{\text{nonlinear}} < CE_{\text{linear}},$$

which could determine the lowest nonlinear-correction order automatically necessary for a specified simulation. Such a strategy could lead to an adaptive nonlinear-flux treatment in which higher-order corrections are activated only when they provide measurable benefit.

6.3 Discontinuities, Limiting, and Shock-Capturing

The high-order methods developed in this book are particularly effective for smooth or moderately resolved solutions. However, many important conservation laws admit shocks, contact discontinuities, material interfaces, or other non-smooth structures. According to Godunov's theorem, a linear monotonicity-preserving scheme cannot achieve an order higher than one. High-order methods must therefore modify their behavior in the vicinity of discontinuities.

The availability of several local moments presents both an opportunity and a challenge. A multi-moment cell contains more information about the local solution than a conventional single-moment cell. This information might be used to distinguish between a genuinely smooth high-frequency structure and a non-smooth discontinuity.

A possible future formulation could have the structure

$$D^{\text{MM}}_{\text{hybrid}} = (1-\chi)D^{\text{high}} + \chi D^{\text{robust}},$$

where χ is a smoothness or discontinuity sensor. Important questions include:

- How should a smoothness sensor be defined when point values and integral moments coexist?
- If one moment is limited, how should the remaining moments be modified so that they remain mutually consistent?
- Can limiting preserve the conservation property of the cell-average equation?
- Can a high-order upwind multi-moment formulation be combined with TVD, ENO, or WENO-type nonlinear reconstruction?
- Can positivity and entropy stability be enforced without destroying the compactness advantage?

The development of a robust, conservative, and moment-consistent shock-capturing multi-moment scheme is therefore an important direction for applications to compressible flow.

6.4 Time Integration for Very-High-Order Spatial Schemes

The analysis in Chapter 2 demonstrates that increasing spatial resolution also increases the magnitude of the largest resolvable numerical wavenumber. The semi-discrete spectrum therefore expands, and the maximum stable time step of an explicit time integrator becomes increasingly restrictive.

For convection,

$$z = -\mathbf{i}uk^*\Delta t,$$

while for diffusion,

$$z = -Kk^{2*}\Delta t.$$

Consequently, improving the spatial operator alone does not guarantee an efficient fully discrete method. Another issue is the accuracy. Once the spatial error becomes extremely small, the temporal integration error may become the dominant contribution. A twelfth-order spatial discretization combined with a conventional fourth-order Runge–Kutta method may therefore behave as a fourth-order fully discrete scheme when the time step is refined together with the grid. This suggests several directions for future work:

1. optimized explicit Runge–Kutta schemes whose stability regions are tailored to the coupled multi-moment spectrum;
2. high-order low-storage Runge–Kutta schemes for large-scale simulations;
3. implicit–explicit methods in which convection remains explicit while diffusion is treated implicitly;
4. exponential or integrating-factor methods for strongly diffusive systems;
5. local time stepping in locally refined meshes;
6. space–time formulations in which temporal and spatial approximation orders are designed together.

Rather than treating the time integrator as an independent numerical component, future multi-moment solvers may benefit from a co-designed spatial–temporal discretization.

6.5 Generalization of Coupled Fourier Analysis

A distinctive feature of the multi-moment formulation is that a single physical Fourier mode can excite multiple numerical eigenmodes. The coupled modified wavenumber analysis introduced in this book provides a framework for understanding this behavior. For convection, the interaction of two numerical branches produces a bounded modulation. For diffusion, the secondary mode typically decays faster, and the solution relaxes toward the physical branch.

The analysis can be extended substantially. For a general system containing m numerical moments, the Fourier evolution may be written as

$$\frac{d\widehat{\boldsymbol{\phi}}}{dt} = \mathbf{A}(kh)\widehat{\boldsymbol{\phi}},$$

where

$$\widehat{\boldsymbol{\phi}} = \left[\widehat{\phi}^{(1)}\ \widehat{\phi}^{(2)}\ \cdots\ \widehat{\phi}^{(m)}\right]^T.$$

All eigenvalues then govern the numerical behavior

$$\lambda_1, \lambda_2, \ldots, \lambda_m$$

and their corresponding eigenvectors.

Several issues deserve investigation:

- identification of physical and parasitic branches for systems with many moments;
- extension of CMWA to multiple spatial dimensions;
- analysis of anisotropy with respect to propagation angle;
- non-normal amplification when the eigenvectors are not orthogonal;
- energy exchange among different moments;
- relationship between eigenmode structure and nonlinear stability.

For multidimensional problems, the transfer function should depend not only on the magnitude of the wavenumber but also on its orientation,

$$\mathbf{k} = (k_x, k_y, k_z).$$

A multidimensional coupled spectral analysis could therefore provide an important tool for designing isotropic high-order multi-moment schemes.

6.6 Energy, Entropy, and Structure-Preserving Formulations

The methods presented in this book are constructed primarily from consistency, accuracy, conservation, and spectral-resolution considerations. For more complex physical systems, additional mathematical structure may need to be preserved.

For example, an inviscid flow may satisfy an energy or entropy balance. A discrete method that violates this structure may accumulate nonphysical energy at unresolved scales even if its local truncation error is small.

An important question is therefore whether the coupled multi-moment operator can be designed to satisfy a weighted discrete identity of the form

$$\mathbf{WA} + \mathbf{A}^T\mathbf{W} = 0$$

for nondissipative problems, where $\mathbf{W}$ defines an appropriate multi-moment inner product.

For nonlinear conservation laws, the analogous goal may involve a discrete entropy inequality,

$$\frac{d}{dt}\mathcal{E}_h \leq 0.$$

Developing such structure-preserving formulations could provide a more rigorous foundation for long-time stability and may be especially important in under-resolved turbulent-flow simulations.

6.7 Boundary Conditions Compatible with Multiple Moments

Boundary treatment remains one of the most important open problems.

Chapter 5 distinguishes two separate questions:

1. How should the numerical moments be reconstructed consistently at a boundary?

2. Which physical information should be permitted to enter or leave the computational domain?

The moment-consistent Simpson reconstruction addresses the first question, whereas characteristic, Orlanski–Sommerfeld, NSCBC, and Giles-type conditions address the second. A complete high-order multi-moment boundary condition should satisfy both.

For a general boundary operator,

$$\mathcal{B}\left(\mathbf{U}, \frac{\partial \mathbf{U}}{\partial n}, \frac{\partial \mathbf{U}}{\partial s}, \ldots\right) = 0,$$

the point, line, surface, and volume moments adjacent to the boundary must remain mutually consistent while incoming and outgoing characteristic information is treated correctly.

Several research directions follow:

- moment-consistent characteristic boundary conditions for nonlinear systems;
- multidimensional non-reflecting treatment for oblique waves;
- high-order wall boundaries for viscous flows;
- interface conditions between different materials or numerical discretizations;
- boundary closures for strongly stretched and curvilinear grids.

The multi-moment representation may actually provide an advantage here because normal and tangential derivative information can be approximated from different geometric moments. Determining how best to exploit this information remains an open problem.

6.8 Extension to Curvilinear Grids

The multi-moment formulations developed in this book have primarily been constructed on uniform Cartesian grids. While these grids provide a convenient setting for deriving high-order approximations and investigating their numerical properties, practical engineering applications frequently involve curved boundaries and nonuniform spatial distributions. Extending the multi-moment method to curvilinear grids is therefore an important step toward its application to realistic computational geometries.

A structured curvilinear grid can be generated through a coordinate transformation from the physical domain (x, y) to a computational domain (ξ, η),

$$x = x(\xi, \eta), \qquad y = y(\xi, \eta). \tag{6.1}$$

The computational domain can then be discretized using a regular Cartesian arrangement, while the corresponding physical grid follows the geometry of the problem. This transformation preserves the structured connectivity of the grid but introduces geometric metric terms into the governing equations.

For example, consider the two-dimensional conservation law

$$\frac{\partial \phi}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0, \tag{6.2}$$

where F and G represent the physical fluxes in the x- and y-directions, respectively. Under a time-independent coordinate transformation, the conservative form becomes

$$\frac{\partial (J\phi)}{\partial t} + \frac{\partial \widetilde{F}}{\partial \xi} + \frac{\partial \widetilde{G}}{\partial \eta} = 0, \tag{6.3}$$

where

$$J = x_\xi y_\eta - x_\eta y_\xi \tag{6.4}$$

is the Jacobian determinant, and the transformed fluxes are defined by

$$\begin{aligned} \widetilde{F} &= y_\eta F - x_\eta G, \\ \widetilde{G} &= -y_\xi F + x_\xi G. \end{aligned} \tag{6.5}$$

Here, subscripts denote partial differentiation with respect to the computational coordinates. A smooth, one-to-one mapping with a positive Jacobian is assumed.

From the perspective of the multi-moment method, the principal advantage of this transformation is that the original structured arrangement of discrete moments can be retained in computational space. However, their physical interpretation must account for the geometric transformation.

For example, the physical volume-average in two dimensions is defined by

$$\overline{\phi}_{ij} = \frac{1}{V_{ij}} \int_{\Omega_{ij}} \phi \, d\Omega = \frac{1}{V_{ij}} \int_{\widehat{\Omega}_{ij}} J\phi \, d\xi \, d\eta, \tag{6.6}$$

where

$$V_{ij} = \int_{\widehat{\Omega}_{ij}} J \, d\xi \, d\eta. \tag{6.7}$$

Consequently, the physical cell average is generally not identical to an unweighted average over the computational cell.

Likewise, the line-averaged moments associated with the physical grid edges must incorporate the corresponding arc-length metric. For an edge of constant ξ, for example,

$$\overline{\phi}^e = \frac{1}{L_e} \int_{\eta_1}^{\eta_2} \phi \sqrt{x_\eta^2 + y_\eta^2} \, d\eta, \tag{6.8}$$

where L_e is the physical length of the edge. The pointwise moments, in contrast, remain evaluations of the solution at the physical grid nodes.

An important issue is the consistency between the transformed governing equations and these different moment definitions. In particular, the geometric metric terms must satisfy appropriate discrete metric identities so that a spatially uniform solution remains uniform under numerical evolution. For stationary grids, the continuous metric identities include

$$\begin{aligned} \frac{\partial y_\eta}{\partial \xi} - \frac{\partial y_\xi}{\partial \eta} &= 0, \\ \frac{\partial x_\eta}{\partial \xi} - \frac{\partial x_\xi}{\partial \eta} &= 0. \end{aligned} \tag{6.9}$$

Their discrete counterparts are essential for free-stream preservation.

The extension therefore involves more than transforming the Cartesian approximation coefficients. Each moment equation must retain its appropriate conservative or pointwise interpretation, while the metric quantities and reconstruction procedures remain mutually consistent.

Several questions deserve further investigation:

- Construction of high-order multi-moment approximations on stretched and nonuniform grids.
- Consistent evaluation of metric terms and Jacobians for different types of moments.
- Preservation of uniform solutions and geometric conservation properties.
- Influence of grid stretching, curvature, and skewness on numerical dispersion, dissipation, and stability.
- Development of high-order boundary closures for curved physical boundaries.

The Taylor Table Method introduced in Chapter 1 provides a possible foundation for constructing geometry-dependent approximation coefficients. However, extending this procedure to a coupled multi-moment system requires additional constraints to preserve consistency among the different moment equations.

A successful curvilinear formulation would retain the compactness and high-order resolution of the Cartesian multi-moment method while substantially expanding its applicability to practical geometries.

6.9 Extension to Unstructured Grids

The extension of multi-moment methods to unstructured grids is an important direction toward the simulation of flows in complex geometries. Unlike Cartesian and structured curvilinear grids, unstructured meshes allow greater flexibility in the arrangement and connectivity of computational elements. However, constructing high-order approximations on arbitrary meshes is considerably more difficult because the neighboring discrete data are not distributed in a regular geometric pattern.

Significant progress in this direction has already been made by Xie et al. [37], who developed a multi-moment finite-volume method for incompressible Navier–Stokes equations on triangular and tetrahedral meshes. Their formulation, referred to as the volume-average/point-value (VPM) method, employs two types of prognostic variables: the volume-integrated average (VIA) within each computational cell and the point value (PV) at each mesh vertex.

A particularly important feature of this work is that the two types of moments are evolved through different forms of the governing equations. The cell-average moment is updated through the finite-volume integral formulation, whereas the pointwise moment is updated through the differential form. This construction is closely related to the multi-discretization interpretation of the multi-moment method adopted throughout this book, in which different types of discrete information are treated as independent dependent variables and coupled through numerical approximations.

To illustrate this connection, consider the scalar convection–diffusion equation in conservative form,

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\mathbf{u}\phi) = K\nabla^2 \phi, \tag{6.10}$$

where $\mathbf{u}$ is a prescribed divergence-free velocity field and K is a constant diffusivity.

For an arbitrary computational cell Ω_i, the cell-average moment is defined as

$$\overline{\phi}_i = \frac{1}{|\Omega_i|} \int_{\Omega_i} \phi \, d\Omega. \tag{6.11}$$

Its evolution equation follows directly from integration over the control volume,

$$\begin{aligned}\frac{d\overline{\phi}_i}{dt} = &- \frac{1}{|\Omega_i|}\int_{\partial\Omega_i} \phi\mathbf{u}\cdot\mathbf{n}\,dS \\ &+ \frac{K}{|\Omega_i|}\int_{\partial\Omega_i} \nabla\phi\cdot\mathbf{n}\,dS.\end{aligned} \tag{6.12}$$

The pointwise moment at vertex a is defined by

$$\phi_a = \phi(\mathbf{x}_a), \tag{6.13}$$

and is governed by the differential equation

$$\frac{d\phi_a}{dt} = -\mathbf{u}_a\cdot(\nabla\phi)_a + K(\nabla^2\phi)_a. \tag{6.14}$$

These two equations represent different discretizations of the same physical conservation law. Their coupling provides additional information for constructing high-order approximations without requiring the wide reconstruction stencils commonly encountered in conventional high-order finite-volume methods.

In the VPM method of Xie et al. [37], a local polynomial is reconstructed using the cell average, the point values at the cell vertices, and approximations of the first derivatives at the cell center. For a triangular element, the reconstruction is expressed in local coordinates (ξ,η) as

$$\begin{aligned}P_i(\xi,\eta) = &\, a_{00} + a_{10}\xi + a_{01}\eta \\ &+ a_{11}\xi\eta + a_{20}\xi^2 + a_{02}\eta^2.\end{aligned} \tag{6.15}$$

The six coefficients are determined from six constraints: three vertex values, one cell average, and two first derivatives at the cell centroid. The centroid derivatives are auxiliary quantities reconstructed from the moments of neighboring cells, rather than additional prognostic variables.

This approach provides a compact reconstruction for triangular elements and can be extended to tetrahedral meshes in three dimensions. Numerical experiments reported by Xie et al. [37] demonstrated approximately third-order convergence for smooth advection problems and improved robustness against mesh distortion compared with conventional finite-volume schemes.

The subsequent work of Xie and Xiao [38] extended the multi-moment concept to arbitrary unstructured hybrid grids through the multi-moment constrained finite-volume method, referred to as MCV-SPCV. The method supports triangular and quadrilateral elements in two dimensions and tetrahedral, hexahedral, prismatic, and pyramidal elements in three dimensions.

An important distinction between the VPM and MCV-SPCV formulations concerns the choice of prognostic variables. While VPM independently evolves both VIA and PV, MCV-SPCV evolves point values located at the cell vertices and cell centroid. The finite-volume conservation equation is imposed as a constraint to determine the evolution of the centroid point value.

For example, on a triangular element, the reconstruction used by Xie and Xiao [38] gives the following relation between the cell average and the four point values:

$$\overline{\phi}_i = \frac{1}{12}\left(\phi_{i1} + \phi_{i2} + \phi_{i3} + 9\phi_{ic}\right), \tag{6.16}$$

where ϕ_{i1}, ϕ_{i2}, and ϕ_{i3} represent the vertex values and ϕ_{ic} denotes the point value at the cell centroid.

Differentiating this relation with respect to time yields

$$\frac{d\phi_{ic}}{dt} = \frac{4}{3}\frac{d\overline{\phi}_i}{dt} - \frac{1}{9}\sum_{a=1}^{3}\frac{d\phi_{ia}}{dt}. \tag{6.17}$$

Thus, although the cell average is not stored as an independent prognostic variable, its conservative evolution equation determines the evolution of the centroid point value.

This example illustrates an alternative interpretation of the multi-moment concept. Rather than independently evolving every moment, one may use selected integral moments as constraints on the evolution of a chosen set of pointwise dependent variables. Both approaches exploit the information contained in multiple types of discrete quantities, but differ in their choice of independent variables and in the construction of the discrete evolution equations.

The MCV-SPCV formulation also incorporates local reconstructions and least-squares approximations to improve the evaluation of derivatives on irregular meshes. Xie and Xiao [38] reported approximately third-order convergence for smooth advection and demonstrated improvements in accuracy and robustness over the earlier VPM formulation. Their numerical results also showed reduced numerical dissipation, which is particularly important for simulations of high-Reynolds-number incompressible flows.

These developments establish that multi-moment methods are not restricted to structured grids. Nevertheless, several important research questions remain, especially for extending the very-high-order multi-discretization formulations developed in this book.

First, the existing VPM and MCV-SPCV formulations primarily demonstrate accuracy near third order. Extending the present fourth-, eighth-, and even higher-order multi-moment approximations to unstructured grids requires the development of new geometric reconstruction and differentiation operators. The regular Taylor-series relationships used on Cartesian grids can no longer be applied directly because the spatial arrangement of neighboring moments varies from cell to cell.

Second, the extension need not be restricted to cell averages and vertex values. An unstructured mesh naturally contains a hierarchy of geometric entities consisting of vertices, edges, faces, and cells. In principle, additional moments may be defined on these entities, including line-averaged and surface-averaged quantities. This opens the possibility of coupling the finite-volume, finite-surface, finite-line, and pointwise differential formulations within a common unstructured framework.

However, the introduction of additional moments raises questions concerning the independence of the discrete equations, consistency between shared geometric entities, and the conditioning of high-order reconstruction operators. These issues become particularly important on hybrid meshes containing elements of different shapes and sizes.

Third, the stability properties of the coupled multi-moment system on unstructured grids require further investigation. The coupled Fourier analysis developed in this book relies on the translational invariance of a uniform grid. Such an assumption is generally unavailable on arbitrary unstructured meshes. Alternative approaches based on eigenvalue analysis of the assembled discrete operators, local linearization, and energy estimates may provide a suitable foundation for studying numerical dispersion, dissipation, and stability.

Finally, the computational efficiency of very-high-order multi-moment methods on unstructured grids remains an important question. Although the use of multiple moments increases the number of dependent variables, it may reduce the spatial extent of the re-

construction stencil. Whether this trade-off leads to superior performance depends on the number of moments, reconstruction cost, mesh connectivity, memory access patterns, and communication requirements in parallel computations.

The works of Xie et al. [37] and Xie and Xiao [38] therefore provide a substantial foundation for extending multi-moment methods to complex geometries. The remaining challenge is not simply to demonstrate that the method can operate on unstructured grids, but to determine whether the very-high-order accuracy, compactness, and computational advantages observed on Cartesian grids can be systematically preserved on general unstructured and hybrid meshes.

6.10 Three-Dimensional Formulations and Complex Flow Physics

Chapter 6 shows that the multidimensional structure extends naturally from points to lines, surfaces, and volumes. In three dimensions, the complete geometric hierarchy becomes

$$\mathrm{PV} \rightarrow \mathrm{LV} \rightarrow \mathrm{SV} \rightarrow \mathrm{CV}.$$

An important advantage is that flux information required by a higher-dimensional moment can be supplied directly by a lower-dimensional moment. Thus, the cell average can obtain face fluxes from surface moments, surface moments can obtain edge information from line moments, and line moments can obtain endpoint information from point values.

The next major step is to evaluate this architecture in realistic three-dimensional problems.

Potential applications include:

- transitional and turbulent flows,
- thermal convection,
- scalar mixing at high Schmidt number,
- acoustics,
- reacting transport,
- atmospheric and environmental transport,
- multiphase transport.

For such problems, the principal question is not merely whether the formal order of convergence can be maintained. More important is whether the increased spectral resolution translates into a measurable reduction in total computational cost for a fixed engineering accuracy.

This should be evaluated using comparisons at equal error rather than equal grid size.

6.11 High-Performance Computing and Exascale Architectures

One of the motivations for explicit multi-moment schemes is their highly local data dependence.

A conventional high-order explicit finite-difference scheme increases stencil width as the order rises. A compact scheme reduces stencil width but introduces a global or directional linear solve. The multi-moment formulation offers a third possibility: increasing the information density locally while retaining an explicit update.

This structure is attractive for massively parallel systems.

However, floating-point operation counts alone are insufficient to predict performance on modern hardware. Computational efficiency increasingly depends on

- memory bandwidth,
- cache reuse,
- vectorization,
- GPU occupancy,
- communication volume,
- synchronization,
- load balancing.

Future studies should therefore evaluate the method using hardware-oriented metrics such as

FLOP/byte, communication/DoF, time-to-solution at fixed error.

The last metric is especially important. A method that costs more per cell may still be substantially faster if it requires fewer cells in each spatial direction.

If the required resolution is reduced by a factor r in each direction, the number of grid entities in d dimensions decreases approximately as

$$r^d.$$

For an explicit method whose allowable time step also increases proportionally with h, the total work reduction can approach

$$r^{d+1}.$$

This dimensional scaling makes high-resolution methods particularly attractive for three-dimensional simulations.

6.12 Multi-Moment Methods for Turbulent Flows

Turbulence is perhaps the most demanding target for the methodology.

A turbulent flow contains a broad range of dynamically interacting scales. The numerical method must resolve a substantial part of the spectrum while avoiding excessive dissipation, dispersion, and nonlinear aliasing.

The high spectral resolution of multi-moment methods suggests potential advantages in direct numerical simulation and large-eddy simulation. However, several issues must be addressed systematically:

1. energy conservation of the inviscid terms;
2. nonlinear aliasing;
3. interaction between physical and parasitic coupled modes;
4. behavior near the grid cut-off;
5. treatment of subgrid-scale models;
6. wall boundary conditions;
7. computational scaling in three dimensions.

Of particular interest is the relationship between nonlinear-flux correction and turbulent energy transfer.

The results presented in Chapter 6 indicate that the formal nonlinear correction may be much smaller than the dominant linear convective contribution over practical resolutions. Whether this remains true across Reynolds number, grid anisotropy, and different turbulent-flow configurations should be investigated systematically.

A useful benchmark would compare methods at a specified error in quantities such as

$$E(k), \qquad \langle u_i' u_j' \rangle, \qquad C_f,$$

rather than merely comparing formal convergence rates.

6.13 Automated Construction of Multi-Moment Schemes

The Taylor Table Method converts approximation design into a linear algebra problem. Given a set of supports,

$$\mathcal{S} = \{\phi_i, \overline{\phi}_{i+1/2}, \phi_{i+1}, \ldots\},$$

the coefficients can be generated automatically by enforcing moment conditions.

This opens the possibility of automated numerical-method generation.

A future software framework could accept:

- the desired derivative,
- the desired order,
- the available moment types,
- the local geometry,
- symmetry constraints,

and automatically produce the approximation coefficients, truncation error, spectral response, and stability characteristics.

For example,

support set $\longrightarrow$ Taylor matrix $\longrightarrow$ coefficients $\longrightarrow$ Fourier analysis $\longrightarrow$ optimized scheme.

Optimization could go beyond formal accuracy. The coefficients could be selected to minimize a spectral objective such as

$$J = \int_0^{k_{\max}} w(k)\, |k^*(k) - k|^2 \, dk,$$

subject to the required consistency constraints.

This would lead naturally to dispersion-relation-preserving multi-moment schemes.

6.14 Verification, Benchmarking, and Reproducibility

As the method develops, systematic benchmarking will become increasingly important. The examples in this book provide initial tests involving sine waves, the Runge function, Gaussian diffusion, rotating scalar distributions, and variable velocity fields. A broader benchmark suite should include problems designed to isolate different numerical mechanisms:

smooth convection	dispersion,
sharp convection	limiting and robustness,
diffusion	dissipation and stability,
vortex transport	multidimensional isotropy,
acoustics	boundary reflection,
turbulence	broadband spectral behavior.

Comparisons should ideally be conducted at equal

error, degrees of freedom, memory, and wall-clock time.

No single comparison is sufficient by itself. Providing reference implementations, coefficient-generation tools, and reproducible test cases would also make it easier for independent researchers to verify the claims and develop alternative formulations.

6.15 Toward a General Multi-Discretization Framework

The broader implication of the multi-moment concept may extend beyond the specific schemes presented in this book. Traditionally, numerical methods are classified according to the form of their primary discrete variable:

$$\begin{aligned} \text{FDM} &\rightarrow \text{point values}, \\ \text{FVM} &\rightarrow \text{cell averages}, \\ \text{FSM} &\rightarrow \text{surface averages}, \\ \text{FLM} &\rightarrow \text{line averages}. \end{aligned}$$

The multi-moment viewpoint suggests that these need not be mutually exclusive choices. They may instead be regarded as different projections of the same underlying physical field.

The more general question is therefore:

> Rather than asking which discretization should be used, can several discretizations be evolved together in such a way that each supplies the information that the others lack?

This is the central research question suggested by the framework. In such a view, a future numerical method may not be identified uniquely as a finite-difference, finite-volume, finite-element, or multi-moment method. Instead, it may be characterized by the collection of discrete representations it evolves and by the mechanism through which those representations exchange information. The multi-moment method described in this book represents one realization of this broader idea.

6.16 Concluding Perspective

The principal argument of this book is that numerical information is valuable not only in quantity but also in diversity. Point values and integral moments describe different aspects of the

same physical field. When evolved independently but coupled consistently, they can provide a local representation that is simultaneously compact, conservative, and highly resolved.

Several results presented throughout the book suggest that this approach is promising. At the same time, many important questions remain unresolved. Among the most immediate research priorities are

1. robust extension to nonlinear systems of conservation laws;
2. moment-consistent shock capturing and limiting;
3. higher-order nonlinear-flux treatment;
4. optimized time integration for broad coupled spectra;
5. multidimensional coupled Fourier and stability analysis;
6. characteristic-compatible high-order boundary conditions;
7. extension to nonuniform, curvilinear, and unstructured grids;
8. three-dimensional high-Reynolds-number simulations;
9. hardware-efficient implementations on massively parallel systems.

Longer-term developments may include adaptive selection of moments, automated scheme generation, structure-preserving formulations, and a more general theory of multi-discretization methods. The methods presented in this book should therefore be regarded not as a final formulation, but as a starting point. The essential idea is simple: different numerical representations of the same physical field need not compete. They can cooperate.

The extent to which this principle can be transformed into more accurate, robust, and computationally efficient numerical methods remains an open and potentially fruitful direction for future research.

Appendix A
Coefficients of Multi-Moment methods

This appendix collects the coefficient and leading truncation-error tables for the central explicit and compact multi-moment approximations of the first and second derivatives. The notation and defining equations follow the convection and diffusion chapters. Values are reproduced from the chapter tables; dashes retain their original meaning.

A.1 Explicit first-derivative coefficients

Order	β_1	β_2	β_3	β_4	β_5	β_6	γ_1	γ_2	γ_3	γ_4	γ_5	γ_6
4	2	—	—	—	—	—	$\frac{1}{2}$	—	—	—	—	—
6	$\frac{9}{4}$	$\frac{1}{36}$	—	—	—	—	$\frac{2}{3}$	—	—	—	—	—
8	$\frac{5}{2}$	$\frac{7}{54}$	—	—	—	—	$\frac{8}{9}$	$\frac{1}{36}$	—	—	—	—
10	$\frac{47}{18}$	$\frac{7}{36}$	$\frac{1}{900}$	—	—	—	1	$\frac{1}{20}$	—	—	—	—
12	$\frac{49}{18}$	$\frac{41}{144}$	$\frac{157}{18000}$	—	—	—	$\frac{9}{8}$	$\frac{9}{100}$	$\frac{1}{600}$	—	—	—
14	$\frac{401}{144}$	$\frac{1241}{3600}$	$\frac{301}{18000}$	$\frac{1}{19600}$	—	—	$\frac{6}{5}$	$\frac{3}{25}$	$\frac{2}{525}$	—	—	—
16	$\frac{205}{72}$	$\frac{3737}{9000}$	$\frac{281}{9000}$	$\frac{199}{343000}$	—	—	$\frac{32}{25}$	$\frac{4}{25}$	$\frac{32}{3675}$	$\frac{1}{9800}$	—	—
20	$\frac{5269}{1800}$	$\frac{2807}{5400}$	$\frac{38267}{617400}$	$\frac{157}{50400}$	$\frac{7633}{200037600}$	—	$\frac{25}{18}$	$\frac{100}{441}$	$\frac{25}{1176}$	$\frac{25}{31752}$	$\frac{1}{158760}$	—
22	$\frac{591}{200}$	$\frac{16477}{29400}$	$\frac{10601}{137200}$	$\frac{56431}{11113200}$	$\frac{1181}{11113200}$	$\frac{1}{7683984}$	$\frac{10}{7}$	$\frac{25}{98}$	$\frac{25}{882}$	$\frac{5}{3528}$	$\frac{1}{48510}$	—
24	$\frac{5369}{1800}$	$\frac{372767}{617400}$	$\frac{474511}{4939200}$	$\frac{1095797}{133358400}$	$\frac{38477}{133358400}$	$\frac{88331}{35500006080}$	$\frac{72}{49}$	$\frac{225}{784}$	$\frac{50}{1323}$	$\frac{1}{392}$	$\frac{2}{29645}$	$\frac{1}{2561328}$

Table A.1: Coefficients of the central explicit multi-moment approximation to the first derivative (Eq.(2.8)) .

A.2 Leading truncation errors for the explicit first derivative

Order	LTE	Order	LTE	Order	LTE
4	$-\frac{1}{360}h^4$	12	$-\frac{1}{33633600}h^{12}$	20	$-\frac{1}{2710264100544}h^{20}$
6	$\frac{1}{5040}h^6$	14	$\frac{1}{504504000}h^{14}$	22	$\frac{1}{41557382875008}h^{22}$
8	$-\frac{1}{113400}h^8$	16	$-\frac{1}{9648639000}h^{16}$	24	$-\frac{1}{750341635243200}h^{24}$
10	$\frac{1}{1663200}h^{10}$	18	$\frac{1}{146659312800}h^{18}$	—	—

Table A.2: Local Truncation Error (LTE) of the central explicit multi-moment approximation to the first derivative (Eq.(2.8)) .

A.3 Compact first-derivative coefficients

Order	a_1	a_2	a_3	a_4	b_1	b_2	b_3	b_4	c_1	c_2	c_3	c_4	$LTE_{COM/EX}$
6	$-\frac{1}{8}$	—	—	—	3	—	—	—	$-\frac{9}{8}$	—	—	—	$\frac{1}{4}$
8	$-\frac{1}{6}$	—	—	—	$\frac{13}{4}$	$-\frac{1}{108}$	—	—	$-\frac{23}{18}$	—	—	—	$\frac{1}{4}$
10	$-\frac{2}{9}$	—	—	—	$\frac{7}{2}$	$-\frac{1}{18}$	—	—	$-\frac{38}{27}$	$\frac{1}{108}$	—	—	$\frac{1}{9}$
12	$-\frac{8}{27}$	$\frac{1}{216}$	—	—	$\frac{15}{4}$	$-\frac{65}{324}$	—	—	$-\frac{40}{27}$	$\frac{25}{432}$	—	—	$\frac{4}{81}$
14	$-\frac{1}{3}$	$\frac{1}{120}$	—	—	$\frac{139}{36}$	$-\frac{7}{24}$	$\frac{1}{9000}$	—	$-\frac{3}{2}$	$\frac{109}{1200}$	—	—	$\frac{1}{36}$
16	$-\frac{3}{8}$	$\frac{3}{200}$	—	—	$\frac{143}{36}$	$-\frac{241}{576}$	$\frac{431}{360000}$	—	$-\frac{3}{2}$	$\frac{279}{2000}$	$-\frac{1}{6000}$	—	$\frac{1}{64}$
18	$-\frac{27}{64}$	$\frac{27}{1000}$	$-\frac{1}{8000}$	—	$\frac{49}{12}$	$-\frac{455}{768}$	$\frac{17549}{2400000}$	—	$-\frac{189}{128}$	$\frac{2079}{10000}$	$-\frac{147}{80000}$	—	$\frac{9}{1600}$
20	$-\frac{9}{20}$	$\frac{9}{250}$	$-\frac{1}{3500}$	—	$\frac{199}{48}$	$-\frac{2123}{3000}$	$\frac{1021}{75000}$	$-\frac{1}{686000}$	$-\frac{291}{200}$	$\frac{633}{2500}$	$-\frac{909}{245000}$	—	$\frac{9}{2500}$
22	$-\frac{12}{25}$	$\frac{6}{125}$	$-\frac{4}{6125}$	—	$\frac{101}{24}$	$-\frac{12643}{15000}$	$\frac{1873}{75000}$	$-\frac{281}{12005000}$	$-\frac{178}{125}$	$\frac{191}{625}$	$-\frac{1578}{214375}$	$\frac{1}{343000}$	$\frac{1}{625}$
24	$-\frac{64}{125}$	$\frac{8}{125}$	$-\frac{64}{42875}$	$\frac{1}{343000}$	$\frac{205}{48}$	$-\frac{50053}{50000}$	$\frac{6769}{150000}$	$-\frac{522313}{2521050000}$	$-\frac{864}{625}$	$\frac{228}{625}$	$-\frac{21408}{1500625}$	$\frac{2283}{48020000}$	$\frac{4}{5625}$

Table A.3: Coefficients of the central compact multi-moment approximation of the first derivative (Eq.(2.49)) and the ratio between the local truncation errors of the compact scheme and the explicit scheme ($LTE_{COM/EX}$).

A.4 Explicit second-derivative coefficients

Order	d_1	b_1	b_2	b_3	b_4	b_5	b_6
2	-6	3	—	—	—	—	—
4	-12	$\frac{15}{2}$	—	—	—	—	—
6	$-\frac{27}{2}$	$\frac{209}{24}$	$\frac{1}{24}$	—	—	—	—
8	-15	$\frac{1439}{144}$	$\frac{31}{144}$	—	—	—	—
10	$-\frac{47}{3}$	$\frac{38083}{3600}$	$\frac{1183}{3600}$	$\frac{1}{900}$	—	—	—
12	$-\frac{49}{3}$	$\frac{201209}{18000}$	$\frac{4417}{9000}$	$\frac{167}{18000}$	—	—	—
14	$-\frac{401}{24}$	$\frac{40642487}{3528000}$	$\frac{2116727}{3528000}$	$\frac{63431}{3528000}$	$\frac{3}{78400}$	—	—
16	$-\frac{205}{12}$	$\frac{586085413}{49392000}$	$\frac{36056101}{49392000}$	$\frac{2493}{5488000}$	$\frac{96}{2}$	—	—
24	$\frac{-5369}{300}$	$\frac{4483308727523}{355000060800}$	$\frac{384765751523}{355000060800}$	$\frac{19002573199}{177500030400}$	$\frac{1172497199}{177500030400}$	$\frac{64252693}{355000060800}$	$\frac{90641}{71000012160}$
Order	—	c_1	c_2	c_3	c_4	c_5	c_6
2	—	—	—	—	—	—	—
4	—	$-\frac{3}{2}$	—	—	—	—	—
6	—	-2	—	—	—	—	—
8	—	$-\frac{8}{3}$	$-\frac{1}{24}$	—	—	—	—
10	—	-3	$-\frac{3}{40}$	—	—	—	—
12	—	$-\frac{27}{8}$	$-\frac{27}{200}$	$-\frac{1}{600}$	—	—	—
14	—	$-\frac{18}{5}$	$-\frac{9}{5}$	$-\frac{2}{525}$	—	—	—
16	—	$-\frac{6}{25}$	$-\frac{32}{3675}$	$-\frac{3}{3675}$	$-\frac{3}{39200}$	—	—
24	—	$-\frac{216}{49}$	$-\frac{675}{1568}$	$-\frac{50}{1323}$	$-\frac{3}{1568}$	$-\frac{6}{148225}$	$-\frac{1}{5122656}$

Table A.4: Coefficients of the explicit multi-moment approximation of the second derivative in Eq.(3.23).

A.5 Compact second-derivative coefficients

Order	a_1	a_2	a_3	a_4	b_1	b_2	b_3	b_4
6	$-\frac{1}{32}$	—	—	—	$\frac{315}{32}$	—	—	—
8	$-\frac{1}{20}$	—	—	—	$\frac{421}{40}$	$-\frac{1}{40}$	—	—
10	$-\frac{1}{12}$	—	—	—	$\frac{1175}{108}$	$-\frac{5}{27}$	—	—
12	$-\frac{34}{261}$	$\frac{7}{8352}$	—	—	$\frac{6486715}{601344}$	$-\frac{330565}{601344}$	—	—
14	$-\frac{146}{899}$	$\frac{53}{28768}$	—	—	$\frac{109664981}{10356480}$	$-\frac{2892473}{3452160}$	$\frac{13}{40455}$	—
16	$-\frac{549}{2696}$	$\frac{45}{10784}$	—	—	$\frac{990197537}{97056000}$	$-\frac{121201213}{97056000}$	$\frac{201103}{48528000}$	—
18	$-\frac{880389}{3506560}$	$\frac{191241}{21916000}$	$-\frac{1781}{87664000}$	—	$\frac{1269639222971}{131496000000}$	$-\frac{459625307183}{262992000000}$	$\frac{5217629221}{262992000000}$	—
20	$-\frac{136593}{478553}$	$\frac{626643}{47855300}$	$-\frac{673}{11963825}$	—	$\frac{18539293107361}{2009922600000}$	$-\frac{4250265215639}{2009922600000}$	$\frac{78511405243}{2009922600000}$	$-\frac{24747}{5359793600}$
22	$-\frac{141399}{435695}$	$\frac{172017}{8713900}$	$-\frac{347}{2178475}$	—	$\frac{156044132981411}{17933206200000}$	$-\frac{45370805292229}{17933206200000}$	$\frac{1350289295393}{17933206200000}$	$-\frac{103199049}{1195547080000}$
24	$-\frac{19878976}{54322025}$	$\frac{17341906}{597542275}$	$-\frac{1732928}{4182795925}$	$\frac{60791}{133849469600}$	$\frac{119368112575631753}{14691317783296000}$	$-\frac{1085976652915034527}{367282944582400000}$	$\frac{5947042234855067}{44073953349888000}$	$-\frac{637842268943069}{1101848833747200000}$

Order	c_1	c_2	c_3	c_4	d	LTE
6	$-\frac{75}{32}$	—	—	—	-15	$-\frac{h^6}{40320}$
8	$-\frac{51}{20}$	—	—	—	$-\frac{159}{10}$	$\frac{h^8}{1039500}$
10	$-\frac{89}{36}$	$\frac{1}{36}$	—	—	$-\frac{33}{2}$	$-\frac{h^{10}}{37065600}$
12	$-\frac{1570}{783}$	$\frac{10855}{100224}$	—	—	$-\frac{1935}{116}$	$\frac{h^{12}}{1519333200}$
14	$-\frac{4310}{2697}$	$\frac{60437}{345216}$	—	—	$-\frac{179695}{10788}$	$-\frac{79h^{14}}{3364509657600}$
16	$-\frac{21255}{21568}$	$\frac{290679}{1078400}$	$-\frac{869}{1617600}$	—	$-\frac{66655}{4044}$	$\frac{137h^{16}}{202765901184000}$
18	$-\frac{2955771}{14026240}$	$\frac{817086123}{2191600000}$	$-\frac{91999649}{26299200000}$	—	$-\frac{5316451}{328740}$	$-\frac{8249h^{18}}{476175629529600000}$
20	$\frac{3580101}{9571060}$	$\frac{2105322849}{4785530000}$	$-\frac{26470957}{3589147500}$	—	$-\frac{182728721}{11485272}$	$\frac{71h^{20}}{120891995451727500}$
22	$\frac{461331990101}{43569500}$	$\frac{434253771}{871390000}$	$-\frac{481310447}{32023582500}$	$\frac{10863}{1067452750}$	$-\frac{32577979}{2091336}$	$-\frac{60791h^{22}}{3592528557882264000000}$
24	$\frac{461331990101}{43569500}$	$\frac{434253771}{871390000}$	$-\frac{481310447}{32023582500}$	$\frac{10863}{1067452750}$	$-\frac{17439603755}{1147281168}$	$\frac{11162299h^{24}}{252202976859623572282500000}$

Table A.5: Coefficients and leading truncation error (LTE) for the compact multi-moment approximation of the second derivative (Eq.(3.25)).

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